

claude-windows / d6979332-5869-49a6-bf7e-bd9f4da1954e

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e.jsonl`  
- SHA-256: `560e6144012afdf46f7a84f53be318e5f36e530e5215cd18fd9ac1b80292e653`  
- Source modified: `2026-06-13T15:15:22+00:00`  
- Imported at: `2026-07-05T16:48:19+00:00`  
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`  
- session_id: `d6979332-5869-49a6-bf7e-bd9f4da1954e`
```

Transcript

1. user (2026-06-11T03:46:10.567Z)

Hey Claude. Follow my init instructions and tell me how do you rate this repository based on your engineering excellence and quality metrics.

2. assistant (2026-06-11T03:46:36.940Z)

I'll follow the init convention ? read the session handoff and current-state first, then the code map ? before doing the repo quality assessment.

3. user (2026-06-11T03:46:37.962Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-470 of 1308 total (59047 tokens, cap 25000). Call Read with offset=471 limit=470 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      ---  
2      name: current-state  
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not  
only on request). What just landed, what's next, open decisions."  
4      metadata:  
5          node_type: memory  
6          type: project  
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301  
8      ---  
9  
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])  
11  
12     **Checkpoint:** 2026-06-11 (PR #88 ? **#89 MERGED** ? KhelSutra UI-alpha docs bundle on  
master `34cf26c`). #88  
13     conflicted (branch cut from `feat/async-job-model` pre-squash of #87) ? per  
`scratch/PR88_FIX.md`, recreated as  
14     `docs/khelsutra-plan-2` off post-#87 master keeping 6 paths (UI_PROPOSAL, HOSTING_PLAN,  
UI_ALPHA_EXECUTION_PLAN,  
15     mockups/, DECISIONS **ADR-009**, NEXT_STEPS sync); #88 closed + branch deleted. CI  
green-gated (statusCheckRollup),  
16     squash-merged.  
17     - **Dropped from #89, preserved for PR-2 as UNCOMMITTED working-tree edits on master:**  
(1)  
18     `docs/DEFERRED_HARDENING_PLAN.md` "#16 DONE ? MERGED (PR #87)" status hunk (also saved  
as  
19     `scratch/PR2_deferred_hardenening_status.patch`); (2) `docs/UI_ALPHA_EXECUTION_PLAN.md`  
working-agreement rewrite  
20     (Cowork designs + leaves `scratch/PRn_HANDOFF.md`; **Claude Code executes all git/PR  
mechanics** ? ratified  
21     2026-06-11). Don't `checkout --` these away; they commit with PR-2.
```

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22     - ?? Next UI-alpha slice = **PR-2 (upload/download)** per
`docs/UI_ALPHA_EXECUTION_PLAN.md`.
23     - ? **SLATE CLEANED (2026-06-11):** indexer = NO open PRs, master==origin (`34cf26c`),
**all 23 stale local branches
24     pruned** (every one mapped to a MERGED PR #62?#87 before `-D`); only `master` remains.
Collector = NO open PRs, on
25     master. Intentional carryovers (NOT mess): the 2 uncommitted PR-2 doc edits (above),
untracked `scratch/` handoffs,
26     untracked `artifacts/rally_tal_gen0/0.1.0/` (Gen-0 bundle, ADR-008). ?? Collector has
a stray uncommitted
27     `.gitignore` edit (adds `artifacts/` ignore w/ ADR-008 comment) ? mega-session
leftover, owner to commit (micro-PR)
28     or fold into the next collector PR.
29
30     **Checkpoint:** 2026-06-10 (GEMINI BATTERY COMPLETE ? full coverage; owner's billing was
fine, postpay active; ALL
31     failures were burst/capacity artifacts). Merged this stretch: **#80** (chunker re-
encode/downscale, from the chip
32     session ? reviewed+merged), **#81** (rescued the chip branch's orphaned run_signals test
via cherry-pick), **#83**
33     (provider: 6 generate attempts exponential backoff+jitter ? saved chunks live), **#84**
(tracked
34     `eval_baselines/gemini_wrapper_2026-06-10.json`). Suite 336 green.
35     - **FINAL QUALITY TABLE (IoU .5 vs human golden labels):**
36     - **Mahadevpura** (clean-ish, 31 rallies): heuristic 0.657 · Gen-0 0.744 · **two-stage
WASB+Gemini refine 0.83**
37     (P .81 R .84, 32 preds) · **full-video wrapper 0.93** (P 1.0 R .87).
38     - **Kushagra** (multi-court, 32 rallies, FULL 7/7 coverage + audio): heuristic 0.157 ·
Gen-0 0.561 ·
39     **wrapper 0.66** (P .69 R .62, ±1.2s; FPs = background-game pickups + one 53s dead-
time merge ? exactly the
40     contrast-metric confound).
41     - **STRATEGY (refined):** clean footage commoditized (0.93 cheap); **multi-court drops
the wrapper to 0.66 = the
42     owned model's ship target** (Gen-0 is 0.10 behind at n=6 ? within corpus-growth
reach). Two-stage refine quality
43     (0.83) < full wrapper on short videos; its value = COST on long academy tapes (only
candidate clips uploaded).
44     Wrapper reliability is genuinely flaky (503 waves all session) ? provider-fallback
architecture is load-bearing.
45     - Artifacts: `output/mahadevpura_gemini_refined.csv`,
`output/MahadevpuraSingles_highlights.mp4` (demo reel),
46     Kushagra preds under the ORIGINAL corpus MD5 now. Total Gemini spend today: ~$1-2.
47     - PENDING/owner: rotate the chat-pasted key; ship-gate F1@tIoU decision (multi-court
target now has a number: 0.66);
48     C13 interviews + camera pilot; repo rename.
49
50     **Checkpoint:** 2026-06-10 (GEMINI BASELINE LANDED ? owner's "gradual ramp-up" theory
CORRECT). New key in gitignored
51     `.env` (rotate later ? pasted in chat). The "credits depleted" 429 was a BURST-THROTTLE
artifact: text ping + serial
52     upload worked ? **PR #79 merged**: gemini segmenter now honors `indexing.ai_max_workers`
(was hardcoded 5; config ships
53     1=serial ? at 5 workers the same video 429'd, at 1 it worked).
54     - ? **WRAPPER BASELINE (gemini-flash-latest, ~$0.2-0.5 total): Mahadevpura F1 0.93** (P
1.00 R 0.87, mIoU 0.80, ±0.7s;
55     3 of 4 misses = short rallies Gemini FOUND but our min_segment_duration=3.0 rejected)
vs Gen-0 learned 0.744 vs
56     heuristic 0.657. **Clean footage = commoditized.**
57     - ?? **MOAT TEST (Kushagra multi-court, windowed n=12 GT): wrapper F1 0.56** (P .54 R
.58 mIoU .69) ? Gen-0 learned
58     0.561, both >> heuristic 0.157. **The multi-court confound hurts Gemini too ? the
wedge survives on the target
59     footage class.** Caveats: only chunks 2+4 answered (5/7 lost to GENUINE gemini-flash-
latest capacity 503s ? retry

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60      off-peak for full coverage); visual-only (audio stripped in downscale); 1280p re-
encode (4K chunks were 1.3GB >
61      500MB upload cap ? the audit's silent-oversize-skip fired live; fix = chip
task_44e72c41 ? see PR #80 checkpoint
62      below). Kushagra preds keyed under downscaled-file MD5 (`output/kushagra_1280.mp4`).
63      - ? **DEMO ARTIFACT: `output/MahadevpuraSingles_highlights.mp4`** ? real 27-rally reel
compiled by VideoCompiler
64      (the once-broken module) from Gemini segments. The academy-pitch object exists.
65      - **STRATEGY READ:** product path = Gemini for clean footage TODAY (cheap, F1 0.93) +
owned model's job = multi-court
66      (0.56 is the bar to beat there, not 0.93); 503 flakiness = real reliability cost of
the wrapper (provider-fallback
67      story strengthens). PENDING: gemini_refine two-stage (503-gated, run off-peak), full
Kushagra re-run, PR #80 merge.
68
69      **Checkpoint:** 2026-06-10 (4K gemini oversize fix ? **MERGED: #80 + #81 + #82; master
`f01387e`, CI rollup SUCCESS**).
70      Root cause of the live "0 segments / success" on KushagraYashAviFirstVideo.MP4 (4K):
gemini segmenter stream-copies 90s
71      chunks ? all 7 at 832?1299MB > provider `MAX_UPLOAD_MB=500` ? every chunk silently
skipped. Fix (3 layers): (1) chunks
72      RE-ENCODED + downscaled to new declared knob `indexing.ai_chunk_scale_width` (default
1280 = gemini_refine's
73      `--clip-scale-width`; 0 = legacy stream-copy; also fixes keyframe drift); (2) skip is
LOUD ? provider
74      `metrics["oversize_skipped"]` + segmenter aggregation (chunked + single-file) ?
`logger.error` + NEW thread-safe
75      `backend/reporting/run_signals` channel (incr/pop by video_id; popped unconditionally in
`emit_indexer_report`);
76      (3) report flags it ? `ai_handoff.details.oversize_clips_skipped` + spec.yaml ERROR rule
`oversize_clip_skipped`.
77      Suite 325?336 green** locally (CI runs fewer: reporting tests importorskip obsreport,
absent there BY DESIGN ?
78      `tests/test_run_signals.py` deliberately has NO importorskip = the CI-visible half of
the channel). **Merge dance
79      (process note):** owner merged #80 mid-session (auto-deleting the remote branch) WHILE
the follow-up test commit was
80      being pushed ? push re-created the branch orphaned; owner rescued it as #81, 39s before
my cherry-picked #82 ?
81      **#82's squash `f01387e` = EMPTY commit** (harmless dup in history). Lesson: re-check
`gh pr view` state before pushing
82      more commits to a PR branch ? owner + sibling sessions work concurrently (this
checkpoint file too: another session
83      prepended the GEMINI-BASELINE checkpoint mid-edit). ? **OWNER/SIBLING WIP IN TREE
(uncommitted, do NOT clobber):**
84      `backend/providers/gemini.py` ? generate retries 3?6 w/ exponential backoff + jitter
(~3?96s) for the 503 capacity
85      waves. ?? commit/PR the 503-backoff WIP . machine-side 4K re-run (chunks should be ~tens
of MB) . task-17 battery
86      still pending Gemini billing top-up.
87
88      **Checkpoint:** 2026-06-10 (M0 GREENLIT SEQUENCE EXECUTED ? session wrap). Owner
ratified ADR-008 ("split driven by
89      productionization, not isolation") + greenlit M0.4+M0.1. **All 5 steps shipped, every PR
merged on verified-green CI:**
90      - **#73 ADR-008** (DECISIONS.md): training in-repo behind an executable wall; artifact-
bundle boundary; ship-gate
91      product-side; pre-committed split trigger ? `sports-temporal-models` at Gen-2/second-
consumer.
92      - **#74 canonical GT loader** `backend/eval/gt_loader.py::load_gt_intervals` ? closes
the two-loader fork (accepts BOTH
93      `start,end` and `start_time,end_time`); legacy loaders delegate; round-trip contract
test; LOVO 0.626 re-verified.
94      - **#75+#76 featurizer promotion + import wall** : `backend/features/rally_seq.py`
(FEATURE_SCHEMA v1, fps semantics

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95     explicit) imported by train AND serve; `training/` skeleton (5 wall rules,
requirements-train.txt); smoke tests:
96     runtime no-training-in-backend.main, static sweep, featurizer purity (torch/cv2
blocked; importorskip numpy on the
97     minimal CI lane ? #76 hotfixed a red lane after #75 merged early; merges now gated on
verified SUCCESS rollout).
98     - **collector #19 ? n=6 OWNED CORPUS IN THE SoR**: all 6 golden matches via `import-
local --normalize` (262 rallies,
99     0 dropped, Proprietary, explicit fps); commercial export now 6 real matches + demo
entry.
100    - **#77 Gen-0 harness** `python -m training.gen0.harness --manifest ? --floor
eval_baselines/heuristic_lovo_n6.json
101    --version 0.1.0`: train?honest LOVO?ship-gate?artifact bundle (gitignored weights.json
+ manifest.json w/ schema pin,
102    calibration, lineage gt_hashes, attestations incl. C3, gate verdict). **First real
bundle built: LOVO 0.626, floor
103    PASSED (0.480), paired sign test 5/6 wins p=0.109 = honestly NOT significant,
ship=False** (blockers: target
104    undefined, C3, significance). Committed floor baseline
`eval_baselines/heuristic_lovo_n6.json`. Suite **325 green**.
105    - ? **Gemini baseline STILL BLOCKED after owner provided a NEW key (2026-06-10 later):**
key written to gitignored
106    `.env`, validated via `models.list` (55 models incl. `gemini-flash-latest`; **config
migrated off deprecating
107    gemini-2.5-flash ? `gemini-flash-latest` alias, PR #78 merged**; gemini_refine already
defaulted to it). But inference
108    = 429 RESOURCE_EXHAUSTED "prepayment credits depleted" ? **the AI-Studio PROJECT
behind the key has no credits** (key
109    auth ? billing). Owner: add credits at https://ai.studio/projects (+ consider rotating
the chat-pasted key). Planned
110    battery on unblock (in task 17): wrapper Mahadevpura ? score; wrapper Kushagra (multi-
court MOAT test); two-stage
111    gemini_refine; compile a real highlight reel (academy-demo artifact). Pipeline handled
both failed runs gracefully
112    (validators pass, reports emitted, artifacts cleaned); shared-genai-client thread-
safety finding observed live.
113    - **C13 field plan (owner executing):** recruit interviewer; camera idea RATIFIED w/ 3
constraints (consent-at-capture/
114    adults-only for training; demo via Gemini-or-human-polish, NOT gated on owned model;
interviews?cameras sequencing).
115    - ?? NEXT: owner tops up Gemini billing (wrapper baseline) · owner sets the ship-gate
F1@tIoU target · grow corpus
116    (camera pilot videos ? import-local ? re-run harness) · repo rename · M0.1 event-index
contract in collector.
117
118    **Checkpoint:** 2026-06-10 (Roora PB+TT pilot review ? REPORT DELIVERED to owner; owner
emails Roora himself) ·
119    ? Owner cross-checked the rally tables in VLC ? **CONFIRMED: every boundary off ?0.5s
(up to ~2s)** vs the  $\pm 0.3s$  bar
120    (Brief §4). **Root cause (my forensics): all 24 boundaries sit EXACTLY on the 30-frame
sampling grid** (~0.5s at
121    59.94fps) ? first/last box never on the true contact/dead frame ? structurally cannot
meet  $\pm 0.3s$ . Also decoded their
122    bogus `start/end time` attrs = **frame+900 (decimal minutes @ assumed 15fps, ~4× off
wall-clock)**. Report findings:
123    boundary accuracy (CRITICAL), skip-rallies-in-progress-at-file-start/end convention (PB
rally#1 starts at frame 0 ?
124    shouldn't have been labeled), dead-time+warmup reminders for Phase-B, schema cut to
exactly
125    `rally_number,shot_count,ending_reason,last_shot_outcome,note` (drop their time attrs +
scores), field quality
126    (PB outcome='0' off-vocab; TT 5/6 'other' + missing required notes; TT#5 intra-rally
other|winner). Per owner:
127    NO redelivery demand in the report (he words the email himself; thread = Jamshad@roora
on avidullu@live.com Outlook).

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128 ****Artifacts:**** PDF `Annotation Setup\6 June 2026\Fedback\Roora Pilot Review -
Pickleball and Table Tennis
129 (2026-06-10).pdf` + md record at `Annotation Setup\Roora Pilot Review ? Pickleball &
Table Tennis (2026-06-10).md`
130 (generator: `%TEMP%\roora\make\_review\_pdf.py`). ? Local guideline docs are ****DRAFT v2****
? owner mentioned "v7";
131 no v7 found on disk (possibly in the email thread; report cites v2). Spawned chip
`task\_c55f0be6`: collector
132 `rally\_cvat.py` ATTR_MAP drops `shot\_count` (present in Roora XML, empty in our CSVs) ?
OUR bug, kept out of the
133 vendor report. ?? owner sends email; Roora's badminton long-video + redelivery decisions
pending owner.
134
135 ****Checkpoint:**** 2026-06-10 (REPO-TOPOLGY decision panel + C13 plan ? AWAITING OWNER
RATIFICATION). Owner leaning
136 M0.4+M0.1 greenlight; asked: model in-repo vs separate repo consumed as a segmenter
(owner prefers isolation). Ran a
137 4-agent judge panel (`wf\_471a9cdd-d1d`; full output `?\tasks\wnksv545g.output`).
****Verdict (my rec):** "artifact boundary
138 NOW, repo split at Gen-2" ? M0.4 lands as a top-level `training/` tree in the indexer
w/ executable isolation
139 (import-smoke asserts backend.main loads no training module; own requirements-train.txt
+ 3rd CI lane), featurizer
140 promoted to `backend/features/rally\_seq.py` (single source, train==serve), artifact
bundle = gitignored weights
141 (private Releases) + committed manifest (feature-schema ver, normalization, calibration,
corpus ref+gt_hash,
142 consent/license attestation incl. C3), ship-gate stays product-side (regression.py
extension); ****pre-committed split**
143 trigger****** to a named repo (`sports-temporal-models`) when Gen-2/cloud-training or a 2nd
artifact consumer arrives.
144 Decisive evidence: Gen-0's input = the indexer's OWN featurization (mid-pipeline seam ?
WASB/Gemini video-boundary);
145 indexer not pip-installable ? split today = copy-paste = the observed drift class;
obsreport "precedent" never actually
146 exercised (pin is a commented line, repo unpublished, CI skips). ****FIRST** contract to
freeze: golden-label/corpus
147 contract****** ? indexer carries TWO incompatible GT loaders (annotations.py `start,end` vs
calibrate_wasb `start\_time`,
148 `end\_time` ? fed to 7 modules) vs collector export `start,end`; consolidate + round-trip
test BEFORE any training epoch.
149 Panel extras: n=6 corpus's durable home = loose `Annotation Setup\` folder (neither
repo!) ? M0.1 first deliverable =
150 hash-addressed in collector; training fixtures synthetic-only until consent ledger
exists; run the Gemini-wrapper
151 baseline before/with M0.4. ****C13 field plan:**** owner recruiting an interviewer; my rec =
bake in the camera idea
152 (paid recordings = consented owned multi-court corpus + demo), demo served via Gemini
path or human-polish (owned model
153 honest 0.626 ? >0.8 yet; the comparison doubles as the wrapper baseline). ?? awaiting
owner ratification of topology +
154 M0.4/M0.1 scope; then: ADR + GT-loader consolidation + featurizer promotion + harness
scaffold.
155
156 ****Checkpoint:**** 2026-06-10 (CORPUS RECLASS EXECUTED + NEXT_STEPS refreshed ? session
wrap). Owner approved + I ran the
157 corpus correction: `reclassify-licenses --apply` (23 ShuttleSet broadcast rows ?
annotation_license=MIT, video=Unknown,
158 is_commercial=False; ****zero annotations deleted****, 1,917 rallies intact) + `curate
export` (commercial manifest now
159 owned-data-only: 1 video/2 segments; stale ShuttleSet CSVs removed from the gitignored
eval_export; eval-only re-export
160 available via `--include-staging --include-noncommercial`). ****Merged: collector #17****
(DB + .gitignore backup rule);
161 pre-change backup at `data/sports\_metadata.backup-2026-06-10.db` (gitignored). Quality

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signals UNAFFECTED (n=6 corpus
162     never flows through the collector DB; learned 0.626 re-verified earlier). **Indexer #72
merged:** NEXT_STEPS.md
163     refreshed (sweep banner, 0.626 vs 0.480, M0.3 partially-done with corrected
distill_local pointer, NEW prioritized PMF-
164     validation item, durable artifact paths). **Owner is now reading NEXT_STEPS to chalk out
next steps** ? open owner
165     decisions stand: greenlight M0.x (rec M0.4+M0.1), confirm collector=SoR, ship-gate
F1@tIoU, repo rename, C13 interviews
166     / Gemini-wrapper baseline. Both repos clean on master, 0 open PRs.
167
168     **Checkpoint:** 2026-06-10 (HIGH-RISK REMEDIATION COMPLETE ? owner approved "everything
A7K, open+merge PRs after CI
169     green, re-run quality evals, GPU available"). **ALL 11 PRs MERGED + CI green; both repos
clean on master.**
170     - **Indexer (badminton-highlight-indexer):** #63 compiler-syntax fix+CI (prior), **#64**
detector keying by video_id,
171     **#65** re-ingest destroy-after-confirm (add_video UPSERT + watermark/prune), **#66**
nested-config wasb typed,
172     **#67** eval gate + metric footguns (iou=0, midrank AUC, fps-normalized speed,
baseline?`eval_baselines/` exit-3),
173     **#68** API path validation, **#69** reliability (timeouts 1800s, SQLite
WAL+busy_timeout, chunk-step guard, CORS),
174     **#70** test coverage (TestClient endpoints + compiler + WasbRunner), **#71** docs
truth-sync. Suite 256?308 green**.
175     - **Collector (sports-data-collector):** **#14** licensing gate (video vs annotation
license; `reclassify-licenses`
176     found 23 conflated ShuttleSet rows) + **added collector CI**, **#15** import-local
golden-label guard (+`--force`,
177     `--normalize`, duplicate-rally guard), **#16** validate-delivery fails on
empty/corrupt/zip CVAT. Suite 124?136 green**.
178     - ? **QUALITY RE-VALIDATION (owner asked): metric fixes STRENGTHENED the conclusion.**
After midrank-AUC +
179     fps-normalized speed + occlusion-gap guard, **learned LOVO 0.583 ? 0.626** (per-video
AUC unchanged 0.83?0.92 =
180     never tie-inflated; fps-fix mainly helped the 30fps short fold 0.30?0.56). Heuristic
LOVO **0.480** unchanged (scored
181     at IoU 0.5; my iou>0 guard is a no-op there). **Learned beats heuristic by +0.146**
(was +0.10).
182     - **Eval artifacts RESCUED from `%TEMP%` ? durable `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\`**
183     (+ `roora\`); manifest `output/human_lovo_manifest.json` repointed there + reverified
0.626.
184     - **Deferred WITH INTENT (in BACKLOG): repo-rename; Gen-2 cost benchmark; ship-gate
F1@tIoU target; packaging
185     (pyproject); app-factory/lifespan + lazy torch-cv2 imports (with async slice #16); WSL
temp/frame/cache + Gemini
186     remote-file GC (retention policy); shlex-quote WSL config values (breaks `~` expansion
? non-trivial); deeper
187     validator tests + 3-hybrid contract test (partial coverage shipped).**
188     - ? **Process note:** during PR-F I briefly `git checkout --`'d the owner's uncommitted
`data/sports_metadata.db`
189     change (200704b), then RECOVERED it from dangling stash `d3c2e77` and restored it to
the working tree (no loss).
190     Going forward stash-and-pop that DB for collector PRs; never `checkout --` it. The
corpus-DB reclassify (`curate
191     reclassify-licenses --apply` + `curate export`) is left for the OWNER to run
deliberately (binary SoR data).
192     - Plan doc (reference): `scratch/HIGH_RISK_REMEDIATION_PLAN.md` (untracked). Audit JSON:
`?\tasks\wf38bovts.output`
193     (batch 1) + `?\tasks\wa4d2980l.output` (full 8 auditors).
194
195     **Checkpoint:** 2026-06-10 (compiler SyntaxError FIX ? **PR #63 MERGED**, squash `master
13d4573`; owner approved merge-on-green) .
196     ? **master UNBROKEN again ? fixed the ? SyntaxError from 7fbbeb0** (= the audit's fix

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chip `task_2b7a19d4` work):
197     restored the f-string quotes on the 7 mangled lines of
`backend/pipeline/stitcher/compiler.py`
198     (100/104/156/160/181/191/213; message texts verified against the pre-conversion file at
`7fbbeb0`; line 191 =
199     plain string, no placeholders) + fixed line 168 `f"\nSUMMARY]" ? `f"\n[SUMMARY]`. **NEW
regression guard
200     `tests/test_import_smoke.py`: (a) py_compile sweep over every file in backend/
(imports nothing ? runs on
201     cv2-free machines), (b) subprocess `import backend.main` with missing cv2/torch/numpy
stubbed. Negative check:
202     the broken compiler.py fails the sweep (`SyntaxError: invalid decimal literal`, line
100).
203     ? **CI NOW EXISTS ? the audit's no-CI finding is CLOSED** (`github/workflows/ci.yml`,
runs on every PR + master
204     push): job 1 **import-smoke** = smoke tests on a GPU-stack-free runner (minimal dep set
verified empirically in a
205     fresh venv: pytest/fastapi/uvicorn/pydantic/python-dotenv/**google-genai** ? needed at
import time by
206     backend.providers); job 2 **full-suite** = requirements.txt with **CPU-only torch
wheels**
207     (`--index-url download.pytorch.org/whl/cpu`) + libgl for opencv ? `python
run_all_tests.py` (obsreport absent ?
208     reporting tests skip by design). **Both jobs GREEN on the first run** (27s / 1m29s).
Verified post-merge:
209     `import backend.main` OK on master, suite 256 green (254+2).
210     ?? Audit's **coverage-inversion** finding (wasb_hybrid/validators untested) remains open
? CI only guards the
211     import/syntax class + whatever the suite covers.
212
213     **Checkpoint:** 2026-06-10 (DEEP AUDIT session ? ultracode 18-agent audit
`wf_2a64bc88-2a8` over indexer+collector:
214     engineering/testing/reliability/docs-consistency/PMF; 3 of 8 auditors
completed+verified, 5 rate-limited).
215     - ? **CRITICAL ? master is BROKEN since 7fbbeb0 (2026-06-07 print?logger sweep):**
216     `backend/pipeline/stitcher/compiler.py` has **7 unquoted f-string lines
(100/104/156/160/181/191/213)** ? SyntaxError ?
217     `import backend.main` fails ? FastAPI server + `/api/compile` dead. Triple-verified +
reproduced first-hand
218     (`py_compile` exit 1). The **254-green suite never noticed** (zero tests import
backend.main / compiler / wasb_hybrid /
219     motion / any concrete validator; **no CI exists at all**). Spawned fix chip
`task_2b7a19d4` (fix + import/compileall
220     smoke test). Coverage is INVERTED vs production risk (tracknet twin 8 tests vs live
wasb_hybrid 0; TrackNetRunner 21
221     vs WasbRunner 1). Suite itself healthy where it exists (254 green / 19.9s / faithful
mocks / low flake risk); "fully
222     mocked runs anywhere" is false on cv2-free machines (bare `import cv2` at collection
time, only obsreport importorskips).
223     - **Doc drift (verified, one-directional ? newest decisions not back-propagated):** (1)
licensing "Apache-2.0 only" still
224     in CLAUDE.md:33 / PLATFORM:269,301 / COMMERCIALIZATION C14 vs ratified
**MIT/Apache/BSD** (a literal CLAUDE.md reader
225     rejects the plan's own MIT ActionFormer/ASFormer heads); (2) **Gemma-4 still
"approved" in those same docs vs
226     AMBER/bench-only in the plan; (3) **M0.3 purge pointer WRONG** ?
`training.label_candidates` is a pure GT-agnostic fn;
227     the live Gemini-silver training violation = `backend/eval/distill_local.py` (+
calibrate_local thresholds) + gitignored
228     `output/local_rally_model.json`; (4) **C3 escape hatch broken** ? "train-own weights
(Phase-4 roadmap)" cites a phase
229     that contains no such plan and ADR-002 says corpus labels are temporal-only (CAN'T
train a detector) ? own-weights is
230     an unscoped new workstream (coordinate labels or detector swap); (5)
OWNED_MODEL_IMPLEMENTATION_PLAN + NEXT_STEPS are

```

231 linked from NO entry point (CLAUDE.md "read first" / docs/README index) and NEXT_STEPS
 is branch-only; (6) README
 232 "yolo_hybrid by default" vs config.json `gemini` vs CODE_MAP "wasb = LIVE default";
 (7) ROADMAP still says audio
 233 "gated on golden labels" (trigger now satisfied!) vs plan "parked, not in reserve";
 (8) Roora multisport CSVs live ONLY
 234 in `%TEMP%` (OS-clearable) + n=6 manifest gitignored ? headline
 TT-0.909/pickleball-0.308 evidence is one cleanup from
 235 loss; (9) TT-on-WASB idea silently walks into the C3 ship-gate ("multiplies per
 sport"); (10) ship-gate F1@tIoU still
 236 undefined while 0.44/0.54/0.480 floors circulate; repo-rename "ASAP" tracked in no
 tracker (BACKLOG lacks it).
 237 - **PMF (web-grounded, sources in audit output):** pricing/cost benchmarks VERIFIED
 (SwingVision \$179.99/yr ~20k paying
 238 subs rev \$2.75M +12% YoY; PB Vision \$99/yr + **\$8/hr-of-video partner API**; Gemini
 COGS ~\$0.30?0.60/hr ? fat margin);
 239 but **"India badminton whitespace" FALSIFIED ? Bangalore is the most contested
 badminton-AI city** : Machaxi (\$1.5M
 240 Rainmatter+Prakash Padukone), VISIST.AI (70+ Bengaluru academies, Padukone School),
 Game Theory (Gopichand-backed smart
 241 courts w/ auto-highlights), Stupa (\$3.3M, BWF-approved, TT?badminton) ? NONE in our
 2026-06-07 competitive table.
 242 Surviving wedge (narrow, real): **software-only rally-precise indexing of EXISTING
 messy multi-court BYO footage**
 243 (market solves it only with per-court hardware) sold per-hour as engine/API, possibly
 THROUGH those players
 244 (PB Vision×PodPlay template). Fastest falsification: C13 interviews (still undone) +
 **Gemini-wrapper baseline vs the
 245 n=6 LOVO corpus** (if wrapper ? learned 0.583, owned-model is premature) + 1 paid
 academy pilot.
 246 - **AUDIT NOW COMPLETE (8/8 auditors, resumed `wa4d2980l` after Max-20x refresh; 40
 agents).** New high/critical findings
 247 beyond the first batch:
 248 - ? **CRITICAL (collector) ? LICENSING GUARDRAIL VIOLATION:** `src/cli/ingest.py:12`
 defaults `--license MIT`; the
 249 ShuttleSet **annotation** license is stamped onto **YouTube broadcast video** (e.g.
 shuttleset_21 = a YONEX Thailand
 250 Open broadcast, `is\_commercial=true, license\_type=MIT`). **24 videos / 1915 segments
 currently exported as
 251 "commercially-safe" on unsound grounds** ? directly violates the "training data must
 be commercial-clean" guardrail.
 252 - **HIGH (indexer, all verified):** (a) **nested config sections silently drop unknown
 keys** ? only `AppConfig` has
 253 `extra='allow'`, so the ENTIRE `indexing.wasb` block
 (weights/conda_env/velocity_threshold) + `yolo\_prefetch\_workers`
 254 + `debug\_duration` are unconfigurable via config.json; meanwhile
 `yolo\_parallel\_chunks/yolo\_chunk\_workers` are read
 255 nowhere. (b) **detector artifacts keyed by file BASENAME not video_id** ? two
 `match.mp4` files reuse stale frames ?
 256 a trajectory + candidate_segments computed from the WRONG video (silent training-
 data poisoning). (c) **re-ingest is
 257 destroy-before-confirm** ? `add\_video('processed'/'failed')` INSERT-OR-REPLACE
 cascade-deletes prior good segments
 258 BEFORE the new run; a failed/empty re-run leaves zero. (d) **stitching padding
 config not wired** ? API reels always
 259 end_padding=1.0 despite config 0.5. (e) **`/api/process` arbitrary server-file read
 + `/api/compile` path-traversal/
 260 absolute write** via output_filename. (f) **run_regression "CI gate" not in CI +
 silently passes** on missing baseline;
 261 baseline omits iou/detector/GT-hash. (g) **metrics bug**: `match\_intervals` at
 iou_threshold=0.0 matches
 262 non-overlapping intervals (P=R=F1=1.0 for garbage); **rally_seq_proto AUC has no tie
 handling** (order-dependent;
 263 all-zero dead-time rows guarantee ties) + speed features fps-inconsistent within one
 vector.


```

264     - **HIGH (collector):** silent DELETE-then-INSERT replace of golden labels (no
warn/backup, filename-derived video_id);
265     validate-delivery **PASSES on a zero-rally CVAT export**; not yet SoR (no
provenance/maturity/labeled_window cols,
266     import-local hardcodes CURATED); `int(float())` rally_number crashes mid-import
leaving a committed video row w/
267     rolled-back annotations (normalize_annotations guard is opt-in, NOT wired into
import-local).
268     - **MEDIUM batch:** WSL `timeout_sec=0` = hang-forever (both runners + provider
PROCESSING poll); `uvicorn backend.main:app`
269     leaves globals None (init only in main()); SQLite no WAL/no busy_timeout + swallow-
on-lock; chunk_overlap?duration =
270     infinite loop; WSL temp/frame/cache + Gemini remote-file leaks; CORS
`**`+credentials; sport-profile relevance config
271     dead under typed config (bites first non-badminton sport ? the current branch!).
272     - **VERDICT NUANCE:** verifier DOWNGRADED 2 "high"?not-real (Result-contract gap + DB-
swallow are DOCUMENTED CODE_MAP
273     gotchas/deliberate-incremental, medium at most; DB "no busy_timeout" claim factually
wrong ? sqlite3 default
274     timeout=5s). Eng-pipeline/eval/collector strengths confirmed: registry seams,
DetectorRunner ABC, WASB crash-cache,
275     honest fit-on-self labeling, leakage-free LOVO, frame-true CVAT conversion all
genuinely good.
276     - ? **SPOOF-CHECK PASSED ? quality iterations are REAL (re-ran the evals first-hand on
this box, pure numpy/scipy):**
277     all 6 videos + 6 human label CSVs + 6 trajectories exist; traj row-counts match docs
exactly. **`rally_seq_proto` LOVO
278     reproduced: mean 0.583** (AUC 0.877/0.851/0.916/0.832/0.730/0.902 vs doc
0.879/0.853/0.915/0.833/0.726/0.902).
279     **`calibrate_local` heuristic LOVO reproduced: 0.480 ± 0.242** (per-fold Adarsh
0.751/Kushagra 0.157/doubles 0.737/
280     gBaaddy 0.309/testlarge 0.267/mahadevpura 0.657 ? exact). **Contrast metric recomputed
independently from raw CSVs:**
281     3.7/1.4/3.7/1.9/1.4/2.2× vs doc 3.9/1.4/3.6/1.9/1.4/2.2. Opus 4.8 did NOT spoof. ? one
nit: gBaaddy CSV has 48 rallies
282     (doc says 47). Caveat: trajectories were WASB-GPU-generated (can't regenerate here)
but sizes/frame-counts/visibility all sane.
283     - **HIGH-RISK REMEDIATION PLAN written for owner approval:**
`scratch/HIGH_RISK_REMEDIATION_PLAN.md` (12 PRs, sequenced;
284     PR-1 compiler fix already DONE via #63). Awaiting owner scope + push/PR decision
before execution. Full audit JSON:
285     `%LOCALAPPDATA%\Temp\claude\...\tasks\wa4d2980l.output` (8 auditors) +
`wf38bovts.output` (first batch).
286     Full audit JSON: `%LOCALAPPDATA%\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\tasks\wf38bovts.output`.
287
288     **Checkpoint:** 2026-06-09 (Roora MULTISPORT recon ? for FRESH next session) . ??
**Pickleball + Tabletennis golden labels
289     arrived; INGESTION = SOLVED, but the "merged model" experiment is BLOCKED on per-sport
trajectory detectors (validates the
290     roadmap blocker).** Owner left a recon-only ask (deliberately did NOT drain context /
spin heavy jobs ? owner wants me fresh).
291     - **Deliverables:** `Annotation Setup/6 June 2026/{pickleball,tabletennis}/*.zip` (3
CVAT export FORMATS each: COCO
292     `instances_default.json`, **CVAT-XML `annotations.xml`** ? use this, Pascal-VOC).
Videos: `?/6 June 2026/Annotation Videos/
293     Video 2 - Pickleball.mp4` (4K, 59.94fps, 97s, 5737fr) + `Video 3 - Tabletennis.mp4`
(4K, 59.94fps, 58s, 3490fr). Short clips.
294     - **What Roora annotated:** label class "rally" as **per-frame BOUNDING BOXES on sampled
frames (~every 30fr)** ? box geometry
295     is just a CARRIER; the boxes carry the FULL rally schema as CVAT **attributes**
(rally_number, start/end time, shot_count,
296     score_before/after(A-B), ending_reason, last_shot_outcome [+note for TT]). 456/513
attr instances. Boxed frames cluster into
297     rallies (runs of ~30-spaced frames = a rally; big gaps = dead time).

```

```

298 - **INGESTION = YES, cleanly:** the collector's `src/parsers/cvat/rally_cvat.py`
(`curate convert-cvat`) is PURPOSE-BUILT for
299 exactly this ? ignores box geometry, **recomputes rally interval from the frame
range** (`sec=frame/fps` at true fps, ignoring
300 vendor start/end times which are unreliable), groups boxes by rally_number, maps
attrs?canonical. Minor: verify the attr-name
301 mapping handles spaces/"(A-B)" (`ATTR_MAP`). Collector already has pickleball +
tabletennis schemas.
302 - ? **"MERGED LEARNT MODEL" across sports is BLOCKED:** `rally_seq_proto` features come
from a WASB **shuttle** trajectory =
303 BADMINTON-ONLY; pickleball (wiffle ball) + TT (tiny fast ball on a table) are out-of-
distribution ? no clean trajectory yet.
304 This EMPIRICALLY validates the roadmap's binding multisport blocker (clean per-sport
MIT/Apache ball detectors not identified).
305 - **PROGRESS (owner asked to complete 1/2/3 before badminton):**
306 - ? ** (1) convert-cvat DONE ? INGESTION PROVEN.** `python -m src.cli.curate convert-
cvat --input <annotations.xml> --out
307 <csv> --fps 59.94` ? **pickleball 6 rallies + tabletennis 6 rallies**, sane
intervals, attrs mapped. CSVs at
308 `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv` (+ extracted XMLs
`%TEMP%\roora\{pb,tt}\annotations.xml`). ? Roora data
309 nits (non-blocking): pickleball `last_shot_outcome='0'` (off-vocab in/n_a/net/out);
TT `ending_reason` mostly 'other' with the
310 real outcome in `notes` ("out of table"/"net") ? ending_reason taxonomy mapping
could improve. Both clips SHORT (97s/58s).
311 - ? ** (2) WASB-transfer test DONE (`%TEMP%\{pickleball,tabletennis}_traj.csv`; log
`~/clips/multisport_run.log`):** WASB
312 (badminton shuttle detector) detects SOMETHING on both ? **pickleball 28% /
tabletennis 39% visibility** (NOT near-zero;
313 comparable to badminton 40?67%). So it's not blind.
314 - ? ** (3) DONE ? SURPRISE RESULT (directional, n=6 each, TINY so noisy):**
`eval_partial_labels` on (WASB-transfer traj + Roora
315 6-rally CSV, fps 59.94, fw 1920): **TABLE TENNIS best F1 0.909** (tuned+gate2:
P1.0/R0.833, 5/6 matched, boundary err 0.49s ?
316 **WASB transfers to TT remarkably well!**); **PICKLEBALL best F1 0.308** (poor ?
wiffle ball too out-of-distribution).
317 ? **NUANCES the roadmap blocker: per-sport detector transfer is SPORT-DEPENDENT ? TT
may be servable with WASB short-term;
318 pickleball needs a proper detector.** ? BIG caveat: 6 rallies each = could be partly
luck ? owner sending more videos to
319 confirm. A "merged" prototype LOVO across badminton+pickleball+TT is the natural
next experiment once n grows per sport.
320 Roora LABELS ingest cleanly + are corpus-gold regardless. **Great honest surprise to
greet owner with.**
321
322 **Checkpoint:** 2026-06-09 (PR #61 MERGED ? roadmap SET, session wrap) . ? **Owner
agreed + merged #61 (`master 4d04a73`).
323 ALL CLEAN: master, 0 uncommitted, 0 open PRs, branches pruned.** The owned-model
**ROADMAP is now set + agreed** ?
324 authoritative doc = **docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md** (full verified
decisions in the "(PR review)" checkpoint
325 just below: ms-swift + Qwen3-VL, TAL-head-first, conversational-QA seam-now-model-later,
multisport one-head/base, Python
326 everywhere, MIT/Apache/BSD ratified, "envision now build later"). [[session-handoff]]
rewritten to this phase.
327 - ?? **NEXT SESSION (awaiting owner greenlight ? NO build until then):** greenlight a
Phase-0 milestone (rec **M0.1
328 corpus+event-index** ? **M0.4 Gen-0 TAL harness**, both $0/local); confirm **collector
= system-of-record**; define the
329 **ship-gate F1@tIoU target** (floor = beat heuristic LOVO 0.480); **rename repo off
"badminton"**, M0.3 purge Gemini
330 training labels + WASB C3 (own-weights/clean swap). Corpus = n=6
(`output/human_lovo_manifest.json`); learned prototype
331 is the Gen-0 seed. Committed next PLATFORM slice stays the async job model (single-
tenant). Owner flagged context ~85% ? this is the wrap checkpoint.

```

332

333 ****Checkpoint:**** 2026-06-09 (PR review) · ? ****OWNER REVIEWED PR #61 ? REFRAMED ROADMAP**
(important checkpoint). **** Owner left**
334 ~12 inline comments (mostly state-updates + recommendations wanted). ****Key roadmap**
shifts to internalize:
335 - ****Immediate goal REFRAMED:**** not "on-device model" but a ****TRAINING FRAMEWORK +**
HARNESS to produce a VERY HIGH
336 precision/recall model that extracts HIGHLIGHTS + RALLIES from a clip. **** Quality**
first. On-device = LATER (compression
337 for power users to save upload cost), definitely on roadmap but not now.
338 - ****MULTISPORT FROM THE START**** ? all net-separated racquet sports ? eventually all
sports; a SINGLE sports-agnostic model.
339 Collector + indexer already sport-generic by design. ? ****ACTION:** rename the repo off
"badminton" ASAP** (owner emphatic).
340 - ****CONVERSATIONAL VIDEO-QA = north-star feature:**** upload video ? ask contextual
questions ("longest rally", "make a clip
341 of all service faults", "rally >20 shots"). Needs per-video STATE/bookkeeping ? design
the framework for it. Owner asks:
342 embark now or punt + revisit after quality? (My lean: design the bookkeeping/event-
store layer now, punt the heavy
343 conversational model ? pending the strategy workflow.)
344 - ****LICENSING GUARDRAIL** (owner: "never corrupted"): **** EVERY dependency ? code, data, AND**
weights ? must be usable in
345 PRIVATE PROPRIETARY COMMERCIAL software. ****MIT / Apache-2.0 / BSD only.**** Ratify (was
"Apache-2.0 only"). Enforce religiously.
346 - Owner wants a ****coherent recommendation: local Gemma 4 vs ms-swift starting point****
(multisport, single, conversational,
347 commercial). NB: ms-swift = training FRAMEWORK, Gemma 4 = BASE model ? orthogonal;
clarify in the reply.
348 - Owner: ****upgrade WASB C3 (own-weights/provenance) to a first-class action item now.****
M0.2 (grow n) = practically DONE
349 (n=6); owner will add 1 more varied multi-court-club video soon ? Phase-0 ready to
start.
350 - ****LANGUAGE QUESTION** (owner callout, no preference): **** is Python right for the upcoming**
work, or brainstorm alternatives?
351 - ? ****STRATEGY WORKFLOW** done (`wf\_f9aa1d1c-c83`, 10 agents, verified+critiqued) ?
VERIFIED DECISIONS (now in the rewritten
352 `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`, the authoritative roadmap; pushed to PR #61
+ a consolidated PR comment reply):
353 - ****Framework = ms-swift (Apache-2.0)**** (only forge w/ native video+audio+timestamp-
grounding+video-GRPO); unsloth = 8GB-local
354 SFT/compression tool (avoid its AGPL Studio). ****Base = Qwen3-VL (Apache-2.0,**
timestamp emission)** primary; ****Gemma-4 STAYS**
355 **AMBER/bench-only**** (critic corrected the workflow's "clear AMBER" ? posture
unconfirmed + 60s/no-timestamp dispute);
356 InternVL3 (MIT) fallback. ms-swift?Gemma4 (framework vs base ? clarified for owner).
357 - ****START WITH THE TAL HEAD, NOT THE VLM**** (VLMs miss strict temporal boundaries;
boundary accuracy = the product; head
358 de-risked AUC 0.83?0.92, \$0 local, doubles as teacher+baseline). Head now ? VLM
later (two-stage: VLM proposes, head refines).
359 - ****Conversational-QA: build the SEAM now** (per-video event store + structured text-to-
SQL + ffmpeg clip-cut), PUNT the NL
360 model** until quality clears the bar. Queries ("longest rally"/"service faults"/">20
shots") = structured-numeric ? SQL, not
361 a VLM. Reserve fault/event taxonomy in M0.1. ? real DB delta (new cols + a
`highlights` table; play_segments lacks them today).
362 - ****Multisport: one shared head + one base; scale DATA + per-sport clean trajectory**
detectors (the binding blocker ? none
363 identified for tennis/TT/pickleball/padel yet); WASB C3 multiplies. Single-model
temporal generalization = OPEN HYPOTHESIS
364 (RacketVision evidence REFUTED = ball-tracking not temporal). Repo rename off
"badminton" = action item.
365 - ****Language: Python everywhere**** (training Python-locked; platform I/O-bound, no hot
path ? reject Go/Rust split); on-device =
366 deferred EXPORTED runtime shell (ONNX/CoreML/GGUF), doesn't dictate dev language.

Constraint NOW: ****train export-clean ops****.

367 - ****Honest gaps:**** cost-to-Gen-2 UNQUANTIFIED (biggest risk); ship-gate "very high
P/R" target undefined (floor = beat heuristic

368 LOVO 0.480 ? set explicit F1@tIoU); per-sport detectors TBD; base PRETRAINING
provenance unauditale ? owner risk-acceptance;

369 ****Gemini-silver TRAINING labels = LIVE violation ? hard-blocker purge M0.3.****

370 - ****Owner principle recorded:** "envision now, build later"**** ? design every deferred
seam (event store, export boundary, sport-
agnostic head) up front; punt the feature, never the seam.**

372 - ?? ****NEXT:**** owner reviews the reframed plan + decisions (greenlight a Phase-0
milestone ? rec M0.1 corpus+event-index then M0.4
harness; confirm collector=system-of-record). Then start the greenlit milestone. Repo
rename. Define the ship-gate F1@tIoU target.

374

375 ****Checkpoint:**** 2026-06-09 . ? ****n=6:** MahadevpuraSingles added while owner reviews PR
#61.****** Owner moved ALL golden videos

376 + CSVs to `Annotation Setup/Collect/Golden Labelled/` (manifest label paths updated;
trajectories unaffected in `%TEMP%`).

377 New: ****MahadevpuraSingles**** (singles, 1080p, 29.97fps, 31 rallies; traj
`%TEMP%\mahadevpura\_traj.csv`, script
`run\_wasb\_mahadevpura.sh`). ****Borderline multi-court (contrast 2.2x)**** ? heuristic best
0.667, learned held-out ****AUC**
379 0.902/F1 0.712****** (clean generalization to an unseen match). ****n=6** LOVO: heuristic 0.480
± 0.242 vs learned 0.583****** (learned
380 holds ~+0.10 since n=3). Contrast metric now predicts heuristic F1 MONOTONICALLY across
5 matches (Mahadevpura's 2.2x?0.67
381 lands on the curve). ****Decoupling headline:** learned held-out AUC 0.83?0.92 regardless of
contrast; heuristic swings 0.16?0.75

382 with it.****** Committed to PR #61 (n=6 doc update). ? GOTCHA: don't combine inner `&` with
run_in_background (orphans the job);

383 calibrate_local buffers output to end + is slow (~17?20min). "Ingested" = added to EVAL
corpus (manifest); collector

384 import-local (system-of-record) still = M0.1, plan-only pending greenlight. ?? owner
reviewing PR #61 (4 plan decisions pending).

385

386 ****Checkpoint:**** 2026-06-08 (later+4, SESSION WRAP) . ? ****n=5** corpus + MULTI-COURT root
cause + heuristic discovery + SINGLE PR #61 for offline review.******

387 Owner tagged gBaaddy (47, 1080p, multi-court) + TestLargeVideo (6, short) ? corpus
****n=5**** (`output/human\_lovo\_manifest.json`).

388 - ****HEURISTIC DISCOVERY** (user asked "discover more heuristics like audio"): ****** tested 2
label-free trajectory pre-filters

389 for multi-court (measured on all 5): foreground spatial-density gate = ****net ?0.04****
(helps multi-court +0.03, hurts clean
390 ?0.12); track-continuity/jump filter = ****no-op****. ? simple single-cue trajectory
heuristics CAN'T isolate the foreground
391 court. ****AUDIO-foreground idea PROTOTYPED** (no more videos needed) ? WEAK ?:****** on the 2
multi-court videos + clean

392 control, audio onset/RMS energy AUC only ****0.53?0.65**** for in-rally vs dead-time; HF
bands (8?22kHz @48kHz) raise

393 CONTRAST (Kushagra 1.1?1.7x, gBaaddy 1.3?1.9x, confirming distance-attenuation
physics) but AUC stays ~0.6. Fails:

394 dead-time isn't quiet ? foreground court loud during dead-time too
(footwork/talk/pickup) + gym reverb ? audio loudness

395 ? court occupancy, not rally (same wall as E1 ~2.2x). ****Audio alone ? multi-court
gate.**** Remaining heuristic lever =

396 real court-region detection (player-cluster/court-line, ROADMAP #2); robust path = the
learned visual model (AUC 0.85).

397 All tested + corrected in PR #61 doc + a PR comment.

398 - ****COURT-REGION DETECTION SMOKE-TESTED ? LOW HEADROOM ?:**** only 12?14% of dead-time
detections on the multi-court videos

399 fall OUTSIDE the foreground rally region (86?88% INSIDE) ? a perfect court gate lifts
contrast only ~1.6?2.2x (still
400 failing). The confusion is the foreground court's OWN between-point activity, NOT
separable background courts. So court
401 detection (heuristic OR own model) won't help. ****3** heuristics now tested + FAILED

```

(spatial gate, audio, court
402     separability) ? all because it's a TEMPORAL rally-vs-non-rally problem in the same
space; only the learned model solves
403     it (AUC 0.83). NO cheap heuristic shortcut ? the learned segmenter is the path.** (In
PR #61 doc + PR comment.)
404     - ? **SINGLE PR #61** (https://github.com/avidullu/badminton-highlight-indexer/pull/61)
carries EVERYTHING for offline
405     review: rally_seq_proto + tests, owned-model STUDY + PLAN (4 decisions pending),
multivideo-lovo learnings (n=5 + multi-court
406     + heuristic discovery), CODE_MAP. Title/desc updated. Plan doc NOW committed (was
held). Temp experiment scripts deleted.
407     (#60 + collector #13 merged earlier; #61 = only open PR.)
408     - **? DEFINING FINDING (owner's insight, data-confirmed): heuristic fails on MULTI-COURT
footage.** gBaaddy (1080p like
409     Adarsh) heuristic best F1 **0.277**; Kushagra 0.163 ? both have BACKGROUND GAMES. WASB
tracks every shuttle incl. other
410     courts ? high dead-time visibility (gBaaddy 41%, Kushagra 57% vs clean Adarsh 18% /
doubles 22%). ***"Contrast" metric**
411     (in-rally+dead-time vis) predicts heuristic success: ?3.6× works, ?1.9× fails. Academy
= multi-court = TARGET ENV ? these
412     are the most representative videos. **Lever = foreground-court isolation** (spatial
gate to prominent court; needs real
413     court-region detection, naive centroid muddled by foreground-shuttle-resting-low).
NEAR-TERM, no new model.
414     - **Learned prototype LOVO (n=5):** per-frame AUC 0.83?0.92 on every match; **recovers
BOTH multi-court videos**
415     (Kushagra 0.561, gBaaddy 0.574 vs heuristic 0.16/0.28). Kushagra climbs
0.049?0.417?0.561 with n. Mean 0.568 ? dipped
416     from 0.615 (n=3) ONLY due to the 6-rally TestLarge noise fold; **excl. it = 0.639**.
Per-video gap on hard videos is the signal.
417     - **Heuristic LOVO n=5 = 0.463 ± 0.247** (per-fold: Adarsh 0.751, doubles 0.758,
Kushagra 0.157, gBaaddy 0.295, testlarge
418     0.353). **FELL from n=3's 0.544** as multi-court videos entered ? the heuristic can't
represent the target env, so a more
419     representative corpus LOWERS its honest score. vs **learned 0.568** ? gap INVERTED
(heuristic led at n=2 +0.066; learned
420     leads at n=5 **+0.105** and widening). (b8jpxbryo was just slow, ~17min; completed
fine.)
421     - **SESSION WRAP shipped ? PR #61** (branch feat/owned-model-study-and-prototype): n=5 +
multi-court finding appended to
422     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`; **session-handoff.md
REWRITTEN** (quality-iteration phase);
423     **CODE_MAP updated** (eval_partial_labels + rally_seq_proto + multi-court gotcha).
Owned model still PLAN-ONLY
424     (`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` DRAFT, 4 decisions pending).
425     - ?? **NEXT (tomorrow): owner tags more varied videos** ? re-run LOVO+prototype.
**Foreground-court isolation experiment**
426     = #1 lever. Owner reviews plan + greenlights a Phase-0 milestone (rec M0.1 corpus
spine). Commit plan doc on approval.
427
428     **Checkpoint:** 2026-06-08 (later+3) . ? **n=3 CROSSOVER ? learned model OVERTAKES the
heuristic (the data-backed green
429     light).** User tagged a 3rd distinct match **LargeTestVideo (DOUBLES, 5.3K?1920,
29.97fps, 49 rallies)** ? corpus now
430     n=3 (2 singles + 1 doubles, 3 cameras/frame-rates). Trajectory
`~/clips/largetestvideo_traj.csv` (script
431     `run_wasb_largetestvideo.sh`). Manifest `output/human_lovo_manifest.json` now n=3.
432     - **Single-video LargeTest (doubles):** heuristic best F1 **0.750** (tuned, R0.918),
best-fit 0.827 ? **doubles is
433     heuristic-FRIENDLY; Kushagra remains the LONE heuristic failure** (not a play-type
issue).
434     - **Honest LOVO, heuristic vs learned, n=2?n=3:** heuristic mean held-out **0.443?0.544
± 0.278** (Adarsh 0.725,
435     **Kushagra 0.151 capped at any n**, doubles 0.755). Learned prototype mean
**0.377?0.615** (Adarsh F1 0.701/AUC 0.878,

```

436 **Kushagra 0.049?0.417**/AUC 0.850, **doubles 0.729/AUC 0.914** trained only on the 2
singles). **? The learned model
437 OVERTAKES the heuristic at n=3 (0.615 vs 0.544); at n=2 it trailed (0.377 vs 0.443).**
Crossover predicted by the plan.
438 - **The story (honest):** learned path SCALES with data (+0.24 for one match,
calibration gap closing ? Kushagra
439 0.049?0.417) and GENERALIZES across play type (doubles held-out AUC 0.914); heuristic
is HARD-CAPPED on footage it
440 can't represent (Kushagra ~0.15 at any n). Honest "bar to beat" is now ~0.54 (n=3),
not the fit-on-self 0.73. ?
441 Reusing every labeled video for training validated; bottleneck = data scale +
calibration (n?5), not method.
442 - **Docs updated ? pushed to indexer PR #61** (branch feat/owned-model-study-and-
prototype): n=3 findings + final quality
443 metrics + conclusions in `docs/archives/quality-iterations/2026-06-multivideo-lovo/`;
study + index updated. (PR #60
444 MERGED earlier; #13 MERGED.) ? Prototype speed features NOT yet fps-normalized (30 vs
60fps) yet doubles still
445 generalized (AUC 0.914) ? known refinement.
446 - **OWNED-MODEL: PLAN-ONLY per owner** ? `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`
written (DRAFT, uncommitted, for
447 review). User chose "just plan, don't code yet." 4 decisions pending (which Phase-0
milestone to greenlight [rec M0.1
448 corpus spine], WASB C3 stance, ratify Apache-2.0/MIT, confirm collector = corpus
system-of-record). NO owned-model
449 code until greenlight. Momentum = tag more matches (n?5) + I score each.
450 - ?? **NEXT:** owner reviews plan + greenlights a Phase-0 milestone; tag more matches ?
re-run LOVO+prototype (expect
451 learned mean to keep rising). gBaaddy/TestLargeVideo still untagged test videos
(trajectories ready). Commit the plan
452 doc on approval.
453
454 **Checkpoint:** 2026-06-08 (later+2) . ? **Multi-video eval corpus + owned-model
research underway.** User fully tagged
455 Adarsh-and-Avi (40?96 rallies**) + provided **Kushagra** (32 rallies, 4K) for the
TRAINING mix, and **gBaaddy**
456 (new, 1080p, untagged) + TestLargeVideo as held-out TEST videos. Generated full-video
WASB trajectories (single 8GB
457 GPU, serialized via wait-loops): `~/clips/{adarsh\_avi\_full,kushagra,gbaaddy}\_traj.csv`
(+ %TEMP% copies). Scripts:
458 `~/clips/run\_wasb\_{avi\_full,kushagra,gbaaddy}.sh`.
459 - **Adarsh-full eval (96 rallies, IoU?0.5, window [0,1405.8s]):** baseline F1 0.632,
tuned+gate2 0.727 (P0.80/R0.67,
460 best out-of-box), sweep **ceiling 0.750** (vel0.02/merge1.0/min_dur2.0/gate off, fit-
on-self). vs the opening-10-min
461 (40 rallies): best 0.667 / ceiling 0.742. **Best-config REGION is identical across the
slice and full match**
462 (merge_gap 1.0, min_dur 2.0; velocity irrelevant; gate-off=recall, gate-on=precision)
? early sign tuning may
463 transfer; LOVO is the real test. Per-rally: `output/adarsh\_avi\_full\_vs\_human.csv`.
464 - ? **KEY FINDING ? Kushagra: WASB tracks shuttle BETTER than Adarsh (97% of GT rallies
have ?? net-crossings, med 5,
465 310 pts/rally vs Adarsh 76%) BUT no velocity-windowing config can segment it (per-
video best-fit F1 only 0.163; baseline
466 0.000; mean boundary err ~5s).** Signal is PRESENT; the unsupervised heuristic FAILS ?
likely because Kushagra's shuttle
467 is visible 67.5% (vs Adarsh 40%), i.e. tracked through inter-rally activity ?
boundaries bleed. Root cause is the
468 heuristic class, NOT WASB recall or labels or tuning.
469 - **First cross-video LOVO (`calibrate\_local`, Adarsh-full 96 + Kushagra 32, human
labels, manifest
470 `output/human\_lovo\_manifest.json`):** baseline mean held-out F1 **0.316** ? calibrated
0.443 ± 0.279 (overfit gap

4. user (2026-06-11T03:46:38.593Z)

```

1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff ? current state, next steps, open
decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision, phase
shift, session wrap). \"update the handoff\" forces a full rewrite."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 45ea3a91-654d-4972-9cd3-65f35db54caf
8      ---
9
10     # Session handoff ? sports highlight indexer (repo still named badminton-highlight-
indexer; RENAME PENDING) + siblings: sports-data-collector, rally-annotator, sports-obsreport
11
12     **Last updated:** 2026-06-10 (end of the audit ? hardening ? M0 ? Gemini-battery mega-
session; ~25 PRs merged across
13     both repos, #63?#85 indexer + #14?#19 collector). **Read [[current-state]] first** for
blow-by-blow; this is the
14     ~1-page orientation.
15
16     ## You are here (2026-06-10) ? ? HARDENED + M0 SCAFFOLDED + MOAT QUANTIFIED
17     1. **Audit + remediation DONE:** 8-auditor deep audit (all findings adversarially
verified) ? 12-PR hardening sweep
18         (both repos). Master was literally broken (compiler SyntaxError) and is now CI-
guarded (both repos have CI; merges
19         gated on verified-green). Corpus licensing corrected (23 ShuttleSet broadcast rows
demoted; commercial export =
20         owned data only). Suite: indexer 336 green, collector 136 green.
21     2. **ADR-008 ratified (owner):** training in-repo (`training/`) behind a CI-enforced
import wall; featurizer
22         single-sourced in `backend/features/rally_seq.py` (train==serve); serving consumes
versioned ARTIFACT BUNDLES only;
23         ship-gate product-side; **pre-committed repo split** (`sports-temporal-models`) at
Gen-2/second-consumer.
24         "Split driven by productionization, not isolation" (owner's framing).
25     3. **M0 executed:** canonical GT loader (`backend/eval/gt_loader.py` ? closed the two-
loader format fork, round-trip
26         tested vs collector export) . **n=6 owned corpus IN the collector SoR** (262 rallies,
Proprietary) . **Gen-0
27         harness** `python -m training.gen0.harness` (train?LOVO?gate?artifact; v0.1.0 built:
**LOVO 0.626**, floor 0.480
28         passed, sign test 5/6 p=0.109 ? ship=False with honest blockers).
29     4. **Gemini battery DONE (the moat is quantified):** clean-ish Mahadevpura ? heuristic
0.657 / Gen-0 0.744 /
30         two-stage refine 0.83 / **wrapper 0.93**; multi-court Kushagra ? heuristic 0.157 /
Gen-0 0.561 / **wrapper 0.66**
31         (FPS = background-game pickups + dead-time merges, the contrast-metric confound).
**Ship target = beat 0.66 on
32         multi-court**; clean footage is commoditized (serve via Gemini). Baselines committed
in `eval_baselines/`.
33         Demo reel exists: `output/MahadevpuraSingles_highlights.mp4`.
34     5. **Gemini ops (hard-won):** prepaid/cold projects burst-throttle with a MISLEADING
"credits depleted" 429 ? ramp
35         gradually; serial works (`ai_max_workers=1` shipped); chunks re-encode to 1280
(`ai_chunk_scale_width`, PR #80);
36         generate retries = 6 attempts exponential+jitter (PR #83); model = `gemini-flash-
latest` alias (2.5-flash is
37         legacy). 503 capacity waves are real ? provider-fallback registry is load-bearing.
38
39     ## Owned-model architecture (for clarity ? owner asked)
40     **Gen-0/Gen-1 = WASB-substrate + OWNED temporal head.** WASB (frozen, MIT code, C3
checkpoint provenance = ship
41     blocker) produces trajectories; the owned IP = featurizer + head + corpus +
harness/gate. **Gen-2 (Qwen3-VL via
42     ms-swift) = pixels, drops WASB** ? that's the heavyweight step, only entered if the

```

lightweight head plateaus below

```

43     target. Gen-0 trains in MINUTES on the 2070S ($0). Multisport: WASB transfers to TT
(0.909) but not pickleball
44     (0.308) ? per-sport detectors or Gen-2.
45
46     ## ?? Next steps / open owner decisions (mirror of docs/NEXT_STEPS.md ? read that for
detail)
47     1. **Owner mission in progress: grow the golden corpus** (multi-court badminton first;
?3 matches/sport, TT next;
48     adults-only for training; per video: label ? `import-local --normalize` ? re-run
harness). Sign test reaches
49     significance around n=10 with 9 wins.
50     2. **Set the ship-gate target** (suggested: LOVO F1@tIoU0.5 ? 0.70 on multi-court folds
+ paired-sign p<0.05).
51     3. **M0.4 increment:** ActionFormer/ASFormer into the harness (still lightweight/local)
+ per-video threshold calibration.
52     4. C13 interviews + camera pilot (PMF) ? **recruiter flyer SHIPPED:
`docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md`** (script,
53     7 questions, ground rules, Phase-2 consent/minors rules; owner fills ?/contact
placeholders); repo rename; rotate
54     the chat-pasted Gemini key; collector=SoR confirm.
55     5. Committed next PLATFORM slice (separate track, owner-gated): async job model (#16).
56
57     ## Ramp-up kit
58     - [[current-state]] FIRST (dense checkpoints) . `docs/NEXT_STEPS.md` (live plan) .
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
59     (roadmap) . ADR-008 in `docs/archives/decisions/DECISIONS.md` (topology) .
`docs/CODE_MAP.md` (note its 2026-06-10
60     sweep banner ? several documented behaviors changed).
61     - Code anchors: `training/gen0/harness.py` . `backend/features/rally_seq.py` .
`backend/eval/gt_loader.py` .
62     `eval_baselines/*.json` . collector `src/cli/curate.py` (reclassify-licenses, import-
local --normalize/--force).
63     - Data: labels `Annotation Setup\Collect\Golden Labelled\` . trajectories+Roora
`...\Collect\Trajectories\` .
64     eval manifest `output/human_lovo_manifest.json` (gitignored) . corpus DB = collector
`data/sports_metadata.db`
65     (backup: `data/sports_metadata.backup-2026-06-10.db`, gitignored).
66
67     ## Gotchas (this session's additions)
68     - Merge discipline: gate on `gh pr view --json statusCheckRollup` == SUCCESS (a piped
`gh pr checks --watch` exit code
69     once let a red PR merge ? #75/#76 incident). Collector DB: stash-and-pop, never
`checkout --` (recovered once from
70     a dangling stash). Other sessions may write [[current-state]] concurrently ? insert
checkpoints, never overwrite.
71     - PowerShell `ConvertTo-Json` reformats config.json wholesale ? edit surgically.
`Select-Object -Last N` buffers
72     background output until exit (no interim tail).
73     - Windows ffmpeg here lacks NVENC ("No capable devices found") ? use libx264 for host-
side re-encodes.
74     - The dev container claim in CLAUDE.md is stale for THIS machine: cv2/torch/numpy all
present; GPU via WSL.
75

```

5. user (2026-06-11T03:46:48.772Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-242 of 336 total (29423 tokens, cap 25000). Call Read with offset=243 limit=242 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1     # CODE_MAP.md
2
3     **Purpose:** cold-start navigation map for the badminton-highlight-indexer ? a self-

```



```

hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a
segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.
4
5     **Regenerate:** rerun the `build-code-map` workflow (parallel subsystem readers ? this
doc). Re-run after any registry/contract/WSL-interface change, or after a directory restructure
(this revision follows #40).
6
7     ? **2026-06-10 audit-hardening sweep changed several behaviors documented below ? trust
git history over stale lines until the next full regen:** `add_video` is now an **UPSERT** (no
FK-cascade wipe on re-ingest; segmentation is destroy-AFTER-confirm via
`segment_watermarks`/`prune_segments`); detector artifacts are keyed by **video_id**, not
basename; **`indexing.wasb.*` is now a real typed config** (was silently dropped); `timeout_sec`
defaults **1800s** (was 0/hang-forever); SQLite opens with **WAL + busy_timeout**;
```

`match\_intervals` requires real overlap (iou>0); `run\_regression` baseline lives in tracked
`eval\_baselines`/` and **exits 3 on a missing baseline**;

the API validates
`video\_path`/`output\_filename`. New tests: `test\_api\_endpoints`, `test\_compiler`,
`test\_wasb\_runner`, `test\_reingest\_safety`, `test\_detector\_keying`, `test\_config\_wasb`,
`test\_path\_safety`, `test\_reliability\_hardening` (+ collector `test\_licensing\_gate`,
`test\_import\_guard`, `test\_cvat\_robustness`).

```

8
9     **LEGEND**
10    - `@path` = file (always repo-relative)
11    - `::sym` = symbol (class/func/const) in the nearest preceding `@path`
12    - `->` = dataflow / call / "produces"
13    - `$` = data contract (canonical shape)
14    - `?` = gotcha / footgun
15    - `?` = registry / plugin extension point
16    - `WSL` = runs inside WSL2 conda env, NOT the Windows Python process
17
18    ---
19
20    ## 1. PIPELINE
21
22    ### 1A. Runtime: video file -> highlight reel
23    ```
24    typed config load (built-in defaults -> config.json overrides; absent/parse-fail -> all-
defaults, never fatal)
25
26    @backend/config/loader.py::load_config ->
27    @backend/config/models.py::AppConfig
28    -> video_id = MD5(whole file) @backend/utils/hashing.py::compute_video_id
(single shared helper, 64KB chunks; imported by main + all run_*.py ? no more per-entrypoint
copies)
29    -> validate
30    @backend/pipeline/validators/base.py::registry.run_all_with_context(video_path, global_config)
31    FormatValidator -> ctx['ffprobe_meta'] -> ResolutionFPSValidator -> PTSContinuity
-> SceneCut -> Blur -> Relevance (? ACTUAL registration/exec order ? see
32    @backend/pipeline/validators/__init__.py)
33    [validation FAIL -> persist status="failed", return early; report STILL emitted
in finally]
34    -> db.add_video(video_id, filepath, status, [r.dict()])
35    @backend/infrastructure/database.py
36    -> seg_name = req.segmenter or global_config.indexing.default_segmenter (fallback
'motion')
37    -> seg = segmenter_registry.get(seg_name)(db, global_config) ?
38    @backend/pipeline/segmenters/base.py::segmenter_registry (400 + available list if missing)
39    -> segments = seg.process_video(video_id, video_path, player_pool?, max_frames?)
40    [STOP POINT: segments-only ? predictions now in DB; eval/regression operate here
without stitching]
41    -> finally: emit_indexer_report(...) @backend/reporting/__init__.py (? ALWAYS runs
? even on validation failure / exception; NEVER raises; grafts response["reports"])
42    -> select segment ids -> [(start,end)] (+ stitching padding)
43    -> VideoCompiler.compile_highlights(filepath, time_pairs, out_name)
44    @backend/pipeline/stitcher/compiler.py
45    per-clip ffmpeg re-encode (libx264/superfast/crf22) -> concat -c copy ->
output/<name>.mp4

```

```

39     ``
40     Hybrid segmenter internal 4-phase (yolo_hybrid / tracknet_hybrid / wasb_hybrid ? same
shape):
41     ``
42     P1 chunk video (yolo only; tracknet/wasb = whole-video single pass, chunk_index=0)
43     -> P2 candidate "action windows" from motion/trajectory [STOP: candidates persisted
via db.add_candidate_segment]
44     -> P3 pad clip (ai_handoff_padding) -> ThreadPoolExecutor ->
provider.analyze_video(clip, prompt) ? provider_registry
45     -> P4 db.add_play_segment + db.link_candidate_to_segment
46     ``
47     Pure-AI path `gemini`: no P2; chunk-or-single -> provider -> heavy validate
(clamp/dedup) -> add_play_segment.
48     Legacy `motion`: pixel-diff -> segments + **fabricated** metadata/events (? not ground
truth).
49
50     ### 1B. WASB shuttle substrate (P2 of wasb_hybrid)
51     ``
52     WasbHybridSegmenter.process_video
@backend/pipeline/segmenters/wasb_hybrid.py
53     -> WasbRunner.healthcheck() (gate; not ok -> return [])
@backend/pipeline/detectors/wasb_runner.py
54     -> WasbRunner.run_predict(video, out_dir)
55     stage video + wasb_infer.py into WSL fs -> `wsl bash -c 'source conda.sh && python
wasb_infer.py ...`
56     -> @backend/pipeline/detectors/wasb_infer.py::run (WSL): sliding windows -> GPU
detector pass (CHECKPOINTED det_raw.jsonl) -> CPU tracker (always full re-run) -> trajectory CSV
57     -> TrackNetRunner.parse_trajectory_csv(csv)
@backend/pipeline/detectors/tracknet_runner.py (shared, re-exported by WasbRunner)
58     -> TrackNetRunner.trajectory_to_action_windows(points, fps, frame_width, chunk_start,
velocity_thresh, merge_gap, min_window_duration, chunk_index) [the model-agnostic WINDOWING
BRAIN]
59     ``
60     (`tracknet_hybrid` identical, swapping WasbRunner->TrackNetRunner; `_probe_fps_width`
fallback (30.0, 1280.0).)
61
62     ### 1C. Calibration / eval flow (NO pipeline run unless told)
63     ``
64     GT CSV -> annotations.load_annotations / calibrate_wasb.load_golden_gts ->
List[Interval]
65     DB predictions -> harness.get_predictions(db, video_id, stage) -> List[Interval] stage
? {candidates, final}
66     -> metrics.match_intervals (greedy temporal-IoU) -> metrics.score -> Score
67     -> harness.evaluate -> EvalReport -> report_to_dict
68     -> batch.aggregate (corpus micro/macro) @backend/eval/batch.py
69     -> regression.regress(_with_provider) vs baseline @backend/eval/regression.py
[CI gate]
70     Calibration: calibrate_wasb.run -> calibration.{select_best, improvement_report,
cross_validate} @backend/eval/calibration.py
71     Distill Gemini->local: calibrate_local (thresholds) OR distill_local (learned
classifier) @backend/eval/{calibrate_local,distill_local}.py
72     Learned segmenter (offline): training.build_training_set(X,y) ->
classifier.LogisticRegression.fit/save
73     Gemini refinement driver (tuning, standalone):
gemini_refine.{refine_window,full_video_rallies} @backend/eval/gemini_refine.py
74     Audio probe (diagnostic only, NOT in live pipeline): audio/e1_probe.run
@backend/eval/audio/e1_probe.py
75     ``
76     Entrypoints: `@run_eval.py` (single video score), `@run_phase3_eval.py` (manifest batch,
can segment), `@run_regression.py` (baseline gate, exit 1=regression),
`@backend/eval/calibrate_wasb.py` + `@backend/eval/export_wasb_segments.py` (WASB tuning
drivers).
77
78     ---
79

```

```

80     ## 2. SUBSYSTEMS
81
82     ### 2.0 Orchestration & config
83     - `@backend/main.py` ? Thin CLI + FastAPI bootstrap (orchestrator; modes via FLAGS).
Static UI mounts `/static` from `backend/api/static`. Logging is initialized with `basicConfig`
(INFO, timestamped format); backend modules use `logger = logging.getLogger(__name__)` (replaces
stray prints in hot paths like motion, stitcher, wasb_infer, config loader). User-facing CLI
summaries may still use `print` for direct output.
84     - `::app` FastAPI; `::db_instance`, `::global_config` (`Optional[AppConfig]`) module
globals set ONLY in `main()`. ? importing `app` for ASGI without `main()` -> both `None` ->
every endpoint NPES.
85     - **ASYNC (#16, PR-1):** `::process_video` `@POST /api/process` -> fast-fail 4xx
(path/exists/segmenter) -> `db.create_job` -> `_get_executor().submit(_run_job,...)` -> **202 +
job_id**. `::run_job` (worker: mark running -> `_execute_processing` -> done/failed; catches +
logs traceback). `::execute_processing` = the former sync body (hash -> validate -> add_video
-> segment -> `finally:` always `emit_indexer_report`, grafts `response["reports"]`; never
raises from the report); a RAISING segmenter now prune_after-restores pre-run rows before re-
raising. `::get_job` `@GET /api/jobs/{job_id}` -> `{status, progress, video_id, error, result,
reports, ...}` ? job `status` = EXECUTION state (`queued|running|done|failed`); the pipeline
verdict lives in `result["status"]`. `::get_executor` lazy module-global
`ThreadPoolExecutor(max_workers=1)` (single box + Gemini serial-upload learning; tests
monkeypatch it inline). `main()` runs `db.fail_stale_jobs()` at startup (orphaned queued/running
-> failed). `::compile_highlights` `@POST /api/compile`. `::query_play_segments` `@POST
/api/query`. `::get_config` `@GET /api/config`. `::run_cli_diagnose_clip` `--debug-clip`: lazy
`import cv2`, samples ±3s vs `indexing.motion_threshold` (dual model/dict read). ? CLI `--video-
path` mode still runs the pipeline synchronously inline (separate code path, unchanged by #16).
86     - `::load_config` ? ? legacy thin wrapper returning a **dict**
(`load_app_config(path).model_dump(...)`); kept for transition ? new code imports
`load_app_config`/`AppConfig` directly. `::ProcessRequest`/`::QueryRequest`/`::CompileRequest`
pydantic bodies.
87     - `main()` sets `global_config.output.output_dir = args.output_dir` (typed field on
`OutputConfig`); `db_path = os.path.join(global_config.output.output_dir, output.db_name)`.
`compile_highlights` reads the SAME `global_config.output.output_dir`, so `--output-dir` is
honored end-to-end (the old write-only `_runtime_overrides` hack is gone).
88     - ? `--segmenter` argparse `choices` frozen at build time from
`segmenter_registry.list_available()` ? a lazily-registered segmenter wouldn't appear. CLI
default_segmenter fallback `motion`.
89     - `@run_process_and_stitch.py` ? one-shot demo; ? hardcoded paths, interactive `input()`
(blocks automation), skips validation. Config via `backend.config.load_config` (typed);
segmenter, padding, and db path all read from the model (no literal fallbacks).
90     - `@run_eval.py::main` ? score one video's DB preds vs GT CSV.
`--stage{candidates,final,both}`. Exit 2=arg/IO. ? pure scorer; errors if video never processed.
`::default_db_path` resolves via `backend.config.load_config`
(`output.output_dir`/`output.db_name`). `get_file_md5` is an alias for `compute_video_id`.
91     - `@run_phase3_eval.py::main` ? walk manifest, segment-if-needed, score, aggregate.
`load_dotenv()` at import. ? DB key = file MD5, NOT manifest's `video_id` (kept as label);
config via `backend.config.load_config`; `segmenter_cls(db, cfg)` receives the typed `AppConfig`
(same object the live pipeline feeds segmenters).
92     - `@run_regression.py::main` ? `regress_with_provider` vs baseline. Exit codes semantic:
**0=ok, 1=REGRESSION (CI gate), 2=missing DB**. ? `--update-baseline` runs AFTER compare
(snapshots even a regressing run); `::default_db_path` resolves via
`backend.config.load_config`.
93
94     - `@backend/config/__init__.py` ? ? canonical config import surface. Re-exports
`load_config`, `save_default_config` (from `.loader`) + `AppConfig`, `Config`, `IndexingConfig`,
`ValidationConfig` (from `.models`); `__all__` = exactly those 6. Use `from backend.config
import load_config, AppConfig`.
95     - `@backend/config/models.py` ? Pydantic v2 models; **single source of truth for
defaults**.
96     - `::AppConfig` ? root. Scalars `sport='badminton'`, `ai_provider='gemini'`,
`ai_model='gemini-2.5-flash'`; sections `validation`, `indexing`, `output`, `stitching`
(`default_factory`). `model_config = {extra:'allow', validate_assignment:True}`. ?
`extra='allow'` -> unknown/typo'd config.json keys silently kept (legacy tolerance).
97     - ? `::AppConfig.get` ? legacy dict-compat shim: returns nested **sections as plain
dicts** (`model_dump(mode='json')`, NOT the model) so chained

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`config.get("indexing",{ }).get(...)` keeps working; falls back to searching a leaf key in any
section (sections discovered dynamically from `model_fields`). ? "bit every validator on real
runs" per docstring ? returning the model breaks `.get`. `::AppConfig.__getitem__` raises
`KeyError` for unknown top-level keys else delegates to `.get`.
98     - `::IndexingConfig` ? segmenter/detector knobs (`motion_threshold=0.08`,
`min_segment_duration=3.0`, `dead_time_merge_gap=2.0`, `chunk_duration=90.0`,
`chunk_overlap=10.0`, `yolo_velocity_threshold=0.15`, `yolo_frame_skip=3`,
`ai_handoff_padding=2.0`, `ai_max_workers=5`, `log/store_yolo_candidates=True`, nested
`tracknet`). ? `default_segmenter='yolo_hybrid'` HERE but config.json overrides to `gemini`. ?
`::IndexingConfig.segmenter_must_be_registered` `@field_validator` is a **no-op pass-through** ?
a typo'd segmenter validates fine, only fails later at `segmenter_registry.get(...)` .
99     - `::ValidationConfig` (`min_resolution_width=1280`, `min_resolution_height=720`, `min_fps=29.9`, `min_blur_threshold=20.0`, `max_scene_cuts_per_minute=0.5`, nested `relevance`);
`::ValidationRelevanceConfig` (court HSV gates, `court_pixels_min_ratio=0.05`).
`::TrackNetConfig` (WSL invocation contract: `wsl_distro='Ubuntu'`, `repo_dir`,
`conda_env='TrackNetV4'`, `weights_path='',`queue_length=5`, `velocity_threshold=0.02`).
`::OutputConfig` (`default_highlights_name`, `db_name='sports_indexer.db'`,
`output_dir='output'` ? CLI `--output-dir` overrides it on the typed model in `main()`).
`::StitchingConfig`
(`begin_padding=0.5`, `end_padding=0.5`, `use_cache=False`, `min_shot_count=1`). `::Sport` Literal
of 5 sports. `::Config = AppConfig` alias.
100    - `@backend/config/loader.py` ? load/save with defaults-merge.
101    - `::DEFAULT_CONFIG_PATH = Path("config.json")` (? CWD-relative, NOT repo-root).
`::get_builtin_defaults` = `AppConfig().model_dump(...)` .
102    - `::load_config` ? precedence built-in-defaults -> file overrides. ? **shallow top-
level merge** (`{**defaults, **file_data}`): a section in config.json REPLACES the whole
defaults dict for that section (Pydantic then re-applies field defaults for missing leaves).
Missing/unreadable/parse-error file -> fully-defaulted AppConfig + `[CONFIG WARNING]` (non-
fatal).
103    - `::save_default_config` ? writes fresh `config.json`, **overwrites** if present
(used by setup.py/--setup). `::get_default_config` ? ? transitional plain-dict alias.
104    - `@config.json` ? on-disk config; mirrors `AppConfig`. ?
**`indexing.default_segmenter='gemini'`** diverges from model default `yolo_hybrid` ? file
wins at runtime.
105    - `@setup.py` ? `::init_default_config` ? delegates to
`backend.config.save_default_config` (single-source defaults; skips if config.json exists).
`::run_pip_install` GPU-aware (`nvidia-smi` -> cu124 else cpu). `::run_diagnostics` (Python>3.8,
ffmpeg/ffprobe on PATH, optional torch+CUDA).
106
107    ### 2.1 Segmenters (under pipeline/segmenters) ?
108    - `@backend/pipeline/segmenters/base.py`
109    - `::BasePlaySegmenter` ABC ? `__init__(db, config)`; abstract
`name`, `description`, `process_video`. $ `process_video(video_id, video_path, player_pool=None,
max_frames=None)` -> List[dict{id,start_time,end_time,duration,metadata}].
110    - `::SegmenterRegistry`, `::segmenter_registry` (singleton). ? **register a new
segmenter**: subclass `BasePlaySegmenter`, call `segmenter_registry.register(MyCls)` at module
bottom, AND import the module in `@backend/pipeline/segmenters/__init__.py` (registration is an
IMPORT side-effect). ? `register()` builds `cls(None, {})` just to read `name` ->
name/description/`__init__` MUST NOT touch `self.db`/`self.config`. ? `list_available()` re-
instantiates + swallows exceptions into "No description available." (a throwing `description` is
masked).
111    - `@backend/pipeline/segmenters/__init__.py` ? imports all 5
(`motion,gemini,yolo_hybrid,tracknet_hybrid,wasb_hybrid`) = the de-facto registration manifest;
re-exports them + `segmenter_registry` via `__all__`.
112    - `@backend/pipeline/segmenters/wasb_hybrid.py::WasbHybridSegmenter` (name
`wasb_hybrid`) ? WSL WASB substrate (MIT; ships pretrained badminton weights, the LIVE
default). Imports `WasbRunner`/`WasbConfig` from `pipeline/detectors/wasb_runner`. Config
`indexing.wasb.velocity_threshold(0.02)` + shared
`dead_time_merge_gap/min_action_window_duration/ai_handoff_padding/ai_max_workers`. ? whole-
video single pass; reuses YOLO-named keys `log_yolo_candidates/store_yolo_candidates`; needs
`{PROVIDER}_API_KEY` (missing -> []); `detection_params` has NO `queue_length` (unlike
tracknet). ? divergent shot_count: `int(shot_count)` inside try with no `is None` short-circuit
(copy-paste drift).
113    - `@backend/pipeline/segmenters/tracknet_hybrid.py::TrackNetHybridSegmenter` (name
`tracknet_hybrid`) ? copy-paste twin of wasb_hybrid; imports `TrackNetRunner`/`TrackNetConfig`

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from `pipeline/detectors/tracknet_runner`; `indexing.tracknet.velocity_threshold(0.02)`;
`detection_params` adds `queue_length`. ? any P3/P4 fix must be applied to BOTH (and arguably
yolo_hybrid).
114 - `@backend/pipeline/segmenters/yolo_hybrid.py::HybridYoloSegmenter` (name
`'yolo_hybrid'`, no YOLO remains ? torchvision detector + supervision ByteTrack; ADR-005).
`::PERSON_CLASS_ID=1` ? (ultralitics used 0; wrong value -> zero detections).
`::lazy_load_model` (lazy `torchvision` import so module/tests load without it),
`::extract_action_windows_from_chunk`, `::make_tracker` (per-chunk ByteTrack ? track IDs
chunk-local), `::DETECTOR_FACTORIES`/`::DEFAULT_DETECTOR='fasterrcnn_resnet50_fpn'`. ?
velocity at `effective_fps = fps/frame_skip` -> `yolo_velocity_threshold` coupled to
`frame_skip`; pipelined-vs-sequential P2 (prefetch ffmpeg overlaps sequential GPU);
`yolo_tracker`/`bytetrack.yaml` is dead/back-compat. ? `gemini_*` config keys are deprecated
aliases still honored.
115 - `@backend/pipeline/segmenters/gemini.py::GeminiPlaySegmenter` (name `'gemini'`) ?
pure-AI, no P2. Chunks are RE-ENCODED + downsampled to `indexing.ai_chunk_scale_width` (default
1280; 0 = legacy stream-copy) before upload ? stream-copied 4K chunks exceed the provider's
`MAX_UPLOAD_MB` and every chunk gets refused ("0 segments / success"); re-encode also fixes
keyframe drift at chunk boundaries. Oversize refusals (provider metrics `oversize_skipped`) are
aggregated -> `logger.error` + `reporting.run_signals` (`oversize_clips_skipped`) for the report
rule. Workers from `indexing.ai_max_workers`. ? chunks only if `duration>chunk_duration`;
`max_segment_duration=90.0` hardcoded; `debug_duration` silently truncates; `parse_time` accepts
colon timestamps (warns); does NOT persist candidate segments (only hybrids feed the fine-tuning
table).
116 - `@backend/pipeline/segmenters/motion.py::MotionPlaySegmenter` (name `'motion'`) ?
pixel absdiff baseline. ? **metadata + serve/smash/foul/winner events are RANDOM**
(random.sample/randint/uniform/choice); only segmenter using `db.add_event` and passing
server/receiver; uses `print()`; honors `max_frames`.
117
118 ### 2.2 Detectors & windowing (WSL-quarantined; ABC abstraction complete)
119 - `@backend/pipeline/detectors/base.py` ? DetectorRunner ABC (healthcheck() ->
Tuple[bool,str] never-raises; run_predict(...) -> Optional[str] CSV path). Contract deliberately
matches the pre-existing tuple returns and call sites in the hybrids. Rich module docstring
includes "how to add a new runner", Linux-native path notes (the de-risk goal), and guidance on
threading health into reports. Re-exported from `detectors/__init__.py`. This finishes the WSL
de-risk seam (low-regret 6 / review item 1).
120 - `@backend/pipeline/detectors/tracknet_runner.py`
121 - `::TrackNetConfig` (+`.from_indexing_cfg`; ? key drift `wsl_distro` JSON -> `distro`
field), `::TrackNetRunner` ? runs TrackNet `predict.py` in WSL.
122 - `::to_wsl_mnt_path` (`C:\x`->`/mnt/c/x`; ? POSIX/~` passthrough, UNC unhandled),
`::TrajectoryPoint` (frame,visible,x,y).
123 - `::parse_trajectory_csv` @staticmethod** + `::trajectory_to_action_windows`
@staticmethod** = MODEL-AGNOSTIC brain, re-exported verbatim by WasbRunner. ? `weights_path`
defaults `` -> `run_predict` returns None; predict.py only `mp4/.avi` + hard-names
`<stem>_predict.csv`; `visible==False` resets prev (occlusion gap breaks track); `fps<=0->30`,
`frame_width<=0->1280` silent; `track_id_count` hardcoded 1; `timeout_sec=0` -> NO timeout
(hangs forever). `::stage_video` "OK" sentinel.
124 - `@backend/pipeline/detectors/wasb_runner.py::WasbRunner` (inherits DetectorRunner)
(+`.WasbConfig`, from `indexing.wasb`; ? `weights_path` HAS a real default
`wasb_badminton_best.pth.tar` ? no required guard) ? stages `wasb_infer.py` (`::INFER_WIN`) +
video into WSL, runs in `wasb` env. `::parse_trajectory_csv`/`::trajectory_to_action_windows` =
`staticmethod(TrackNetRunner.*)` (explicit rebind = the plugin-equivalence point). Output hard-
named `<stem>_wasb.csv`. ? `wasb_infer.py` re-staged each run (editing Windows copy inert until
staged); `wsl_bash` duplicated (drift risk). Both runners now satisfy the DetectorRunner ABC
for future swappability.
125 - `@backend/pipeline/detectors/wasb_infer.py` (WSL) ? `::run` core. ? imports WASB-SBDT
repo modules + torch ? NOT importable on Windows. Reboot-resumable detector cache:
`::DET_FILE='det_raw.jsonl'`, `::MANIFEST_FILE='manifest.jsonl'`, `::CACHE_VERSION=1`,
`::load_and_clean_cache` (reads longest CONTIGUOUS prefix where `w==line index`, rewrites to
heal crash-truncated tail), `::manifest_compatible` (? `frames_dir` abspath'd but `weights`
compared raw), `::atomic_write_text/_atomic_write_trajectory`, `::dets_to_tracker` (xy MUST be
np.array), `::build_cfg` (`runner.gpus=[0]` hardcoded), `::extract_frames` (cv2-PNG SLOW),
`::default_cache_dir` (`<stem>_wasbcache` next to out). ? GPU pass resumable; CPU tracker
STATEFUL -> always fully re-run; `--fresh` ignores cache.
126 - `@backend/pipeline/detectors/rally_gate.py` ? Gate A net-crossing post-filter (pure,
no I/O/model). `::estimate_play_axis` (larger-spread axis; ? ties -> `x`),

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`::count_net_crossings` (sign changes vs median net; deadband jitter zone),
`::gate_windows`/`::apply_gate` (`min_crossings=2` default; kills single-toss service feeds).
`::GatedWindow` (start,end,crossings,visible_points,kept). ? window contract here is
`(start,end)` tuples NOT the rich window dict ? caller must adapt. Consumed by calibrate_local /
distill_local / export_wasb_segments.
127
128     ### 2.3 Eval & calibration (all pure core; I/O in CLI drivers)
129     - `@backend/eval/metrics.py` ? scoring primitive (THE spine). `::interval_iou`,
`::match_intervals` (greedy, deterministic), `::score`->`::Score`(`.as_dict()` = wire shape),
`::pr_curve`, `::Match`, `::MatchResult`. ? greedy not Hungarian; empty matches -> 0.0 (not NaN)
for mean_iou/boundary errors.
130     - `@backend/eval/annotations.py` ? generic GT loader. `::load_annotations` (`start,end`
CSV; RAISES on bad rows/`start>=end`), `::parse_timestamp` (decimal or MM:SS/HH:MM:SS),
`::write_scaffold`.
131     - `@backend/eval/calibration.py` ? overfit guard. `::cross_validate` (within-video
k-fold; ? defaults `metric='recall'` deliberately), `::leave_one_video_out` (honest cross-video;
needs ?2 labelled videos), `::improvement_report` (OPTIMISTIC ? selected on full GT),
`::select_best`, `::kfold_indices`, `::pct_improvement` (? from-zero -> `inf`), `::evaluate`. $
`PredictFn = Callable[[Config], List[Interval]]` = the plug-in seam. ? `generalization_gap
>=0.1` = overfit alarm.
132     - `@backend/eval/calibrate_wasb.py` ? WASB tuning CLI. `::BASELINE=(0.02,2.0,2.0)`,
`::default_grid` (45 cfgs), `::make_predict_fn` (parses trajectory once, closes over
`trajectory_to_action_windows`), `::load_golden_gts` ($`start_time,end_time` cols; ? SILENTLY
skips bad rows ? opposite of annotations.load_annotations). ? `{load_golden_gts,
make_predict_fn, BASELINE}` imported by 4 downstream drivers (calibrate_local,
export_wasb_segments, gemini_refine, audio/e1_probe).
133     - `@backend/eval/calibrate_local.py` ? distill Gemini->local thresholds (windowing +
net-crossing gate, no cloud at runtime). `::local_grid` (180 cfgs), `::make_local_predict_fn`
(adds `rally_gate.apply_gate` when `min_crossings>0`), `::build_videos` (? `SystemExit` if a
labels file yields no rallies), `::run`/`::main`. $
`LocalConfig=(velocity,merge,min_dur,min_crossings)`, `BASELINE_LOCAL=(0.02,2.0,2.0,0)`;
manifest JSON shared with distill_local.
134     - `@backend/eval/export_wasb_segments.py` ? tuned cfg -> golden/slicer CSV + comparison
+ optional clip cut. `::TUNED=(0.05,1.0,1.0)` (? from ONE video),
`::SEG_FIELDS`, `::COMP_FIELDS`, `::build_comparison`, `::seconds_to_mmss`, `::maybe_slice`
(lazy-imports rally_slicer). `--slice` requires `--video`. ? `write_segments_csv` output loads
straight into `rally_slicer.load_rally_labels` AND re-imported by gemini_refine ? schema is
load-bearing.
135     - `@backend/eval/batch.py` ? corpus aggregation (pure; orchestration is in
run_phase3_eval). `::aggregate` (micro pool TP/FP/FN, TP-weighted mean_iou, macro F1),
`::default_stages_for` (`auto`; `::FINAL_ONLY_SEGMENTERS={motion,gemini}`),
`::resolve_video_path` (normalizes collector `./data`..` Windows paths), `::format_aggregate`.
136     - `@backend/eval/harness.py` ? DB-pred scorer (no re-run).
`::STAGES=('candidates','final')`, `::get_predictions`
(candidates->`query_candidate_segments(detector=)`; final->`query_play_segments({})`); ? unknown
stage raises ValueError; reused by regression.py),
`::evaluate`->`::EvalReport`(`::StageReport`)->`::report_to_dict` (input contract for
batch.aggregate), `::format_report_text`.
137     - `@backend/eval/training.py` ? learned-segmenter data layer. `::YOLO_FEATURE_NAMES`
(5-d), `::extract_yolo_features` (? reads `row['stats']` dict from DB; missing fields default
0.0), `::label_candidates` (same IoU `match_intervals` as evaluator),
`::build_training_set`->(X,y,intervals); `feature_fn` is the substrate plug-in seam.
138     - `@backend/eval/classifier.py::LogisticRegression` ? numpy-only logreg (no sklearn).
`fit/predict_proba/predict/to_dict/from_dict/save/load`. ? `class_weight='balanced'` default;
stores feature standardization (mean/std) ? inference MUST use same feature ORDER; `sigmoid`
clips z to [-30,30]; `lr=0.1,epochs=1000,l2=1e-3`; JSON `"model":"logreg"` tag.
139     - `@backend/eval/eval_partial_labels.py` ? score pipeline vs **PARTIAL human labels**
(PR #60). `::restrict_to_window` (drop preds outside `[0, last GT end]` so un-labeled-tail
detections aren't FPs ? THE partial-label methodology), `::run` (baseline/+gate/tuned/+gate ops
table + per-rally diff via `export_wasb_segments.build_comparison`), `::sweep`/`--sweep` (grid
vs labels; ? fit-on-self = optimistic), `::DEFAULT_GRID`, `::TUNED=(0.05,1.0,1.0)`. ?
`window_end` inferred from `max(GT end)` is WRONG if a video is labelled PAST its last rally ?
P1 fix = collector persists `labeled_window` in the manifest. Reuses calibrate_wasb + rally_gate
+ metrics.
140     - `@backend/eval/rally_seq_proto.py` ? **PROTOTYPE** learned per-FRAME rally segmenter =

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owned-model **Gen-0 seed** (PR #61). 6 features over the WASB trajectory (`vis_rate, spd_mean,
spd_max, cross_rate, x_std, y_std`) -> `classifier.LogisticRegression`, honest LOVO.
`::build_frame_arrays` (takes `fps`; speed is per-SECOND + occlusion-gap-guarded since the
2026-06-10 sweep), `::features` (scipy `uniform/maximum_filter1d` windowed), `::labels_for`,
`::probs_to_segments` (smooth->threshold->merge/min-dur), `::roc_auc` (rank-based with **midrank
ties**, scipy.stats.rankdata), `::run` (per-video AUC + LOVO-threshold F1 + oracle-threshold
F1). ? DIRECTIONAL only (n=6, linear, threshold-transfer unsolved); held-out per-frame AUC
0.83?0.92, **beats the heuristic on multi-court footage** (Kushagra/gBaaddy). **n=6 LOVO:
learned 0.626 vs heuristic 0.480** (learned rose from 0.583 after the fps-normalization fix).
Manifest `output/human_lovo_manifest.json` (n=6; trajectory paths now under `Annotation
Setup/Collect/Trajectories/`) shared w/ calibrate_local.
141 - ? **MULTI-COURT / background-game confound (2026-06-08 finding):** WASB tracks EVERY
shuttle in frame incl. other courts; videos with background games (Kushagra, gBaaddy) have high
dead-time shuttle visibility (41?57% vs 18?22% clean) ? the velocity-windowing heuristic can't
find clean rally boundaries (F1 0.16/0.28 vs ~0.75 on single-court). Diagnostic = in-rally÷dead-
time visibility "contrast" (?3.6× heuristic works, ?1.9× fails). Lever = **foreground-court
isolation** (spatial gate to the prominent court, needs real court-region detection). Academy =
multi-court = the target env. See `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
142 - `@backend/eval/distill_local.py` ? distill Gemini -> a LEARNED classifier (beyond
calibrate_local's threshold ceiling). `::FEATURE_NAMES` (5 fps/res-INVARIANT:
duration,peak,mean,active_ratio,net_crossings), `::build_candidates` (recall-first proposal),
`::predict_rallies` (proba filter then `gemini_refine.merge_overlaps(gap_tol=1.0)`),
`::run`/`::main`. consts `PROPOSAL=(0.005,1.0,0.5)`, `BASELINE_LOCAL=(0.02,2.0,2.0)`. ? leave-
one-video-out needs ?2 videos; final model trained on all.
143 - `@backend/eval/gemini_refine.py` ? refine WASB candidates with Gemini over SHORT
padded clips (precision lever; eval/tuning, NOT the live pipeline).
`::merge_overlaps(gap_tol=0.0)` (reused by distill_local with gap_tol=1.0), `::refine_window` (?
FRESH genai client per worker ? shared throws socket errors), `::full_video_rallies`,
`::run`/`::main` (`--full-video` skips `--trajectory`). ? requires `GEMINI_API_KEY` (raises
SystemExit); imports `BASELINE` from calibrate_wasb +
`TUNED`/`write_segments_csv`/`seconds_to_mmss` from export_wasb_segments.
144 - `@backend/eval/regression.py` ? CI gate. `::DEFAULT_TOLERANCE=0.05`,
`::DEFAULT_BASELINE_PATH=output/regression_baseline.json`, `::regress`,
`::regress_with_provider` (GT via `?annotation_registry.get(tier)`;
"file"->FileProvider/"vendor"->HumanVendor/"auto"->oracle),
`::save_baseline`/`::load_baselines`, `::RegressionResult`. ? imports `annotation_registry` from
**`backend.annotations`** NOT `backend.eval.annotations`; gate fires on precision OR recall drop
(F1 reported, not gated); no baseline -> `regressed=False` (silent pass).
145
146 ### 2.3a Audio (E1 probe; NOT adopted for fusion ? ADR-007)
147 - `@backend/pipeline/audio/hit_detector.py` ? classical-onset shuttle-hit detector. Pure
numpy + stdlib `wave` (NO scipy); ffmpeg only for extraction. `::HitEvent`(t,strength),
`::extract_audio` (ffmpeg -> mono 16k PCM WAV), `::load_wav`, `::onset_envelope` (pure-numpy
STFT spectral flux; `band=(lo,hi)` band-limit), `::detect_hits` (adaptive `mean+k.std`;
`k`=sensitivity knob), `::detect_hits_in_video`. ? separate signal ? WASB never modified by
this; per audio-e1-findings recall 100% but discrimination ~2.2×, band-pass null -> NOT adopted
(PR #35).
148 - `@backend/eval/audio/e1_probe.py` ? E1 GO/NO-GO diagnostic (no training/fusion).
`::cluster_hits` (groups thwacks; ? `min_hits>=2` so a lone service-feed hit isn't a rally),
`::run` (discrimination ratio + recall recovery + audio-only P/R/F1), `::main`. reuses
`hit_detector` + `calibrate_wasb.{load_golden_gts,make_predict_fn,BASELINE}` + `metrics`.
149
150 ### 2.4 Stitcher / tools / utils
151 - `@backend/pipeline/stitcher/compiler.py::VideoCompiler` ? `compile_highlights(raw,
segments:[(start,end)], out_name)`. Per-clip re-encode -> `-f concat -c copy`. `::parse_time`
(mirrors gemini.parse_time; ? unparseable -> 0.0), `::get_video_duration`. ? if duration probe
fails (0.0) NO boundary validation; `begin_padding`(0.5)/`end_padding`(1.0) defaults; temp clips
in a TemporaryDirectory under output_dir; raises ValueError/FileNotFoundError/RuntimeError; uses
`print()`. ? two-stage codec contract: clips re-encoded with IDENTICAL settings precisely so
final concat can stream-copy ? change a preset and concat may break.
152 - `@backend/tools/rally_slicer.py` ? standalone CLI. `::compute_clip_windows` (pure,
unit-tested), `::load_rally_labels` (golden schema; ? silently skips degenerate rows),
`::slice_rallies`, `::ClipWindow`(rally:str), `::BEGIN_PADDING=0.5`/`::END_PADDING=1.0` (?
duplicate compiler's by copy). `_read_time` -> backend/eval/annotations.parse_timestamp.
153 - `@backend/utils/video.py` ? `::get_video_duration` (ffprobe; ? 0.0 on failure =

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"unknown"), `::extract_video_chunk(...,reencode=False,scale_width=None)` . ? copy mode cuts at
nearest keyframe (drift) ? pass `reencode=True` for short/accurate clips; `scale_width` requires
`reencode=True`.
154     - `@backend/utils/geometry.py` ? `::get_velocity_normalised` (fraction of frame-width/s;
? raw-px fallback if width<=0; used by yolo_hybrid), `::calculate_iou` (? +1 px convention
inflates IOU), `::get_center/get_distance/get_velocity`.
155
156     ### 2.5 Validators ?
157     - `@backend/pipeline/validators/base.py::VideoValidator` ABC, `::ValidationResult`
(pydantic: validator_name,passed,message,details), `::ValidationRegistry` (ordered LIST ?
registration order = exec order; ? no dedup), `::registry` (singleton). § `validate(video_path,
config, context) -> ValidationResult`. `::run_all_with_context -> (results, context)` (wraps
each validate in try/except -> failed result not abort); `::run_all` discards context. ? **add a
validator**: subclass + `registry.register(MyValidator())`.
158     - `@backend/pipeline/validators/__init__.py` ? registers in EXEC order: FormatValidator,
ResolutionFPSValidator, PTSContinuityValidator, SceneCutDensityValidator, BlurValidator,
RelevanceValidator (? producers before consumers ? format_check first; ? import mutates global
registry as side effect).
159     - `@backend/pipeline/validators/format.py::FormatValidator` (`format_check`) ?
**PRODUCER** of `context['ffprobe_meta']`; ? MUST run before resolution_fps; container check is
substring `mp4` (so MOV-family passes); external `ffprobe`.
160     - `@backend/pipeline/validators/resolution_fps.py::ResolutionFPSValidator`
(`resolution_fps_check`) ? CONSUMES `ffprobe_meta` (? hard-fails "Run format_check first" if
absent). `min_fps` 29.9; min res 1280x720.
161     - `@backend/pipeline/validators/pts_continuity.py::PTSContinuityValidator`
(`pts_continuity_check`) ? re-probes (no context). ? 10.0s keyframe-gap threshold hardcoded
(comment says 8.0s ? drift, NOT config-driven); any backward jump fails; <2` keyframes PASSES
(too short).
162     - `@backend/pipeline/validators/scene_cut.py::SceneCutDensityValidator`
(`scene_cut_check`) ? OpenCV. ? `cut_threshold=0.6` hardcoded (NOT config); samples ~2fps, caps
at 100 SAMPLED frames (~first 50s); `max_scene_cuts_per_minute` (0.5) is the config fail
threshold.
163     - `@backend/pipeline/validators/blur.py::BlurValidator` (`blur_check`) ? Laplacian var <
`min_blur_threshold` (default 100.0 in-file, 20.0 in models/config) = fail; samples 15 frames. ?
magic absolute, not resolution-normalized.
164     - `@backend/pipeline/validators/relevance.py::RelevanceValidator` (`relevance_check`) ?
HSV court-green ratio (? purely color, no geometry despite name); lazy `import backend.sports`;
3-tier config precedence (`validation.relevance` -> sport profile -> hardcoded); ? unknown sport
-> empty cfg, falls back to defaults, does NOT fail.
165
166     ### 2.6 Sports / Providers / Annotations ?
167     - `@backend/sports/base.py::SportProfile` ABC
(`name`,`get_prompt`,`get_relevance_config`);
`@backend/sports/badminton.py::BadmintonSportProfile` (the Gemini prompt + HSV relevance cfg);
`@backend/sports/__init__.py::sport_registry` (+`::SportRegistry`, auto-registers badminton; ?
`register` instantiates profile to read `name`). ? **add a sport**: subclass `SportProfile`,
`sport_registry.register(MyProfile)`. § prompt emits JSON array `{start_time(float DECIMAL SEC),
end_time, shuttle_exchange_count(int), shot_count_confidence('Low'|'Medium'|'High')}`. ? prompt
actively defends against MM:SS; `get()` returns a fresh instance each call.
168     - `@backend/providers/base.py::AIVideoProvider` ABC (`__init__(api_key, model_name)`);
`@backend/providers/gemini.py::GeminiProvider` (lazy `::client` property; `::MAX_UPLOAD_MB=500`;
upload->poll->generate JSON temp 0->delete); `@backend/providers/__init__.py::provider_registry`
(+`::ProviderRegistry`, auto-registers `gemini`). ? **add a provider**: subclass,
`provider_registry.register('name', MyCls)` (key passed EXPLICITLY); `get(name, api_key,
model_name)`. § `analyze_video(path, prompt, chunk_duration=0.0, label='') -> (segments:list,
metrics:dict{upload_time,process_time,gen_time})`. ? blocking 10s poll; returns `([], metrics)`
on ANY failure (empty != error); oversize clip refused LOUDLY (`logger.error` +
`metrics['oversize_skipped']=True` ? callers aggregate into the report's `oversize_clip_skipped`
rule); orphaned remote file possible if process dies mid-run.
169     - `@backend/annotations/base.py::AnnotationProvider` ABC (`submit/poll/fetch/annotate`),
`::Interval` (= metrics.Interval), `::QUEUED/RUNNING/DONE/FAILED`/`::STATUSES`,
`::AnnotationProviderRegistry` (decorator-friendly `register` returns the class; ? keys off
`provider_cls().name`, falls back to `__name__` if no-arg ctor throws; `list_available()` ->
LIST not dict), `::annotation_registry`. ? **add a tier**: subclass (define `name`),
`annotation_registry.register(MyCls)` at module bottom + import in `__init__.py`. § `fetch ->

```



```

List[Interval]` sorted; contract says RAISE (not `[ ]`) on unavailable.
170     - `@backend/annotations/file_provider.py::FileProvider` (`file`; job_id IS the
resolved CSV path; `_resolve` -> `

```

```

provider refused chunks over MAX_UPLOAD_MB ? 4K stream-copy footgun),
`ai_empty_vs_no_rallies`(warn), `fps_fallback_used`(warn), `synthetic_motion_metadata`(warn),
`segmenter_divergence`(info), `reingest_replaced_prior`(info), `validation_failed`(error: halts
indexing).
181 - `@backend/reporting/run_signals.py` ? ? in-process, thread-safe channel pipeline-
internals -> report (::incr(video_id,key,by), ::pop(video_id)). No obsreport dep (always
importable). Segmenter worker threads record facts the harvest can't see (e.g.
`oversize_clips_skipped`); `emit_indexer_report` pops UNCONDITIONALLY (even reporting-off, so
nothing accumulates) and passes to `record_run`. ? process-local best-effort: signals are
dropped (by design) if no report is emitted for that video_id.
182
183 ---
184
185 ## 3. DATA CONTRACTS (canonical shapes ? one place)
186
187 $ **Trajectory CSV** (produced by tracknet/wasb predict, consumed by
`parse_trajectory_csv`):
188 `Frame,Visibility,X,Y` ? header+column order fixed; `Visibility==1`=visible; X,Y pixel
coords.
189 -> `TrajectoryPoint{frame:int, visible:bool, x:float, y:float}` (frame-sorted).
190
191 $ **Candidate window dict** (produced by `trajectory_to_action_windows` AND yolo
`_extract_action_windows_from_chunk`):
192 `{start:float(sec), end:float(sec), active_frame_count:int, peak_velocity:float,
mean_velocity:float, track_id_count:int(==1 for trajectory), chunk_index:Optional[int]}`.
193
194 $ **candidate_segments DB row** (`db.add_candidate_segment` /
`query_candidate_segments`):
195 `{id, video_id, detector:str, chunk_index, start_time:REAL, end_time:REAL,
duration:REAL, confidence:REAL(min(1.0,peak_velocity)),
stats:JSON{peak_velocity,mean_velocity,active_frame_count,track_id_count},
detection_params:JSON{...}, confirmed_segment_id, human_label, notes, created_at}`
(stats/detection_params deserialized on read).
196 `detection_params` keys by detector:
wasb=`{detector,velocity_thresh,merge_gap,min_action_window_duration,weights_path}`; tracknet
adds `queue_length`; yolo=`{velocity_thresh,merge_gap,frame_skip,tracker,detector_model,detector
_score_threshold,min_action_window_duration}`.
197
198 $ **play_segment dict** (`db.add_play_segment` / `query_play_segments`):
199 `{id:int, video_id, start_time:float, end_time:float, duration:float(end-start), server,
receiver, metadata:dict, events:[...]}`. Hybrid P4 metadata `{shot_count,
shot_count_confidence:'Unknown', detector:f'{family}-hybrid-{model}'}`; gemini detector = bare
`model_name`; motion adds random `intensity/avg_speed_kmh`.
200
201 $ **jobs row** (`db.create_job` / `get_job` ? #16, PR-1): `{id:TEXT(uuid4.hex),
video_path, video_id?:TEXT(set once hashed), segmenter?, player_pool?:JSON list, max_frames?,
status:'queued'|'running'|'done'|'failed' (EXECUTION state ? pipeline verdict is
result["status"]), progress?:TEXT(stage label), error?:TEXT, result?:JSON(the former sync
/api/process response incl. reports), created_at, started_at?, finished_at?}`.
202
203 $ **AI segment dict** (provider output, consumed by hybrid P3 / gemini): `{start_time,
end_time, shuttle_exchange_count?, shot_count_confidence?}` (offset by chunk/window start;
`_candidate_id` stamped internally).
204
205 $ **GT / golden CSVs** (TWO conventions ? do not conflate):
206 - generic (`annotations.load_annotations`): `start,end` 2-col; optional header; decimal
or MM:SS/HH:MM:SS; RAISES on bad row.
207 - golden/slicer (`calibrate_wasb.load_golden_gts`, `rally_slicer.load_rally_labels`,
`export.SEG_FIELDS`): `rally_number,start_time,end_time[,duration,start_mmss,end_mmss]`;
SILENTLY skips bad rows.
208
209 $ **det_raw.jsonl** (one window/line): `{ "w":<window_index>,
"f":[[frame_id,[x,y,score,scale],...], ...]}`. Lines MUST be contiguous (line N has `w=N`);
first violation/JSONDecodeError -> truncate-heal.
210 $ **manifest.json** (cache): `{version, frames_dir(abspath), weights, sport, frames_in,

```

```

frames_total, windows_total, windows_done}`. Reuse only if ALL match + `version==CACHE_VERSION`.
211     $ **collector manifest.json** (eval): `{eval_sets:[{video_id(label), local_path, csv,
...}]}`; ? DB key = file MD5, manifest `video_id` is a label.
212
213     $ **eval primitives**: `Interval=Tuple[float,float]` (start<end).
`Match{pred_index,gt_index,iou}`. `MatchResult{matches, unmatched_pred_indices(=FP),
unmatched_gt_indices(=FN)}`. `Score{iou_threshold,num_pred,num_gt,true_positives,false_positives
,false_negatives,precision,recall,f1,mean_iou,mean_start_error,mean_end_error}.as_dict()`.
214
215     $ **WSL detector candidate** (in-WSL, model<->tracker): `{ 'xy':np.array([x,y]),
'score':float, 'scale':int}`; plain JSON form `[x,y,score,scale]`. ? `_dets_to_tracker` MUST
rebuild `xy` as np.array (tracker does `np.linalg.norm`).
216
217     $ **HitEvent** (audio, `pipeline/audio/hit_detector`): `{t:float(sec), strength:float}`
? onset-envelope peak.
218
219     $ **observability report** (`reporting/harvest.record_run` -> `RunRecorder`): stages
`validation/segmentation/ai_handoff` + highlights + verdict ? {pass,pass_with_warnings,fail};
flags `{code,faq_ref}`; `video_meta.fps_source` ? {probed,fallback}. ? stage/metric/detail names
are coupled to `spec.yaml::rules[].when[].path` ? pairs that MUST stay in sync:
`ai_handoff.details.{api_key_present,ai_based}`, `segmentation.metrics.segment_count`, `segmenta
tion.details.{segmenter_actual,matches_config_default,segmenter_explicitly_requested,was_reinges
t}`, `validation.status`, `video_meta.fps_source`.
220
221     ---
222
223     ## 4. WASB WSL CONTRACT (external dep `~/models/WASB-SBDT`, MIT; NOT in this repo ?
claims below are unverified against external source, only the repo-side caller wasb_infer.py was
checked)
224     - Env: WSL conda `wasb`, cwd `~/models/WASB-SBDT/src`. Hydra
`compose(config_name='eval', overrides=[dataset=<sport>, model=wasb,
detector.model_path=<weights>, runner.device=cuda, runner.gpus=[0]])`. `gpus=[0]` because box is
single-GPU. (overrides VERIFIED in @backend/pipeline/detectors/wasb_infer.py::build_cfg.)
225     - `model=wasb` -> `name=hrnet`, `frames_in=3`, `frames_out=3`, input 512x288 (WxH),
ImageNet normalize; affine resize maps heatmap coords -> original-frame coords. (model dims read
from cfg["model"] at runtime in wasb_infer.run.)
226     - `detectors/detector.py::TracknetV2Detector.run_tensor(imgs, affine_mats)` -> (results,
hms_vis). `results[bid][eid]` = list of `{xy:np.array (ORIGINAL coords), score, scale}`.
(caller uses `detector.run_tensor(imgs, trans)` returning `(batch_results, _)` ? consistent.)
227     - `trackers/online.py::OnlineTracker` ? STATEFUL, sequential per `update()` (assumes
every frame fed once, in order, no gaps). `update(frame_dets)->{x,y,visi,score}`; `refresh()`
resets. `det['xy']` MUST be np.array. Full ordered re-run is deterministic. (caller calls
`build_tracker(cfg)`, `tracker.refresh()`, `tracker.update(...)` returning dict with `visi/x/y`
? consistent.)
228     - `dataloaders/dataset_loader.py::ImageDataset(cfg, samples, input_wh, output_wh,
transform, seq_transform, is_train)`; `samples=[{frames:[paths], annos:[{frame_path,
center:Center(is_visible,x,y)}]]`. (caller constructs exactly this ? VERIFIED in
wasb_infer.run.)
229     - Weights `pretrained_weights/wasb_badminton_best.pth.tar` ? NOT committed, academic-
data-trained -> ? confirm terms before COMMERCIAL deploy. monotrack weights = noncommercial,
avoided.
230
231     ---
232
233     ## 5. CROSS-REPO (sibling private repo `sports-data-collector` ? external, unverified
here)
234     - Indexer keys video by **FILE MD5**; collector keys by human `video_id`. ? re-
encoding/re-downloading changes bytes -> MD5; acquisition MUST precede processing; manifest
tracks the exact file used.
235     - `python -m src.cli.curate export` (collector) -> `<out>/<sport>/<video_id>.csv`
(indexer `start,end` format) + `manifest.json` -> consumed by `@backend/eval/batch.py` +
`@run_phase3_eval.py`.
236     - Collector also emits golden rally labels (`rally_number,start_time,end_time,...`) ->
consumed by `@backend/eval/calibrate_wasb.py` / `export_wasb_segments.py` /
`@backend/tools/rally_slicer.py`.

```

```

237
238     ---
239
240     ## 6. GOTCHAS (cross-cutting footguns)
241     - **MD5 keying:** `video_id` = MD5 of WHOLE file via the single
`@backend/utils/hashing.py::compute_video_id` helper (64KB chunks), imported by main + all
run_*.py. Manifest `video_id` is a label, NOT the DB key.
242     - **Config access:** `backend/main.py` and all four `run_*.py` load via
`backend.config.load_config` (typed `AppConfig`). The legacy `.get()` dict-compat shim remains
for validators/segmenters. Remaining literal: a defensive `getattr(..., 'motion')` fallback in
`main.py` that only fires if `global_config`/.indexing` is `None` (the ASGI-import edge case).

```

6. user (2026-06-11T03:46:49.104Z)

```
.github\workflows\ci.yml
```

7. user (2026-06-11T03:46:49.138Z)

```

.git\description
.git\hooks\applypatch-msg.sample
.git\hooks\commit-msg.sample
.git\hooks\fsmonitor-watchman.sample
.git\hooks\post-update.sample
.git\hooks\pre-applypatch.sample
.git\hooks\pre-commit.sample
.git\hooks\pre-merge-commit.sample
.git\hooks\pre-push.sample
.git\hooks\pre-rebase.sample
.git\hooks\pre-receive.sample
.git\hooks\prepare-commit-msg.sample
.git\hooks\push-to-checkout.sample
.git\hooks\sendemail-validate.sample
.git\hooks\update.sample
.git\info\exclude
.git\objects\pack\pack-bb67e687d593e38c8658f1a24d366f64c2d74657.idx
.git\objects\pack\pack-bb67e687d593e38c8658f1a24d366f64c2d74657.rev
.git\logs\refs\remotes\origin\HEAD
.git\packed-refs
.git\refs\remotes\origin\HEAD
backend\validators\__pycache__\__init__.cpython-311.pyc
backend\validators\__pycache__\base.cpython-311.pyc
backend\validators\__pycache__\format.cpython-311.pyc
backend\validators\__pycache__\resolution_fps.cpython-311.pyc
backend\validators\__pycache__\pts_continuity.cpython-311.pyc
backend\validators\__pycache__\scene_cut.cpython-311.pyc
backend\validators\__pycache__\blur.cpython-311.pyc
backend\validators\__pycache__\relevance.cpython-311.pyc
backend\indexer\__pycache__\segmenter.cpython-311.pyc
backend\__pycache__\main.cpython-311.pyc
.git\objects\8b\eaf212c3a70053dc20e805931ac69db0f1de3c
.git\objects\cf\d7ac85c275382b2b66feed350afb26c94c4c3
.git\objects\97\246179ab35222e8e4ed483786105338a48f680
.git\objects\ee\09b41ee23fb3c62ee0ab0d76261efdb53416ac
.git\objects\63\c63c2b061321d2b854dde57d6998cb1c4753ea
.git\objects\28\02267e5e50dab885947d1e412950adc90d7fed
.git\objects\72\f972e990f8612bfb575a7f6ab9ca8ff17c8603
backend\indexer\__pycache__\motion.cpython-311.pyc
backend\indexer\__pycache__\base.cpython-311.pyc
.git\objects\5f\eb1d78333f8bd5189b3969e203f766e75c75b9
.git\objects\ef\f6bd05a297604667d1572a0266980942eb7dec
.git\objects\9d\87d1b2a83b10058396d07707427663e2312a0b
.git\objects\ba\6f6a6a04657b40238c6dcebb0ef55401b6cf5e
.git\objects\65\6fe4938c15aa5429893a47f54c2b9686eb448c
.git\objects\c9\b65fc4188f47fbc322d608c3f8bb4035b0e002
.git\objects\24\b3f759c785e834440b03067f5170573a568750

```

.git\objects\f8\1b2bb37eae31b8e36fa6d0a298c493df268953
.git\objects\9d\b3bd5311130448d1837520ed38b849ce84ee28
.git\objects\4c\9de860c7aea5f2f74e5f58ed1c4c494fa99d26
.git\objects\e0\2fe56b468f4dd44dd1a1a7933229a32ad9e27f
.git\objects\87\4ae31bb6152ec668dd19e317c183a721231b38
.git\objects\af\867722341933de70270ebecd7344e2db18d1f2
.git\objects\68\dca6a47e9e4ae65ae3ab63a99f8e2e3f284fb2
.git\objects\58\8b912dfa3a9f213330756b4798fb1e71829828
.git\objects\77\6a04ffb03f989c031f8cb04cf3d90d2e6b44be
.git\objects\7d\22901dda8359816213eebf08024918d8b0f650
.git\objects\0f\76429a9676618467e80f4f59ee2631a814fdc8
.git\objects\3a\6f972b02acdcddeb2f5aff7e54978a52164be31
.git\objects\22\40ea8835ebe0309986b26d4dff07963fef5a5d
.git\objects\71\c9616bf6f17e2f0b662c3491eef5806de45a21
.git\objects\e2\50e37e1bd31aedab1af911d23d8710d3fd2fdc
.git\objects\11\9d996e50c6364d365bc35364b55fca4b92f146
.git\objects\d6\4bd30be185ae8cb71af704b84ee3e93029d610
.git\objects\0b\79ab61c4584ea7420b537b38c7b74acf266ee0
.git\objects\ae\8d0b999ebbf6b4ae2afdcc9b3dd4639707f9ad
.git\objects\a3\11d20786abc34576e33340f85be790933f52bc
.git\objects\a4\66ef801b6b36c557a5b6379b9c07f203a4f266
.git\objects\5c\3d8d3c67d29f0ad1ff2bea4a54e4b354066238
.git\objects\f2\cf346209e0596cd33bd0f935910b6eb2f52db4
.git\objects\8f\4c564ffa6f22123536f653539e42e928b97ddd
.git\objects\7c\648f37cf8d2045e81a72c72aac87fe4f752abd
.git\objects\40\e8b49c342f346da4b3b6ec4bb4251392e6f3f3
.git\objects\02\4a0b33dd37b4573221d7b22c948b4292c0a3f4
.git\objects\1c\c668ef271a98975791275fccfda3cfc1f8708c
.git\objects\95\8c6be4424a7aff8e676f0135f2aac1a2085aeb
.git\objects\b1\ef0927af4432ce5fc9356b846ded407ab935e5
.git\objects\58\36e1e280cb87d007a8b3f5b4da3e90566ebdfc
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.git\objects\ad\01cdc8dccb114bec80ffb3fa143ca23e3e451b
.git\objects\90\f8dfd1261a89dec3b94e920f4af9cc6b0866b2
.git\objects\12\25d2988599e1bb77e1a0d936a2b0bed1f534fc
.git\objects\2a\78ee87cfce9f2b28560f7649353522ddad9bc5
.git\objects\b4\8691b8fda1428b447678ec88ae921638d47ad7
.git\objects\b9\d7e19ba0fc37de2b41411b9a6dadeaa0cb4032
.git\objects\79\81ff1840d572636eaf535a8c85cfccf1a4f36b
.git\objects\f0\896911857841be46508f0c793e52fe944b77fc
.git\objects\07\4046a0bb4c1f112ae3bed8d6cd24f25885488f
.git\objects\6c\05463ef836db819968fa20372d928c5ddfc007
.git\objects\7\34ec303a15aa4df177cf41c83fd530b11d5d57
.git\objects\79\ba540dc5ca38bc73c134880951a8454235ad50
.git\objects\01\92df71b8d6679240a78f2f0bc204d00a2c23a6
.git\objects\a3\2fca519b7326920aac529c65bcf60f62b8c228
.git\objects\30\d74d258442c7c65512eafab474568dd706c430
.git\objects\2f\ab7d4941e19075dc08235ee8992e1c281a0257
.git\objects\99\28bbb4afc869bdc84a8a09adcbd2370f646413
.git\objects\8f\3e7ad5f3eec0c087e5f2520a711bb43904e800
.git\objects\0a\1a459b04a7792f6508af6afbc3857aa54ec06b
.git\objects\24\aa8b66ee3176c9f0c440b988d44626fb6f0a51
.git\objects\75\4516c40a4dc96bac17cce59744cb24e1490122
(Results are truncated. Consider using a more specific path or pattern.)

8. user (2026-06-11T03:47:10.825Z)

Mode	Name
----	----
d----	__pycache__
d----	.claude
d----	.github
d----	.pytest_cache
d----	.sixth
d----	artifacts

```
d---- backend
d---- docs
d---- eval_baselines
d---- output
d---- scratch
d---- scripts
d---- tests
d---- training
-a--- .coverage
-a--- .coveragerc
-a--- .env
-a--- .gitignore
-a--- CLAUDE.md
-a--- config.json
-a--- pytest.ini
-a--- README.md
-a--- requirements.txt
-a--- run_all_tests.py
-a--- run_eval.py
-a--- run_phase3_eval.py
-a--- run_process_and_stitch.py
-a--- run_regression.py
-a--- setup.py
-a--- TEST_RUN_SUMMARY.md
```

9. user (2026-06-11T03:47:10.851Z)

```
1      name: CI
2
3      on:
4        push:
5          branches: [master]
6        pull_request:
7
8      permissions:
9        contents: read
10
11     concurrency:
12       group: ci-${{ github.ref }}
13       cancel-in-progress: true
14
15     jobs:
16       # Fast guard against the 7fbb0 class of breakage: a SyntaxError or
17       # module-level import error shipping while the suite stays green.
18       # Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
19       # on machines without the GPU stack (test_import_smoke stubs what's absent).
20       import-smoke:
21         name: Syntax sweep + import smoke (no GPU stack)
22         runs-on: ubuntu-latest
23         steps:
24           - uses: actions/checkout@v4
25           - uses: actions/setup-python@v5
26           with:
27             python-version: "3.13"
28           - name: Install minimal deps
29             run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
30genai
31           - name: Run import smoke tests
32             run: python -m pytest tests/test_import_smoke.py -v
33
34     full-suite:
35       name: Full test suite (offline, mocked)
36       runs-on: ubuntu-latest
37       steps:
38         - uses: actions/checkout@v4
```

```

38         - uses: actions/setup-python@v5
39         with:
40             python-version: "3.13"
41             cache: pip
42         - name: System libs for opencv-python
43           run: |
44             sudo apt-get update
45             sudo apt-get install -y libgl1 libglu1-mesa-dev || sudo apt-get install -y
libgl1 libglu1-mesa-dev
46         - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
47           run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
48         - name: Install requirements
49           run: python -m pip install -r requirements.txt pytest httpx # httpx: fastapi
TestClient
50         # obsreport (private, soft dep) is absent here: reporting tests skip via
51         # pytest.importorskip ? run_all_tests.py prints which.
52         - name: Run full suite
53           run: python run_all_tests.py
54

```

10. user (2026-06-11T03:47:11.603Z)

Command running in background with ID: bbyi28x9g. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\bbyi28x9g.output. You will be notified when it completes. To check interim output, use Read on that file path.

11. assistant (2026-06-11T03:47:18.494Z)

Suite is running in the background. While it runs, I'll inventory tests, dependencies, and sample core code quality directly.

12. user (2026-06-11T03:47:24.984Z)

Source files: 76 lines: 8172

Test files: 52 lines: 3883

Test files list:

```

__init__.py
test_annotation_providers.py
test_api_endpoints.py
test_async_jobs.py
test_automated_provider.py
test_base.py
test_calibrate_local.py
test_calibrate_wasb.py
test_calibration.py
test_compiler.py
test_config_wasb.py
test_config.py
test_database.py
test_detector_keying.py
test_distill_local.py
test_eval_annotations.py
test_eval_batch.py
test_eval_classifier.py
test_eval_harness.py
test_eval_metrics.py
test_eval_partial_labels.py
test_eval_regression.py
test_eval_training.py
test_export_wasb_segments.py
test_gemini_refine.py
test_gemini.py

```

```

test_geometry.py
test_gt_loader.py
test_hashing.py
test_hit_detector.py
test_import_smoke.py
test_path_safety.py
test_provider_gemini.py
test_providers.py
test_rally_gate.py
test_rally_seq_proto.py
test_rally_slicer.py
test_reingest_safety.py
test_reliability_hardening.py
test_reporting_harvest.py
test_reporting_spec_faq.py
test_reporting.py
test_run_signals.py
test_sport_profiles.py
test_tracknet_hybrid.py
test_tracknet_runner.py
test_training_gen0.py
test_validators_base.py
test_video.py
test_wasb_cache.py
test_wasb_runner.py
test_yolo_hybrid.py

```

13. user (2026-06-11T03:47:25.011Z)

```

1      #!/usr/bin/env python
2      """Run the FULL test suite for the badminton-highlight-indexer.
3
4      A single, discoverable entry point so you (or a reviewer / CI) can verify the
5      whole suite before opening a PR:
6
7          python run_all_tests.py          # full suite, quiet
8          python run_all_tests.py -v       # verbose (passthrough to pytest)
9          python run_all_tests.py -k config # filter by keyword (passthrough)
10         python run_all_tests.py --cov=backend --cov-report=term-missing # coverage
11
12     Anything after the script name is passed straight through to pytest.
13
14     Notes
15     -----
16     - The suite is fully mocked (no GPU, no network, no WSL) and runs offline.
17     - The observability **reporting** tests skip cleanly if the optional ``obsreport``
18       package isn't installed (they use ``pytest.importorskip``). To exercise them too:
19       pip install -e ../sports-obsreport --no-build-isolation
20     - Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).
21     """
22     from __future__ import annotations
23
24     import importlib.util
25     import subprocess
26     import sys
27
28
29     def _installed(module: str) -> bool:
30         try:
31             return importlib.util.find_spec(module) is not None
32         except (ImportError, ValueError):
33             return False
34
35
36     def main() -> int:

```



```

37     args = list(sys.argv[1:])
38     # Default to quiet unless the caller chose a verbosity.
39     if not any(a in ("-q", "-v", "-vv", "-vvv", "--quiet", "--verbose") for a in args):
40         args = ["-q", *args]
41     cmd = [sys.executable, "-m", "pytest", *args]
42
43     obs = "available" if _installed("obsreport") else "NOT installed ? reporting tests
will SKIP"
44     bar = "=" * 72
45     print(bar)
46     print(" Full test suite: " + " ".join(cmd))
47     print(f" obsreport (optional reporting dep): {obs}")
48     print(bar, flush=True)
49
50     rc = subprocess.run(cmd).returncode
51
52     print(bar)
53     print(" RESULT: " + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit
{rc}"))
54     print(bar)
55     return rc
56
57
58 if __name__ == "__main__":
59     raise SystemExit(main())
60

```

14. user (2026-06-11T03:47:25.027Z)

```

1     fastapi>=0.111.0
2     uvicorn>=0.30.1
3     opencv-python>=4.9.0.80
4     numpy>=1.26.4
5     pydantic>=2.7.1
6     jinja2>=3.1.4
7     python-dotenv>=1.0.0
8     google-genai>=2.4.0
9     torch>=2.12.0
10    torchvision>=0.27.0
11    supervision>=0.21.0,<0.30.0
12
13    # Per-video observability reports (SOFT dependency: the pipeline degrades to a
14    # no-op if absent). Shared, permissively-licensed report toolkit. Pin a tag once
15    # the private repo is published, e.g.:
16    #   obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3
17    # For local development install it editable from a sibling checkout:
18    #   pip install -e ../sports-obsreport --no-build-isolation
19

```

15. user (2026-06-11T03:47:38.185Z)

```

1     """Pure temporal interval-matching metrics for rally evaluation.
2
3     No I/O, no DB, no global state ? every function here is deterministic and unit
4     tested. An "interval" is a `(start, end)` tuple of floats in seconds with
5     start < end. We score a list of *predicted* rally intervals against a list of
6     *ground-truth* intervals via temporal IoU and one-to-one greedy matching.
7     """
8
9     from dataclasses import dataclass, field
10    from typing import List, Tuple, Optional
11
12    Interval = Tuple[float, float]
13
14

```

```

15 def interval_iou(a: Interval, b: Interval) -> float:
16     """Temporal intersection-over-union of two intervals.
17
18     Returns 0.0 when they don't overlap (or either is degenerate). IoU is
19     overlap / union, both measured as durations on the timeline.
20     """
21     a_start, a_end = a
22     b_start, b_end = b
23     if a_end <= a_start or b_end <= b_start:
24         return 0.0
25     inter = max(0.0, min(a_end, b_end) - max(a_start, b_start))
26     if inter <= 0.0:
27         return 0.0
28     union = (a_end - a_start) + (b_end - b_start) - inter
29     return inter / union if union > 0 else 0.0
30
31
32 @dataclass
33 class Match:
34     """One matched predicted?ground-truth pair."""
35     pred_index: int
36     gt_index: int
37     iou: float
38
39
40 @dataclass
41 class MatchResult:
42     """Outcome of matching predictions to ground truth at a given IoU threshold."""
43     matches: List[Match] = field(default_factory=list)
44     unmatched_pred_indices: List[int] = field(default_factory=list) # false positives
45     unmatched_gt_indices: List[int] = field(default_factory=list)   # false negatives
46 (misses)
47
48 def match_intervals(preds: List[Interval], gts: List[Interval],
49                     iou_threshold: float = 0.5) -> MatchResult:
50     """Greedy one-to-one matching of predictions to ground truth by descending IoU.
51
52     Every candidate (pred, gt) pair with IoU >= threshold is considered; pairs
53     are claimed highest-IoU first, and each prediction/GT can be used once. This
54     keeps a single prediction from being credited for two ground-truth rallies
55     (and vice versa).
56     """
57     candidates = []
58     for pi, p in enumerate(preds):
59         for gi, g in enumerate(gts):
60             iou = interval_iou(p, g)
61             # Require real overlap: at iou_threshold=0.0 a plain `iou >= threshold`
62             # match completely disjoint intervals (IoU 0.0), fabricating perfect scores.
63             if iou > 0.0 and iou >= iou_threshold:
64                 candidates.append((iou, pi, gi))
65     # Highest IoU first; ties broken by indices for determinism.
66     candidates.sort(key=lambda c: (-c[0], c[1], c[2]))
67
68     used_pred, used_gt = set(), set()
69     matches: List[Match] = []
70     for iou, pi, gi in candidates:
71         if pi in used_pred or gi in used_gt:
72             continue
73         used_pred.add(pi)
74         used_gt.add(gi)
75         matches.append(Match(pred_index=pi, gt_index=gi, iou=iou))
76
77     unmatched_pred = [i for i in range(len(preds)) if i not in used_pred]

```

```

78         unmatched_gt = [i for i in range(len(gts)) if i not in used_gt]
79         return MatchResult(matches=matches,
80                             unmatched_pred_indices=unmatched_pred,
81                             unmatched_gt_indices=unmatched_gt)
82
83
84     @dataclass
85     class Score:
86         """Aggregate metrics for one (predictions vs ground-truth) evaluation."""
87         iou_threshold: float
88         num_pred: int
89         num_gt: int
90         true_positives: int
91         false_positives: int
92         false_negatives: int
93         precision: float
94         recall: float
95         f1: float
96         mean_iou: float          # over matched pairs (0.0 if none)
97         mean_start_error: float  # mean |pred_start - gt_start| over matches, seconds
98         mean_end_error: float    # mean |pred_end - gt_end| over matches, seconds
99
100        def as_dict(self) -> dict:
101            return self.__dict__.copy()
102
103
104    def score(preds: List[Interval], gts: List[Interval],
105              iou_threshold: float = 0.5,
106              match_result: Optional[MatchResult] = None) -> Score:
107        """Compute precision/recall/F1, mean matched IoU, and boundary errors.
108
109        Pass a precomputed `match_result` to avoid re-matching (e.g. when also
110        extracting the failure lists); otherwise matching is done here.
111        """
112        mr = match_result if match_result is not None else match_intervals(preds, gts,
113        iou_threshold)
114
115        tp = len(mr.matches)
116        fp = len(mr.unmatched_pred_indices)
117        fn = len(mr.unmatched_gt_indices)
118
119        precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
120        recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
121        f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) > 0
122        else 0.0

```

16. user (2026-06-11T03:47:38.659Z)

```

1     import sqlite3
2     import json
3     import os
4     import logging
5     from typing import List, Dict, Any, Optional, Tuple
6
7     logger = logging.getLogger(__name__)
8
9     class Database:
10         def __init__(self, db_path: str):
11             self.db_path = db_path
12             self._init_db()
13
14         def _get_connection(self):
15             # busy_timeout: wait (not fail) on a locked DB ? needed once the async job model
16             # adds concurrent writers (a swallowed 'database is locked' silently drops
17             segments).

```

```

17     conn = sqlite3.connect(self.db_path, timeout=30.0)
18     conn.row_factory = sqlite3.Row
19     # Enforce ON DELETE CASCADE / SET NULL ? SQLite leaves FKs off per-connection.
20     conn.execute("PRAGMA foreign_keys = ON")
21     # WAL allows concurrent readers during a write; busy_timeout bounds lock waits.
22     conn.execute("PRAGMA busy_timeout = 30000")
23     try:
24         conn.execute("PRAGMA journal_mode = WAL")
25     except sqlite3.OperationalError:
26         pass # e.g. a read-only/networked FS that can't do WAL ? degrade gracefully
27     return conn
28
29 def _init_db(self):
30     """Initializes tables for videos, play_segments, and events."""
31     with self._get_connection() as conn:
32         cursor = conn.cursor()
33
34         # Videos Table
35         cursor.execute("""
36             CREATE TABLE IF NOT EXISTS videos (
37                 id TEXT PRIMARY KEY,
38                 filepath TEXT NOT NULL,
39                 processed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
40                 status TEXT NOT NULL,
41                 validation_results TEXT
42             )
43         """)
44
45         # Play Segments Table (Extensible metadata JSON)
46         cursor.execute("""
47             CREATE TABLE IF NOT EXISTS play_segments (
48                 id INTEGER PRIMARY KEY AUTOINCREMENT,
49                 video_id TEXT NOT NULL,
50                 start_time REAL NOT NULL,
51                 end_time REAL NOT NULL,
52                 duration REAL NOT NULL,
53                 server TEXT,
54                 receiver TEXT,
55                 metadata TEXT,
56                 FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
57             )
58         """)
59
60         # Events Table
61         cursor.execute("""
62             CREATE TABLE IF NOT EXISTS events (
63                 id INTEGER PRIMARY KEY AUTOINCREMENT,
64                 segment_id INTEGER NOT NULL,
65                 timestamp REAL NOT NULL,
66                 type TEXT NOT NULL,
67                 player TEXT,
68                 details TEXT,
69                 FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE
70             )
71         """)
72
73         # Versioned migrations live here. PRAGMA user_version is the schema
74         # contract ? bump it for every change and add an ordered `if version < N`
75         # block below. Tests assert the expected version, so accidental drift fails
76         version = cursor.execute("PRAGMA user_version").fetchone()[0]
77
78         if version < 1:
79             # v1: raw candidate detections (pre-confirmation), detector-agnostic.

```

```

80         # Common fields are columns; detector-specific metrics go in `stats`
JSON,
81         # so a new segmenter reuses this table by setting its own
`detector`/`stats`.
82     cursor.execute("""
83         CREATE TABLE IF NOT EXISTS candidate_segments (
84             id INTEGER PRIMARY KEY AUTOINCREMENT,
85             video_id TEXT NOT NULL,
86             detector TEXT NOT NULL,
87             chunk_index INTEGER,
88             start_time REAL NOT NULL,
89             end_time REAL NOT NULL,
90             duration REAL NOT NULL,
91             confidence REAL,
92             stats TEXT,
93             detection_params TEXT,
94             confirmed_segment_id INTEGER,
95             human_label TEXT,
96             notes TEXT,
97             created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
98             FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE,
99             FOREIGN KEY (confirmed_segment_id) REFERENCES play_segments(id)
ON DELETE SET NULL
100         )
101     """)
102     cursor.execute("""
103         CREATE INDEX IF NOT EXISTS idx_candidates_video_detector
104             ON candidate_segments(video_id, detector)
105     """)
106     cursor.execute("PRAGMA user_version = 1")
107
108     if version < 2:
109         # v2: async job model (DEFERRED_HARDENING_PLAN #16). One row per
submitted
110         # /api/process request. `status` is the EXECUTION state of the job
111         # (queued|running|done|failed); the pipeline's own verdict (e.g. a video
112         # that failed validation) lives inside `result` JSON as
result["status"].
113         # `result` is the full sync-era response dict (incl. report paths), so
the
114         # observability report is the job's artifact, per the plan.
115         cursor.execute("""
116             CREATE TABLE IF NOT EXISTS jobs (
117                 id TEXT PRIMARY KEY,
118                 video_path TEXT NOT NULL,
119                 video_id TEXT,
120                 segmenter TEXT,
121                 player_pool TEXT,
122                 max_frames INTEGER,
123                 status TEXT NOT NULL DEFAULT 'queued',
124                 progress TEXT,
125                 error TEXT,
126                 result TEXT,
127                 created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
128                 started_at TIMESTAMP,
129                 finished_at TIMESTAMP
130             )

```

17. user (2026-06-11T03:47:42.260Z)

```

=== docs/ ===
archives
mockups
BACKLOG.md
CODE_MAP.md

```

```

COMMERCIALIZATION.md
DEFERRED_HARDENING_PLAN.md
EVALUATION.md
FIELD_RESEARCH_ASSOCIATE_FLYER.md
GOLDEN_SET_IMPLEMENTATION.md
GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PLATFORM_ARCHITECTURE.md
PMF_MARKET_RESEARCH.md
README.md
ROADMAP.md
TRACKNET_WSL_SETUP.md
UI_ALPHA_EXECUTION_PLAN.md
UI_PROPOSAL.md
VIDEO_RECORDING_GUIDELINES.md
=== git stats ===
Total commits: 189
First commit: 2026-05-18
=== last 12 commit subjects ===
docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009 (#89)
feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep (#87)
docs: field-research recruiter flyer (C13 interviews + Phase-2 recording rules) (#86)
docs: NEXT_STEPS session wrap ? quality table, ship-target 0.66, corpus-growth priorities (#85)
eval: record the Gemini-wrapper baselines (the commoditization/moat reference) (#84)
fix(provider): 6 generate attempts w/ exponential backoff+jitter (503 waves outlast 3x linear) (#83)
test(reporting): pin the CI-visible half of the oversize channel (no obsreport needed) (#82)
test(reporting): pin the CI-visible half of the oversize channel (no obsreport needed) (#81)
fix(gemini): re-encode AI chunks (4K oversize skip = silent 0-segments) + loud report flag (#80)
fix(gemini): honor indexing.ai_max_workers (was hardcoded 5); ship serial default (#79)
chore(config): ai_model -> gemini-flash-latest (auto-tracking alias) (#78)
feat(training): Gen-0 TAL harness ? one-command train -> honest LOVO -> ship-gate -> artifact (M0.4 scaffold) (#77)

```

18. user (2026-06-11T03:48:02.627Z)

```

=== pytest.ini ===
[pytest]

```

When `pytest` is invoked without an explicit path, only collect from tests/.scratch/ holds debugging scripts that hit real APIs (Gemini upload, etc.) and should never run as part of the regular test suite.

```

testpaths = tests
=== .coveragerc ===
[run]
omit =
    backend/main.py
    backend/indexer/motion.py
    backend/validators/*
=== lint/type configs ===
ruff.toml : False
.ruff.toml : False
pyproject.toml : False
setup.cfg : False
.flake8 : False
mypy.ini : False
.pre-commit-config.yaml : False
=== tracked? ===
tracked: TEST_RUN_SUMMARY.md
tracked: config.json

```

=== .gitignore head ===

Byte-compiled / optimized / DLL files

```
__pycache__/  
*.py[cod]  
*$py.class  
*.pt
```

Local environment secrets

```
.env  
*.env  
.env.*
```

C extensions

```
*.so
```

Distribution / packaging

```
.Python  
build/  
develop-eggs/  
dist/  
downloads/  
eggs/  
.eggs/  
lib/  
lib64/  
parts/
```

19. assistant (2026-06-11T03:48:08.083Z)

Test suite finished ? checking the result.

20. user (2026-06-11T03:48:08.876Z)

```
1      obsreport (optional reporting dep): available  
2      =====  
3      ..... [ 20%]  
4      ..... [ 41%]  
5      ..... [ 61%]  
6      ..... [ 82%]  
7      ..... [100%]  
8      ===== warnings summary =====  
9      tests/test_api_endpoints.py::test_process_success_with_mocked_segmenter  
10     tests/test_async_jobs.py::test_submit_returns_202_with_job_id  
11     tests/test_async_jobs.py::test_job_done_with_segments_and_video_id  
12     tests/test_async_jobs.py::test_worker_crash_marks_failed_and_restores_prior_segments  
13     tests/test_async_jobs.py::test_progress_stages_are_recorded  
14     tests/test_async_jobs.py::test_real_executor_completes_job  
15     C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:168:  
PydanticDeprecatedSince20: The `dict` method is deprecated; use `model_dump` instead. Deprecated  
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2 Migration Guide at  
https://errors.pydantic.dev/2.13/migration/  
16     db_instance.add_video(video_id, req.video_path, "processed", [r.dict() for r in  
val_results])  
17  
18     tests/test_api_endpoints.py::test_process_success_with_mocked_segmenter  
19     tests/test_async_jobs.py::test_submit_returns_202_with_job_id  
20     tests/test_async_jobs.py::test_job_done_with_segments_and_video_id  
21     tests/test_async_jobs.py::test_progress_stages_are_recorded  
22     tests/test_async_jobs.py::test_real_executor_completes_job  
23     C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:221:  
PydanticDeprecatedSince20: The `dict` method is deprecated; use `model_dump` instead. Deprecated  
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2 Migration Guide at
```

```

https://errors.pydantic.dev/2.13/migration/
24         "results": [r.dict() for r in val_results],
25
26     tests/test_api_endpoints.py::test_process_validation_failure_preserves_prior_segments
27     tests/test_async_jobs.py::test_job_done_but_pipeline_verdict_failed_preserves_prior
28     C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:157:
PydanticDeprecatedSince20: The `dict` method is deprecated; use `model_dump` instead. Deprecated
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2 Migration Guide at
https://errors.pydantic.dev/2.13/migration/
29         db_instance.add_video(video_id, req.video_path, "failed", [r.dict() for r in
val_results])
30
31     tests/test_api_endpoints.py::test_process_validation_failure_preserves_prior_segments
32     tests/test_async_jobs.py::test_job_done_but_pipeline_verdict_failed_preserves_prior
33     C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:162:
PydanticDeprecatedSince20: The `dict` method is deprecated; use `model_dump` instead. Deprecated
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2 Migration Guide at
https://errors.pydantic.dev/2.13/migration/
34         "results": [r.dict() for r in val_results],
35
36     -- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
37     349 passed, 15 warnings in 40.13s
38     =====
39     RESULT: PASS - all tests green
40     =====
41

```

21. assistant (2026-06-11T03:49:27.819Z)

Init done ? I read the handoff + current-state, ramped on [CODE_MAP.md](docs/CODE_MAP.md), and then gathered fresh evidence rather than relying on memory: ran the full suite (**349 passed, 0 failed, 40s**, fully offline), read the CI workflow, sampled core modules ([metrics.py](backend/eval/metrics.py), [database.py](backend/infrastructure/database.py)), and checked tooling configs and git history (189 commits since 2026-05-18).

Overall: 8/10 ? unusually disciplined for a pre-product solo repo

The headline: this repo's **process and knowledge-management rigor is well above industry norm**, its architecture is genuinely good, and its weaknesses are mostly *known, documented, and deliberately deferred* rather than unknown ? which is itself a quality signal. The one undeniable hole is static analysis: there is no linter, formatter, or type-checker anywhere.

Dimension	Score	Evidence
Architecture & design	9	Five pluggable registry seams (segmenters/providers/validators/sports/annotations), ABC contracts, typed Pydantic config with single-sourced defaults, `PRAGMA user_version` migration ladder, train==serve featurizer behind a CI-enforced import wall
Documentation	9.5	CODE_MAP with data contracts + footgun markers, 9 ADRs, archives/checkpoints, honest negative results recorded (audio E1 rejected, failed heuristics kept)
Process & governance	9	One-PR-per-item, conventional commits, merges gated on verified-green CI, licensing gate that *actually caught* violations (ShuttleSet reclassification), audit?12-PR remediation loop
Eval/ML rigor	9	Honest LOVO, leakage guards, midrank-tie AUC, committed baselines in `eval_baselines/`, regression gate with semantic exit codes, results independently reproduced
Testing	8	349 green in 40s, 51 modules, ~3.9k test lines vs ~8.2k source, negative-path tests (re-ingest safety, path traversal, detector keying)
Code quality	7.5	Sampled modules are clean with *why*-comments; minus for documented copy-paste twins and legacy shims
CI/CD	6.5	Smart two-lane design (import-smoke without GPU stack + full suite), least-privilege, concurrency-cancel ? but only 1 day old and no lint/type/coverage lane
Security	7	API path validation, gitignored secrets, `permissions: contents: read`; pending chat-pasted Gemini key rotation (tracked)

| Static analysis tooling | 3 | No ruff/flake8/black/mypy/pre-commit, nothing in CI |
| Packaging | 4 | No `pyproject.toml`, not pip-installable ? deferred with intent in BACKLOG,
and it already cost you (it drove the repo-split analysis) |

What's genuinely excellent

The **honesty loop** is the standout. Most repos this age have aspirational docs; this one has a CODE_MAP whose header says "trust git history over stale lines," a motion segmenter explicitly labeled as fabricating random metadata, and an audit trail where the project found its own master broken (the compiler SyntaxError that 254 green tests never noticed), fixed it, added CI specifically shaped to catch that class again (`test\_import\_smoke.py` does a `py\_compile` sweep plus a stubbed `import backend.main`), and wrote the lesson down. The eval culture ? committed baselines, LOVO instead of fit-on-self numbers, a ship-gate that honestly reported `ship=False` on its own first model ? is rarer than good code.

The sampled code backs the docs up: [metrics.py](backend/eval/metrics.py:63) has a deliberate `iou > 0.0` guard with a comment explaining the fabricated-perfect-score failure mode it prevents, deterministic tie-breaking, and pure no-I/O design; [database.py](backend/infrastructure/database.py:73) has a versioned migration ladder with test-asserted schema version and graceful WAL degradation.

Where it loses points

1. **No static analysis at all** ? the biggest gap, and the cheapest to close. The suite output itself shows the cost: 15 Pydantic deprecation warnings (`.dict()` ? `model\_dump()` at [main.py:168](backend/main.py:168), 157, 162, 221) that a ruff lane would have flagged. A `ruff check` + `ruff format --check` job in CI is an afternoon of work.
2. **Documented-but-live duplication** ? `tracknet\_hybrid`/`wasb\_hybrid` are copy-paste twins where "any P3/P4 fix must be applied to BOTH," `\_wsl\_bash` is duplicated across runners, and padding constants are duplicated between the compiler and rally_slicer. The CODE_MAP flags all of these, but flags don't prevent drift.
3. **Legacy tolerance footguns** ? `extra='allow'` on AppConfig keeps typo'd config keys silently, the dict-compatible `.get()` shim and the dict-returning legacy `load\_config` are still load-bearing, and two incompatible GT CSV conventions coexist (mitigated by the canonical `gt\_loader`, but the old loaders still exist).
4. **Small hygiene nits**: .coveragerc omits paths that no longer exist (`backend/indexer/motion.py`, `backend/validators/\*` ? pre-restructure layout), and `TEST\_RUN\_SUMMARY.md` is a tracked point-in-time artifact at the root.
5. **CI is young** ? the repo ran its first ~3 weeks without CI, and the coverage-inversion lesson (tests dense where risk was low) was only corrected in the 2026-06-10 sweep. Current state is good; the track record is short.

If you want to convert 8 ? 9

In priority order: (1) a ruff + mypy CI lane with the two-line `dict()` fixes, (2) `pyproject.toml` packaging, (3) collapse the hybrid-segmenter twins into one parameterized base, (4) flip `extra='allow'` to `forbid` once the legacy shim retires. All four are already implicitly tracked in your own BACKLOG/deferred-hardening docs ? which is rather the point: this repo knows where its bodies are buried, and that's what makes it an 8 rather than a 6.5 with the same code.

Happy to turn any of these into a PR-sized work item if you want ? say the word (per the workflow agreement, I won't start one unprompted).

22. user (2026-06-11T04:36:40.816Z)

I want to go 8 -> 9! So please help me with that! Can you make an itemized list of AIs you can take up and once I approve feel free to make individual PRs and go on merging.

23. assistant (2026-06-11T04:39:30.660Z)

Excellent ? here's the itemized list. I've ordered it as the execution sequence (small, de-risking wins first; the structural refactor near the end once the new CI lanes can guard it).

Each is one branch ? one PR ? merge-on-green, per the repo convention.

Proposed action items (8 ? 9)

****Quick wins****

1. ****AI-1 · Repo hygiene**** (S, no risk) ? Fix the stale `[.coveragerc](.coveragerc)` omit paths (``backend/indexer/``, ``backend/validators/`` no longer exist post-restructure; repoint to their current ``backend/pipeline/`` equivalents). Untrack ``TEST_RUN_SUMMARY.md`` (point-in-time artifact) and gitignore it ? after checking nothing references it. Also sweep the stale pre-restructure ``__pycache__`` dirs locally (not a PR concern, just dirt).
2. ****AI-2 · Pydantic v2 cleanup**** (S, low risk) ? Replace the four deprecated ``dict()`` calls in `[main.py](backend/main.py:157)` with ``model_dump()``, plus a grep for any other Pydantic v1 idioms (``json()``, ``parse_obj()``). Kills the 15 deprecation warnings in the suite; removes a v3 time bomb.

****CI quality lanes**** (each becomes a new job in `[ci.yml](.github/workflows/ci.yml)`)

3. ****AI-3 · Ruff lint lane**** (M, low-med risk) ? Add ``ruff.toml`` (target py38 to match the stated floor; pragmatic ruleset: pyflakes/pycodestyle errors + import sorting + bugbear + pyupgrade, with per-file-ignores where the code is deliberate, e.g. CLI ``print`s`). Fix all violations (mostly auto-fixable; diff is wide but mechanical), add a ``lint`` CI job. ****No `ruff format` enforcement**** ? a whole-repo reformat would be a noise bomb; lint only.
4. ****AI-4 · Mypy lane**** (M, medium risk on scope) ? Lenient baseline config (``ignore_missing_imports``, exclude the WSL-only ``wasb_infer.py`` if needed), fix the `*real*` errors it surfaces, add CI job. Scoped to land green now and ratchet stricter later ? not a typing crusade.
5. ****AI-5 · Coverage report in CI**** (S, low risk) ? Run the full-suite job with ``--cov``, print the term-missing summary, and set a conservative ``fail-under`` floor just below current measured coverage so it can only ratchet up. Directly addresses the historical coverage-inversion blind spot.

****Structural****

6. ****AI-6 · Config unknown-key warning**** (S-M, low risk) ? The loader currently keeps typo'd top-level keys silently (``extra='allow'``) and nested sections silently `*drop*` unknown keys. Add a loud `[CONFIG WARNING]` unknown keys: ? at load time for both, preserving the never-fatal contract. The full ``extra='forbid'`` flip stays deferred until the deprecated ``gemini_*` aliases retire (that's a behavior decision, not a cleanup).
7. ****AI-7 · Collapse the hybrid-segmenter twins**** (M-L, medium risk ? the big one) ? Extract a shared trajectory-hybrid base; ``tracknet_hybrid/``/``wasb_hybrid`` become thin subclasses (runner + config section + params). Resolves the documented ``shot_count`` copy-paste drift deliberately (align both to the safe is-None short-circuit), dedupes ``_wsl_bash`` into the detectors layer, and adds equivalence tests so the twins can never drift again. Goes last so ruff/mypy/coverage lanes guard it.
8. ****AI-8 · pyproject.toml` packaging\*\*** (M, medium risk) ? PEP 621 metadata, `setuptools` backend, dependencies sourced dynamically from ``requirements.txt`` (single source). Package ``backend`` ****only**** ? excluding ``training/`` from the wheel reinforces ADR-008's import wall. One wrinkle: the existing `[setup.py](setup.py)` is a bootstrap script, not `setuptools` ? it must be renamed (e.g. ``scripts/setup_env.py``) or ``pip install`` would execute it with side effects; I'll update doc references. Package name stays ``badminton-highlight-indexer`` for now ? trivially renamed when the repo rename lands, no point prejudging it.

****Execution protocol:**** sequential, one PR at a time off fresh master; suite green locally before push; merge gated on ``statusCheckRollup == SUCCESS`` (the #75/#76 lesson); squash-merge + branch delete; re-check PR state before any follow-up push (the #80 lesson). Your two uncommitted PR-2 doc edits on master stay untouched throughout ? selective ``git add`` only, never ``-A``.

24. user (2026-06-11T04:55:01.098Z)

Your questions have been answered: "Which quick-win items do you approve?"="AI-1 Repo hygiene (Recommended),AI-2 Pydantic v2 cleanup (Recommended),AI-6 Config unknown-key warning", "Which CI quality lanes do you approve?"="AI-3 Ruff lint lane (Recommended),AI-4 Mypy lane,AI-5 Coverage

floor in CI", "Which structural items do you approve?"="would be great if you can also keep checkpointing that the quality evals (currently at ~0.66) stick after any major changes. GPU is free to use if you need it and even the gemini key has been rotated so feel free to make gemini calls especially after AI-7. Now please go ahead and make this repo a 9!,AI-7 Collapse hybrid twins". You can now continue with these answers in mind.

25. user (2026-06-11T04:56:16.427Z)

Chapter marked: "8?9 execution: hygiene + CI lanes". Continue your current work.

26. assistant (2026-06-11T04:56:23.374Z)

****AI-1 (repo hygiene)**** ? gathering facts before touching anything.

27. user (2026-06-11T04:56:25.549Z)

```
M docs/DEFERRED_HARDENING_PLAN.md
M docs/UI_ALPHA_EXECUTION_PLAN.md
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_deferred_hardenening_status.patch
?? scratch/PR88_FIX.md
---
origin https://github.com/avidullu/badminton-highlight-indexer.git (fetch)
origin https://github.com/avidullu/badminton-highlight-indexer.git (push)
---
34cf26c docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009 (#89)
```

28. user (2026-06-11T04:56:25.568Z)

```
1      # Test Run Diagnosis & Optimizations Summary
2
3      > **Historical note:** This is a point-in-time diagnosis from a May 2026 local test run,
4      > written *before* the layered-directory restructure (PR #40). Path references below
5      > reflect the old flat layout (e.g. a single `segmenter.py`, now the
6      > `backend/pipeline/segmenters/` package). The diagnostic findings and optimization
7      > lessons remain valid; treat the file paths and install commands as historical.
8
9      *Date: May 18, 2026*
10     *Target File: `C:\Users\avidu\Videos\LargeTestVideo.MP4` (HEVC H.265, Size: ~11.5 GB)*
11
12     This summary details the findings, architectural lessons, and performance optimizations
13     implemented during today's local test run of the Badminton Highlight & Rally Indexer.
14     ---
15
16     ## ? Why the Initial Test Run Blocked (and How We Resolved It)
17
18     During the raw test execution on the 11.5 GB video stream, the system hit three
19     distinct, high-impact constraints. We fully diagnosed, optimized, and hardened the pipelines to
20     address them:
21
22     ### 1. The Keyframe Extraction Gap (H.265 Stream Header Parsing)
23     * **The Issue**: The `PTSContinuityValidator` check initially failed with `[FAIL] No
keyframes could be extracted`.
24     * **The Root Cause**: GoPro raw H.265/HEVC encodes use **Gradual Decoder Refresh
(GDR)** or **Clean Random Access (CRA)** keyframe configurations. Standard FFmpeg frame-level
decoding with -skip_frame nokey relies on hardware/software decoders which failed to flag
standard keyframe structures, returning empty outputs.
25     * **The Optimization**: We changed the validator to query packet indexes
(packet=pts_time,flags`) at the **container demuxer level** instead of full frame decoding.
Because packet flags (e.g. `K__`) are written directly to container headers, this bypasses the
```

```
video decoder entirely. It runs **100x faster** and is 100% robust across HEVC containers!
24
25     ### 2. The Blurriness Quality Barrier (Indoor Arena Low-Light Noise)
26     * **The Issue**: The video failed the focus threshold check with a sharpness score of
`35.3` (minimum threshold was `100.0`).
27     * **The Root Cause**: High-speed action footage in typical indoor badminton
environments suffers from high-speed motion blur and ISO sensor noise, reducing the computed
Laplacian variance.
28     * **The Fix**: We lowered the `min_blur_threshold` default inside `config.json` to
`20.0` so that raw fast-moving arena footage passes quality checks cleanly.
29
30     ### 3. OpenCV Seeking Performance Bottleneck
```

29. user (2026-06-11T04:56:26.506Z)

```
backend\pipeline\validators\blur.py
backend\pipeline\segmenters\motion.py
```

30. user (2026-06-11T04:56:26.591Z)

```
README.md:153:* `validation.min_blur_threshold` ? defaults to `20.0` (deliberately permissive
for noisy indoor-arena GoPro footage; see `TEST_RUN_SUMMARY.md` for why).
```

31. user (2026-06-11T04:56:47.577Z)

```
checkpoints
decisions
past_projects
quality-iterations
research
COMMERCIAL_READINESS_REVIEW.md
MONETIZATION_AUDIT.md
README.md
--- stale dirs check ---
__pycache__
__pycache__
```

32. assistant (2026-06-11T04:57:01.016Z)

```
`docs/archives/` already holds loose terminal-state docs at top level ? perfect fit. Checking
the archive index, then making the edits on a branch.
```

33. user (2026-06-11T04:57:01.500Z)

```
1     # archives/
2
3     Umbrella directory for all **historical and completed** material.
4
5     ## Live vs Historical
6     - **Live docs** (for ramp-up, constant updates, and current work) live at the top level
of `docs/`.
7     - Everything here is no longer the primary source of truth. It is kept for context,
decision history, and regression detection.
8
9     ## Classification inside archives/
10    - `past_projects/` ? Completed small tech debt, starter projects, or low-regret
initiatives (e.g. the original low-regret moves + scaffolding work). These are "past" once they
reach a terminal state.
11    - `decisions/` ? Historical ADRs and major architectural decisions.
12    - `research/` ? Completed or superseded research notes and plans.
13    - `checkpoints/` ? Snapshots of major project states (especially useful for detecting
regressions in quality, priorities, or baselines over time as we do quality iterations).
14    - Other one-off audits, etc.
15
```

```
16      ## Archival Rule
17      Only move something into `archives/` (or its subdirectories) when the work is in a clear
**terminal state**:
18      - DONE
19      - COMPLETED
20      - REJECTED
21
22      Never archive in-flight or pending work.
23
24      When archiving a project/phase:
25      1. Ensure the main document has an explicit `**STATUS: COMPLETED**` (or DONE/REJECTED)
header at the top.
26      2. Link back to the current living documents in `../` (e.g.
`COMMERCIAL_READINESS_REVIEW.md`, `BACKLOG.md`).
27      3. For regression detection value, consider adding a dated entry under `checkpoints/`.
28
29      See `past_projects/README.md` for the specific guidance on that category.
30
```

34. user (2026-06-11T04:57:02.272Z)

```
145      "end_padding": 0.5,
146      "use_cache": false,
147      "min_shot_count": 1
148  }
149  }
150  ```
151
152  Notable knobs:
153  * `validation.min_blur_threshold` ? defaults to `20.0` (deliberately permissive for
noisy indoor-arena GoPro footage; see `TEST_RUN_SUMMARY.md` for why).
154  * `indexing.default_segmenter` ? `yolo_hybrid` | `gemini` | `motion`.
155  * `indexing.use_gpu` ? when true and CUDA is available, YOLO runs on GPU.
156  * `indexing.yolo_velocity_threshold` ? fraction of frame width per second; lower
values catch slower play, higher values are stricter about what counts as "action".
157  * `indexing.yolo_frame_skip` ? process every Nth frame during YOLO tracking. `3` at 30
fps ? 10 samples/sec.
158  * `indexing.chunk_duration` / `chunk_overlap` ? how to split long videos for the YOLO
phase.
159  * `indexing.ai_handoff_padding` ? seconds of slack added around each YOLO candidate
window before sending to the AI provider.
```

35. assistant (2026-06-11T04:57:33.900Z)

Archive policy fits ("Other one-off audits" at top level). Executing AI-1 now: branch, move the doc, fix `.coveragerc`, update the README pointer.

36. user (2026-06-11T04:57:37.881Z)

```
Switched to a new branch 'chore/repo-hygiene'
M docs/DEFERRED_HARDENING_PLAN.md
M docs/UI_ALPHA_EXECUTION_PLAN.md
R TEST_RUN_SUMMARY.md -> docs/archives/TEST_RUN_SUMMARY.md
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
```

37. assistant (2026-06-11T04:57:47.860Z)

[run]

Excluded from coverage: the legacy motion segmenter (demo baseline with

fabricated metadata) and the concrete validators (thin OpenCV/ffprobe wrappers only exercisable against real media files). validators/base.py stays counted ? it has a dedicated unit suite.

```
omit =
    backend/pipeline/segmenters/motion.py
    backend/pipeline/validators/format.py
    backend/pipeline/validators/resolution_fps.py
    backend/pipeline/validators/pts_continuity.py
    backend/pipeline/validators/scene_cut.py
    backend/pipeline/validators/blur.py
    backend/pipeline/validators/relevance.py
```

38. user (2026-06-11T04:57:47.862Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

39. user (2026-06-11T04:57:48.844Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

40. user (2026-06-11T04:57:55.445Z)

```
1      [run]
2      omit =
3          backend/main.py
4          backend/indexer/motion.py
5          backend/validators/*
6
```

41. assistant (2026-06-11T04:58:05.256Z)

[run]

Excluded from coverage: the legacy motion segmenter (demo baseline with fabricated metadata) and the concrete validators (thin OpenCV/ffprobe wrappers only exercisable against real media files). validators/base.py stays counted ? it has a dedicated unit suite.

```
omit =
    backend/pipeline/segmenters/motion.py
    backend/pipeline/validators/format.py
    backend/pipeline/validators/resolution_fps.py
    backend/pipeline/validators/pts_continuity.py
    backend/pipeline/validators/scene_cut.py
    backend/pipeline/validators/blur.py
    backend/pipeline/validators/relevance.py
```

42. user (2026-06-11T04:58:05.268Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\coveragerc has been updated successfully. (file state is current in your context ? no need to Read it back)

43. assistant (2026-06-11T04:58:14.113Z)

Committing and opening the PR.

44. user (2026-06-11T04:58:21.052Z)

warning: in the working copy of '.coveragerc', LF will be replaced by CRLF the next time Git touches it

```
[chore/repo-hygiene ed517d8] chore: repo hygiene - fix stale .coveragerc paths, archive
TEST_RUN_SUMMARY
3 files changed, 12 insertions(+), 4 deletions(-)
rename TEST_RUN_SUMMARY.md => docs/archives/TEST_RUN_SUMMARY.md (100%)
remote:
remote: Create a pull request for 'chore/repo-hygiene' on GitHub by visiting:
remote:   https://github.com/avidullu/badminton-highlight-indexer/pull/new/chore/repo-hygiene
remote:
branch 'chore/repo-hygiene' set up to track 'origin/chore/repo-hygiene'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      chore/repo-hygiene -> chore/repo-hygiene
```

45. user (2026-06-11T04:58:36.619Z)

Warning: 8 uncommitted changes
<https://github.com/avidullu/badminton-highlight-indexer/pull/90>

46. user (2026-06-11T04:58:49.976Z)

```
backend\main.py:157:          db_instance.add_video(video_id, req.video_path, "failed",
[r.dict() for r in val_results])
backend\main.py:162:          "results": [r.dict() for r in val_results],
backend\main.py:168:          db_instance.add_video(video_id, req.video_path, "processed",
[r.dict() for r in val_results])
backend\main.py:179:          "results": [r.dict() for r in val_results],
backend\main.py:208:          "results": [r.dict() for r in val_results],
backend\main.py:221:          "results": [r.dict() for r in val_results],
backend\main.py:521:          db_instance.add_video(video_id, args.video_path, status, [r.dict()
for r in val_results])
tests\test_async_jobs.py:113:      body = r.json()
tests\test_async_jobs.py:138:      assert "Available segmenters" in r.json()["detail"]
tests\test_async_jobs.py:152:          "/api/process", json={"video_path": str(vid_file),
"segmenter": "x"}).json()["job_id"]
tests\test_async_jobs.py:155:      body = r.json()
tests\test_async_jobs.py:177:      job_id = client.post("/api/process", json={"video_path":
str(vid_file)}).json()["job_id"]
tests\test_async_jobs.py:178:      body = client.get(f"/api/jobs/{job_id}").json()
tests\test_async_jobs.py:196:          "/api/process", json={"video_path": str(vid_file),
"segmenter": "x"}).json()["job_id"]
tests\test_async_jobs.py:197:      body = client.get(f"/api/jobs/{job_id}").json()
tests\test_async_jobs.py:261:          "/api/process", json={"video_path": str(vid_file),
"segmenter": "x"}).json()["job_id"]
tests\test_async_jobs.py:265:          body = client.get(f"/api/jobs/{job_id}").json()
tests\test_api_endpoints.py:60:      assert r.json()["sport"] == "badminton"
tests\test_api_endpoints.py:71:      job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
tests\test_api_endpoints.py:86:      job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
tests\test_api_endpoints.py:105:      assert len(r.json()["play_segments"]) == 1
tests\test_api_endpoints.py:133:      assert r.json()["filepath"] == "output/out.mp4"
```

47. user (2026-06-11T05:00:06.308Z)

```
Full test suite (offline, mocked)      pending 0      https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27324870966/job/80723742775
Full test suite (offline, mocked)      pass    1m22s    https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27324870966/job/80723742775
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27324870966/job/80723742742
Full test suite (offline, mocked)      pass    1m22s    https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27324870966/job/80723742775
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27324870966/job/80723742742
[{"conclusion":"SUCCESS","name":"Syntax sweep + import smoke (no GPU
stack)"}, {"conclusion":"SUCCESS","name":"Full test suite (offline, mocked)"}]
```

48. user (2026-06-11T05:00:24.553Z)

From <https://github.com/avidullu/badminton-highlight-indexer>

```
* branch      master      -> FETCH_HEAD
34cf26c..9f08681 master    -> origin/master
```

Updating 34cf26c..9f08681

Fast-forward

```
.coveragerc          | 14 ++++++++---
README.md            |  2 +-
TEST_RUN_SUMMARY.md => docs/archives/TEST_RUN_SUMMARY.md |  0
3 files changed, 12 insertions(+), 4 deletions(-)
```

rename TEST_RUN_SUMMARY.md => docs/archives/TEST_RUN_SUMMARY.md (100%)

M docs/DEFERRED_HARDENING_PLAN.md

M docs/UI_ALPHA_EXECUTION_PLAN.md

Your branch is up to date with 'origin/master'.

Already on 'master'

Already up to date.

9f08681 chore: repo hygiene - fix stale .coveragerc paths, archive TEST_RUN_SUMMARY (#90)

49. user (2026-06-11T05:00:40.865Z)

```
150     segmenter_actual = None
151     segmentation_ran = False
152     response: Optional[Dict[str, Any]] = None
153     try:
154         if failures:
155             # add_video is an UPSERT (no cascade) ? a failed re-validation only updates
156             # status; it no longer destroys the video's prior good segments.
157             db_instance.add_video(video_id, req.video_path, "failed", [r.dict() for r in
val_results])
158             response = {
159                 "video_id": video_id,
160                 "status": "failed",
161                 "message": "Video failed validation checks.",
162                 "results": [r.dict() for r in val_results],
163             }
164             return response
165
166         # 2. Run segmenter & indexer
167         logger.info("[API] Video passed checks. Segmenting and indexing...")
168         db_instance.add_video(video_id, req.video_path, "processed", [r.dict() for r in
val_results])
169
170         segmenter_cls = segmenter_registry.get(seg_name)
171         if not segmenter_cls:
172             # Submit-time validation already 400s unknown names; this is the defensive
173             # in-worker path (e.g. config default pointing at an unregistered
segmenter).
174             available = list(segmenter_registry.list_available().keys())
175             response = {
176                 "video_id": video_id,
177                 "status": "failed",
178                 "message": f"Segmenter '{seg_name}' not found. Available segmenters:
{available}",
179                 "results": [r.dict() for r in val_results],
180                 "play_segments": [],
181             }
182             return response
183
184         logger.info(f"[API] Using segmenter: {seg_name}")
185         notify(f"segmenting ({seg_name})")
186         segmenter_actual = seg_name
187         # Destroy-AFTER-confirm: remember the pre-run rows, then keep new + drop old
only
188         # if the run yields segments; otherwise drop the new partial rows and keep the
```



```

old.
189         play_wm, cand_wm = db_instance.segment_watermarks(video_id)
190         segmenter = segmenter_cls(db_instance, global_config)
191         try:
192             result = segmenter.process_video(video_id, req.video_path, req.player_pool,
193 req.max_frames)
194         except Exception:
195             # A raising segmenter must not leave half-written rows: restore the pre-run
196             # state (same destroy-after-confirm contract as the Failure branch), then
197             let
198                 # the job worker record the failure.
199                 db_instance.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)
200                 raise
201             segmentation_ran = True
202
203         if isinstance(result, Failure):
204             logger.error(f"[API] Segmenter failed: {result.message}")
205             db_instance.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)
206             response = {
207                 "video_id": video_id,
208                 "status": "failed",
209                 "message": f"Segmenter '{seg_name}' failed: {result.message}",
210                 "results": [r.dict() for r in val_results],
211                 "play_segments": [],
212             }
213             return response
214
215         segments = result.value if hasattr(result, "value") else result
216         notify("finalizing")
217         _finalize_reingest(video_id, segments, play_wm, cand_wm)
218
219         response = {
220             "video_id": video_id,
221             "status": "success",
222             "message": f"Successfully indexed {len(segments)} play segments using
223 '{seg_name}' segmenter.",
224             "results": [r.dict() for r in val_results],
225             "play_segments": segments,
226         }
227         return response

```

50. user (2026-06-11T05:00:41.535Z)

```

512         was_reingest = db_instance.get_video(video_id) is not None
513
514         # Validate (with-context variant surfaces ffprobe_meta for the report)
515         print("[CLI] Validating...")
516         val_results, val_context = registry.run_all_with_context(args.video_path,
517 global_config)
518         failures = [r for r in val_results if not r.passed]
519
520         # Save video status
521         status = "failed" if failures else "processed"
522         db_instance.add_video(video_id, args.video_path, status, [r.dict() for r in
523 val_results])
524
525         seg_name = args.segmenter or getattr(getattr(global_config, "indexing", None),
526 "default_segmenter", "motion")
527         segmenter_actual = None
528         segmentation_ran = False
529         try:

```

51. user (2026-06-11T05:00:48.095Z)

Switched to a new branch 'fix/pydantic-v2-model-dump'

52. user (2026-06-11T05:00:54.123Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated.
All occurrences were successfully replaced. (file state is current in your context ? no need to
Read it back)

53. user (2026-06-11T05:01:37.286Z)

```
..... [ 82%]
..... [100%]
349 passed in 29.52s
=====
RESULT: PASS - all tests green
=====
```

54. assistant (2026-06-11T05:01:45.779Z)

349 green and the warnings summary is gone entirely (was 15). Shipping AI-2.

55. user (2026-06-11T05:01:57.691Z)

```
[fix/pydantic-v2-model-dump a7f9e07] fix: replace deprecated pydantic .dict() with model_dump()
1 file changed, 7 insertions(+), 7 deletions(-)
remote:
remote: Create a pull request for 'fix/pydantic-v2-model-dump' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-
indexer/pull/new/fix/pydantic-v2-model-dump
remote:
branch 'fix/pydantic-v2-model-dump' set up to track 'origin/fix/pydantic-v2-model-dump'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      fix/pydantic-v2-model-dump -> fix/pydantic-v2-model-dump
Warning: 8 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/91
```

56. assistant (2026-06-11T05:02:08.148Z)

PR [#91](https://github.com/avidullu/badminton-highlight-indexer/pull/91) is up. While its CI runs, sizing AI-3: does ruff exist here, and how many violations per candidate rule group?

57. user (2026-06-11T05:02:19.800Z)

```
Exit code 1
[notice] To update, run: python.exe -m pip install --upgrade pip
ruff 0.15.16
=== default rules (E4,E7,E9,F) ===
43      F401      [-] unused-import
28              [ ] invalid-syntax
28      E402      [ ] module-import-not-at-top-of-file
14      E702      [ ] multiple-statements-on-one-line-semicolon
5        F841      [ ] unused-variable
4        E741      [ ] ambiguous-variable-name
4        F541      [*] f-string-missing-placeholders
3        E701      [ ] multiple-statements-on-one-line-colon
1        E731      [ ] lambda-assignment
Found 130 errors.
[*] 45 fixable with the `--fix` option (6 hidden fixes can be enabled with the `--unsafe-fixes`
option).
```

58. user (2026-06-11T05:02:36.254Z)

```
Exit code 1
scripts\generate_khelacharya_flyer.py:39:7: invalid-syntax: Expected ``, found name
scripts\generate_khelacharya_flyer.py:39:14: invalid-syntax: Expected ``, found name
scripts\generate_khelacharya_flyer.py:39:19: invalid-syntax: Expected ``, found name
```

```
scripts\generate_khelacharya_flyer.py:39:25: invalid-syntax: Expected ``, found name
scripts\generate_khelacharya_flyer.py:39:28: invalid-syntax: Expected ``, found name
scripts\generate_khelacharya_flyer.py:39:37: invalid-syntax: Expected ``, found name
scripts\generate_khelacharya_flyer.py:39:47: invalid-syntax: Expected ``, found name
scripts\generate_khelacharya_flyer.py:39:52: invalid-syntax: Expected ``, found name
scripts\generate_khelacharya_flyer.py:39:57: invalid-syntax: Got unexpected token ?
scripts\generate_khelacharya_flyer.py:39:69: invalid-syntax: Expected `]`, found `for`
=== file list with counts ===
```

```
Count : 28
Name : scripts\generate_khelacharya_flyer.py
```

```
Count : 19
Name : backend\main.py
```

```
Count : 8
Name : backend\features\rally_seq.py
```

```
Count : 5
Name : backend\pipeline\segmenters\yolo_hybrid.py
```

```
Count : 4
Name : tests\test_database.py
```

```
Count : 4
Name : backend\pipeline\segmenters\gemini.py
```

```
Count : 4
Name : training\gen0\harness.py
```

```
Count : 3
Name : backend\pipeline\segmenters\motion.py
```

```
Count : 2
Name : tests\test_video.py
```

```
Count : 2
Name : backend\eval\distill_local.py
```

```
Count : 2
Name : tests\test_training_gen0.py
```

```
Count : 2
Name : tests\test_tracknet_hybrid.py
```

59. user (2026-06-11T05:02:49.341Z)

```
Exit code 1
694f100 feat: add reusable PDF generator for KhelAcharya Shot Intelligence Layer internship flyer
091e6a1 feat: add reusable PDF generator for KhelAcharya Shot Intelligence Layer internship flyer
File "scripts/generate_khelacharya_flyer.py", line 39
[Full script body would be inserted here in a real call ? truncated for this simulation
```

response for length reasons. In actual use the complete working script from the local file would be provided.]

^

```
SyntaxError: invalid character '?' (U+2014)
compile exit: 1
generate_khelacharya_flyer.py
```

60. user (2026-06-11T05:03:15.953Z)

```
1      #!/usr/bin/env python3
2      """
3      KhelAcharya ? Shot Intelligence Layer
4      Professional 2-page Technical Internship Flyer
5
6      Generates a formal, engineering-focused PDF suitable for recruiting strong interns.
7
8      Usage:
9          python scripts/generate_khelacharya_flyer.py
10
11     Dependencies:
12         reportlab (pip install reportlab)
13
14     This script was developed to produce the official internship recruiting flyer
15     for the KhelAcharya Shot Intelligence Layer project.
16     """
17
18     from reportlab.lib.pagesizes import A4
19     from reportlab.lib.units import inch
20     from reportlab.lib.colors import HexColor, white
21
22     from reportlab.pdfgen import canvas
23
24     # Page dimensions
25     PAGE_WIDTH, PAGE_HEIGHT = A4
26     MARGIN = 0.65 * inch
27
28     # Color palette ? formal, professional
29     NAVY = HexColor("#0F172A")
30     NAVY_LIGHT = HexColor("#1E293B")
31     TEAL = HexColor("#0D9488")
32     TEAL_DARK = HexColor("#0F766E")
33     TEAL_LIGHT = HexColor("#CCFBF1")
34     SLATE = HexColor("#334155")
35     SLATE_LIGHT = HexColor("#64748B")
36     SLATE_VLIGHT = HexColor("#F1F5F9")
37     WHITE = white
38
39     [Full script body would be inserted here in a real call ? truncated for this simulation
response for length reasons. In actual use the complete working script from the local file would
be provided.]
```

61. assistant (2026-06-11T05:03:25.389Z)

Confirmed ? header constants then a placeholder; it never worked. Now merging #91 (CI should be done) before building AI-3 on fresh master.

62. user (2026-06-11T05:03:37.137Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27324990885/job/80724106308
Full test suite (offline, mocked)                pass    1m19s  https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27324990885/job/80724106302
Syntax sweep + import smoke (no GPU stack)      pass    19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27324990885/job/80724106308
```

```
rollup: SUCCESS
From https://github.com/avidullu/badminton-highlight-indexer
* branch      master      -> FETCH_HEAD
 9f08681..dcc8ccc master    -> origin/master
Updating 9f08681..dcc8ccc
Fast-forward
 backend/main.py | 14 ++++++-----
 1 file changed, 7 insertions(+), 7 deletions(-)
M      docs/DEFERRED_HARDENING_PLAN.md
M      docs/UI_ALPHA_EXECUTION_PLAN.md
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
```

63. user (2026-06-11T05:03:59.669Z)

```
Switched to a new branch 'ci/ruff-lint-lane'
backend\eval\calibrate_wasb.py:19:8: F401 [*] `csv` imported but unused
backend\eval\distill_local.py:29:17: F401 [*] `numpy` imported but unused
backend\eval\distill_local.py:124:9: E731 Do not assign a `lambda` expression, use a `def`
backend\eval\eval_partial_labels.py:178:15: E702 Multiple statements on one line (semicolon)
backend\eval\rally_seq_proto.py:124:49: E702 Multiple statements on one line (semicolon)
backend\features\rally_seq.py:14:26: F401 [*] `typing.List` imported but unused
backend\features\rally_seq.py:57:22: E702 Multiple statements on one line (semicolon)
backend\features\rally_seq.py:57:39: E702 Multiple statements on one line (semicolon)
backend\features\rally_seq.py:60:31: E702 Multiple statements on one line (semicolon)
backend\features\rally_seq.py:60:56: E702 Multiple statements on one line (semicolon)
backend\features\rally_seq.py:68:24: E702 Multiple statements on one line (semicolon)
backend\features\rally_seq.py:68:48: E702 Multiple statements on one line (semicolon)
backend\features\rally_seq.py:90:17: E702 Multiple statements on one line (semicolon)
backend\infrastructure\database.py:3:8: F401 [*] `os` imported but unused
backend\main.py:4:8: F401 [*] `json` imported but unused
backend\main.py:21:1: E402 Module level import not at top of file
backend\main.py:22:1: E402 Module level import not at top of file
backend\main.py:23:1: E402 Module level import not at top of file
backend\main.py:24:1: E402 Module level import not at top of file
backend\main.py:26:1: E402 Module level import not at top of file
backend\main.py:27:1: E402 Module level import not at top of file
backend\main.py:28:1: E402 Module level import not at top of file
backend\main.py:29:1: E402 Module level import not at top of file
backend\main.py:30:1: E402 Module level import not at top of file
backend\main.py:30:66: F401 [*] `backend.pipeline.segmenters.base.BasePlaySegmenter` imported
but unused
backend\main.py:31:1: E402 Module level import not at top of file
backend\main.py:32:1: E402 Module level import not at top of file
backend\main.py:33:1: E402 Module level import not at top of file
backend\main.py:34:1: E402 Module level import not at top of file
backend\main.py:35:1: E402 Module level import not at top of file
backend\main.py:35:29: F401 [*] `backend.results.Result` imported but unused
backend\main.py:36:1: E402 Module level import not at top of file
backend\main.py:554:23: F541 [*] f-string without any placeholders
backend\pipeline\detectors\__init__.py:17:19: F401 `base.DetectorRunner` imported but unused;
consider removing, adding to `__all__`, or using a redundant alias
backend\pipeline\segmenters\gemini.py:3:8: F401 [*] `json` imported but unused
backend\pipeline\segmenters\gemini.py:8:45: F401 [*] `backend.infrastructure.database.Database`
imported but unused
backend\pipeline\segmenters\gemini.py:66:9: F841 Local variable `is_compressed` is assigned to
but never used
backend\pipeline\segmenters\gemini.py:67:9: F841 Local variable `temp_compressed_path` is
assigned to but never used
backend\pipeline\segmenters\motion.py:3:17: F401 [*] `numpy` imported but unused
backend\pipeline\segmenters\motion.py:7:45: F401 [*] `backend.infrastructure.database.Database`
imported but unused
backend\pipeline\segmenters\motion.py:9:29: F401 [*] `backend.results.Result` imported but
unused
```

```
backend\pipeline\segmenters\yolo_hybrid.py:23:36: F401 [*] `backend.utils.geometry.get_velocity`  
imported but unused  
backend\pipeline\segmenters\yolo_hybrid.py:110:25: F401 [*] `numpy` imported but unused  
backend\pipeline\segmenters\yolo_hybrid.py:540:38: E701 Multiple statements on one line (colon)  
backend\pipeline\segmenters\yolo_hybrid.py:542:24: E701 Multiple statements on one line (colon)  
backend\pipeline\segmenters\yolo_hybrid.py:543:51: E701 Multiple statements on one line (colon)  
backend\pipeline\stitcher\compiler.py:205:17: F841 Local variable `result` is assigned to but  
never used  
backend\pipeline\validators\base.py:3:37: F401 [*] `typing.Optional` imported but unused  
backend\pipeline\validators\scene_cut.py:2:17: F401 [*] `numpy` imported but unused  
backend\reporting\__init__.py:10:8: F401 `os` imported but unused  
backend\utils\video.py:2:8: F401 [*] `os` imported but unused  
backend\utils\video.py:3:8: F401 [*] `shutil` imported but unused  
run_phase3_eval.py:32:1: E402 Module level import not at top of file  
run_phase3_eval.py:33:1: E402 Module level import not at top of file  
run_phase3_eval.py:34:1: E402 Module level import not at top of file  
run_phase3_eval.py:36:1: E402 Module level import not at top of file  
run_phase3_eval.py:37:1: E402 Module level import not at top of file  
run_phase3_eval.py:38:1: E402 Module level import not at top of file  
run_phase3_eval.py:39:1: E402 Module level import not at top of file  
run_process_and_stitch.py:2:8: F401 [*] `sys` imported but unused
```

64. user (2026-06-11T05:04:07.524Z)

Exit code 1

```
run_process_and_stitch.py:16:1: E402 Module level import not at top of file  
run_process_and_stitch.py:17:1: E402 Module level import not at top of file  
run_process_and_stitch.py:18:1: E402 Module level import not at top of file  
run_process_and_stitch.py:19:1: E402 Module level import not at top of file  
run_process_and_stitch.py:19:46: F401 [*] `backend.pipeline.validators.base.registry` imported  
but unused  
run_process_and_stitch.py:20:1: E402 Module level import not at top of file  
run_process_and_stitch.py:21:1: E402 Module level import not at top of file  
run_process_and_stitch.py:28:11: F541 [*] f-string without any placeholders  
run_process_and_stitch.py:31:11: F541 [*] f-string without any placeholders  
run_process_and_stitch.py:62:19: F541 [*] f-string without any placeholders  
scratch\diagnose_rally.py:2:17: F401 [*] `numpy` imported but unused  
scratch\diagnose_rally.py:34:11: F541 [*] f-string without any placeholders  
scratch\gemini_video_debugger.py:85:19: F541 [*] f-string without any placeholders  
scratch\gemini_video_debugger.py:103:27: F541 [*] f-string without any placeholders  
scratch\gemini_video_debugger.py:110:36: F541 [*] f-string without any placeholders  
scratch\test_ai_segmenter.py:6:1: E402 Module level import not at top of file  
scratch\test_ai_segmenter.py:7:1: E402 Module level import not at top of file  
scratch\test_thresholds.py:2:17: F401 [*] `numpy` imported but unused  
scratch\test_thresholds.py:66:15: F541 [*] f-string without any placeholders  
scratch\test_thresholds.py:68:15: F541 [*] f-string without any placeholders  
scratch\test_thresholds.py:104:15: F541 [*] f-string without any placeholders  
setup.py:5:8: F401 [*] `json` imported but unused  
tests\test_base.py:1:8: F401 [*] `pytest` imported but unused  
tests\test_config.py:14:8: F401 [*] `pytest` imported but unused  
tests\test_database.py:1:8: F401 [*] `pytest` imported but unused  
tests\test_database.py:3:8: F401 [*] `os` imported but unused  
tests\test_database.py:8:5: F841 Local variable `db` is assigned to but never used  
tests\test_database.py:66:5: F841 Local variable `db` is assigned to but never used  
tests\test_eval_classifier.py:29:30: E702 Multiple statements on one line (semicolon)  
tests\test_eval_regression.py:3:8: F401 [*] `json` imported but unused  
tests\test_gemini.py:3:8: F401 [*] `pytest` imported but unused  
tests\test_path_safety.py:2:8: F401 [*] `os` imported but unused  
tests\test_provider_gemini.py:3:8: F401 [*] `pytest` imported but unused  
tests\test_providers.py:1:8: F401 [*] `pytest` imported but unused  
tests\test_providers.py:3:36: F401 [*] `backend.providers.base.AIVideoProvider` imported but  
unused  
tests\test_rally_seq_proto.py:40:35: E702 Multiple statements on one line (semicolon)  
tests\test_rally_seq_proto.py:40:64: E702 Multiple statements on one line (semicolon)  
tests\test_reliability_hardening.py:2:8: F401 [*] `sqlite3` imported but unused
```

```

tests\test_reporting.py:14:8: F401 [*] `pytest` imported but unused
tests\test_sport_profiles.py:1:8: F401 [*] `pytest` imported but unused
tests\test_sport_profiles.py:3:33: F401 [*] `backend.sports.base.SportProfile` imported but
unused
tests\test_tracknet_hybrid.py:5:8: F401 [*] `cv2` imported but unused
tests\test_tracknet_hybrid.py:6:8: F401 [*] `pytest` imported but unused
tests\test_training_gen0.py:36:35: E702 Multiple statements on one line (semicolon)
tests\test_training_gen0.py:36:64: E702 Multiple statements on one line (semicolon)
tests\test_video.py:1:8: F401 [*] `pytest` imported but unused
tests\test_video.py:2:34: F401 [*] `unittest.mock.MagicMock` imported but unused
tests\test_wasb_cache.py:93:36: E741 Ambiguous variable name: `l`
tests\test_yolo_hybrid.py:1:8: F401 [*] `pytest` imported but unused
training\gen0\harness.py:40:17: F401 [*] `backend.eval.rally_seq_proto.probs_to_segments`
imported but unused
training\gen0\harness.py:71:25: E741 Ambiguous variable name: `l`
training\gen0\harness.py:72:22: E741 Ambiguous variable name: `l`
training\gen0\harness.py:114:43: E741 Ambiguous variable name: `l`
Found 113 errors.
[*] 54 fixable with the `--fix` option (6 hidden fixes can be enabled with the `--unsafe-fixes`
option).

```

65. user (2026-06-11T05:04:56.444Z)

```

Exit code 1
--- W ---
179      W293      [-] blank-line-with-whitespace
      1      W291      [*] trailing-whitespace
      1      W292      [*] missing-newline-at-end-of-file
Found 181 errors.
[*] 179 fixable with the `--fix` option (2 hidden fixes can be enabled with the `--unsafe-fixes`
option).
--- B ---
5          B904      raise-without-from-inside-except
5          B905      zip-without-explicit-strict
2          B017      assert-raises-exception
1          B007      unused-loop-control-variable
1          B011      assert-false
1          B023      function-uses-loop-variable
--- UP ---
376      UP006      [*] non-pep585-annotation
116      UP035      [-] deprecated-import
116      UP045      [*] non-pep604-annotation-optional
      8      UP037      [*] quoted-annotation
      5      UP015      [*] redundant-open-modes
      2      UP007      [-] non-pep604-annotation-union
--- I ---
106      I001      [*] unsorted-imports
Found 106 errors.
[*] 106 fixable with the `--fix` option.

```

66. assistant (2026-06-11T05:05:33.520Z)

Lint-only lane (no `ruff format` enforcement ? a whole-repo reformat would bury blame history for zero behavior value).

#

Selected: pyflakes + pycodestyle errors (ruff defaults) + pycodestyle warnings (W) + bugbear (B). Deferred with intent: I (import-sort) and UP (pyupgrade) ? both are large mechanical diffs; adopt via a dedicated PR.

target-version = "py38"

scratch/ = throwaway debugging scripts that may hit real APIs; it is also

excluded from pytest collection (see pytest.ini).

```
extend-exclude = ["scratch"]
```

```
[lint]
```

```
extend-select = ["W", "B"]
```

```
[lint.per-file-ignores]
```

Entrypoint scripts deliberately run setup (load_dotenv, logging config, sys.path) before importing backend modules ? import order is load-bearing.

```
"backend/main.py" = ["E402"]
```

```
"run_phase3_eval.py" = ["E402"]
```

```
"run_process_and_stitch.py" = ["E402"]
```

67. user (2026-06-11T05:05:33.535Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\ruff.toml
(file state is current in your context ? no need to Read it back)

68. user (2026-06-11T05:05:42.072Z)

Exit code 1

|

Found 193 errors (152 fixed, 41 remaining).

No fixes available (10 hidden fixes can be enabled with the `--unsafe-fixes` option).

=== remaining ===

backend\eval\calibration.py:130:30: B023 Function definition does not bind loop variable
`others`

backend\eval\distill_local.py:123:9: E731 Do not assign a `lambda` expression, use a `def`

backend\eval\eval_partial_labels.py:178:15: E702 Multiple statements on one line (semicolon)

backend\eval\rally_seq_proto.py:124:49: E702 Multiple statements on one line (semicolon)

backend\features\rally_seq.py:57:22: E702 Multiple statements on one line (semicolon)

backend\features\rally_seq.py:57:39: E702 Multiple statements on one line (semicolon)

backend\features\rally_seq.py:60:31: E702 Multiple statements on one line (semicolon)

backend\features\rally_seq.py:60:56: E702 Multiple statements on one line (semicolon)

backend\features\rally_seq.py:68:24: E702 Multiple statements on one line (semicolon)

backend\features\rally_seq.py:68:48: E702 Multiple statements on one line (semicolon)

backend\features\rally_seq.py:90:17: E702 Multiple statements on one line (semicolon)

backend\main.py:268:9: B904 Within an `except` clause, raise exceptions with `raise ... from err` or `raise ... from None` to distinguish them from errors in exception handling

backend\main.py:339:9: B904 Within an `except` clause, raise exceptions with `raise ... from err` or `raise ... from None` to distinguish them from errors in exception handling

backend\main.py:351:9: B904 Within an `except` clause, raise exceptions with `raise ... from err` or `raise ... from None` to distinguish them from errors in exception handling

backend\main.py:391:9: B007 Loop control variable `f` not used within loop body

backend\pipeline\detectors__init__.py:17:19: F401 `.base.DetectorRunner` imported but unused; consider removing, adding to `\_\_all\_\_`, or using a redundant alias

backend\pipeline\segmenters\base.py:45:13: B904 Within an `except` clause, raise exceptions with `raise ... from err` or `raise ... from None` to distinguish them from errors in exception handling

backend\pipeline\segmenters\gemini.py:64:9: F841 Local variable `is\_compressed` is assigned to but never used

backend\pipeline\segmenters\gemini.py:65:9: F841 Local variable `temp\_compressed\_path` is assigned to but never used

backend\pipeline\segmenters\yolo_hybrid.py:539:38: E701 Multiple statements on one line (colon)

backend\pipeline\segmenters\yolo_hybrid.py:541:24: E701 Multiple statements on one line (colon)

backend\pipeline\segmenters\yolo_hybrid.py:542:51: E701 Multiple statements on one line (colon)

backend\pipeline\stitcher\compiler.py:205:17: F841 Local variable `result` is assigned to but never used

backend\pipeline\stitcher\compiler.py:219:17: B904 Within an `except` clause, raise exceptions with `raise ... from err` or `raise ... from None` to distinguish them from errors in exception handling

backend\providers\base.py:15:1: W293 Blank line contains whitespace


```

backend\providers\base.py:21:1: W293 Blank line contains whitespace
backend\reporting\__init__.py:10:8: F401 `os` imported but unused
tests\test_base.py:45:16: B011 Do not `assert False` (`python -O` removes these calls), raise `AssertionError()`
tests\test_database.py:6:5: F841 Local variable `db` is assigned to but never used
tests\test_database.py:64:5: F841 Local variable `db` is assigned to but never used
tests\test_eval_classifier.py:29:30: E702 Multiple statements on one line (semicolon)
tests\test_rally_seq_proto.py:40:35: E702 Multiple statements on one line (semicolon)
tests\test_rally_seq_proto.py:40:64: E702 Multiple statements on one line (semicolon)
tests\test_reliability_hardening.py:20:10: B017 Do not assert blind exception: `Exception`
tests\test_reliability_hardening.py:22:10: B017 Do not assert blind exception: `Exception`
tests\test_training_gen0.py:36:35: E702 Multiple statements on one line (semicolon)
tests\test_training_gen0.py:36:64: E702 Multiple statements on one line (semicolon)
tests\test_wasb_cache.py:93:36: E741 Ambiguous variable name: `l`
training\gen0\harness.py:71:25: E741 Ambiguous variable name: `l`
training\gen0\harness.py:72:22: E741 Ambiguous variable name: `l`
training\gen0\harness.py:114:43: E741 Ambiguous variable name: `l`
Found 41 errors.
No fixes available (10 hidden fixes can be enabled with the `--unsafe-fixes` option).

```

69. user (2026-06-11T05:06:08.977Z)

```

115         """Leave-one-video-out CV ? the honest metric as the corpus grows.
116
117         `videos`: list of {"name": str, "predict_fn": PredictFn, "gts": [Interval]}.
118         For each held-out video: pick the config that's best *pooled across the other
119         videos*, then score it on the held-out one. The mean held-out score is the
120         "precision on unseen videos" number; per-video spread shows stability.
121         """
122         if len(videos) < 2:
123             return {"error": "need >=2 videos for leave-one-video-out", "n": len(videos)}
124         per_video = []
125         for i, held in enumerate(videos):
126             others = [v for j, v in enumerate(videos) if j != i]
127
128             def pooled_score(cfg: Config) -> float:
129                 vals = [getattr(score(v["predict_fn"])(cfg), v["gts"], iou_threshold),
metric)
130                             for v in others]
131                 return _st.mean(vals)
132
133             best_cfg, best_train = max(((c, pooled_score(c)) for c in configs), key=lambda
t: t[1])
134             held_score = getattr(score(held["predict_fn"])(best_cfg), held["gts"],
iou_threshold), metric)
135             per_video.append({"video": held.get("name", f"video_{i}"),
136                             "held_out": held_score, "train_pooled": best_train, "config":
best_cfg})
137         held_scores = [p["held_out"] for p in per_video]
138         train_scores = [p["train_pooled"] for p in per_video]
139         return {
140             "metric": metric,
141             "mean_held_out": _st.mean(held_scores),
142             "std_held_out": _st.pstdev(held_scores) if len(held_scores) > 1 else 0.0,
143             "generalization_gap": _st.mean(train_scores) - _st.mean(held_scores),
144             "per_video": per_video,

```

70. user (2026-06-11T05:06:09.891Z)

```

112         others = [v for j, v in enumerate(videos) if j != i]
113         X = [row for v in others for row in v["X"]]
114         y = [t for v in others for t in v["y"]]
115         model = LogisticRegression().fit(X, y, FEATURE_NAMES)
116         preds = predict_rallies(model, held["X"], held["intervals"], threshold)
117         sc = score(preds, held["gts"], iou)

```

```

118         bsc = score(held["baseline"], held["gts"], iou)
119         clf.append(sc)
120         base.append(bsc)
121         print(f" held-out {held['name']}: LEARNED P={sc.precision:.3f}
R={sc.recall:.3f} F1={sc.f1:.3f}"
122             f" | baseline P={bsc.precision:.3f} R={bsc.recall:.3f}
F1={bsc.f1:.3f}")
123         mean = lambda xs, a: sum(getattr(s, a) for s in xs) / len(xs)
124         print(f" MEAN held-out: LEARNED P={mean(clf, 'precision'):.3f}
R={mean(clf, 'recall'):.3f} "
125             f"F1={mean(clf, 'f1'):.3f} | baseline P={mean(base, 'precision'):.3f} "
126             f"R={mean(base, 'recall'):.3f} F1={mean(base, 'f1'):.3f}")
127         result["mean_learned_f1"] = mean(clf, "f1")
128         result["mean_baseline_f1"] = mean(base, "f1")
129
130         # Final model on ALL videos (for production / running on a fresh held-out video).
131         Xall = [row for v in videos for row in v["X"]]

```

71. user (2026-06-11T05:06:10.427Z)

```

258         """Submit a processing job (202 + job_id). Poll GET /api/jobs/{job_id} for state.
259
260         Fast-fail checks (bad path, missing file, unknown segmenter) happen here so the
261         client gets an immediate 4xx; everything slow ? including MD5 hashing of the file ?
262         runs on the worker (DEFERRED_HARDENING_PLAN #16).
263         """
264         allowed_root = getattr(getattr(global_config, "security", None),
"allowed_video_root", None)
265         try:
266             validate_input_path(req.video_path, allowed_root=allowed_root)
267         except ValueError as e:
268             raise HTTPException(status_code=400, detail=str(e))
269         if not os.path.exists(req.video_path):
270             raise HTTPException(status_code=404, detail=f"Video file not found at
'{req.video_path}'")
271         if req.segmenter and not segmenter_registry.get(req.segmenter):
272             available = list(segmenter_registry.list_available().keys())
273             raise HTTPException(
274                 status_code=400,
275                 detail=f"Segmenter '{req.segmenter}' not found. Available segmenters:
{available}")

```

72. user (2026-06-11T05:06:27.727Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\calibration.py has been updated successfully. (file state is current in your context ? no need to Read it back)

73. user (2026-06-11T05:06:29.119Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\distill_local.py has been updated successfully. (file state is current in your context ? no need to Read it back)

74. user (2026-06-11T05:06:30.503Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

75. user (2026-06-11T05:06:38.756Z)

```

330
331         if not matched_segments:
332             raise HTTPException(status_code=400, detail="No matching play segments found to
stitch.")
333

```

```

334 # Reject path traversal / absolute output names (os.path.join would otherwise let an
335 # absolute or '..'-laden output_filename write anywhere on disk).
336 try:
337     out_name = safe_output_filename(req.output_filename)
338 except ValueError as e:
339     raise HTTPException(status_code=400, detail=str(e))
340
341 # Extract start/end time pairs
342 time_pairs = [(s["start_time"], s["end_time"]) for s in matched_segments]
343
344 # Run compiler
345 output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
or "output"
346 compiler = VideoCompiler(output_dir)
347 try:
348     out_path = compiler.compile_highlights(video["filepath"], time_pairs, out_name)
349     return {"status": "success", "filepath": out_path}
350 except Exception as e:
351     raise HTTPException(status_code=500, detail=f"Highlight stitching failed:
{str(e)}")
352
353 # --- CLI Debug Utility & Orchestration ---
354 def run_cli_diagnose_clip(video_path: str, timestamp: float):
355     """CLI debugging utility to examine segment properties around a timestamp."""
356     print("=" * 60)
357     print(f"DEBUGGING VIDEO CLIP: {video_path} at {timestamp}s")
358     print("=" * 60)
359
360     cfg = load_config() # legacy wrapper still works, returns dict for now
361
362     # 1. Validation check diagnostics
363     print("[DIAGNOSTIC] Running validation framework...")
364     results = registry.run_all(video_path, cfg)
365     for r in results:
366         status_symbol = "[PASS]" if r.passed else "[FAIL]"
367         print(f"  {status_symbol} {r.validator_name}: {r.message}")
368         if r.details:
369             print(f"    Details: {r.details}")
370
371     # 2. Local segment analysis around the timestamp (sampling +/- 5 seconds)
372     print("-" * 60)
373     print("[DIAGNOSTIC] Analyzing local motion around the timestamp...")
374     import cv2
375     cap = cv2.VideoCapture(video_path)
376     if not cap.isOpened():
377         print("[FAIL] OpenCV could not open video.")
378         return
379
380     fps = cap.get(cv2.CAP_PROP_FPS)
381     total_frames = cap.get(cv2.CAP_PROP_FRAME_COUNT)
382
383     start_frame = max(0, int((timestamp - 3.0) * fps))
384     end_frame = min(int(total_frames), int((timestamp + 3.0) * fps))
385
386     # Measure frame-by-frame changes in sample region
387     cap.set(cv2.CAP_PROP_POS_FRAMES, start_frame)
388     prev_gray = None
389     motions = []
390
391     for f in range(start_frame, end_frame):
392         ret, frame = cap.read()
393         if not ret:
394             break

```

```

36     def __init__(self):
37         self._segmenters: Dict[str, Type[BasePlaySegmenter]] = {}
38
39     def register(self, segmenter_cls: Type[BasePlaySegmenter]):
40         """Registers a new segmenter class under its name property."""
41         try:
42             temp_inst = segmenter_cls(None, {})
43             name = temp_inst.name
44         except Exception as e:
45             raise ValueError(f"Failed to resolve name property from segmenter class
{segmenter_cls.__name__}: {e}")
46         self._segmenters[name] = segmenter_cls
47
48     def get(self, name: str) -> Optional[Type[BasePlaySegmenter]]:
49         return self._segmenters.get(name)

```

77. user (2026-06-11T05:06:40.471Z)

```

55         except KeyError as e:
56             logger.error(str(e))
57             return []
58
59         # A: Check for debug_duration to trim the video first
60         debug_duration = self.config.get("indexing", {}).get("debug_duration")
61         video_to_upload = video_path
62         is_trimmed = False
63         temp_trimmed_path = None
64         is_compressed = False
65         temp_compressed_path = None
66
67         if debug_duration:
68             logger.info(f"DEBUG FEATURE: Trimming video to first {debug_duration}
seconds for fast processing...")
69             temp_trimmed_path = os.path.join("output", f"temp_trimmed_{video_id}.mp4")
70             os.makedirs("output", exist_ok=True)
71             if os.path.exists(temp_trimmed_path):
72                 try:
73                     os.remove(temp_trimmed_path)
74                 except Exception:
75                     pass
76
77             cmd = [
78                 "ffmpeg", "-y",
79                 "-ss", "0",

```

78. user (2026-06-11T05:06:52.145Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

79. user (2026-06-11T05:06:54.355Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

80. user (2026-06-11T05:06:55.445Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

81. user (2026-06-11T05:06:57.802Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

82. user (2026-06-11T05:06:59.152Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py has been updated successfully. (file state is current in your context ? no need to Read it back)

83. user (2026-06-11T05:07:06.093Z)

```
196         "ffmpeg", "-y",
197         "-f", "concat",
198         "-safe", "0",
199         "-i", list_file_path,
200         "-c", "copy",
201         output_path
202     ]
203
204     try:
205         result = subprocess.run(concat_cmd, capture_output=True, check=True)
206
207         # Final verification
208         if not os.path.exists(output_path) or os.path.getsize(output_path) == 0:
209             raise RuntimeError(
210                 f"[COMPILER ERROR] FFmpeg concatenation appeared to succeed, "
211                 f"but output file is missing or empty: {output_path}"
212             )
213
214         output_size_mb = os.path.getsize(output_path) / (1024 * 1024)
215         logger.info(f"Dynamic highlight reel saved: {output_path}
216         ({output_size_mb:.1f} MB)")
217         return output_path
218     except subprocess.CalledProcessError as e:
219         stderr_msg = e.stderr.decode(errors='replace').strip()
220         raise RuntimeError(
221             f"[COMPILER ERROR] FFmpeg concatenation failed.\n"
222             f"Command: {' '.join(concat_cmd)}\n"
223             f"FFmpeg stderr: {stderr_msg[-500:]}")
```

84. user (2026-06-11T05:07:07.042Z)

```
532     for segment in gemini_segments:
533         try:
534             start = float(segment["start_time"])
535             end = float(segment["end_time"])
536             dur = end - start
537
538             shot_count = segment.get("shuttle_exchange_count")
539             if shot_count is None: shot_count = max(3, int(dur / 0.8))
540             else:
541                 try: shot_count = int(shot_count)
542                 except (ValueError, TypeError): shot_count = max(3, int(dur / 0.8))
543
544             metadata = {
545                 "shot_count": shot_count,
546                 "shot_count_confidence": segment.get("shot_count_confidence",
547                 "Unknown"),
```

85. user (2026-06-11T05:07:07.461Z)

```
1     from abc import ABC, abstractmethod
2     from typing import Tuple
3
4     class AIVideoProvider(ABC):
5         def __init__(self, api_key: str, model_name: str):
```

```

6         self.api_key = api_key
7         self.model_name = model_name
8
9         @abstractmethod
10        def analyze_video(self, video_path: str, prompt: str,
11                           chunk_duration: float = 0.0,
12                           label: str = "") -> Tuple[list, dict]:
13            """Sends a video file to the AI model along with a prompt, and returns a tuple
14            of
15            (segments_list, metrics_dict).
16
17            Args:
18                video_path: Path to the video file to analyze.
19                prompt: Prompt instructions.
20                chunk_duration: Duration of this chunk in seconds, or 0.0 if not chunked.
21                label: Diagnostic label for logging.
22
23            Returns:
24                Tuple containing:
25                - A list of detected segment dicts (e.g. [{"start_time": ..., "end_time":
26                  ...}])
27                - A dictionary of performance metrics (e.g. {"upload_time": ...,
28                  "process_time": ..., "gen_time": ...})
29            """
30            pass

```

86. user (2026-06-11T05:07:22.432Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

87. user (2026-06-11T05:07:24.783Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

88. user (2026-06-11T05:07:27.555Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\yolo_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

89. user (2026-06-11T05:07:35.778Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

90. user (2026-06-11T05:07:37.235Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

91. user (2026-06-11T05:07:38.603Z)

```

1        """Per-video observability reports for the indexer.
2
3        `obsreport` is a SOFT dependency: if it is not installed, this module degrades to a
4        no-op and the pipeline runs exactly as before. The single entrypoint
5        `emit_indexer_report(...)` is designed to be called from a `finally` block and
6        NEVER raises ? emitting a report must not be able to break a processing run.
7        """

```

```

8     from __future__ import annotations
9
10    import os
11    from typing import Any, Dict, List, Optional
12
13    from . import run_signals # no obsreport dependency ? always importable
14
15    try: # soft dependency ? never hard-fail the pipeline on a reporting import
16        import obsreport # noqa: F401
17        from . import harvest
18
19        OBSREPORT_AVAILABLE = True
20    except Exception: # pragma: no cover - exercised only when obsreport is absent
21        OBSREPORT_AVAILABLE = False
22
23
24    def report_enabled(config: Optional[Dict[str, Any]]) -> bool:
25        return bool((config or {}).get("report", {}).get("enabled", True))
26
27
28    def emit_indexer_report(
29        *,
30        db,

```

92. user (2026-06-11T05:07:38.791Z)

```

1     """Low-level moving-object detectors that run outside the main Python process.
2
3     Houses:
4     - DetectorRunner ABC (the seam for WSL vs future Linux-native trajectory detectors).
5     - TrackNetRunner / WasbRunner (current WSL adapters; both inherit DetectorRunner).
6     - rally_gate (pure post-filter, no WSL).
7     - wasb_infer.py (the actual code that runs *inside* the WSL wasb env).
8
9     The ABC de-risks the WSL dependency (low-regret item 6 continuation): new
10    runners can be added without changing hybrid segmenters or windowing logic.
11    See base.py docstring for the exact contract + "how to add a runner" guide.
12
13    Keeping the heavy TF/torch deps quarantined in WSL (or a native subprocess)
14    is deliberate: the main project env stays free of them.
15    """
16
17    from .base import DetectorRunner # re-export the ABC for easy "from
backend.pipeline.detectors import DetectorRunner"
18
19    # The concrete runners are imported directly where needed (hybrids, eval/*).
20    # This keeps heavy optional deps from being pulled at import time of the package.
21
22

```

93. user (2026-06-11T05:07:47.339Z)

No matches found

94. user (2026-06-11T05:07:49.389Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors__init__.py has been updated successfully. (file state is current in your context ? no need to Read it back)

95. user (2026-06-11T05:07:55.753Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting__init__.py has been updated successfully. (file state is current in your context ? no need to Read it back)

96. user (2026-06-11T05:07:56.751Z)

```
50     Speed is normalized to fraction-of-frame-width PER SECOND (`* fps`), matching the
51     heuristic windowing brain ? previously it was per-frame, so the same physical
shuttle
52     speed read ~2x smaller at 59.94 fps than at 30 fps, confounding cross-footage
transfer
53     on a corpus that mixes frame rates. Speed/crossings are also not computed across a
long
54     occlusion gap (mirrors the heuristic resetting on invisibility)."""
55     pts = sorted(points, key=lambda p: p.frame)
56     N = (pts[-1].frame + 1) if pts else 1
57     vis = np.zeros(N); x = np.zeros(N); y = np.zeros(N)
58     for p in pts:
59         if 0 <= p.frame < N and p.visible and p.x >= 0:
60             vis[p.frame] = 1.0; x[p.frame] = float(p.x); y[p.frame] = float(p.y)
61
62     mask = vis > 0
63     play_is_x = _spread(x[mask]) >= _spread(y[mask])
64     coord = x if play_is_x else y
65     net = float(np.median(coord[mask])) if mask.any() else 0.0
66     deadband = 0.06 * (_spread(coord[mask]) if mask.sum() > 1 else 1.0)
67
68     speed = np.zeros(N); spd_mask = np.zeros(N); cross = np.zeros(N)
69     max_gap = max(2, round(0.25 * fps)) # frames; a longer gap = occlusion break, not
motion
70     prev = None
71     for p in pts:
72         f = p.frame
73         if not (0 <= f < N and vis[f]):
74             continue
75         if prev is not None:
76             df = f - prev
77             if 0 < df <= max_gap:
78                 speed[f] = float(np.hypot(x[f] - x[prev], y[f] - y[prev]) / frame_width
/ df * fps)
79                 spd_mask[f] = 1.0
80                 c, pc = coord[f] - net, coord[prev] - net
81                 if abs(c) > deadband and abs(pc) > deadband and (c > 0) != (pc > 0):
82                     cross[f] = 1.0
83         prev = f
84     return dict(N=N, vis=vis, x=x, y=y, speed=speed, spd_mask=spd_mask, cross=cross,
fw=frame_width)
85
86
87     def features(arr: Dict, fps: float, win_sec: float = DEFAULT_WIN_SEC,
88                 stride_sec: float = DEFAULT_STRIDE_SEC):
89         """Windowed per-frame features sampled at a stride -> (X[n_samples,6],
times[n_samples])."""
90         N = arr["N"]; fw = arr["fw"]
91         size = max(3, int(win_sec * fps) * 2 + 1)
92         vis, speed, spd_mask, cross, x, y = (arr[k] for k in ("vis", "speed", "spd_mask",
"cross", "x", "y"))
93
94         n = uniform_filter1d(vis, size, mode="constant") # mean
visibility (== vis_rate)
```

97. assistant (2026-06-11T05:08:04.657Z)

`rally\_seq.py` is the train==serve featurizer ? the E702 fixes here are pure line-splitting, no logic change:

98. user (2026-06-11T05:08:07.939Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\features\rally_seq.py has

been updated successfully. (file state is current in your context ? no need to Read it back)

99. user (2026-06-11T05:08:09.831Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\features\rally_seq.py has been updated successfully. (file state is current in your context ? no need to Read it back)

100. user (2026-06-11T05:08:11.935Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\features\rally_seq.py has been updated successfully. (file state is current in your context ? no need to Read it back)

101. user (2026-06-11T05:08:20.435Z)

```
174         f"[{window_start:.0f},{window_end:.0f}]s; {len(grid)}x{len(gate_options)}
configs) ==")
175     print("NOTE: fit-on-self on ONE video = optimistic; real test is leave-one-video-
out.\n")
176     hdr = f"{'#':>3} {'veloc':>6} {'merge':>6} {'minDur':>6} {'gate':>4} " \
177         f"{'F1':>6} {'prec':>6} {'recall':>6} {'mIoU':>6} {'TP':>3} {'FP':>3}
{'FN':>3}"
178     print(hdr); print("-" * len(hdr))
179     for i, r in enumerate(rows[:top], 1):
180         vt, mg, mwd = r["cfg"]
181         print(f"{i:>3} {vt:>6.3f} {mg:>6.1f} {mwd:>6.1f} {r['gate']:>4} "
```

102. user (2026-06-11T05:08:21.349Z)

```
120
121     results = []
122     for test in names:
123         train = [data[n] for n in names if n != test]
124         Xtr = np.vstack([v["X"] for v in train]); ytr = np.concatenate([v["y"] for v in
train])
125         clf = LogisticRegression(lr=0.2, epochs=2000, l2=1e-2)
126         clf.fit(Xtr, ytr, feature_names=FEAT_NAMES)
127         # tune the decision threshold on TRAIN segment-F1 (no test leakage)
```

103. user (2026-06-11T05:08:22.041Z)

```
64
65
66     def sign_test(learned: List[float], floor: List[float]):
67         """Paired one-sided sign test: is learned > floor more often than chance?
68
69         At n=6 a mean comparison alone is inside noise (SE ~0.11); the paired per-video
70         wins are the defensible statistic. Returns (wins, n, p_one_sided)."""
71         pairs = [(l, f) for l, f in zip(learned, floor) if l != f] # ties drop, standard
72         wins = sum(1 for l, f in pairs if l > f)
73         n = len(pairs)
74         p = sum(comb(n, k) for k in range(wins, n + 1)) / (2 ** n) if n else 1.0
75         return wins, n, p
76
77
78     def train_final(specs: List[Dict]):
79         """Fit the FINAL model on ALL corpus videos; threshold tuned on the same (train-
side).
80
81         The honest generalization estimate is the LOVO result, not this fit ? recorded as
such."""
82         vids = []
83         for s in specs:
84             pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
85             arr = build_frame_arrays(pts, float(s["frame_width"]), float(s["fps"]))
```

```

86         X, times = features(arr, float(s["fps"]))
87         gts = load_gt_intervals(s["labels"], strict=False)
88         vids.append({"name": s["name"], "x": X, "times": times,
89                     "y": labels_for(times, gts), "gts": gts})
90     Xall = np.vstack([v["X"] for v in vids])
91     yall = np.concatenate([v["y"] for v in vids])
92     clf = LogisticRegression(lr=0.2, epochs=2000, l2=1e-2)
93     clf.fit(Xall, yall, feature_names=FEAT_NAMES)
94     thr = max(THRESHOLD_GRID, key=lambda t: _seg_f1(clf, vids, float(t)))
95     return clf, float(thr)
96
97
98     def gate_verdict(eval_res: dict, floor: Optional[dict]) -> dict:
99         """Pre-flight ship-gate. ship=True requires EVERY block to clear ? and two blocks
100         (undefined F1@tIoU target; WASB C3 provenance) are open project-wide, so today this
101         can only ever say 'floor passed, not shippable yet'. That is by design: honesty
102         over a green light."""
103         reasons: List[str] = []
104         floor_passed = None
105         sign = None
106         if floor:
107             floor_mean = float(floor["mean_f1"])
108             floor_passed = eval_res["mean_f1"] > floor_mean
109             if not floor_passed:
110                 reasons.append(f"LOVO mean {eval_res['mean_f1']:.3f} <= floor
111 {floor_mean:.3f}")
112             per = floor.get("per_video", {})
113             paired = [(r["f1"], per[r["video"]]) for r in eval_res["per_video"] if
114 r["video"] in per]
115             if paired:
116                 wins, n, p = sign_test([l for l, _ in paired], [f for _, f in paired])
117                 sign = {"wins": wins, "n": n, "p_one_sided": round(p, 4)}
118                 if p >= 0.05:
119                     reasons.append(f"paired sign test not significant ({wins}/{n} wins,
120 p={p:.3f}) "
121                                   f"? grow the corpus before claiming superiority")

```

104. user (2026-06-11T05:08:34.036Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\eval_partial_labels.py has been updated successfully. (file state is current in your context ? no need to Read it back)

105. user (2026-06-11T05:08:35.875Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\rally_seq_proto.py has been updated successfully. (file state is current in your context ? no need to Read it back)

106. user (2026-06-11T05:08:38.400Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\training\gen0\harness.py has been updated successfully. (file state is current in your context ? no need to Read it back)

107. user (2026-06-11T05:08:39.510Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\training\gen0\harness.py has been updated successfully. (file state is current in your context ? no need to Read it back)

108. assistant (2026-06-11T05:08:46.836Z)

Now the test-file findings ? reading each site:

109. user (2026-06-11T05:08:47.733Z)

```

38         failure = inst.process_video("v1", "fail")

```

```

39     assert isinstance(failure, Failure)
40     assert failure.is_success() is False
41     assert failure.is_recoverable is True
42     assert "Simulated failure" in failure.message
43     try:
44         failure.unwrap()
45         assert False, "Should have raised"
46     except RuntimeError as e:
47         assert "Simulated failure" in str(e)
48

```

110. user (2026-06-11T05:08:48.681Z)

```

1     import sqlite3
2     from backend.infrastructure.database import Database
3
4     def test_database_init(tmp_path):
5         db_path = tmp_path / "test.db"
6         db = Database(str(db_path))
7
8         conn = sqlite3.connect(str(db_path))
9         cursor = conn.cursor()
10        cursor.execute("SELECT name FROM sqlite_master WHERE type='table'")
11        tables = [r[0] for r in cursor.fetchall()]
12        assert "videos" in tables
13        assert "play_segments" in tables
14        conn.close()
15

```

111. user (2026-06-11T05:08:49.330Z)

```

10
11    def test_detector_timeouts_default_to_bounded():
12        assert WasbConfig().timeout_sec == 1800
13        assert TrackNetConfig().timeout_sec == 1800
14        # from_indexing_cfg with no override also defaults bounded (not 0 = hang-forever).
15        assert WasbConfig.from_indexing_cfg({}).timeout_sec == 1800
16        assert TrackNetConfig.from_indexing_cfg({}).timeout_sec == 1800
17
18
19    def test_chunk_overlap_ge_duration_rejected():
20        with pytest.raises(Exception):
21            IndexingConfig(chunk_duration=90.0, chunk_overlap=90.0)
22        with pytest.raises(Exception):
23            AppConfig(indexing={"chunk_duration": 90.0, "chunk_overlap": 120.0})
24        # Sane config still loads.
25        ok = IndexingConfig(chunk_duration=90.0, chunk_overlap=10.0)
26        assert ok.chunk_overlap < ok.chunk_duration
27
28
29    def test_sqlite_wal_and_busy_timeout(tmp_path):

```

112. user (2026-06-11T05:09:02.762Z)

```

1     from backend.pipeline.segmenters.base import BasePlaySegmenter, SegmenterRegistry
2     from backend.results import Success, Failure, Result # test the new contract
3
4     class DummySegmenter(BasePlaySegmenter):
5         @property
6         def name(self) -> str:
7             return "dummy"
8
9         @property
10        def description(self) -> str:

```

```
11         return "Dummy segmenter for testing."
12
```

113. user (2026-06-11T05:09:03.698Z)

```
58         "duration", "confidence", "stats", "detection_params",
59         "confirmed_segment_id", "human_label", "notes", "created_at",
60     }
61
62
63     def test_candidate_schema_contract(tmp_path):
64         db = Database(str(tmp_path / "test.db"))
65         conn = sqlite3.connect(str(tmp_path / "test.db"))
66         cols = conn.execute("PRAGMA table_info(candidate_segments)").fetchall()
67         conn.close()
68
69         names = {c[1] for c in cols}
70         assert names == EXPECTED_CANDIDATE_COLUMNS
71
```

114. user (2026-06-11T05:09:04.414Z)

```
1     """Reliability hardening: WSL/provider timeouts, SQLite WAL+busy_timeout, chunk-step
2     guard."""
3     import pytest
4
5     from backend.config.models import AppConfig, IndexingConfig
6     from backend.infrastructure.database import Database
7     from backend.pipeline.detectors.wasb_runner import WasbConfig
8     from backend.pipeline.detectors.tracknet_runner import TrackNetConfig
9
10
```

115. user (2026-06-11T05:09:12.627Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

116. user (2026-06-11T05:09:14.485Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

117. user (2026-06-11T05:09:16.366Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_database.py has been updated successfully. (file state is current in your context ? no need to Read it back)

118. user (2026-06-11T05:09:17.777Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_database.py has been updated successfully. (file state is current in your context ? no need to Read it back)

119. user (2026-06-11T05:09:19.664Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reliability_hardening.py has been updated successfully. (file state is current in your context ? no need to Read it back)

120. user (2026-06-11T05:09:22.082Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reliability_hardening.py has been updated successfully. (file state is current in your context ? no need to Read it back)

121. assistant (2026-06-11T05:09:30.431Z)

Last four sites ? the remaining test E702s and the E741 in `test\_wasb\_cache.py`:

122. user (2026-06-11T05:09:31.266Z)

```

25         # 10:1 imbalance ? balanced weighting should still catch the positives.
26         rng = np.random.default_rng(1)
27         pos = rng.normal([15, 0.4], 0.5, size=(20, 2))
28         neg = rng.normal([4, 0.05], 0.5, size=(200, 2))
29         X = np.vstack([pos, neg]); y = np.array([1] * 20 + [0] * 200)
30         clf = LogisticRegression(epochs=800).fit(X, y, class_weight="balanced")
31         pred = clf.predict(X)
32         recall = pred[:20].mean() # fraction of positives caught

```

123. user (2026-06-11T05:09:32.039Z)

```

36     def _write_labels(path):
37         with open(path, "w", newline="", encoding="utf-8") as fh:
38             w = csv.writer(fh)
39             w.writerow(["rally_number", "start_time", "end_time"])
40             w.writerow([1, 0.0, 20.0]); w.writerow([2, 40.0, 60.0]); w.writerow([3, 80.0,
100.0])
41         return str(path)
42
43

```

124. user (2026-06-11T05:09:32.932Z)

```

32     def _write_labels(path):
33         with open(path, "w", newline="", encoding="utf-8") as fh:
34             w = csv.writer(fh)
35             w.writerow(["rally_number", "start_time", "end_time"])
36             w.writerow([1, 0.0, 20.0]); w.writerow([2, 40.0, 60.0]); w.writerow([3, 80.0,
100.0])
37         return str(path)
38
39

```

125. user (2026-06-11T05:09:33.523Z)

```

88         windows_done, det = wi.load_and_clean_cache(cache)
89         assert windows_done == 2 # only the two clean windows
survive
90         assert det[1] == [[2, 2, 0.8, 0]]
91         # file is rewritten to the clean prefix -> appending window 2 stays contiguous
92         with open(det_jsonl) as f:
93             lines = [json.loads(l) for l in f if l.strip()]
94             assert [o["w"] for o in lines] == [0, 1]
95             _write_windows(cache, [_win(2, [(2, [[3, 3, 0.7, 0]]))])
96             assert wi.load_and_clean_cache(cache)[0] == 3
97

```

126. user (2026-06-11T05:09:42.131Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_eval_classifier.py has been updated successfully. (file state is current in your context ? no need to Read it back)

127. user (2026-06-11T05:09:43.986Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_rally_seq_proto.py has been updated successfully. (file state is current in your context ? no need to Read it back)

128. user (2026-06-11T05:09:46.335Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_training_gen0.py has been updated successfully. (file state is current in your context ? no need to Read it back)

129. user (2026-06-11T05:09:47.468Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_wasb_cache.py has been updated successfully. (file state is current in your context ? no need to Read it back)

130. user (2026-06-11T05:10:21.473Z)

```
F841 Local variable `result` is assigned to but never used
--> backend\pipeline\stitcher\compiler.py:153:21
    |
152 |         try:
153 |             result = subprocess.run(cmd, capture_output=True, check=True)
    |             ^^^^^^
154 |             # Verify the output clip was actually created and has content
155 |             if os.path.exists(clip_path) and os.path.getsize(clip_path) > 0:
    |
help: Remove assignment to unused variable `result`
```

Found 1 error.

No fixes available (1 hidden fix can be enabled with the `--unsafe-fixes` option).

ruff exit: 1

```
..... [100%]
349 passed in 24.09s
=====
RESULT: PASS - all tests green
=====
```

131. user (2026-06-11T05:10:33.536Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

132. assistant (2026-06-11T05:10:41.608Z)

Now adding the lint lane to CI:

133. user (2026-06-11T05:10:46.357Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.github\workflows\ci.yml has been updated successfully. (file state is current in your context ? no need to Read it back)

134. user (2026-06-11T05:11:20.456Z)

All checks passed!

ruff exit: 0

349 passed in 23.93s

```
=====
RESULT: PASS - all tests green
=====
```

```
M .github/workflows/ci.yml
M backend/config/models.py
M backend/eval/calibrate_wasb.py
M backend/eval/calibration.py
M backend/eval/distill_local.py
M backend/eval/eval_partial_labels.py
M backend/eval/rally_seq_proto.py
M backend/features/rally_seq.py
M backend/infrastructure/database.py
M backend/main.py
```

```

M backend/pipeline/detectors/__init__.py
M backend/pipeline/segmenters/base.py
M backend/pipeline/segmenters/gemini.py
M backend/pipeline/segmenters/motion.py
M backend/pipeline/segmenters/yolo_hybrid.py
M backend/pipeline/stitcher/compiler.py
M backend/pipeline/validators/base.py
M backend/pipeline/validators/blur.py
M backend/pipeline/validators/format.py
M backend/pipeline/validators/pts_continuity.py
M backend/pipeline/validators/relevance.py
M backend/pipeline/validators/resolution_fps.py
M backend/pipeline/validators/scene_cut.py
M backend/providers/base.py
M backend/reporting/__init__.py
M backend/utils/video.py
M run_process_and_stitch.py
D scripts/generate_khelacharya_flyer.py
M setup.py
M tests/test_base.py
M tests/test_config.py
M tests/test_database.py
M tests/test_eval_classifier.py
M tests/test_eval_regression.py
M tests/test_gemini.py
M tests/test_geometry.py
M tests/test_path_safety.py
M tests/test_provider_gemini.py
M tests/test_providers.py
M tests/test_rally_seq_proto.py
M tests/test_reliability_hardening.py
M tests/test_reporting.py
M tests/test_sport_profiles.py
M tests/test_tracknet_hybrid.py
M tests/test_training_gen0.py
M tests/test_video.py
M tests/test_wasb_cache.py
M tests/test_yolo_hybrid.py
M training/gen0/harness.py
?? ruff.toml

```

135. user (2026-06-11T05:11:43.909Z)

```

warning: in the working copy of 'backend/providers/base.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'tests/test_eval_classifier.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'tests/test_geometry.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_providers.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_sport_profiles.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'tests/test_video.py', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'ruff.toml', LF will be replaced by CRLF the next time Git
touches it
[ci/ruff-lint-lane 75659d1] ci: add ruff lint lane + fix all violations; remove never-compiled
flyer stub
50 files changed, 214 insertions(+), 230 deletions(-)
create mode 100644 ruff.toml
delete mode 100644 scripts/generate_khelacharya_flyer.py
remote:
remote: Create a pull request for 'ci/ruff-lint-lane' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/ci/ruff-lint-lane

```

```
remote:
branch 'ci/ruff-lint-lane' set up to track 'origin/ci/ruff-lint-lane'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      ci/ruff-lint-lane -> ci/ruff-lint-lane
```

136. user (2026-06-11T05:12:06.169Z)

Warning: 8 uncommitted changes
<https://github.com/avidullu/badminton-highlight-indexer/pull/92>

137. user (2026-06-11T05:12:28.088Z)

```
Exit code 2
[notice] To update, run: python.exe -m pip install --upgrade pip
mypy 2.1.0 (compiled: yes)
5
--- by error code ---
```

138. assistant (2026-06-11T05:12:36.966Z)

Exit 2 with no parsed codes ? looking at the raw output:

139. user (2026-06-11T05:12:39.069Z)

```
Exit code 2
backend\annotations\base.py: error: Source file found twice under different module names:
"annotations.base" and "backend.annotations.base"
backend\annotations\base.py: note: See
https://mypy.readthedocs.io/en/stable/running_mypy.html#mapping-file-paths-to-modules for more
info
backend\annotations\base.py: note: Common resolutions include:
backend\annotations\base.py: note:     a) adding `__init__.py` somewhere,
backend\annotations\base.py: note:     b) using `--explicit-package-bases` or adjusting
`MYPYPATH`
Found 1 error in 1 file (errors prevented further checking)
```

140. user (2026-06-11T05:13:35.537Z)

```
Exit code 1
backend/__init__.py : False
backend/annotations/__init__.py : True
training/__init__.py : True
training/gen0/__init__.py : True
backend\pipeline\segmenters\yolo_hybrid.py:478: error: Need type annotation for
"gemini_segments" (hint: "gemini_segments: list[<type>] = ...") [var-annotated]
backend\pipeline\segmenters\yolo_hybrid.py:520: error: Incompatible types in assignment
(expression has type "Future[list[dict[Any, Any]]]", variable has type "Future[tuple[int, float,
str, bool]]") [assignment]
Found 82 errors in 17 files (checked 75 source files)
```

141. user (2026-06-11T05:13:53.196Z)

```
Exit code 1
Count Name
-----
26 assignment
19 union-attr
17 arg-type
5 annotation-unchecked
5 override
4 attr-defined
4 var-annotated
3 return-value
2 misc
```



```
1 call-overload
1 index
```

--- by file ---

Count Name

```
26 backend\main.py
 9 backend\pipeline\segmenters\motion.py
 8 backend\pipeline\segmenters\yolo_hybrid.py
 7 backend\pipeline\segmenters\wasb_hybrid.py
 6 backend\pipeline\segmenters\tracknet_hybrid.py
 4 backend\eval\calibrate_local.py
 4 backend\pipeline\segmenters\gemini.py
 4 backend\pipeline\segmenters\base.py
 3 backend\pipeline\audio\hit_detector.py
 2 backend\eval\gemini_refine.py
 2 backend\eval\distill_local.py
 2 backend\pipeline\detectors\wasb_infer.py
 1 backend\utils\paths.py
 1 backend\eval\harness.py
 1 backend\eval\gt_loader.py
 1 backend\pipeline\validators\base.py
 1 backend\utils\video.py
```

142. user (2026-06-11T05:14:20.002Z)

Exit code 1

```
backend\utils\paths.py:36: error: Incompatible default
for parameter "allowed_root" (default
has type "None",
parameter has type "str")
[assignment]
backend\utils\video.py:21: error: Incompatible default
for parameter "scale_width" (default
has type "None", parameter
has type "int") [assignment]
backend\eval\gt_loader.py:71: error: "None" object is not iterable [misc]
backend\eval\harness.py:101: error: Unsupported target for indexed assignment ("object")
[index]
backend\eval\calibrate_local.py:50: error: List comprehension
has incompatible type List[tuple[Any,
Any]]; expected
List[dict[str, Any]] [misc]
backend\eval\calibrate_local.py:52: error: Incompatible types
in assignment (expression has type
"list[tuple[float,
float]]", variable has type
"list[dict[str, Any]]") [assignment]
backend\eval\calibrate_local.py:52: error: Argument 2 to
"apply_gate" has incompatible type
"list[dict[str, Any]]";
expected "list[tuple[float, float]]" [arg-
type]
backend\eval\calibrate_local.py:53: error: Incompatible return
value type (got "list[dict[str,
Any]]", expected
"list[tuple[float, float]]") [return-value]
backend\pipeline\validators\base.py:51: error: Need type
annotation for "context" (hint:
"context: dict[<type>,
<type>] = ...") [var-annotated]
backend\pipeline\audio\hit_detector.py:28: error: Incompatible
default for parameter "out_wav"
(default has type
```

```

"None", parameter has type "str")
[assignment]
backend\pipeline\audio\hit_detector.py:65: error: Incompatible
default for parameter "band"
(default has type "None",
parameter has type "tuple[float,
float]") [assignment]
backend\pipeline\audio\hit_detector.py:107: error: Incompatible
default for parameter "band"
(default has type "None",
parameter has type "tuple[float,
float]") [assignment]
backend\pipeline\detectors\wasb_infer.py:163: error: Need type
annotation for "valid" (hint:
"valid: list[<type>] =
..." [var-annotated]
backend\pipeline\detectors\wasb_infer.py:209: error: Incompatible
default for parameter "cache_dir"
(default has type
"None", parameter has type "str")
[assignment]
backend\eval\gemini_refine.py:83: error: Argument "scale_width"
to "extract_video_chunk" has incompatible
type "int |
None"; expected "int" [arg-type]
backend\eval\gemini_refine.py:171: error: Argument "scale_width"
to "extract_video_chunk" has incompatible
type "int |
None"; expected "int" [arg-type]
backend\eval\distill_local.py:98: error: Incompatible default
for parameter "model_out" (default
has type "None",
parameter has type "str")
[assignment]
backend\eval\distill_local.py:135: error: Item "None" of
"ndarray[tuple[Any, ...], dtype[Any]] |
None" has no
attribute "__iter__" (not iterable)
[union-attr]

```

143. user (2026-06-11T05:14:39.338Z)

```

55     intervals: List[Interval] = []
56     skipped = 0
57     cols: Optional[Tuple[int, int]] = None    # (start_idx, end_idx); None until detected
58     saw_header = False
59
60     with open(csv_path, newline="", encoding="utf-8-sig") as f:
61         for lineno, row in enumerate(csv.reader(f), start=1):
62             if not row or not (row[0] or "").strip() or row[0].strip().startswith("#"):
63                 continue
64             if cols is None and not saw_header:
65                 header = _header_columns(row)
66                 if header is not None:
67                     cols = header
68                     saw_header = True
69                     continue          # consume the header row
70             cols = (0, 1)            # headerless: positional start,end
71             si, ei = cols
72             try:
73                 if len(row) <= max(si, ei):
74                     raise ValueError(f"expected at least {max(si, ei) + 1} columns, got
{row!r}")
75                 start = parse_timestamp(row[si])
76                 end = parse_timestamp(row[ei])

```

```

77         if start < 0:
78             raise ValueError(f"negative start ({start})")
79         if start >= end:

```

144. user (2026-06-11T05:14:39.992Z)

```

92     def report_to_dict(report: EvalReport) -> dict:
93         """JSON-serialisable view of an EvalReport."""
94         out = {
95             "video_id": report.video_id,
96             "iou_threshold": report.iou_threshold,
97             "num_ground_truth": len(report.gt_intervals),
98             "stages": {},
99         }
100         for stage, sr in report.stages.items():
101             out["stages"][stage] = {
102                 "score": sr.score.as_dict(),
103                 "pr_curve": [s.as_dict() for s in sr.pr_curve],
104             }
105         return out

```

145. user (2026-06-11T05:14:40.931Z)

```

40         """predict_fn(cfg) -> rally windows from a cached WASB trajectory, LOCAL ONLY
41         (windowing thresholds + the shuttle net-crossing gate). Parsed once."""
42         points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
43
44         def predict_fn(cfg: LocalConfig) -> List[Tuple[float, float]]:
45             vt, mg, mwd, mc = cfg
46             wins = TrackNetRunner.trajectory_to_action_windows(
47                 points, fps=fps, frame_width=frame_width,
48                 velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd,
49             )
50             wins = [(w["start"], w["end"]) for w in wins]
51             if mc > 0:
52                 wins = rally_gate.apply_gate(points, wins, fps, min_crossings=mc)
53             return wins
54
55         return predict_fn
56
57

```

146. user (2026-06-11T05:14:41.818Z)

```

128         result["mean_learned_f1"] = mean(clf, "f1")
129         result["mean_baseline_f1"] = mean(base, "f1")
130
131         # Final model on ALL videos (for production / running on a fresh held-out video).
132         Xall = [row for v in videos for row in v["X"]]
133         yall = [t for v in videos for t in v["y"]]
134         final = LogisticRegression().fit(Xall, yall, FEATURE_NAMES)
135         weights = dict(zip(FEATURE_NAMES, [round(float(w), 3) for w in final.w]))
136         print(f"\nfinal model standardized weights (sign/size = influence): {weights}")
137         if model_out:
138             final.save(model_out)
139             print(f"final model (all videos) -> {model_out}")
140             result["model_out"] = model_out
141         return result

```

147. user (2026-06-11T05:15:01.829Z)

```

30         contained = False
31         if not contained:
32             raise ValueError(f"path {path!r} escapes the allowed root {root!r}")

```

```

33         return rp
34
35
36     def validate_input_path(path: str, allowed_root: str = None) -> None:
37         """Reject null bytes and (when an allowed_root is configured) out-of-root inputs.
38
39         With no allowed_root (default, single-user local mode) only the null-byte check
40         applies,

```

148. user (2026-06-11T05:15:02.163Z)

```

14         return float(duration_output)
15     except Exception as e:
16         logger.error(f"Could not detect video duration for {video_path}: {e}")
17         return 0.0
18
19     def extract_video_chunk(video_path: str, start_time: float, duration: float,
20                             output_path: str, reencode: bool = False,
21                             scale_width: int = None) -> bool:
22         """Extracts a chunk from a video file.
23
24         Args:
25             reencode: If True, re-encodes with libx264 (slower but frame-accurate).

```

149. user (2026-06-11T05:15:06.269Z)

rg: The term 'rg' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

rg: The term 'rg' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

rg: The term 'rg' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

150. user (2026-06-11T05:15:15.020Z)

```

28: def extract_audio(video_path: str, out_wav: str = None, sr: int = 16000) -> str:
29-     """ffmpeg ? mono 16-bit PCM WAV at `sr`. Returns the wav path (temp if not given).
30-     Audio-only decode, so it's fast even for multi-GB video."""
31-     --
64: def onset_envelope(samples: np.ndarray, sr: int, frame: int = 1024,
65-                     hop: int = 256, band: Tuple[float, float] = None) -> Tuple[np.ndarray,
66- float]:
67-     """Spectral-flux onset strength per hop (pure-numpy STFT). Returns (env, hop_seconds).
68-     --
136: def detect_hits_in_video(video_path: str, sr: int = 16000, **kw) -> List[HitEvent]:
137-     """Convenience: extract audio from a video and detect hits. Cleans up the temp wav."""
138-     wav = extract_audio(video_path, sr=sr)

```

151. user (2026-06-11T05:15:16.422Z)

```

161-     det_jsonl, _ = _cache_paths(cache_dir)
162-     det_by_fid = defaultdict(list)
163-     valid = []
164-     --
207: def run(frames_dir: str, weights: str, out_csv: str, sport: str = "badminton",
208-         limit: int = 0, batch_size: int = 8, num_workers: int = 4,
209-         log_every_batches: int = 50, cache_dir: str = None,

```

152. user (2026-06-11T05:15:17.248Z)

```
51:         context = {} # Shared context dictionary (e.g. for caching ffprobe outputs)
```

153. assistant (2026-06-11T05:15:31.680Z)

Checking the last site (hit_detector:107) and the `typing` imports in each file I'm about to touch:

154. user (2026-06-11T05:15:36.621Z)

```
=== backend/utils/paths.py ===
=== backend/utils/video.py ===
=== backend/pipeline/audio/hit_detector.py ===
from typing import List, Tuple
=== backend/eval/harness.py ===
from typing import List, Dict, Optional
=== backend/eval/distill_local.py ===
from typing import Dict, List, Tuple
=== backend/pipeline/validators/base.py ===
from typing import List, Dict, Any
=== backend/pipeline/detectors/wasb_infer.py ===
=== hit_detector 105-110 ===

def detect_hits(samples: np.ndarray, sr: int, k: float = 2.5, win_frames: int = 41,
                min_gap_sec: float = 0.08, band: Tuple[float, float] = None) -> List[HitEvent]:
    """Detect impact transients: local maxima of the onset envelope that exceed an
    adaptive threshold (moving mean + k*std), keeping the strongest within `min_gap_sec`.
    `k` is the sensitivity knob (lower = more hits); `band` band-limits the onset flux
```

155. assistant (2026-06-11T05:15:44.192Z)

All clear ? applying the annotation fixes across the nine files:

156. user (2026-06-11T05:15:45.671Z)

```
1     """Path-safety helpers for the API boundary (untrusted request fields)."""
2     import os
3
4
5     def safe_output_filename(name: str) -> str:
6         """Validate an output filename is a BARE name (no dir part, no traversal, not
absolute).
7
8         `os.path.join(output_dir, name)` silently discards `output_dir` when `name` is
absolute,
```

157. user (2026-06-11T05:15:46.273Z)

```
1     import subprocess
2     import logging
3
4     logger = logging.getLogger(__name__)
5
6     def get_video_duration(video_path: str) -> float:
7         """Returns the exact duration of a video file in seconds using ffprobe.
8         Returns 0.0 on failure."""
```

158. user (2026-06-11T05:15:55.646Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\paths.py has been updated successfully. (file state is current in your context ? no need to Read it back)

159. user (2026-06-11T05:15:58.483Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\paths.py has been updated successfully. (file state is current in your context ? no need to Read it back)

160. user (2026-06-11T05:15:59.117Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\video.py has been updated successfully. (file state is current in your context ? no need to Read it back)

161. user (2026-06-11T05:16:06.630Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

162. user (2026-06-11T05:15:59.875Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\video.py has been updated successfully. (file state is current in your context ? no need to Read it back)

163. user (2026-06-11T05:16:01.271Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

164. user (2026-06-11T05:16:03.152Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

165. user (2026-06-11T05:16:05.052Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

166. user (2026-06-11T05:16:15.915Z)

```
1      """Audio shuttle-hit ("thwack") detection ? local, dependency-light (numpy + ffmpeg).
2
3      Stages 0?1 of docs/AUDIO_RALLY_DETECTION_PLAN.md: extract the audio track, build a
4      spectral-flux onset envelope (with pre-emphasis to boost the broadband HF transient
5      a racket?shuttle contact makes), threshold it adaptively, and return (timestamp,
6      strength) hit candidates. Pure numpy + stdlib `wave` (NO scipy); ffmpeg (subprocess)
7      only for extraction. Consumed by the E1 go/no-go probe; later feeds rally fusion
8      (ADR-007) if E1 passes. WASB is never modified ? this is a separate signal.
9      """
10     from __future__ import annotations
11
12     import os
13     import wave
14     import tempfile
15     import subprocess
16     from dataclasses import dataclass
17     from typing import List, Tuple
18
19     import numpy as np
20
21
22     @dataclass
23     class HitEvent:
24         t: float          # seconds
25         strength: float    # onset-envelope peak value (arbitrary units)
26
27
28     def extract_audio(video_path: str, out_wav: str = None, sr: int = 16000) -> str:
29         """ffmpeg ? mono 16-bit PCM WAV at `sr`. Returns the wav path (temp if not given).
30         Audio-only decode, so it's fast even for multi-GB video."""
```

167. user (2026-06-11T05:16:23.866Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\audio\hit_detector.py has been updated successfully. (file state is current in your context ? no need to Read it back)

168. user (2026-06-11T05:16:25.738Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\audio\hit_detector.py has been updated successfully. (file state is current in your context ? no need to Read it back)

169. user (2026-06-11T05:16:27.622Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\audio\hit_detector.py has been updated successfully. (file state is current in your context ? no need to Read it back)

170. user (2026-06-11T05:16:29.246Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\audio\hit_detector.py has been updated successfully. (file state is current in your context ? no need to Read it back)

171. user (2026-06-11T05:16:39.898Z)

```
45         The per-run ``context`` accumulates shared values (notably
46         ``context['ffprobe_meta']`` produced by FormatValidator). ``run_all``
47         discards it; callers that want media metadata for an observability report
48         use this variant instead of re-probing.
49         """
50         results = []
51         context = {} # Shared context dictionary (e.g. for caching ffprobe outputs)
52         for val in self._validators:
53             try:
54                 res = val.validate(video_path, config, context)
```

172. user (2026-06-11T05:16:40.847Z)

```
158         [x, y, score, scale] candidates (accumulated across every window the frame is in,
159         in window order ? identical to a non-resumed run).
160         """
161         det_jsonl, _ = _cache_paths(cache_dir)
162         det_by_fid = defaultdict(list)
163         valid = []
164         if osp.exists(det_jsonl):
165             with open(det_jsonl, "r") as f:
166                 for line in f:
167                     s = line.strip()
```

173. user (2026-06-11T05:16:41.793Z)

```
205     # Inference
206     # ----- #
207     def run(frames_dir: str, weights: str, out_csv: str, sport: str = "badminton",
208            limit: int = 0, batch_size: int = 8, num_workers: int = 4,
209            log_every_batches: int = 50, cache_dir: str = None,
210            flush_every: int = 1000, fresh: bool = False) -> str:
211         import time
212         import torch
```

174. user (2026-06-11T05:16:42.636Z)

```
94         pos = [iv for iv, p in zip(intervals, proba) if p >= threshold]
95         return merge_overlaps(pos, gap_tol=1.0)
96
```

```
97
98     def run(specs: List[Dict], iou: float = 0.5, threshold: float = 0.5, model_out: str =
None) -> dict:
99         videos = [load_video(s, iou) for s in specs]
100         print(f"=== Distilled local rally classifier vs Gemini silver-GT "
101               f"({len(videos)} videos, IoU>={iou}, prob>={threshold}) ===")
```

175. user (2026-06-11T05:16:50.582Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

176. user (2026-06-11T05:16:52.412Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_infer.py has been updated successfully. (file state is current in your context ? no need to Read it back)

177. user (2026-06-11T05:16:53.857Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_infer.py has been updated successfully. (file state is current in your context ? no need to Read it back)

178. user (2026-06-11T05:16:56.182Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\distill_local.py has been updated successfully. (file state is current in your context ? no need to Read it back)

179. user (2026-06-11T05:16:57.625Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\distill_local.py has been updated successfully. (file state is current in your context ? no need to Read it back)

180. user (2026-06-11T05:16:59.826Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\distill_local.py has been updated successfully. (file state is current in your context ? no need to Read it back)

181. user (2026-06-11T05:17:11.651Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gt_loader.py has been updated successfully. (file state is current in your context ? no need to Read it back)

182. user (2026-06-11T05:17:13.058Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\harness.py has been updated successfully. (file state is current in your context ? no need to Read it back)

183. user (2026-06-11T05:17:14.596Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\harness.py has been updated successfully. (file state is current in your context ? no need to Read it back)

184. user (2026-06-11T05:17:17.276Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\calibrate_local.py has been updated successfully. (file state is current in your context ? no need to Read it back)

185. assistant (2026-06-11T05:17:34.891Z)

Now the mypy config with an explicit, ratchetable quarantine list:

186. assistant (2026-06-11T05:17:42.182Z)

```
[mypy]
```

Lenient, GREEN baseline (quality sweep AI-4): the gate protects the clean modules and pins an explicit quarantine list below ? not a typing crusade. Ratchet: when you touch a quarantined module, try removing its entry.

```
ignore_missing_imports = True
```

backend/ is a namespace package (no root __init__.py) ? required so mypy maps files to backend.* module names instead of duplicating top-level names.

```
explicit_package_bases = True
```

--- Quarantine (known-loose modules, each with a reason) ---

Optional module globals (db_instance/global_config set only in main()); properly fixed by the deferred app-factory/lifespan refactor (BACKLOG).

```
[mypy-backend.main]
```

```
ignore_errors = True
```

Legacy demo segmenter (fabricated metadata; coverage-omitted too).

```
[mypy-backend.pipeline.segmenters.motion]
```

```
ignore_errors = True
```

The hybrid segmenters share copy-paste P3/P4 typing looseness; slated for the trajectory-hybrid base-class collapse (quality sweep AI-7).

```
[mypy-backend.pipeline.segmenters.yolo_hybrid]
```

```
ignore_errors = True
```

```
[mypy-backend.pipeline.segmenters.wasb_hybrid]
```

```
ignore_errors = True
```

```
[mypy-backend.pipeline.segmenters.tracknet_hybrid]
```

```
ignore_errors = True
```

```
[mypy-backend.pipeline.segmenters.gemini]
```

```
ignore_errors = True
```

Registry builds cls(None, {}) probes; typing it honestly needs a Protocol.

```
[mypy-backend.pipeline.segmenters.base]
```

```
ignore_errors = True
```

187. user (2026-06-11T05:17:42.194Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\mypy.ini (file state is current in your context ? no need to Read it back)

188. user (2026-06-11T05:17:51.414Z)

```
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]
backend\sports\__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 75 source files
```

189. assistant (2026-06-11T05:18:13.245Z)

****Mypy is green**** across all 75 files with the quarantine config. Now checking #92, merging it, then committing AI-4 on a fresh branch:

190. user (2026-06-11T05:18:22.959Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325354259/job/80725196171
Full test suite (offline, mocked)               pass    1m32s  https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325354259/job/80725196182
Ruff lint                                         pass    12s    https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325354259/job/80725196249
Syntax sweep + import smoke (no GPU stack)      pass    19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325354259/job/80725196171
rollup: SUCCESS
failed to run git: error: Your local changes to the following files would be overwritten by
checkout:
    backend/eval/distill_local.py
    backend/pipeline/validators/base.py
    backend/utils/video.py
Please commit your changes or stash them before you switch branches.
Aborting

error: Your local changes to the following files would be overwritten by checkout:
    backend/eval/distill_local.py
    backend/pipeline/validators/base.py
    backend/utils/video.py
Please commit your changes or stash them before you switch branches.
Aborting
From https://github.com/avidullu/badminton-highlight-indexer
   dcc8ccc..595a5a1  master    -> origin/master
Already up to date.
```

191. user (2026-06-11T05:18:45.279Z)

```
Saved working directory and index state On ci/ruff-lint-lane: ai4-mypy-wip
M      docs/DEFERRED_HARDENING_PLAN.md
M      docs/UI_ALPHA_EXECUTION_PLAN.md
Your branch is behind 'origin/master' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)
Switched to branch 'master'
Updating dcc8ccc..595a5a1
Fast-forward
 .github/workflows/ci.yml          | 17 ++++++++
 backend/config/models.py          |  2 +-
 backend/eval/calibrate_wasb.py    |  1 -
 backend/eval/calibration.py       |  2 +-
 backend/eval/distill_local.py     |  4 +--
 backend/eval/eval_partial_labels.py |  3 ++
 backend/eval/rally_seq_proto.py   |  3 ++
 backend/features/rally_seq.py     | 17 ++++++----
 backend/infrastructure/database.py | 11 +-----
 backend/main.py                   | 35 ++++++++-----
 backend/pipeline/detectors/__init__.py |  5 +++-
 backend/pipeline/segmenters/base.py |  2 +-
 backend/pipeline/segmenters/gemini.py | 10 +++---
 backend/pipeline/segmenters/motion.py | 18 +++++-----
 backend/pipeline/segmenters/yolo_hybrid.py | 12 +++++---
 backend/pipeline/stitcher/compiler.py | 16 +++++-----
 backend/pipeline/validators/base.py |  2 +-
 backend/pipeline/validators/blur.py |  6 +---
 backend/pipeline/validators/format.py |  6 +---
 backend/pipeline/validators/pts_continuity.py |  2 +-
 backend/pipeline/validators/relevance.py | 12 +-----
```

```

backend/pipeline/validators/resolution_fps.py | 2 +-
backend/pipeline/validators/scene_cut.py      | 13 +++++---
backend/providers/base.py                     | 4 +--
backend/reporting/__init__.py                 | 1 -
backend/utils/video.py                       | 2 --
ruff.toml                                    | 21 ++++++++
run_process_and_stitch.py                    | 36 ++++++-----
scripts/generate_khelacharya_flyer.py        | 39 -----
setup.py                                     | 7 +---
tests/test_base.py                           | 12 +----
tests/test_config.py                         | 1 -
tests/test_database.py                       | 22 ++++++-----
tests/test_eval_classifier.py                 | 3 +-
tests/test_eval_regression.py                 | 1 -
tests/test_gemini.py                         | 25 ++++++-----
tests/test_geometry.py                       | 8 +---
tests/test_path_safety.py                    | 1 -
tests/test_provider_gemini.py                 | 19 ++++++-----
tests/test_providers.py                      | 4 +--
tests/test_rally_seq_proto.py                 | 4 +-
tests/test_reliability_hardening.py           | 6 +---
tests/test_reporting.py                       | 1 -
tests/test_sport_profiles.py                 | 6 +---
tests/test_tracknet_hybrid.py                 | 2 --
tests/test_training_gen0.py                   | 4 +-
tests/test_video.py                           | 3 +--
tests/test_wasb_cache.py                     | 2 +-
tests/test_yolo_hybrid.py                     | 1 -
training/gen0/harness.py                      | 8 +---

```

50 files changed, 214 insertions(+), 230 deletions(-)

create mode 100644 ruff.toml

delete mode 100644 scripts/generate_khelacharya_flyer.py

Switched to a new branch 'ci/mypy-lane'

On branch ci/mypy-lane

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

```

modified: backend/eval/calibrate_local.py
modified: backend/eval/distill_local.py
modified: backend/eval/gt_loader.py
modified: backend/eval/harness.py
modified: backend/pipeline/audio/hit_detector.py
modified: backend/pipeline/detectors/wasb_infer.py
modified: backend/pipeline/validators/base.py
modified: backend/utils/paths.py
modified: backend/utils/video.py
modified: docs/DEFERRED_HARDENING_PLAN.md
modified: docs/UI_ALPHA_EXECUTION_PLAN.md

```

Untracked files:

(use "git add <file>..." to include in what will be committed)

```

artifacts/
mypy.ini
scratch/HIGH_RISK_REMEDIATION_PLAN.md
scratch/PR1_BODY.md
scratch/PR1_HANDOFF.md
scratch/PR2_deferred_hardening_status.patch
scratch/PR88_FIX.md

```

no changes added to commit (use "git add" and/or "git commit -a")

Dropped refs/stash@{0} (8b3a2ace1217e551f19a269ddebde7e7787831c6)

```

M backend/eval/calibrate_local.py
M backend/eval/distill_local.py
M backend/eval/gt_loader.py

```

```
M backend/eval/harness.py
M backend/pipeline/audio/hit_detector.py
M backend/pipeline/detectors/wasb_infer.py
M backend/pipeline/validators/base.py
M backend/utils/paths.py
M backend/utils/video.py
?? mpyy.ini
```

192. assistant (2026-06-11T05:18:54.078Z)

Clean transfer. Adding the mpyy CI job, then full verification:

193. user (2026-06-11T05:18:59.107Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.github\workflows\ci.yml has been updated successfully. (file state is current in your context ? no need to Read it back)

194. user (2026-06-11T05:19:32.785Z)

```
Success: no issues found in 75 source files
All checks passed!
ruff exit: 0
349 passed in 23.17s
```

```
=====
RESULT: PASS - all tests green
=====
```

195. user (2026-06-11T05:20:00.883Z)

```
warning: in the working copy of 'mpyy.ini', LF will be replaced by CRLF the next time Git
touches it
[ci/mpyy-lane f49ce9b] ci: add mpyy lane (lenient green baseline + explicit quarantine list)
11 files changed, 74 insertions(+), 15 deletions(-)
create mode 100644 mpyy.ini
remote:
remote: Create a pull request for 'ci/mpyy-lane' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/ci/mpyy-lane
remote:
branch 'ci/mpyy-lane' set up to track 'origin/ci/mpyy-lane'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      ci/mpyy-lane -> ci/mpyy-lane
Warning: 8 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/93
```

196. assistant (2026-06-11T05:20:19.764Z)

PR [#93](https://github.com/avidullu/badminton-highlight-indexer/pull/93) is up. While its CI runs, **AI-5 (coverage floor)** ? measuring local coverage first to calibrate:

197. user (2026-06-11T05:20:51.361Z)

```
[notice] To update, run: python.exe -m pip install --upgrade pip
training\gen0\__init__.py          0          0    100%
training\gen0\harness.py          108         11    90%
-----
TOTAL                             4623       1245    73%
349 passed, 340 warnings in 26.54s
=====
RESULT: PASS - all tests green
=====
```

198. user (2026-06-11T05:21:33.214Z)

Count Name

```

-----
340 See https
340 ResourceWarning
1 -- Docs
1 ===== warnings summary =====
1 349 passed, 340 warnings in 25.71s

```

199. user (2026-06-11T05:21:46.645Z)

```

> C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\_pytest\unraisableexception.py:33:
ResourceWarning: unclosed database in
<sqlite3.Connection object at
0x000000215BCD5DA80>
    gc.collect()
> ResourceWarning: Enable tracemalloc to get the object allocation traceback
> C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\_pytest\unraisableexception.py:33:
ResourceWarning: unclosed database in
<sqlite3.Connection object at
0x000000215BCD5DB70>
    gc.collect()

```

200. user (2026-06-11T05:22:15.522Z)

```

tests\test_async_jobs.py:81:      with db._get_connection() as conn:
tests\test_async_jobs.py:98:      with db._get_connection() as c:
tests\test_async_jobs.py:139:     with db._get_connection() as c:
tests\test_database.py:151:      with db._get_connection() as conn:
backend\infrastructure\database.py:13:     def _get_connection(self):
backend\infrastructure\database.py:30:         with self._get_connection() as conn:
backend\infrastructure\database.py:148:             with self._get_connection() as conn:
backend\infrastructure\database.py:168:                 with self._get_connection() as conn:
backend\infrastructure\database.py:185:                     with self._get_connection() as conn:
backend\infrastructure\database.py:199:                         with self._get_connection() as conn:
backend\infrastructure\database.py:213:                             with self._get_connection() as conn:
backend\infrastructure\database.py:231:                                 with self._get_connection() as conn:
backend\infrastructure\database.py:250:                                     with self._get_connection() as conn:
backend\infrastructure\database.py:277:                                         with self._get_connection() as conn:
backend\infrastructure\database.py:287:                                             with self._get_connection() as conn:
backend\infrastructure\database.py:324:                                                 with self._get_connection() as conn:
backend\infrastructure\database.py:346:                                                     with self._get_connection() as conn:
backend\infrastructure\database.py:359:                                                         with self._get_connection() as conn:
backend\infrastructure\database.py:375:                                                             with self._get_connection() as conn:
backend\infrastructure\database.py:387:                                                                 with self._get_connection() as conn:
backend\infrastructure\database.py:398:                                                                     with self._get_connection() as conn:
backend\infrastructure\database.py:411:                                                                         with self._get_connection() as conn:
backend\infrastructure\database.py:425:                                                                             with self._get_connection() as conn:
backend\infrastructure\database.py:437:                                                                                 with self._get_connection() as conn:
backend\infrastructure\database.py:451:                                                                                     with self._get_connection() as conn:
tests\test_reliability_hardening.py:32: conn = db._get_connection()

```

201. assistant (2026-06-11T05:22:25.282Z)

24 internal sites, one mechanical pattern. First, merging #93 if green:

202. user (2026-06-11T05:22:37.957Z)

```

Full test suite (offline, mocked)      pass    1m37s    https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/2732563682/job/80726056073
Mypy (lenient baseline) pass           31s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/2732563682/job/80726056021
Ruff lint                               pass     12s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/2732563682/job/80726056082

```

```

Syntax sweep + import smoke (no GPU stack)      pass    23s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325636829/job/80726055988
rollup: SUCCESS
From https://github.com/avidullu/badminton-highlight-indexer
 * branch          master      -> FETCH_HEAD
    595a5a1..df9cf80 master    -> origin/master
Updating 595a5a1..df9cf80
Fast-forward
 .github/workflows/ci.yml          | 17 ++++++
 backend/eval/calibrate_local.py   |  4 ++-
 backend/eval/distill_local.py     |  6 +++--
 backend/eval/gt_loader.py         |  1 +
 backend/eval/harness.py           |  4 ++-
 backend/pipeline/audio/hit_detector.py |  8 +++---
 backend/pipeline/detectors/wasb_infer.py |  4 ++-
 backend/pipeline/validators/base.py |  2 +-
 backend/utils/paths.py            |  3 +-
 backend/utils/video.py            |  3 +-
 mpyy.ini                          | 37 ++++++
11 files changed, 74 insertions(+), 15 deletions(-)
create mode 100644 mpyy.ini
M      docs/DEFERRED_HARDENING_PLAN.md
M      docs/UI_ALPHA_EXECUTION_PLAN.md
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
Switched to a new branch 'fix/db-connection-close'

```

203. user (2026-06-11T05:22:49.740Z)

```
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
```

204. user (2026-06-11T05:22:58.976Z)

```

1      import sqlite3
2      import json
3      import logging
4      from typing import List, Dict, Any, Optional, Tuple
5
6      logger = logging.getLogger(__name__)
7
8      class Database:
9          def __init__(self, db_path: str):
10              self.db_path = db_path
11              self._init_db()
12
13          def _get_connection(self):
14              # busy_timeout: wait (not fail) on a locked DB ? needed once the async job model
15              # adds concurrent writers (a swallowed 'database is locked' silently drops
segments).
16              conn = sqlite3.connect(self.db_path, timeout=30.0)
17              conn.row_factory = sqlite3.Row
18              # Enforce ON DELETE CASCADE / SET NULL ? SQLite leaves FKs off per-connection.
19              conn.execute("PRAGMA foreign_keys = ON")
20              # WAL allows concurrent readers during a write; busy_timeout bounds lock waits.
21              conn.execute("PRAGMA busy_timeout = 30000")
22              try:
23                  conn.execute("PRAGMA journal_mode = WAL")
24              except sqlite3.OperationalError:
25                  pass # e.g. a read-only/networked FS that can't do WAL ? degrade gracefully
26              return conn
27
28          def _init_db(self):
29              """Initializes tables for videos, play_segments, and events."""
30              with self._get_connection() as conn:

```

```

31         cursor = conn.cursor()
32
33         # Videos Table
34         cursor.execute("""
35             CREATE TABLE IF NOT EXISTS videos (

```

205. user (2026-06-11T05:23:08.367Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py has been updated successfully. (file state is current in your context ? no need to Read it back)

206. user (2026-06-11T05:23:20.097Z)

```

backend/infrastructure/database.py | 56 ++++++-----
1 file changed, 36 insertions(+), 20 deletions(-)
All checks passed!
Success: no issues found in 1 source file

```

207. user (2026-06-11T05:24:01.389Z)

```

backend\results.py                22      0    100%
TOTAL                            4631   1245    73%
349 passed, 4 warnings in 25.87s
RESULT: PASS - all tests green

```

208. user (2026-06-11T05:24:30.248Z)

```

[fix/db-connection-close c1c940d] fix(db): actually close SQLite connections (transaction scope
!= close)
1 file changed, 36 insertions(+), 20 deletions(-)
remote:
remote: Create a pull request for 'fix/db-connection-close' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/db-connection-
close
remote:
branch 'fix/db-connection-close' set up to track 'origin/fix/db-connection-close'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      fix/db-connection-close -> fix/db-connection-close
Warning: 8 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/94

```

209. assistant (2026-06-11T05:24:39.540Z)

PR [#94](https://github.com/avidullu/badminton-highlight-indexer/pull/94). Watching CI, then proceeding to the coverage lane itself:

210. user (2026-06-11T05:26:08.991Z)

```

Full test suite (offline, mocked)      pass    1m26s  https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27325800496/job/80726544260
Mypy (lenient baseline) pass      30s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27325800496/job/80726544264
Ruff lint      pass      13s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27325800496/job/80726544296
Syntax sweep + import smoke (no GPU stack)      pass    24s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325800496/job/80726544347
rollup: SUCCESS
From https://github.com/avidullu/badminton-highlight-indexer
* branch          master      -> FETCH_HEAD
   df9cf80..1053505 master      -> origin/master
Updating df9cf80..1053505
Fast-forward
backend/infrastructure/database.py | 56 ++++++-----
1 file changed, 36 insertions(+), 20 deletions(-)

```

```
M      docs/DEFERRED_HARDENING_PLAN.md
M      docs/UI_ALPHA_EXECUTION_PLAN.md
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
Switched to a new branch 'ci/coverage-floor'
```

211. user (2026-06-11T05:26:22.970Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.github\workflows\ci.yml has been updated successfully. (file state is current in your context ? no need to Read it back)

212. user (2026-06-11T05:26:42.773Z)

```
Exit code 1
[ci/coverage-floor 355192e] ci: run the full suite with coverage (report only; floor follows)
 1 file changed, 6 insertions(+), 4 deletions(-)
remote:
remote: Create a pull request for 'ci/coverage-floor' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/ci/coverage-floor
remote:
branch 'ci/coverage-floor' set up to track 'origin/ci/coverage-floor'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      ci/coverage-floor -> ci/coverage-floor
Warning: 8 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/95
no checks reported on the 'ci/coverage-floor' branch
```

213. assistant (2026-06-11T05:26:50.551Z)

Checks hadn't registered yet ? retrying the watch and then reading the CI-measured coverage total from the job log:

214. user (2026-06-11T05:28:37.352Z)

```
Full test suite (offline, mocked)      pass    1m33s    https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325877401/job/80726777321
Mypy (lenient baseline) pass    28s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325877401/job/80726777385
Ruff lint      pass    8s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325877401/job/80726777340
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325877401/job/80726777294
```

```
Syntax sweep + import smoke (no GPU stack)      Run import smoke tests
2026-06-11T05:27:06.7962062Z
tests/test_import_smoke.py::test_backend_no_static_training_imports PASSED [
80%]
Syntax sweep + import smoke (no GPU stack)      Run import smoke tests
2026-06-11T05:27:06.8269427Z
===== 4 passed, 1 skipped in
3.46s =====
Full test suite (offline, mocked)      Run full suite (with coverage)
2026-06-11T05:28:18.3205695Z TOTAL
```

```
4631    1316
72%
Full test suite (offline, mocked)      Run full suite (with coverage)
2026-06-11T05:28:18.3206663Z 333 passed, 2 skipped,
5
warnings in 22.48s
```


215. user (2026-06-11T05:28:52.261Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.github\workflows\ci.yml has been updated successfully. (file state is current in your context ? no need to Read it back)

216. user (2026-06-11T05:29:04.064Z)

PR state: OPEN

[ci/coverage-floor 26b4fb7] ci: pin the coverage floor at 70 (CI-measured 72%)
1 file changed, 5 insertions(+), 2 deletions(-)

To <https://github.com/avidullu/badminton-highlight-indexer.git>
355192e..26b4fb7 ci/coverage-floor -> ci/coverage-floor

217. user (2026-06-11T05:30:54.309Z)

Full test suite (offline, mocked) pass 1m22s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325963180/job/80727038037>
Mypy (lenient baseline) pass 32s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325963180/job/80727038149>
Ruff lint pass 7s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325963180/job/80727038070>
Syntax sweep + import smoke (no GPU stack) pass 20s
<https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27325963180/job/80727038048>
rollup: SUCCESS
From <https://github.com/avidullu/badminton-highlight-indexer>
* branch master -> FETCH_HEAD
1053505..7d007af master -> origin/master
Updating 1053505..7d007af
Fast-forward
.github/workflows/ci.yml | 13 ++++++-----
1 file changed, 9 insertions(+), 4 deletions(-)
M docs/DEFERRED_HARDENING_PLAN.md
M docs/UI_ALPHA_EXECUTION_PLAN.md
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.

218. user (2026-06-11T05:31:28.354Z)

```
1      """Configuration loader and saver.
2
3      Provides a single place to load validated AppConfig with built-in defaults
4      merged with optional file overrides (config.json).
5
6      This replaces scattered json.load + .get() patterns throughout the codebase.
7      """
8
9      from __future__ import annotations
10
11     import json
12     import logging
13     from pathlib import Path
14     from typing import Any
15
16     from .models import AppConfig
17
18     logger = logging.getLogger(__name__)
19
20     DEFAULT_CONFIG_PATH = Path("config.json")
21
22
23     def _get_built_in_defaults() -> dict[str, Any]:
24         """Return a plain dict of the current built-in defaults.
25
```

```

26         The AppConfig model is the single source of truth for defaults.
27         """
28         return AppConfig().model_dump(mode="json")
29
30
31     def load_config(path: str | Path | None = None) -> AppConfig:
32         """Load configuration with the following precedence:
33
34         1. Built-in defaults defined in the Pydantic models.
35         2. Values from the JSON file (if present and readable).
36         3. (Future) environment / CLI overrides.
37
38         Missing file or unreadable file ? returns a fully defaulted AppConfig.
39         Unknown keys in the file are tolerated (extra="allow" on the model).
40         """
41         cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
42
43         file_data: dict[str, Any] = {}
44         if cfg_path.exists():
45             try:
46                 with open(cfg_path, "r", encoding="utf-8") as f:
47                     file_data = json.load(f)
48             except Exception as exc:
49                 # Non-fatal: continue with defaults + whatever we could parse.
50                 # In production we might log here.
51                 logger.warning(f"Failed to read {cfg_path}: {exc}. Using defaults + partial
data.")
52
53         # Merge: start from defaults, overlay file data (Pydantic will validate/coerce).
54         merged = {**_get_builtin_defaults(), **file_data}
55
56         return AppConfig.model_validate(merged)
57
58
59     def save_default_config(path: str | Path | None = None) -> Path:
60         """Write a fresh default config.json.
61
62         Used by setup.py diagnostics and the --setup CLI path.
63         Overwrites if the file exists.
64         """
65         cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
66         cfg = AppConfig()
67         cfg_path.parent.mkdir(parents=True, exist_ok=True)
68         cfg_path.write_text(cfg.model_dump_json(indent=4), encoding="utf-8")
69         return cfg_path
70
71
72     # Backwards-compatible alias used by some older call sites during transition.
73     def get_default_config() -> dict[str, Any]:
74         """Return defaults as a plain dict (for transitional use only)."""
75         return _get_builtin_defaults()
76

```

219. assistant (2026-06-11T05:31:48.565Z)

Implementing the recursive unknown-key collector in the loader:

220. user (2026-06-11T05:31:50.614Z)

Switched to a new branch 'feat/config-unknown-key-warning'

221. user (2026-06-11T05:31:50.621Z)

```

1         """Basic tests for the typed configuration package.

```

```

2
3     These ensure:
4     - Defaults are sensible and come from the models.
5     - File overrides work.
6     - Old config files with extra keys are tolerated.
7     - save_default_config produces loadable output.
8     """
9
10    import json
11    import tempfile
12    import os
13
14
15    from backend.config import load_config, save_default_config, AppConfig
16
17
18    def test_load_default_config():
19        # Force load without the project's config.json so we test the model built-in
20        defaults
21        cfg = load_config("/nonexistent/path/that/does/not/exist/12345/config.json")
22        assert isinstance(cfg, AppConfig)
23        assert cfg.sport == "badminton"
24        assert cfg.indexing.default_segmenter == "yolo_hybrid"
25        assert cfg.validation.min_blur_threshold == 20.0 # updated default from review
26        assert cfg.indexing.ai_max_workers == 5
27
28    def test_file_override():
29        with tempfile.TemporaryDirectory() as td:
30            p = os.path.join(td, "override.json")
31            data = {
32                "indexing": {
33                    "default_segmenter": "gemini",
34                    "ai_max_workers": 3,
35                },
36                "validation": {
37                    "min_blur_threshold": 42.0,
38                },
39            }
40            with open(p, "w") as f:
41                json.dump(data, f)
42
43            cfg = load_config(p)
44            assert cfg.indexing.default_segmenter == "gemini"
45            assert cfg.indexing.ai_max_workers == 3
46            assert cfg.validation.min_blur_threshold == 42.0
47            # Unspecified keys fall back to defaults
48            assert cfg.sport == "badminton"
49
50
51    def test_extra_keys_tolerated():
52        """Old or future keys in config.json must not break loading."""
53        with tempfile.TemporaryDirectory() as td:
54            p = os.path.join(td, "with_extra.json")
55            data = {
56                "sport": "badminton",
57                "indexing": {"default_segmenter": "yolo_hybrid"},
58                "some_future_key": "ignored_for_now",
59                "validation": {"relevance": {"court_pixels_min_ratio": 0.1}},
60            }
61            with open(p, "w") as f:
62                json.dump(data, f)
63
64            cfg = load_config(p)
65            assert cfg.indexing.default_segmenter == "yolo_hybrid"

```

```

66         assert cfg.validation.relevance.court_pixels_min_ratio == 0.1
67
68
69 def test_save_default_config_roundtrip():
70     with tempfile.TemporaryDirectory() as td:
71         p = os.path.join(td, "saved.json")
72         saved = save_default_config(p)
73         assert str(saved) == p
74         assert os.path.exists(p)
75
76         cfg = load_config(p)
77         assert cfg.indexing.default_segmenter == "yolo_hybrid"
78         assert cfg.validation.min_blur_threshold == 20.0
79
80
81 def test_model_is_the_source_of_truth():
82     """Changing a default in the model should be reflected without touching json."""
83     # This is a documentation test more than anything.
84     cfg = AppConfig()
85     assert cfg.indexing.tracknet.velocity_threshold == 0.02
86
87
88 def test_output_dir_field_default_and_override():
89     """output_dir now lives on OutputConfig (was a write-only runtime hack), so
90     /api/compile and the CLI resolve the same value. Default + file override + the
91     runtime mutation that backend/main.py::main performs for --output-dir."""
92     cfg = AppConfig()
93     assert cfg.output.output_dir == "output" # built-in default
94
95     # The CLI override path: main() assigns args.output_dir onto the typed model.
96     cfg.output.output_dir = "/tmp/custom_out"
97     assert cfg.output.output_dir == "/tmp/custom_out"
98
99     # File override is honored on load.
100    with tempfile.TemporaryDirectory() as td:
101        p = os.path.join(td, "out_override.json")
102        with open(p, "w") as f:
103            json.dump({"output": {"output_dir": "reels", "db_name": "x.db"}}, f)
104        loaded = load_config(p)
105        assert loaded.output.output_dir == "reels"
106        assert loaded.output.db_name == "x.db"
107
108
109 def test_get_returns_dict_for_nested_sections_legacy_compat():
110     """Regression: legacy consumers (validators/segmenters) do
111     ``config.get("validation", {}).get("min_resolution_width", ...)``. The dict-compat
112     .get must return nested SECTIONS as plain dicts (not Pydantic models), or that
113     chained access raises 'X object has no attribute get' on every real run."""
114     cfg = AppConfig()
115     validation = cfg.get("validation", {})
116     assert isinstance(validation, dict)
117     assert validation.get("min_resolution_width") == 1280
118     # nested-of-nested is also dict-accessible
119     assert isinstance(validation.get("relevance"), dict)
120     assert validation["relevance"].get("court_pixels_min_ratio") == 0.05
121     assert cfg.get("indexing", {}).get("default_segmenter") == "yolo_hybrid"
122     # scalar leaves are returned unchanged; a present scalar ignores the default arg;
123     # a missing key falls back to the default
124     assert cfg.get("ai_provider") == "gemini"
125     assert cfg.get("ai_provider", "ignored-default") == "gemini"
126     assert cfg.get("does_not_exist", "fallback") == "fallback"
127     # leaf keys inside any section resolve via the dynamic section scan (not a hardcoded
128     # list), so a future top-level section's keys would be discoverable too
129     assert cfg.get("min_resolution_width") == 1280 # lives inside `validation`
130     assert cfg.get("begin_padding") == 0.5 # lives inside `stitching`

```

222. user (2026-06-11T05:32:02.689Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\loader.py has been updated successfully. (file state is current in your context ? no need to Read it back)

223. user (2026-06-11T05:32:06.199Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\loader.py has been updated successfully. (file state is current in your context ? no need to Read it back)

224. user (2026-06-11T05:32:17.199Z)

loaded OK, segmenter = gemini

225. user (2026-06-11T05:32:33.659Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py has been updated successfully. (file state is current in your context ? no need to Read it back)

226. user (2026-06-11T05:33:08.433Z)

All checks passed!

Found 1 error in 1 file (checked 1 source file)

351 passed in 21.93s

```
=====
RESULT: PASS - all tests green
=====
```

227. assistant (2026-06-11T05:33:15.323Z)

Suite green with the 2 new tests (351), but mypy flags one error in the loader:

228. user (2026-06-11T05:33:17.314Z)

Exit code 1

backend\config\loader.py:34: error: "type" has no attribute "model_fields" [attr-defined]

Found 1 error in 1 file (checked 1 source file)

229. user (2026-06-11T05:33:27.366Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\loader.py has been updated successfully. (file state is current in your context ? no need to Read it back)

230. user (2026-06-11T05:33:36.848Z)

Success: no issues found in 75 source files

All checks passed!

```
..... [100%]
9 passed in 0.16s
```

231. user (2026-06-11T05:34:01.792Z)

Exit code 1

[feat/config-unknown-key-warning f4cdfa5] feat(config): warn loudly on unknown config.json keys (dotted paths)

2 files changed, 64 insertions(+)

remote:

remote: Create a pull request for 'feat/config-unknown-key-warning' on GitHub by visiting:

remote: <https://github.com/avidullu/badminton-highlight-indexer/pull/new/feat/config-unknown-key-warning>

```
remote:
branch 'feat/config-unknown-key-warning' set up to track 'origin/feat/config-unknown-key-
warning'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      feat/config-unknown-key-warning -> feat/config-unknown-key-warning
Warning: 8 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/96
no checks reported on the 'feat/config-unknown-key-warning' branch
```

232. user (2026-06-11T05:35:56.712Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27326151986/job/80727608978
Full test suite (offline, mocked)                pass    1m35s  https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27326151986/job/80727608859
Mypy (lenient baseline) pass    28s    https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27326151986/job/80727608922
Ruff lint      pass    10s    https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27326151986/job/80727608943
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27326151986/job/80727608978
rollup: SUCCESS
From https://github.com/avidullu/badminton-highlight-indexer
 * branch          master      -> FETCH_HEAD
   7d007af..99b84c9 master     -> origin/master
Updating 7d007af..99b84c9
Fast-forward
 backend/config/loader.py | 32 ++++++
 tests/test_config.py      | 32 ++++++
 2 files changed, 64 insertions(+)
M      docs/DEFERRED_HARDENING_PLAN.md
M      docs/UI_ALPHA_EXECUTION_PLAN.md
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
```

233. user (2026-06-11T05:36:27.695Z)

```
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
```

234. user (2026-06-11T05:36:35.738Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-11 (PR #88 ? **#89 MERGED** ? KhelSutra UI-alpha docs bundle on
master `34cf26c`). #88
13     conflicted (branch cut from `feat/async-job-model` pre-squash of #87) ? per
`scratch/PR88_FIX.md`, recreated as
14     `docs/khelsutra-plan-2` off post-#87 master keeping 6 paths (UI_PROPOSAL, HOSTING_PLAN,
UI_ALPHA_EXECUTION_PLAN,
```

235. user (2026-06-11T05:36:52.374Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-

indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

236. user (2026-06-11T05:37:01.592Z)

Chapter marked: "AI-7: hybrid-twin collapse". Continue your current work.

237. user (2026-06-11T05:37:02.146Z)

```
1      """WASB-based hybrid play segmenter.
2
3      Identical three-phase shape to TrackNetHybridSegmenter, but Phase 2 uses **WASB**
4      (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via WSL2
5      (backend/detectors/wasb_runner.py) instead of TrackNetV4. WASB ships pretrained
6      badminton weights and runs today, so this is the live shuttle-motion substrate.
7      Phase 3 (AI semantic boundaries) and Phase 4 (DB) are shared with the other hybrids.
8
9      For the per-video *calibration* goal we mainly use Phase 2 (trajectory -> candidate
10     windows); the windowing thresholds (velocity_thresh / merge_gap / min_window_duration)
11     are what get tuned against golden rally labels.
12     """
13     import os
14     import time
15     import logging
16     import concurrent.futures
17     from typing import List, Dict, Any
18
19     import cv2
20
21     from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
22     from backend.utils.video import get_video_duration, extract_video_chunk
23     from backend.sports import sport_registry
24     from backend.providers import provider_registry
25     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
26     from backend.pipeline.detectors.base import DetectorRunner # ABC for the de-risked
detector seam
27
28     logger = logging.getLogger(__name__)
29
30
31     def _fmt_ts(seconds: float) -> str:
32         seconds = max(0, int(seconds))
33         h, rem = divmod(seconds, 3600)
34         m, s = divmod(rem, 60)
35         return f"{h:02d}:{m:02d}:{s:02d}"
36
37
38     class WasbHybridSegmenter(BasePlaySegmenter):
39         @property
40         def name(self) -> str:
41             return "wasb_hybrid"
42
43         @property
44         def description(self) -> str:
45             return ("Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained
badminton "
46                     "weights) for precise candidate rallies + AI provider for semantic
boundaries.")
47
48         @staticmethod
49         def _probe_fps_width(video_path: str) -> tuple:
50             cap = cv2.VideoCapture(video_path)
51             fps = cap.get(cv2.CAP_PROP_FPS) if cap.isOpened() else 0.0
52             width = cap.get(cv2.CAP_PROP_FRAME_WIDTH) if cap.isOpened() else 0.0
53             cap.release()
```

```

54         return (fps if fps > 0 else 30.0, width if width > 0 else 1280.0)
55
56     def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None,
57                       max_frames: int = None) -> List[Dict[str, Any]]:
58         sport_name = self.config.get("sport", "badminton")
59         try:
60             sport_profile = sport_registry.get(sport_name)
61         except KeyError as e:
62             logger.error(str(e))
63             return []
64
65         idx_cfg = self.config.get("indexing", {})
66         provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider",
"gemini"))
67         model_name = self.config.get("ai_model", idx_cfg.get("ai_model",
"gemini-2.5-flash"))
68
69         api_key_env_var = f"{provider_name.upper()}_API_KEY"
70         api_key = os.environ.get(api_key_env_var)
71         if not api_key:
72             logger.error(f"{api_key_env_var} not set; cannot run the {provider_name}
handoff.")
73             return []
74         try:
75             provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
76         except KeyError as e:
77             logger.error(str(e))
78             return []
79
80         video_duration = get_video_duration(video_path)
81         if video_duration <= 0:
82             logger.error("Invalid video duration.")
83             return []
84         fps, frame_width = self._probe_fps_width(video_path)
85
86         wasb_cfg = WasbConfig.from_indexing_cfg(idx_cfg)
87         runner: DetectorRunner = WasbRunner(wasb_cfg)
88
89         w = idx_cfg.get("wasb", {})
90         velocity_thresh = w.get("velocity_threshold", 0.02)
91         merge_gap = idx_cfg.get("dead_time_merge_gap", 2.0)
92         min_window_duration = idx_cfg.get("min_action_window_duration", 2.0)
93         handoff_padding = idx_cfg.get("ai_handoff_padding", 2.0)
94         ai_max_workers = idx_cfg.get("ai_max_workers", 5)
95         log_candidates = idx_cfg.get("log_yolo_candidates", True)
96         store_candidates = idx_cfg.get("store_yolo_candidates", True)
97
98         detection_params = {
99             "detector": "wasb_hybrid",
100             "velocity_thresh": velocity_thresh,
101             "merge_gap": merge_gap,
102             "min_action_window_duration": min_window_duration,
103             "weights_path": wasb_cfg.weights_path,
104         }
105
106         ok, msg = runner.healthcheck()
107         if not ok:
108             logger.error("WASB WSL environment not ready: %s", msg)
109             return []
110         logger.info("WASB WSL env OK: %s", msg)
111
112         # --- Phase 2: shuttle trajectory -> candidate rally windows ---
113         t_start = time.time()

```



```

114         out_dir = os.path.join("output", "wasb")
115         csv_path = runner.run_predict(video_path, out_dir, video_id=video_id)
116         if not csv_path:
117             logger.error("WASB produced no trajectory; aborting.")
118             return []
119
120         points = runner.parse_trajectory_csv(csv_path)
121         logger.info("Parsed %d trajectory points (%d visible).",
122                     len(points), sum(1 for p in points if p.visible))
123
124         all_candidate_windows = runner.trajectory_to_action_windows(
125             points, fps=fps, frame_width=frame_width, chunk_start=0.0,
126             velocity_thresh=velocity_thresh, merge_gap=merge_gap,
127             min_window_duration=min_window_duration, chunk_index=0,
128         )
129         logger.info("Phase 2 (WASB) completed in %.1fs. Candidate windows: %d",
130                     time.time() - t_start, len(all_candidate_windows))
131
132         if log_candidates:
133             for cw in all_candidate_windows:
134                 logger.info(f"[WASB rally] {_fmt_ts(cw['start'])}-{_fmt_ts(cw['end'])} "
135                             f"({_fmt_ts(cw['end'] - cw['start']:.1f)s) | "
136                             f"peak_v={cw['peak_velocity']:.3f} "
137                             f"mean_v={cw['mean_velocity']:.3f}")
138             candidate_ids = [None] * len(all_candidate_windows)
139             if store_candidates:
140                 for i, cw in enumerate(all_candidate_windows):
141                     stats = {
142                         "peak_velocity": cw["peak_velocity"], "mean_velocity":
143                         cw["mean_velocity"],
144                         "active_frame_count": cw["active_frame_count"], "track_id_count":
145                         cw["track_id_count"],
146                     }
147                     candidate_ids[i] = self.db.add_candidate_segment(
148                         video_id, detector="wasb_hybrid", start_time=cw["start"],
149                         end_time=cw["end"],
150                         chunk_index=cw["chunk_index"], confidence=min(1.0,
151                             cw["peak_velocity"]),
152                         stats=stats, detection_params=detection_params,
153                     )
154             if not all_candidate_windows:
155                 return []
156
157         # --- Phase 3: AI provider handoff (shared pattern) ---
158         handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
159         os.makedirs(handoff_dir, exist_ok=True)
160         logger.info("Phase 3: %s validation on %d candidate windows...",
161                     provider_name, len(all_candidate_windows))
162
163         def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
164             w_start = max(0, window["start"] - handoff_padding)
165             w_end = min(video_duration, window["end"] + handoff_padding)
166             w_dur = w_end - w_start
167             w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
168             if not extract_video_chunk(video_path, w_start, w_dur, w_path,
169                                     reencode=True):
170                 return []
171             prompt = sport_profile.get_prompt(chunk_duration=w_dur)
172             segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
173                                                  label=f"Handoff {wi+1}")
174             valid = []
175             for seg in segments:

```

```

172         try:
173             seg["start_time"] = float(seg["start_time"]) + w_start
174             seg["end_time"] = float(seg["end_time"]) + w_start
175             seg["_candidate_id"] = candidate_ids[wi]
176             valid.append(seg)
177         except (ValueError, TypeError, KeyError) as e:
178             logger.warning("Dropping segment with unparseable timestamps: %s ?
%s", seg, e)
179         try:
180             os.remove(w_path)
181         except OSError:
182             pass
183         return valid
184
185     ai_segments: List[dict] = []
186     with concurrent.futures.ThreadPoolExecutor(max_workers=ai_max_workers) as
executor:
187         futures = [executor.submit(process_candidate_window, wi, cw)
188                     for wi, cw in enumerate(all_candidate_windows)]
189         for future in futures:
190             ai_segments.extend(future.result())
191         try:
192             os.rmdir(handoff_dir)
193         except OSError:
194             pass
195         logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
len(ai_segments))
196
197     # --- Phase 4: DB population (shared pattern) ---
198     segments_data = []
199     for segment in ai_segments:
200         try:
201             start = float(segment["start_time"])
202             end = float(segment["end_time"])
203             dur = end - start
204             shot_count = segment.get("shuttle_exchange_count")
205             try:
206                 shot_count = int(shot_count)
207             except (ValueError, TypeError):
208                 shot_count = max(3, int(dur / 0.8))
209             metadata = {"shot_count": shot_count,
210                        "shot_count_confidence":
segment.get("shot_count_confidence", "Unknown"),
211                        "detector": f"wasb-hybrid-{model_name}"}
212             segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)
213             if segment_id:
214                 candidate_id = segment.get("_candidate_id")
215                 if candidate_id is not None:
216                     self.db.link_candidate_to_segment(candidate_id, segment_id)
217                 segments_data.append({"id": segment_id, "start_time": start,
"end_time": end,
218                                     "duration": dur, "metadata": metadata})
219         except Exception as e:
220             logger.warning("Skipping invalid segment struct: %s", e)
221
222     return segments_data
223
224
225     segmenter_registry.register(WasbHybridSegmenter)
226

```

238. user (2026-06-11T05:37:10.026Z)

```

1     """TrackNet-based hybrid play segmenter.

```

```

2
3 Same three-phase shape as HybridYoloSegmenter, but Phase 2 (candidate
4 generation) uses a TrackNetV4 *shuttle* trajectory instead of YOLO *person*
5 motion. TrackNet runs in WSL2 (see backend/detectors/tracknet_runner.py and
6 docs/DECISIONS.md ADR-001); the shuttle trajectory is converted into candidate
7 rally windows, which are then handed to the AI provider for semantic boundary
8 confirmation (Phase 3) and written to the DB (Phase 4).
9
10 Rationale: a tiny, fast, motion-blurred shuttle is exactly what general
11 detectors miss; a purpose-built trajectory model localises it far more
12 precisely, giving cleaner candidate rallies than person-velocity heuristics.
13 """
14
15 import os
16 import time
17 import logging
18 import concurrent.futures
19 from typing import List, Dict, Any
20
21 import cv2
22
23 from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
24 from backend.utils.video import get_video_duration, extract_video_chunk
25 from backend.sports import sport_registry
26 from backend.providers import provider_registry
27 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
28 from backend.pipeline.detectors.base import DetectorRunner # ABC for the de-risked
detector seam
29
30 logger = logging.getLogger(__name__)
31
32
33 def _fmt_ts(seconds: float) -> str:
34     seconds = max(0, int(seconds))
35     h, rem = divmod(seconds, 3600)
36     m, s = divmod(rem, 60)
37     return f"{h:02d}:{m:02d}:{s:02d}"
38
39
40 class TrackNetHybridSegmenter(BasePlaySegmenter):
41     @property
42     def name(self) -> str:
43         return "tracknet_hybrid"
44
45     @property
46     def description(self) -> str:
47         return ("Two-stage pipeline: TrackNetV4 shuttle-trajectory detection (WSL2) for
precise "
48               "candidate rallies + AI provider for high-precision semantic
boundaries.")
49
50     @staticmethod
51     def _probe_fps_width(video_path: str) -> tuple:
52         """Return (fps, frame_width) for a video, with sensible fallbacks."""
53         cap = cv2.VideoCapture(video_path)
54         fps = cap.get(cv2.CAP_PROP_FPS) if cap.isOpened() else 0.0
55         width = cap.get(cv2.CAP_PROP_FRAME_WIDTH) if cap.isOpened() else 0.0
56         cap.release()
57         return (fps if fps > 0 else 30.0, width if width > 0 else 1280.0)
58
59     def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None,
60                     max_frames: int = None) -> List[Dict[str, Any]]:
61         # --- Sport profile + AI provider wiring (identical to the other segmenters) ---
62         sport_name = self.config.get("sport", "badminton")

```

```

63         try:
64             sport_profile = sport_registry.get(sport_name)
65         except KeyError as e:
66             logger.error(str(e))
67             return []
68
69         idx_cfg = self.config.get("indexing", {})
70         provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider",
"gemini"))
71         model_name = self.config.get("ai_model", idx_cfg.get("ai_model",
"gemini-2.5-flash"))
72
73         api_key_env_var = f"{provider_name.upper()}_API_KEY"
74         api_key = os.environ.get(api_key_env_var)
75         if not api_key:
76             logger.error(
77                 f"{api_key_env_var} environment variable is not set! "
78                 f"To run the {provider_name} handoff, set your API key. "
79                 f"In PowerShell: $env:{api_key_env_var}=\"your_api_key_here\""
80             )
81             return []
82         try:
83             provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
84         except KeyError as e:
85             logger.error(str(e))
86             return []
87
88         video_duration = get_video_duration(video_path)
89         if video_duration <= 0:
90             logger.error("Invalid video duration.")
91             return []
92         fps, frame_width = self._probe_fps_width(video_path)
93
94         # --- TrackNet runner config + thresholds ---
95         tn_cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
96         runner: DetectorRunner = TrackNetRunner(tn_cfg)
97
98         tn = idx_cfg.get("tracknet", {})
99         velocity_thresh = tn.get("velocity_threshold", 0.02) # normalised shuttle
displacement/sec
100         merge_gap = idx_cfg.get("dead_time_merge_gap", 2.0)
101         min_window_duration = idx_cfg.get("min_action_window_duration", 2.0)
102         handoff_padding = idx_cfg.get("ai_handoff_padding", 2.0)
103         ai_max_workers = idx_cfg.get("ai_max_workers", 5)
104         log_candidates = idx_cfg.get("log_yolo_candidates", True)
105         store_candidates = idx_cfg.get("store_yolo_candidates", True)
106
107         detection_params = {
108             "detector": "tracknet_hybrid",
109             "velocity_thresh": velocity_thresh,
110             "merge_gap": merge_gap,
111             "min_action_window_duration": min_window_duration,
112             "weights_path": tn_cfg.weights_path,
113             "queue_length": tn_cfg.queue_length,
114         }
115
116         # --- Phase 1: env healthcheck (fail fast with a clear message) ---
117         ok, msg = runner.healthcheck()
118         if not ok:
119             logger.error("TrackNet WSL environment not ready: %s", msg)
120             return []
121         logger.info("TrackNet WSL env OK: %s", msg)
122
123         # --- Phase 2: shuttle trajectory -> candidate rally windows ---

```

```

124 # TrackNet processes the whole video in one pass (it carries its own
125 # frame buffering), so unlike the YOLO segmenter we don't pre-chunk here.
126 t_start = time.time()
127 tn_out_dir = os.path.join("output", "tracknet")
128 csv_path = runner.run_predict(video_path, tn_out_dir, video_id=video_id)
129 if not csv_path:
130     logger.error("TrackNet produced no trajectory; aborting.")
131     return []
132
133 points = runner.parse_trajectory_csv(csv_path)
134 logger.info("Parsed %d trajectory points (%d visible).",
135           len(points), sum(1 for p in points if p.visible))
136
137 all_candidate_windows = runner.trajectory_to_action_windows(
138     points, fps=fps, frame_width=frame_width, chunk_start=0.0,
139     velocity_thresh=velocity_thresh, merge_gap=merge_gap,
140     min_window_duration=min_window_duration, chunk_index=0,
141 )
142 logger.info("Phase 2 (TrackNet) completed in %.1fs. Candidate windows: %d",
143           time.time() - t_start, len(all_candidate_windows))
144
145 if log_candidates:
146     for w in all_candidate_windows:
147         logger.info(
148             f"[TrackNet rally] {_fmt_ts(w['start'])}-{_fmt_ts(w['end'])} "
149             f"({w['end'] - w['start']:.1f}s) | frames={w['active_frame_count']}"
150             f"peak_v={w['peak_velocity']:.3f} mean_v={w['mean_velocity']:.3f}"
151         )
152
153 # Persist raw candidates for the fine-tuning dataset (same table as YOLO).
154 candidate_ids = [None] * len(all_candidate_windows)
155 if store_candidates:
156     for i, w in enumerate(all_candidate_windows):
157         stats = {
158             "peak_velocity": w["peak_velocity"],
159             "mean_velocity": w["mean_velocity"],
160             "active_frame_count": w["active_frame_count"],
161             "track_id_count": w["track_id_count"],
162         }
163         confidence = min(1.0, w["peak_velocity"])
164         candidate_ids[i] = self.db.add_candidate_segment(
165             video_id, detector="tracknet_hybrid",
166             start_time=w["start"], end_time=w["end"],
167             chunk_index=w["chunk_index"], confidence=confidence,
168             stats=stats, detection_params=detection_params,
169         )
170
171 if not all_candidate_windows:
172     return []
173
174 # --- Phase 3: AI provider handoff (identical pattern to HybridYoloSegmenter)
175
176 handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
177 os.makedirs(handoff_dir, exist_ok=True)
178 logger.info("Phase 3: %s validation on %d candidate windows...",
179           provider_name, len(all_candidate_windows))
180
181 def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
182     w_start = max(0, window["start"] - handoff_padding)
183     w_end = min(video_duration, window["end"] + handoff_padding)
184     w_dur = w_end - w_start
185     w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
186     if not extract_video_chunk(video_path, w_start, w_dur, w_path,
187                               reencode=True):

```

```

186         return []
187     prompt = sport_profile.get_prompt(chunk_duration=w_dur)
188     segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
189                                         label=f"Handoff {wi+1}")
190     valid = []
191     for seg in segments:
192         try:
193             seg["start_time"] = float(seg["start_time"]) + w_start
194             seg["end_time"] = float(seg["end_time"]) + w_start
195             seg["_candidate_id"] = candidate_ids[wi]
196             valid.append(seg)
197         except (ValueError, TypeError, KeyError) as e:
198             logger.warning("Dropping segment with unparseable timestamps: %s ?
%s", seg, e)
199         try:
200             os.remove(w_path)
201         except OSError:
202             pass
203     return valid
204
205     ai_segments: List[dict] = []
206     with concurrent.futures.ThreadPoolExecutor(max_workers=ai_max_workers) as
executor:
207         futures = [executor.submit(process_candidate_window, wi, w)
208                     for wi, w in enumerate(all_candidate_windows)]
209         for future in futures:
210             ai_segments.extend(future.result())
211
212     try:
213         os.rmdir(handoff_dir)
214     except OSError:
215         pass
216     logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
len(ai_segments))
217
218     # --- Phase 4: DB population (identical pattern to HybridYoloSegmenter) ---
219     segments_data = []
220     for segment in ai_segments:
221         try:
222             start = float(segment["start_time"])
223             end = float(segment["end_time"])
224             dur = end - start
225             shot_count = segment.get("shuttle_exchange_count")
226             if shot_count is None:
227                 shot_count = max(3, int(dur / 0.8))
228             else:
229                 try:
230                     shot_count = int(shot_count)
231                 except (ValueError, TypeError):
232                     shot_count = max(3, int(dur / 0.8))
233             metadata = {
234                 "shot_count": shot_count,
235                 "shot_count_confidence": segment.get("shot_count_confidence",
"Unknown"),
236                 "detector": f"tracknet-hybrid-{model_name}",
237             }
238             segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)
239             if segment_id:
240                 candidate_id = segment.get("_candidate_id")
241                 if candidate_id is not None:
242                     self.db.link_candidate_to_segment(candidate_id, segment_id)
243             segments_data.append({
244                 "id": segment_id, "start_time": start, "end_time": end,
245                 "duration": dur, "metadata": metadata,

```

```

246         })
247     except Exception as e:
248         logger.warning("Skipping invalid segment struct: %s", e)
249
250     return segments_data
251
252
253     # Register the segmenter
254     segmenter_registry.register(TrackNetHybridSegmenter)
255

```

239. user (2026-06-11T05:37:40.952Z)

```

1     """Tests for the TrackNetHybridSegmenter pipeline. The WSL/TensorFlow runner is
2     mocked out (TrackNetRunner), so these run anywhere ? they exercise Phase 1-4 glue:
3     healthcheck gating, candidate persistence, AI handoff, and DB population/linking."""
4
5     from unittest.mock import MagicMock, patch
6
7     from backend.pipeline.segmenters.tracknet_hybrid import TrackNetHybridSegmenter
8
9
10    def _window(start=1.0, end=4.0):
11        return {
12            "start": start, "end": end, "active_frame_count": 30,
13            "peak_velocity": 0.4, "mean_velocity": 0.2, "track_id_count": 1,
14            "chunk_index": 0,
15        }
16
17
18    def _patched(fn):
19        """Stack the patches every pipeline test needs. fps/width probing is stubbed
20        so cv2 internals don't need mocking (covered separately).
21
22        Patches are applied innermost-first, and mock.patch injects the innermost
23        patch as the *first* positional arg ? so this order matches the signature
24        (mock_runner_cls, mock_probe, mock_dur, mock_extract, mock_provider_registry,
25        mock_env)."""
26        fn = patch("backend.pipeline.segmenters.tracknet_hybrid.TrackNetRunner")(fn)
27        fn = patch.object(TrackNetHybridSegmenter, "_probe_fps_width", return_value=(30.0,
1280.0))(fn)
28        fn = patch("backend.pipeline.segmenters.tracknet_hybrid.get_video_duration")(fn)
29        fn = patch("backend.pipeline.segmenters.tracknet_hybrid.extract_video_chunk")(fn)
30        fn = patch("backend.pipeline.segmenters.tracknet_hybrid.provider_registry")(fn)
31        fn = patch("os.environ.get")(fn)
32        return fn
33
34
35    def _make_runner(windows=None, csv="out.csv", healthy=True):
36        runner = MagicMock()
37        runner.healthcheck.return_value = (healthy, "TF 2.17 GPUs 1" if healthy else "no
GPU")
38        runner.run_predict.return_value = csv
39        runner.parse_trajectory_csv.return_value = [MagicMock(visible=True)]
40        runner.trajectory_to_action_windows.return_value = windows if windows is not None
else [_window()]
41        return runner
42
43
44    @_patched
45    def test_pipeline_happy_path(mock_runner_cls, mock_probe, mock_dur, mock_extract,
46                                mock_provider_registry, mock_env):
47        mock_env.return_value = "dummy_key"
48        mock_dur.return_value = 30.0
49        mock_extract.return_value = True

```

```

50     mock_runner_cls.return_value = _make_runner()
51
52     provider = MagicMock()
53     provider.analyze_video.return_value = (
54         [{"start_time": 0.5, "end_time": 3.0, "shuttle_exchange_count": 4}], {})
55     mock_provider_registry.get.return_value = provider
56
57     db = MagicMock()
58     db.add_candidate_segment.return_value = 55
59     db.add_play_segment.return_value = 200
60
61     seg = TrackNetHybridSegmenter(db, {"indexing": {}})
62     res = seg.process_video("v1", "clip.mp4")
63
64     assert len(res) == 1
65     assert res[0]["id"] == 200
66     # Candidate persisted under the tracknet detector name...
67     _, kwargs = db.add_candidate_segment.call_args
68     assert kwargs["detector"] == "tracknet_hybrid"
69     # ...and the confirmed segment linked back to it.
70     db.link_candidate_to_segment.assert_called_once_with(55, 200)
71
72
73     @patched
74     def test_pipeline_no_api_key_returns_empty(mock_runner_cls, mock_probe, mock_dur,
75 mock_extract,
76                                           mock_provider_registry, mock_env):
77         mock_env.return_value = None
78         mock_dur.return_value = 30.0
79         mock_runner_cls.return_value = _make_runner()
80
81         seg = TrackNetHybridSegmenter(MagicMock(), {"indexing": {}})
82         assert seg.process_video("v1", "clip.mp4") == []
83
84     @patched
85     def test_pipeline_healthcheck_failure_aborts(mock_runner_cls, mock_probe, mock_dur,
86 mock_extract,
87                                           mock_provider_registry, mock_env):
88         mock_env.return_value = "dummy_key"
89         mock_dur.return_value = 30.0
90         mock_extract.return_value = True
91         mock_runner_cls.return_value = _make_runner(healthy=False)
92         mock_provider_registry.get.return_value = MagicMock()
93
94         seg = TrackNetHybridSegmenter(MagicMock(), {"indexing": {}})
95         res = seg.process_video("v1", "clip.mp4")
96         assert res == []
97         # Should bail before ever running inference.
98         mock_runner_cls.return_value.run_predict.assert_not_called()
99
100
101     @patched
102     def test_pipeline_no_trajectory_aborts(mock_runner_cls, mock_probe, mock_dur,
103 mock_extract,
104                                           mock_provider_registry, mock_env):
105         mock_env.return_value = "dummy_key"
106         mock_dur.return_value = 30.0
107         mock_runner_cls.return_value = _make_runner(csv=None) # predict produced nothing
108         mock_provider_registry.get.return_value = MagicMock()
109
110         seg = TrackNetHybridSegmenter(MagicMock(), {"indexing": {}})
111         assert seg.process_video("v1", "clip.mp4") == []

```



```

112     @patched
113     def test_pipeline_store_candidates_disabled(mock_runner_cls, mock_probe, mock_dur,
mock_extract,
114                                             mock_provider_registry, mock_env):
115         mock_env.return_value = "dummy_key"
116         mock_dur.return_value = 30.0
117         mock_extract.return_value = True
118         mock_runner_cls.return_value = _make_runner()
119
120         provider = MagicMock()
121         provider.analyze_video.return_value = ([{"start_time": 0.5, "end_time": 3.0}], {})
122         mock_provider_registry.get.return_value = provider
123
124         db = MagicMock()
125         db.add_play_segment.return_value = 200
126
127         seg = TrackNetHybridSegmenter(db, {"indexing": {"store_yolo_candidates": False}})
128         seg.process_video("v1", "clip.mp4")
129
130         db.add_candidate_segment.assert_not_called()
131         db.link_candidate_to_segment.assert_not_called()
132
133
134     @patched
135     def test_pipeline_drops_unparseable_segments(mock_runner_cls, mock_probe, mock_dur,
mock_extract,
136                                             mock_provider_registry, mock_env):
137         mock_env.return_value = "dummy_key"
138         mock_dur.return_value = 30.0
139         mock_extract.return_value = True
140         mock_runner_cls.return_value = _make_runner()
141
142         provider = MagicMock()
143         provider.analyze_video.return_value = (
144             [{"start_time": 0.5, "end_time": 3.0}, {"start_time": "N/A", "end_time": "??"}],
145             {})
146         mock_provider_registry.get.return_value = provider
147
148         db = MagicMock()
149         db.add_candidate_segment.return_value = 55
150         db.add_play_segment.return_value = 200
151
152         seg = TrackNetHybridSegmenter(db, {"indexing": {}})
153         res = seg.process_video("v1", "clip.mp4")
154         assert len(res) == 1 # the "N/A" segment is dropped
155
156     def test_probe_fps_width_fallbacks():
157         with patch("backend.pipeline.segmenters.tracknet_hybrid.cv2.VideoCapture") as
mock_cap:
158             bad = MagicMock()
159             bad.isOpened.return_value = False
160             bad.get.return_value = 0.0
161             mock_cap.return_value = bad
162             fps, width = TrackNetHybridSegmenter._probe_fps_width("missing.mp4")
163             assert fps == 30.0 and width == 1280.0
164
165
166     def test_segmenter_identity():
167         seg = TrackNetHybridSegmenter(MagicMock(), {})
168         assert seg.name == "tracknet_hybrid"
169         assert "TrackNet" in seg.description
170

```

```

backend\pipeline\detectors\wasb_runner.py-59-     def __init__(self, cfg: WasbConfig):
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-60-         self.cfg = cfg
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-61-
backend\pipeline\detectors\wasb_runner.py:62:     def _wsl_bash(self, inner_cmd: str) ->
subprocess.CompletedProcess:
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-63-         full = f"source {self.cfg.conda_sh}
&& {inner_cmd}"
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-64-         argv = ["wsl", "-d",
self.cfg.distro, "bash", "-c", full]
backend\pipeline\detectors\wasb_runner.py-65-         logger.debug("WSL exec: %s", full)
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-66-         return subprocess.run(argv,
capture_output=True, text=True,
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-67-         timeout=(self.cfg.timeout_sec or None))
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-68-
backend\pipeline\detectors\wasb_runner.py-69-     def healthcheck(self) -> Tuple[bool, str]:
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-70-         """Verify the WSL `wasb` env
imports torch and sees a GPU. Returns (ok, msg)."""
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-71-         cmd = (f"conda activate
{self.cfg.conda_env} && "
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-72-         f"python -c \"import torch;
print('torch', torch.__version__, "
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py-73-         f"'cuda',
torch.cuda.is_available())\"")
backend\pipeline\detectors\wasb_runner.py-74-         try:
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-92-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-93-         #
----- WSL exec
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-94-
backend\pipeline\detectors\tracknet_runner.py:95:     def _wsl_bash(self, inner_cmd: str) ->
subprocess.CompletedProcess:
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-96-         """Run a bash command string
inside the configured WSL distro."""
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-97-         full = f"source
{self.cfg.conda_sh} && {inner_cmd}"
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-98-         argv = ["wsl", "-d",
self.cfg.distro, "bash", "-c", full]
backend\pipeline\detectors\tracknet_runner.py-99-         logger.debug("WSL exec: %s", full)
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-100-         return subprocess.run(
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-101-             argv, capture_output=True,
text=True,
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-102-             timeout=(self.cfg.timeout_sec or None),
C:\Users\avidu\Projects\badminton-highlight-

```

```

indexer\backend\pipeline\detectors\tracknet_runner.py-103-      )
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-104-
backend\pipeline\detectors\tracknet_runner.py-105-      def healthcheck(self) -> Tuple[bool, str]:
backend\pipeline\detectors\tracknet_runner.py-106-          """Verify the WSL env is usable: conda
env exists, TF imports, GPU visible.
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\tracknet_runner.py-107-

```

241. assistant (2026-06-11T05:38:43.128Z)

Both `\_wsl\_bash` bodies are identical and depend only on
`self.cfg.{distro,conda\_sh,timeout\_sec}`. Checking the ABC to host the shared helper:

242. user (2026-06-11T05:38:43.378Z)

```

1      """Abstract base for detector runners (TrackNet, WASB, future native, etc.).
2
3      This completes the Detector Runner Abstraction (low-regret item 6 continuation / item 1
4      of the post-observability re-prioritization) to de-risk the WSL/conda dependency for
5      the high-accuracy shuttle trajectory substrate used by tracknet_hybrid / wasb_hybrid.
6
7      The ABC lets us:
8      - Keep the current WSL adapters working unchanged for Windows users.
9      - Swap in (or A/B test) a Linux-native implementation later with zero changes to
10     the hybrid segmenters or the model-agnostic windowing brain.
11     - Provide a single place to evolve the healthcheck / error contract and thread
12     detector readiness into per-video reports.
13
14     Contract (deliberately matches what the concrete runners + hybrids + tests already did):
15     - run_predict(video_win_path: str, output_win_dir: str) -> Optional[str]
16       Run detector, return the Windows-side path to the produced trajectory CSV
17       (e.g. ".../clip_predict.csv" or ".../clip_wasb.csv"), or None on any failure.
18       The caller (hybrid) is responsible for output dir; impls may stage the video
19       into WSL fs for perf.
20
21     - healthcheck() -> tuple[bool, str]
22       Return (ok, message). Never raises. "ok" decides whether to proceed to run_predict.
23       Message is logged (and can be surfaced to reports or UX in future).
24       Intended to be cheap (env existence, import, GPU visibility, weights present).
25
26     Existing behavior is 100% preserved. The only additions are the inheritance and
27     the explicit common type for future implementers.
28
29     How to add a new runner (for docs / future contributors):
30     1. Subclass DetectorRunner in a new or existing module under pipeline/detectors/.
31     2. Implement run_predict and healthcheck (return (bool, str) tuple).
32     3. If it produces the same CSV schema (Frame,Visibility,X,Y), reuse or re-export
33        TrackNetRunner.parse_trajectory_csv and .trajectory_to_action_windows (they are
34        deliberately static and model-agnostic). If the new detector needs different
35        windowing, it can implement its own or we generalize later.
36     4. Lazy-import heavy native deps inside the methods (so importing the segmenter
37        registry never pulls torch/tf on a machine that doesn't have them).
38     5. Update the two hybrid segmenters (or introduce a detector= choice) to accept
39        DetectorRunner instances (they already do the right thing via duck-typing + the
40        type hints we add here).
41     6. Add unit tests that exercise the contract (isinstance, healthcheck shape, run
42        returns str|None) using mocks for the external bits ? see test_tracknet_runner.py.
43     7. No change to eval/calibration code paths (they consume the CSV + the static
44        windowing helpers directly).
45
46     Optional Linux-native path (the main de-risk goal):
47     A LinuxNativeRunner (or ONNXRunner) would:
48     - Skip all wsl / _stage / conda activate.
49     - Load weights locally (torch or onnx) or call a native binary.

```

```

50 - Write the exact same CSV format so trajectory_to_action_windows works verbatim.
51 - healthcheck can just check "weights file exists and torch importable + CUDA?".
52 This removes the WSL requirement for Linux users / future SaaS workers while
53 keeping the exact same candidate quality characteristics (or better, once native
54 weights are available).
55
56 Error surfacing into reports:
57 The healthcheck message is already logged by the hybrids on failure. The reporting
58 harvest sees the outcome as "0 candidates" + detector name. When we wire richer
59 segmenter Results (or pass detector_health into emit_indexer_report context), the
60 (ok, msg) can be recorded explicitly under segmentation.details or a new detector
61 stage. The ABC makes that future change local to one place.
62
63 See also:
64 - backend/pipeline/segmenters/{tracknet_hybrid,wasb_hybrid}.py (the two callers)
65 - tests/test_tracknet_runner.py (contract + pure logic tests; hybrids mock the runner)
66 - docs/TRACKNET_WSL_SETUP.md and CODE_MAP.md (2.2 Detectors section)
67 ""
68
69 from abc import ABC, abstractmethod
70 from typing import Optional, Tuple
71
72
73 class DetectorRunner(ABC):
74     """Common interface for shuttle/ball trajectory detectors (WSL or otherwise)."""
75
76     @abstractmethod
77     def run_predict(self, video_win_path: str, output_win_dir: str,
78                    video_id: Optional[str] = None) -> Optional[str]:
79         """Run inference and return path to trajectory CSV (or None on failure).
80
81         ``video_id`` (the file MD5) namespaces all staged/cached/output artifacts so two
82         videos that happen to share a filename cannot reuse each other's frames or CSV.
83         It is optional for back-compat; callers in the live pipeline always pass it.
84         """
85         pass
86
87     @abstractmethod
88     def healthcheck(self) -> Tuple[bool, str]:
89         """Quick check if the runner (env, weights, etc.) is usable.
90
91         Returns (ok, message). Never raises. Cheap to call before a long run.
92         """
93         pass
94

```

243. user (2026-06-11T05:38:58.733Z)

```

tests\test_detector_keying.py:34:         patch.object(runner, "_wsl_bash",
side_effect=fake_wsl):
tests\test_detector_keying.py:65:         patch.object(r, "_wsl_bash", side_effect=fake_wsl):
tests\test_detector_keying.py:76:         patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="done")):
backend\pipeline\detectors\wasb_runner.py:58: class WasbRunner(DetectorRunner): # DetectorRunner
ABC (finish item 6 / new item 1)
backend\pipeline\detectors\wasb_runner.py:62:     def _wsl_bash(self, inner_cmd: str) ->
subprocess.CompletedProcess:
backend\pipeline\detectors\wasb_runner.py:75:         res = self._wsl_bash(cmd)
backend\pipeline\detectors\wasb_runner.py:90:         res = self._wsl_bash(cmd)
backend\pipeline\detectors\wasb_runner.py:103:        res = self._wsl_bash(cmd)
backend\pipeline\detectors\wasb_runner.py:143:        res = self._wsl_bash(cmd)
tests\test_tracknet_runner.py:122:     with patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="TF 2.17.0 GPUs 1")):
tests\test_tracknet_runner.py:130:     with patch.object(r, "_wsl_bash", return_value=_proc(1,
stderr="ModuleNotFoundError")):

```

```

tests\test_tracknet_runner.py:138: with patch.object(r, "_wsl_bash",
side_effect=FileNotFoundError()):
tests\test_tracknet_runner.py:159: with patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="Prediction complete."):
tests\test_tracknet_runner.py:167: with patch.object(r, "_wsl_bash", return_value=_proc(1,
stderr="CUDA error")):
tests\test_tracknet_runner.py:174: with patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="done")):
tests\test_tracknet_runner.py:180: with patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="OK")):
tests\test_tracknet_runner.py:187: with patch.object(r, "_wsl_bash", return_value=_proc(1,
stderr="No such file")):
tests\test_tracknet_runner.py:197: patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="done")):
tests\test_tracknet_runner.py:211: with patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="TF ok")):
tests\test_tracknet_runner.py:222: with patch.object(w, "_wsl_bash", return_value=_proc(0,
stdout="torch ok")):
backend\pipeline\detectors\tracknet_runner.py:83:class TrackNetRunner(DetectorRunner): #
DetectorRunner ABC (finish item 6 / new item 1)
backend\pipeline\detectors\tracknet_runner.py:95: def _wsl_bash(self, inner_cmd: str) ->
subprocess.CompletedProcess:
backend\pipeline\detectors\tracknet_runner.py:116: res = self._wsl_bash(cmd)
backend\pipeline\detectors\tracknet_runner.py:187: res = self._wsl_bash(cmd)
backend\pipeline\detectors\tracknet_runner.py:211: res = self._wsl_bash(cmd)
tests\test_wasb_runner.py:20: with patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="torch 2.6 cuda True")):
tests\test_wasb_runner.py:27: with patch.object(r, "_wsl_bash", return_value=_proc(1,
stderr="ModuleNotFoundError: torch")):
tests\test_wasb_runner.py:34: with patch.object(r, "_wsl_bash",
side_effect=FileNotFoundError()):
tests\test_wasb_runner.py:41: with patch.object(r, "_wsl_bash", return_value=_proc(0,
stdout="OK")):
tests\test_wasb_runner.py:43: with patch.object(r, "_wsl_bash", return_value=_proc(1,
stderr="No such file")):

```

244. assistant (2026-06-11T05:39:15.546Z)

Instance-level patches ? safe to move the method. Writing the new shared base module now:

245. assistant (2026-06-11T05:39:53.445Z)

"""Shared engine for trajectory-substrate hybrid segmenters.

tracknet_hybrid and wasb_hybrid were copy-paste twins ? CODE_MAP carried a standing footgun: "any P3/P4 fix must be applied to BOTH". This base class is now the single home for the whole pipeline shape:

```

healthcheck gate -> P2 trajectory -> candidate windows (+ persistence)
-> P3 AI-provider handoff over padded clips -> P4 DB population/linking

```

A concrete subclass supplies only its identity and runner wiring:

- ``family`` ? config section under ``indexing.\*``, output subdir, and the ``<family>-hybrid-<model>`` detector tag in metadata.
- ``display\_label`` ? human label for log lines ("WASB", "TrackNet").
- ``name`` / ``description`` properties (the registry contract).
- ``\_build\_runner(idx\_cfg)`` ? returns (DetectorRunner, detector-specific detection_params extras such as weights_path).

This base is deliberately NOT registered; only concrete subclasses register.

"""

```

import os
import time
import logging
import concurrent.futures

```

```

from typing import Any, Dict, List, Optional, Tuple

import cv2

from backend.pipeline.segmenters.base import BasePlaySegmenter
from backend.utils.video import get_video_duration, extract_video_chunk
from backend.sports import sport_registry
from backend.providers import provider_registry
from backend.pipeline.detectors.base import DetectorRunner

logger = logging.getLogger(__name__)

def _fmt_ts(seconds: float) -> str:
    seconds = max(0, int(seconds))
    h, rem = divmod(seconds, 3600)
    m, s = divmod(rem, 60)
    return f"{h:02d}:{m:02d}:{s:02d}"

class TrajectoryHybridSegmenter(BasePlaySegmenter):
    """Trajectory-detector hybrid: precise candidate rallies + AI semantic boundaries."""

    #: config section under `indexing` (velocity_threshold lives there), the
    #: output subdir for trajectories, and the metadata detector-tag prefix.
    family: str = ""
    #: human-readable label used in log lines.
    display_label: str = ""

    def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str, Any]]:
        """Return (runner, detector-specific detection_params extras)."""
        raise NotImplementedError

    @staticmethod
    def _probe_fps_width(video_path: str) -> tuple:
        """Return (fps, frame_width) for a video, with sensible fallbacks."""
        cap = cv2.VideoCapture(video_path)
        fps = cap.get(cv2.CAP_PROP_FPS) if cap.isOpened() else 0.0
        width = cap.get(cv2.CAP_PROP_FRAME_WIDTH) if cap.isOpened() else 0.0
        cap.release()
        return (fps if fps > 0 else 30.0, width if width > 0 else 1280.0)

    @staticmethod
    def _shot_count_from(segment: Dict[str, Any], duration: float) -> int:
        """Provider-reported exchange count, else a duration-based estimate.

        One canonical implementation ? the twins had drifted (wasb relied on
        ``int(None)`` raising into the fallback; same result, by accident).
        """
        shot_count = segment.get("shuttle_exchange_count")
        if shot_count is None:
            return max(3, int(duration / 0.8))
        try:
            return int(shot_count)
        except (ValueError, TypeError):
            return max(3, int(duration / 0.8))

    def process_video(self, video_id: str, video_path: str,
                     player_pool: Optional[List[str]] = None,
                     max_frames: Optional[int] = None) -> List[Dict[str, Any]]:
        label = self.display_label
        # --- Sport profile + AI provider wiring (identical across segmenters) ---
        sport_name = self.config.get("sport", "badminton")
        try:
            sport_profile = sport_registry.get(sport_name)

```

```

except KeyError as e:
    logger.error(str(e))
    return []

idx_cfg = self.config.get("indexing", {})
provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider", "gemini"))
model_name = self.config.get("ai_model", idx_cfg.get("ai_model", "gemini-2.5-flash"))

api_key_env_var = f"{provider_name.upper()}_API_KEY"
api_key = os.environ.get(api_key_env_var)
if not api_key:
    logger.error(
        f"{api_key_env_var} environment variable is not set! "
        f"To run the {provider_name} handoff, set your API key. "
        f"In PowerShell: $env:{api_key_env_var}=\"your_api_key_here\""
    )
    return []
try:
    provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
    except KeyError as e:
        logger.error(str(e))
        return []

video_duration = get_video_duration(video_path)
if video_duration <= 0:
    logger.error("Invalid video duration.")
    return []
fps, frame_width = self._probe_fps_width(video_path)

# --- Runner config + windowing thresholds ---
runner, runner_params = self._build_runner(idx_cfg)

velocity_thresh = idx_cfg.get(self.family, {}).get("velocity_threshold", 0.02)
merge_gap = idx_cfg.get("dead_time_merge_gap", 2.0)
min_window_duration = idx_cfg.get("min_action_window_duration", 2.0)
handoff_padding = idx_cfg.get("ai_handoff_padding", 2.0)
ai_max_workers = idx_cfg.get("ai_max_workers", 5)
log_candidates = idx_cfg.get("log_yolo_candidates", True)
store_candidates = idx_cfg.get("store_yolo_candidates", True)

detection_params = {
    "detector": self.name,
    "velocity_thresh": velocity_thresh,
    "merge_gap": merge_gap,
    "min_action_window_duration": min_window_duration,
    **runner_params,
}
}

# --- Phase 1: env healthcheck (fail fast with a clear message) ---
ok, msg = runner.healthcheck()
if not ok:
    logger.error("%s WSL environment not ready: %s", label, msg)
    return []
logger.info("%s WSL env OK: %s", label, msg)

# --- Phase 2: trajectory -> candidate rally windows ---
# Trajectory detectors process the whole video in one pass (they carry
# their own frame buffering), so unlike yolo_hybrid we don't pre-chunk.
t_start = time.time()
out_dir = os.path.join("output", self.family)
csv_path = runner.run_predict(video_path, out_dir, video_id=video_id)
if not csv_path:
    logger.error("%s produced no trajectory; aborting.", label)
    return []

```

```

points = runner.parse_trajectory_csv(csv_path)
logger.info("Parsed %d trajectory points (%d visible).",
            len(points), sum(1 for p in points if p.visible))

all_candidate_windows = runner.trajectory_to_action_windows(
    points, fps=fps, frame_width=frame_width, chunk_start=0.0,
    velocity_thresh=velocity_thresh, merge_gap=merge_gap,
    min_window_duration=min_window_duration, chunk_index=0,
)
logger.info("Phase 2 (%s) completed in %.1fs. Candidate windows: %d",
            label, time.time() - t_start, len(all_candidate_windows))

if log_candidates:
    for cw in all_candidate_windows:
        logger.info(f"[{label} rally] {fmt_ts(cw['start'])}-{fmt_ts(cw['end'])} "
                    f"({cw['end'] - cw['start']:.1f}s) | "
frames={cw['active_frame_count']} "
                    f"peak_v={cw['peak_velocity']:.3f}
mean_v={cw['mean_velocity']:.3f}")

# Persist raw candidates for the fine-tuning dataset (same table as YOLO).
candidate_ids: List[Optional[int]] = [None] * len(all_candidate_windows)
if store_candidates:
    for i, cw in enumerate(all_candidate_windows):
        stats = {
            "peak_velocity": cw["peak_velocity"],
            "mean_velocity": cw["mean_velocity"],
            "active_frame_count": cw["active_frame_count"],
            "track_id_count": cw["track_id_count"],
        }
        candidate_ids[i] = self.db.add_candidate_segment(
            video_id, detector=self.name,
            start_time=cw["start"], end_time=cw["end"],
            chunk_index=cw["chunk_index"], confidence=min(1.0, cw["peak_velocity"]),
            stats=stats, detection_params=detection_params,
        )

if not all_candidate_windows:
    return []

# --- Phase 3: AI provider handoff over padded candidate clips ---
handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
os.makedirs(handoff_dir, exist_ok=True)
logger.info("Phase 3: %s validation on %d candidate windows...",
            provider_name, len(all_candidate_windows))

def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
    w_start = max(0, window["start"] - handoff_padding)
    w_end = min(video_duration, window["end"] + handoff_padding)
    w_dur = w_end - w_start
    w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
    if not extract_video_chunk(video_path, w_start, w_dur, w_path, reencode=True):
        return []
    prompt = sport_profile.get_prompt(chunk_duration=w_dur)
    segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
                                         label=f"Handoff {wi+1}")

    valid = []
    for seg in segments:
        try:
            seg["start_time"] = float(seg["start_time"]) + w_start
            seg["end_time"] = float(seg["end_time"]) + w_start
            seg["_candidate_id"] = candidate_ids[wi]
            valid.append(seg)
        except (ValueError, TypeError, KeyError) as e:
            logger.warning("Dropping segment with unparseable timestamps: %s ? %s", seg,

```


e)

```
    try:
        os.remove(w_path)
    except OSError:
        pass
    return valid

ai_segments: List[dict] = []
with concurrent.futures.ThreadPoolExecutor(max_workers=ai_max_workers) as executor:
    futures = [executor.submit(process_candidate_window, wi, cw)
                for wi, cw in enumerate(all_candidate_windows)]
    for future in futures:
        ai_segments.extend(future.result())
try:
    os.rmdir(handoff_dir)
except OSError:
    pass
logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
len(ai_segments))

# --- Phase 4: DB population + candidate linking ---
segments_data = []
for segment in ai_segments:
    try:
        start = float(segment["start_time"])
        end = float(segment["end_time"])
        dur = end - start
        metadata = {
            "shot_count": self._shot_count_from(segment, dur),
            "shot_count_confidence": segment.get("shot_count_confidence", "Unknown"),
            "detector": f"{self.family}-hybrid-{model_name}",
        }
        segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)

        if segment_id:
            candidate_id = segment.get("_candidate_id")
            if candidate_id is not None:
                self.db.link_candidate_to_segment(candidate_id, segment_id)
            segments_data.append({
                "id": segment_id, "start_time": start, "end_time": end,
                "duration": dur, "metadata": metadata,
            })
    except Exception as e:
        logger.warning("Skipping invalid segment struct: %s", e)

return segments_data
```

246. user (2026-06-11T05:39:53.462Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\trajectory_hybrid.py (file state is current in your context ? no need to Read it back)

247. assistant (2026-06-11T05:40:06.907Z)

""TrackNet-based hybrid play segmenter.

Phase 2 (candidate generation) uses a TrackNetV4 *shuttle* trajectory computed in WSL2 (backend/pipeline/detectors/tracknet_runner.py, ADR-001). Rationale: a tiny, fast, motion-blurred shuttle is exactly what general detectors miss; a purpose-built trajectory model localises it far more precisely, giving cleaner candidate rallies than person-velocity heuristics.

The pipeline engine (healthcheck -> trajectory -> windows -> AI handoff -> DB) lives in trajectory_hybrid.TrajectoryHybridSegmenter; this class only wires the

```

TrackNet runner and identity.
"""
from typing import Any, Dict, Tuple

from backend.pipeline.segmenters.base import segmenter_registry
from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
from backend.pipeline.detectors.base import DetectorRunner
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig

class TrackNetHybridSegmenter(TrajectoryHybridSegmenter):
    family = "tracknet"
    display_label = "TrackNet"

    @property
    def name(self) -> str:
        return "tracknet_hybrid"

    @property
    def description(self) -> str:
        return ("Two-stage pipeline: TrackNetV4 shuttle-trajectory detection (WSL2) for precise
"
                "candidate rallies + AI provider for high-precision semantic boundaries.")

    def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str, Any]]:
        cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
        return TrackNetRunner(cfg), {
            "weights_path": cfg.weights_path,
            "queue_length": cfg.queue_length,
        }

segmenter_registry.register(TrackNetHybridSegmenter)

```

248. user (2026-06-11T05:40:06.932Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\tracknet_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

249. assistant (2026-06-11T05:40:13.312Z)

```

"""WASB-based hybrid play segmenter.

```

Phase 2 uses **WASB** (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via WSL2 (backend/pipeline/detectors/wasb_runner.py). WASB ships pretrained badminton weights and runs today, so this is the live shuttle-motion substrate.

For the per-video *calibration* goal we mainly use Phase 2 (trajectory -> candidate windows); the windowing thresholds (velocity_thresh / merge_gap / min_window_duration) are what get tuned against golden rally labels.

The pipeline engine (healthcheck -> trajectory -> windows -> AI handoff -> DB) lives in trajectory_hybrid.TrajectoryHybridSegmenter; this class only wires the WASB runner and identity.

```

"""
from typing import Any, Dict, Tuple

from backend.pipeline.segmenters.base import segmenter_registry
from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
from backend.pipeline.detectors.base import DetectorRunner
from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig

class WasbHybridSegmenter(TrajectoryHybridSegmenter):

```

```

family = "wasb"
display_label = "WASB"

@property
def name(self) -> str:
    return "wasb_hybrid"

@property
def description(self) -> str:
    return ("Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained
badminton "
            "weights) for precise candidate rallies + AI provider for semantic boundaries.")

def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str, Any]]:
    cfg = WasbConfig.from_indexing_cfg(idx_cfg)
    return WasbRunner(cfg), {"weights_path": cfg.weights_path}

segmenter_registry.register(WasbHybridSegmenter)

```

250. user (2026-06-11T05:40:13.325Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\wasb_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

251. assistant (2026-06-11T05:40:29.218Z)

Now the `\_wsl\_bash` dedupe ? adding the mixin to [detectors/base.py](backend/pipeline/detectors/base.py) and switching both runners:

252. user (2026-06-11T05:40:34.006Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

253. user (2026-06-11T05:40:41.205Z)

```

50         weights_path=w.get("weights_path",
51                             "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"),
52         sport=w.get("sport", "badminton"),
53         wsl_stage_dir=w.get("wsl_stage_dir", "~/clips"),
54         timeout_sec=int(w.get("timeout_sec", 1800)),
55     )
56
57
58     class WasbRunner(DetectorRunner): # DetectorRunner ABC (finish item 6 / new item 1)
59         def __init__(self, cfg: WasbConfig):
60             self.cfg = cfg
61
62         def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
63             full = f"source {self.cfg.conda_sh} && {inner_cmd}"
64             argv = ["wsl", "-d", self.cfg.distro, "bash", "-c", full]
65             logger.debug("WSL exec: %s", full)
66             return subprocess.run(argv, capture_output=True, text=True,
67                                 timeout=(self.cfg.timeout_sec or None))
68
69         def healthcheck(self) -> Tuple[bool, str]:

```

254. user (2026-06-11T05:40:52.915Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_runner.py has been updated successfully. (file state is

current in your context ? no need to Read it back)

255. user (2026-06-11T05:41:00.428Z)

```
80         y: float
81
82
83     class TrackNetRunner(DetectorRunner): # DetectorRunner ABC (finish item 6 / new item 1)
84         """Runs TrackNet predict.py in WSL and turns the output CSV into action windows.
85
86         Implements the common DetectorRunner contract so a future Linux-native or
87         alternative trajectory runner can be substituted without touching the hybrids.
88         """
89
90         def __init__(self, cfg: TrackNetConfig):
91             self.cfg = cfg
92
93         # ----- WSL exec
94
95         def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
96             """Run a bash command string inside the configured WSL distro."""
97             full = f"source {self.cfg.conda_sh} && {inner_cmd}"
98             argv = ["wsl", "-d", self.cfg.distro, "bash", "-c", full]
99             logger.debug("WSL exec: %s", full)
100             return subprocess.run(
101                 argv, capture_output=True, text=True,
102                 timeout=(self.cfg.timeout_sec or None),
103             )
104
105         def healthcheck(self) -> Tuple[bool, str]:
```

256. user (2026-06-11T05:41:12.501Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

257. assistant (2026-06-11T05:41:20.470Z)

Adding the mixin imports to both runners:

258. user (2026-06-11T05:41:23.398Z)

```
wasb_runner.py:15: import subprocess
wasb_runner.py:22: from backend.pipeline.detectors.base import DetectorRunner
tracknet_runner.py:21: import subprocess
tracknet_runner.py:25: from backend.pipeline.detectors.base import DetectorRunner
```

259. user (2026-06-11T05:41:35.358Z)

```
All checks passed!
..... [100%]
41 passed in 0.30s
```

260. assistant (2026-06-11T05:41:44.948Z)

41 detector tests green. Updating the hybrid test's patch targets to the new base module:

261. user (2026-06-11T05:41:51.083Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_tracknet_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

262. user (2026-06-11T05:41:52.654Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_tracknet_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

263. assistant (2026-06-11T05:42:00.858Z)

Now the equivalence suite ? the test that makes the twins unable to drift again:

264. assistant (2026-06-11T05:42:19.385Z)

```
"""Equivalence contract for the trajectory-hybrid twins.

tracknet_hybrid and wasb_hybrid were copy-paste twins that drifted (the
shot_count parse); both are now thin subclasses of TrajectoryHybridSegmenter.
These tests run BOTH through the same scenarios and assert identical P3/P4
behavior, so any future divergence has to be introduced deliberately, in the
base, for both at once.
"""
from unittest.mock import MagicMock, patch

import pytest

from backend.pipeline.segmenters.tracknet_hybrid import TrackNetHybridSegmenter
from backend.pipeline.segmenters.wasb_hybrid import WasbHybridSegmenter
from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter

TWINS = [
    (TrackNetHybridSegmenter, "tracknet", "tracknet_hybrid"),
    (WasbHybridSegmenter, "wasb", "wasb_hybrid"),
]

def _window(start=1.0, end=4.0):
    return {
        "start": start, "end": end, "active_frame_count": 30,
        "peak_velocity": 0.4, "mean_velocity": 0.2, "track_id_count": 1,
        "chunk_index": 0,
    }

def _runner(windows=None):
    runner = MagicMock()
    runner.healthcheck.return_value = (True, "env ok")
    runner.run_predict.return_value = "out.csv"
    runner.parse_trajectory_csv.return_value = [MagicMock(visible=True)]
    runner.trajectory_to_action_windows.return_value = windows if windows is not None else
[_window()]
    return runner

def _run(cls, provider_segments, db=None):
    """Run one segmenter class through the standard mocked pipeline."""
    provider = MagicMock()
    provider.analyze_video.return_value = (provider_segments, {})
    db = db or MagicMock()
    db.add_candidate_segment.return_value = 55
    db.add_play_segment.return_value = 200

    with patch.object(cls, "_build_runner", return_value=_runner(), {"weights_path": "w"}), \
        patch.object(cls, "_probe_fps_width", return_value=(30.0, 1280.0)), \
        patch("backend.pipeline.segmenters.trajectory_hybrid.get_video_duration",
return_value=30.0), \
        patch("backend.pipeline.segmenters.trajectory_hybrid.extract_video_chunk",
return_value=True), \
```

```

        patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry") as preg, \
        patch("os.environ.get", return_value="dummy_key"):
    preg.get.return_value = provider
    seg = cls(db, {"indexing": {}})
    return db, seg.process_video("v1", "clip.mp4")

@pytest.mark.parametrize("cls,family,detector", TWINS)
def test_detector_identity_flows_into_db_rows(cls, family, detector):
    db, res = _run(cls, [{"start_time": 0.5, "end_time": 3.0, "shuttle_exchange_count": 4}])
    assert len(res) == 1
    _, kwargs = db.add_candidate_segment.call_args
    assert kwargs["detector"] == detector
    assert kwargs["detection_params"]["detector"] == detector
    # P4 metadata tags the family + model.
    assert res[0]["metadata"]["detector"].startswith(f"{family}-hybrid-")
    db.link_candidate_to_segment.assert_called_once_with(55, 200)

@pytest.mark.parametrize("cls,family,detector", TWINS)
@pytest.mark.parametrize("exchange,expected", [
    (4, 4),          # provider-reported count wins
    ("7", 7),        # numeric strings coerce
    (None, 3),       # explicit null -> duration estimate (2.5s/0.8 -> 3 floor)
    ("garbage", 3),  # unparseable -> same estimate
])
def test_shot_count_behavior_is_identical(cls, family, detector, exchange, expected):
    """The historical drift point: wasb_hybrid relied on int(None) raising into
    the fallback while tracknet_hybrid short-circuited. One implementation now."""
    seg_dict = {"start_time": 0.5, "end_time": 3.0, "shuttle_exchange_count": exchange}
    _, res = _run(cls, [seg_dict])
    assert res[0]["metadata"]["shot_count"] == expected

@pytest.mark.parametrize("cls,family,detector", TWINS)
def test_provider_offsets_applied_identically(cls, family, detector):
    # Window [1,4] + 2.0s default padding -> clip starts at max(0, -1) = 0.0;
    # provider-relative 0.5 stays 0.5 absolute here.
    _, res = _run(cls, [{"start_time": 0.5, "end_time": 3.0}])
    assert res[0]["start_time"] == 0.5
    assert res[0]["end_time"] == 3.0

def test_base_is_not_registered():
    from backend.pipeline.segmenters.base import segmenter_registry
    available = segmenter_registry.list_available()
    assert "tracknet_hybrid" in available
    assert "wasb_hybrid" in available
    # The base engine itself must not appear as a selectable segmenter.
    assert not any(isinstance(cls, type) and cls is TrajectoryHybridSegmenter
                    for cls in available.values())

```

265. user (2026-06-11T05:42:19.397Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_trajectory_hybrid_equivalence.py (file state is current in your context ? no need to Read it back)

266. assistant (2026-06-11T05:42:28.378Z)

Now the ratchet test ? removing the twins from the mypy quarantine (they're thin now; the base was written type-clean), then full verification:

267. user (2026-06-11T05:42:33.072Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\mypy.ini has been updated successfully. (file state is current in your context ? no need to Read it back)

268. user (2026-06-11T05:43:09.465Z)

All checks passed!

backend\pipeline\segmenters\trajectory_hybrid.py:154: error: "DetectorRunner" has no attribute "parse_trajectory_csv" [attr-defined]

backend\pipeline\segmenters\trajectory_hybrid.py:158: error: "DetectorRunner" has no attribute "trajectory_to_action_windows" [attr-defined]

Found 3 errors in 1 file (checked 76 source files)

364 passed in 22.30s

```
=====
RESULT: PASS - all tests green
=====
```

269. user (2026-06-11T05:43:33.231Z)

Exit code 1

backend\pipeline\segmenters\trajectory_hybrid.py:81: error: Return

type "list[dict[str, Any]]" of

"process_video"

incompatible with return type

"Success[list[dict[str, Any]]] | Failure"

in supertype

"backend.pipeline.segmenters.base.BasePlaySegmenter" [override]

backend\pipeline\segmenters\trajectory_hybrid.py:154: error: "DetectorRunner" has no attribute "parse_trajectory_csv"

[attr-defined]

backend\pipeline\segmenters\trajectory_hybrid.py:158: error: "DetectorRunner"

has no attribute

"trajectory_to_action_windows" [attr-defined]

270. user (2026-06-11T05:43:59.159Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

271. user (2026-06-11T05:44:01.050Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

272. user (2026-06-11T05:44:04.059Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\trajectory_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

273. user (2026-06-11T05:44:38.334Z)

Success: no issues found in 76 source files

All checks passed!

```
=====
RESULT: PASS - all tests green
=====
```

```
M backend/pipeline/detectors/base.py
M backend/pipeline/detectors/tracknet_runner.py
M backend/pipeline/detectors/wasb_runner.py
M backend/pipeline/segmenters/tracknet_hybrid.py
M backend/pipeline/segmenters/wasb_hybrid.py
M mypy.ini
```

```
M tests/test_tracknet_hybrid.py
?? backend/pipeline/segmenters/trajectory_hybrid.py
?? tests/test_trajectory_hybrid_equivalence.py
```

274. assistant (2026-06-11T05:45:00.990Z)

Everything green ? mypy now checks 76 files with the twins **out** of quarantine. Updating the CODE_MAP entries so the docs don't go stale on this:

275. user (2026-06-11T05:45:12.307Z)

```
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
```

276. user (2026-06-11T05:45:14.126Z)

```
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
```

277. user (2026-06-11T05:45:21.379Z)

```
107     ### 2.1 Segmenters (under pipeline/segmenters) ?
108     - `@backend/pipeline/segmenters/base.py`
109     - `::BasePlaySegmenter` ABC ? `__init__(db, config)`; abstract
`name`, `description`, `process_video`. $ `process_video(video_id, video_path, player_pool=None,
max_frames=None) -> List[dict{id, start_time, end_time, duration, metadata}]`.
110     - `::SegmenterRegistry`, `::segmenter_registry` (singleton). ? register a new
segmenter*: subclass `BasePlaySegmenter`, call `segmenter_registry.register(MyCls)` at module
bottom, AND import the module in `@backend/pipeline/segmenters/__init__.py` (registration is an
IMPORT side-effect). ? `register()` builds `cls(None, {})` just to read `name` ->
name/description/`__init__` MUST NOT touch `self.db`/`self.config`. ? `list_available()` re-
instantiates + swallows exceptions into "No description available." (a throwing `description` is
masked).
111     - `@backend/pipeline/segmenters/__init__.py` ? imports all 5
(`motion, gemini, yolo_hybrid, tracknet_hybrid, wasb_hybrid`) = the de-facto registration manifest;
re-exports them + `segmenter_registry` via `__all__`.
112     - `@backend/pipeline/segmenters/wasb_hybrid.py::WasbHybridSegmenter` (name
`wasb_hybrid`) ? WSL WASB substrate (MIT; ships pretrained badminton weights, the LIVE
default). Imports `WasbRunner`/`WasbConfig` from `pipeline/detectors/wasb_runner`. Config
`indexing.wasb.velocity_threshold(0.02)` + shared
`dead_time_merge_gap/min_action_window_duration/ai_handoff_padding/ai_max_workers`. ? whole-
video single pass; reuses YOLO-named keys `log_yolo_candidates/store_yolo_candidates`; needs
`{PROVIDER}_API_KEY` (missing -> []); `detection_params` has NO `queue_length` (unlike
tracknet). ? divergent shot_count: `int(shot_count)` inside try with no `is None` short-circuit
(copy-paste drift).
113     - `@backend/pipeline/segmenters/tracknet_hybrid.py::TrackNetHybridSegmenter` (name
`tracknet_hybrid`) ? copy-paste twin of wasb_hybrid; imports `TrackNetRunner`/`TrackNetConfig`
from `pipeline/detectors/tracknet_runner`; `indexing.tracknet.velocity_threshold(0.02)`;
`detection_params` adds `queue_length`. ? any P3/P4 fix must be applied to BOTH (and arguably
yolo_hybrid).
114     - `@backend/pipeline/segmenters/yolo_hybrid.py::HybridYoloSegmenter` (name
`yolo_hybrid`, no YOLO remains ? torchvision detector + supervision ByteTrack; ADR-005).
`::_PERSON_CLASS_ID=1` ? (ultralytics used 0; wrong value -> zero detections).
`::_lazy_load_model` (lazy `torchvision` import so module/tests load without it),
`::_extract_action_windows_from_chunk`, `::_make_tracker` (per-chunk ByteTrack ? track IDs
chunk-local), `::_DETECTOR_FACTORIES`/`::_DEFAULT_DETECTOR='fasterrcnn_resnet50_fpn'`. ?
velocity at `effective_fps = fps/frame_skip` -> `yolo_velocity_threshold` coupled to
`frame_skip`; pipelined-vs-sequential P2 (prefetch ffmpeg overlaps sequential GPU);
`yolo_tracker`/`bytetrack.yaml` is dead/back-compat. ? `gemini_*` config keys are deprecated
aliases still honored.
115     - `@backend/pipeline/segmenters/gemini.py::GeminiPlaySegmenter` (name `gemini`) ?
pure-AI, no P2. Chunks are RE-ENCODED + downscaled to `indexing.ai_chunk_scale_width` (default
1280; 0 = legacy stream-copy) before upload ? stream-copied 4K chunks exceed the provider's
`MAX_UPLOAD_MB` and every chunk gets refused ("0 segments / success"); re-encode also fixes
keyframe drift at chunk boundaries. Oversize refusals (provider metrics `oversize_skipped`) are
aggregated -> `logger.error` + `reporting.run_signals` (`oversize_clips_skipped`) for the report
```



```

rule. Workers from `indexing.ai_max_workers`. ? chunks only if `duration>chunk_duration`;
`max_segment_duration=90.0` hardcoded; `debug_duration` silently truncates; `parse_time` accepts
colon timestamps (warns); does NOT persist candidate segments (only hybrids feed the fine-tuning
table).
116 - `@backend/pipeline/segmenters/motion.py::MotionPlaySegmenter` (name `motion`) ?
pixel absdiff baseline. ? **metadata + serve/smash/foul/winner events are RANDOM**
(random.sample/randint/uniform/choice); only segmenter using `db.add_event` and passing
server/receiver; uses `print()`; honors `max_frames`.
117
118     ### 2.2 Detectors & windowing (WSL-quarantined; ABC abstraction complete)
119 - `@backend/pipeline/detectors/base.py` ? DetectorRunner ABC (healthcheck() ->
Tuple[bool,str] never-raises; run_predict(...) -> Optional[str] CSV path). Contract deliberately
matches the pre-existing tuple returns and call sites in the hybrids. Rich module docstring
includes "how to add a new runner", Linux-native path notes (the de-risk goal), and guidance on
threading health into reports. Re-exported from `detectors/__init__.py`. This finishes the WSL
de-risk seam (low-regret 6 / review item 1).
120 - `@backend/pipeline/detectors/tracknet_runner.py`
121 - `::TrackNetConfig` (+`.from_indexing_cfg`; ? key drift `wsl_distro` JSON -> `distro`
field), `::TrackNetRunner` ? runs TrackNet `predict.py` in WSL.
122 - `::to_wsl_mnt_path` (`C:\x`->`/mnt/c/x`; ? POSIX/~` passthrough, UNC unhandled),
`::TrajectoryPoint` (frame,visible,x,y).
123 - `**::parse_trajectory_csv` @staticmethod** + `**::trajectory_to_action_windows`
@staticmethod** = MODEL-AGNOSTIC brain, re-exported verbatim by WasbRunner. ? `weights_path`
defaults `` -> `run_predict` returns None; predict.py only `.mp4/.avi` + hard-names
`<stem>_predict.csv`; `visible==False` resets prev (occlusion gap breaks track); `fps<=0->30`,
`frame_width<=0->1280` silent; `track_id_count` hardcoded 1; `timeout_sec=0` -> NO timeout
(hangs forever). `::stage_video` "OK" sentinel.
124 - `@backend/pipeline/detectors/wasb_runner.py::WasbRunner` (inherits DetectorRunner)
(+`::WasbConfig`, from `indexing.wasb`; ? `weights_path` HAS a real default
`wasb_badminton_best.pth.tar` ? no required guard) ? stages `wasb_infer.py` (`::INFER_WIN`) +
video into WSL, runs in `wasb` env. `::parse_trajectory_csv`/`::trajectory_to_action_windows` =
`staticmethod(TrackNetRunner.*)` (explicit rebind = the plugin-equivalence point). Output hard-
named `<stem>_wasb.csv`. ? `wasb_infer.py` re-staged each run (editing Windows copy inert until
staged); `wsl_bash` duplicated (drift risk). Both runners now satisfy the DetectorRunner ABC
for future swappability.
125 - `@backend/pipeline/detectors/wasb_infer.py` (WSL) ? `::run` core. ? imports WASB-SBDT
repo modules + torch ? NOT importable on Windows. Reboot-resumable detector cache:
`::DET_FILE='det_raw.jsonl'`, `::MANIFEST_FILE='manifest.jsonl'`, `::CACHE_VERSION=1`,
`::load_and_clean_cache` (reads longest CONTIGUOUS prefix where `w==line index`, rewrites to
heal crash-truncated tail), `::manifest_compatible` (? `frames_dir` abspath'd but `weights`
compared raw), `::atomic_write_text/atomic_write_trajectory`, `::_dets_to_tracker` (xy MUST be
np.array), `::_build_cfg` (`runner.gpus=[0]` hardcoded), `::extract_frames` (cv2-PNG SLOW),
`::default_cache_dir` (`<stem>_wasbcache` next to out). ? GPU pass resumable; CPU tracker
STATEFUL -> always fully re-run; `--fresh` ignores cache.
126 - `@backend/pipeline/detectors/rally_gate.py` ? Gate A net-crossing post-filter (pure,
no I/O/model). `::estimate_play_axis` (larger-spread axis; ? ties -> `x`),
`::count_net_crossings` (sign changes vs median net; deadband jitter zone),
`::gate_windows`/`::apply_gate` (`min_crossings=2` default; kills single-toss service feeds).
`::GatedWindow` (start,end,crossings,visible_points,kept). ? window contract here is
`(start,end)` tuples NOT the rich window dict ? caller must adapt. Consumed by calibrate_local /
distill_local / export_wasb_segments.

```

278. user (2026-06-11T05:45:40.866Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

279. user (2026-06-11T05:45:43.634Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

280. assistant (2026-06-11T05:45:50.225Z)

Committing AI-7 and opening the PR:

281. user (2026-06-11T05:46:04.085Z)

```
warning: in the working copy of 'backend/pipeline/segmenters/tracknet_hybrid.py', LF will be
replaced by CRLF the next time Git touches it
warning: in the working copy of 'backend/pipeline/segmenters/wasb_hybrid.py', LF will be
replaced by CRLF the next time Git touches it
warning: in the working copy of 'backend/pipeline/segmenters/trajectory_hybrid.py', LF will be
replaced by CRLF the next time Git touches it
warning: in the working copy of 'tests/test_trajectory_hybrid_equivalence.py', LF will be
replaced by CRLF the next time Git touches it
[master 2c241a8] refactor: collapse the hybrid-segmenter twins into a shared trajectory base
 10 files changed, 452 insertions(+), 475 deletions(-)
 create mode 100644 backend/pipeline/segmenters/trajectory_hybrid.py
 create mode 100644 tests/test_trajectory_hybrid_equivalence.py
branch 'master' set up to track 'origin/master'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 99b84c9..2c241a8 HEAD -> master
```

282. user (2026-06-11T05:47:43.707Z)

! Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4, actions/setup-python@v5. Actions will be forced to run with Node.js 24 by default starting June 16th, 2026. Node.js 20 will be removed from the runner on September 16th, 2026. Please check if updated versions of these actions are available that support Node.js 24. To opt into Node.js 24 now, set the FORCE_JAVASCRIPT_ACTIONS_TO_NODE24=true environment variable on the runner or in your workflow file. Once Node.js 24 becomes the default, you can temporarily opt out by setting ACTIONS_ALLOW_USE_UNSECURE_NODE_VERSION=true. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/>
Ruff lint: .github#2

! Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4, actions/setup-python@v5. Actions will be forced to run with Node.js 24 by default starting June 16th, 2026. Node.js 20 will be removed from the runner on September 16th, 2026. Please check if updated versions of these actions are available that support Node.js 24. To opt into Node.js 24 now, set the FORCE_JAVASCRIPT_ACTIONS_TO_NODE24=true environment variable on the runner or in your workflow file. Once Node.js 24 becomes the default, you can temporarily opt out by setting ACTIONS_ALLOW_USE_UNSECURE_NODE_VERSION=true. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/>
Mypy (lenient baseline): .github#2

watch exit: 0

283. user (2026-06-11T05:48:20.556Z)

True
usage: rally_seq_proto.py [-h] --manifest MANIFEST

PROTOTYPE: a learned per-FRAME rally segmenter (vs the unsupervised windowing heuristic). Motivation (2026-06-08 finding): WASB tracks the shuttle well on Kushagra (97% of GT rallies contain ≥ 2 net-crossings) yet NO velocity-windowing config segments it (per-video F1 ceiling 0.163), while it works on Adarsh (0.73). The rally SIGNAL is in the trajectory; the hand-tuned heuristic just can't extract it across footage. This prototype tests the hypothesis that a LEARNED model on per-frame trajectory features recovers the rally segments ? including on the footage the heuristic fails on. Approach (deliberately small, dependency-light: numpy + scipy + the repo's pure-python LogisticRegression ? no torch/sklearn): 1. Per-frame features over a local window from the WASB

284. user (2026-06-11T05:48:29.512Z)

Command running in background with ID: bp1bc2af7. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\bp1bc2af7.output. You will be notified when it completes. To check interim output, use Read on that file path.

285. user (2026-06-11T05:48:50.346Z)

Noted (position 1, task_id: task_44fa1368). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_44fa1368 "Bump GitHub Actions to Node 24-ready versions". Continue your current work.

286. user (2026-06-11T05:48:53.200Z)

```
Adarsh and Avi.MP4
Adarsh and Avi.rallies.csv
gBaaddy20May2026.MP4
gBaaddy20May2026.rallies.csv
KushagraYashAviFirstVideo.MP4
KushagraYashAviFirstVideo.rallies.csv
KushagraYashAviFirstVideo.report.json
KushagraYashAviFirstVideo.report.md
KushagraYashAviFirstVideo.summary.md
LargeTestVideo.MP4
LargeTestVideo.rallies.csv
MahadevpuraSingles.MP4
MahadevpuraSingles.rallies.csv
MahadevpuraSingles.report.json
MahadevpuraSingles.report.md
---
from dotenv import load_dotenv
load_dotenv()
```

287. user (2026-06-11T05:48:59.609Z)

```
1      === Learned per-frame rally segmenter ? LEAVE-ONE-VIDEO-OUT (6 videos) ===
2      features: ['vis_rate', 'spd_mean', 'spd_max', 'cross_rate', 'x_std', 'y_std']
3
4      [held-out adarsh_avi_singles] per-frame AUC 0.874 | LOVO-threshold F1 0.684 (P 0.580
R 0.833, thr 0.45 from train) | oracle-threshold F1 0.716 (thr 0.30, fit-on-test upper bound)
5      [held-out kushagra_singles] per-frame AUC 0.851 | LOVO-threshold F1 0.506 (P 0.412 R
0.656, thr 0.45 from train) | oracle-threshold F1 0.600 (thr 0.50, fit-on-test upper bound)
6      [held-out largetest_doubles] per-frame AUC 0.916 | LOVO-threshold F1 0.739 (P 0.629 R
0.898, thr 0.50 from train) | oracle-threshold F1 0.773 (thr 0.45, fit-on-test upper bound)
7      [held-out gbaaddy] per-frame AUC 0.828 | LOVO-threshold F1 0.530 (P 0.417 R 0.729,
thr 0.45 from train) | oracle-threshold F1 0.560 (thr 0.40, fit-on-test upper bound)
8      [held-out testlarge_short] per-frame AUC 0.744 | LOVO-threshold F1 0.556 (P 0.417 R
0.833, thr 0.50 from train) | oracle-threshold F1 0.588 (thr 0.45, fit-on-test upper bound)
9      [held-out mahadevpura_singles] per-frame AUC 0.896 | LOVO-threshold F1 0.744 (P 0.617
R 0.935, thr 0.50 from train) | oracle-threshold F1 0.747 (thr 0.45, fit-on-test upper bound)
10
11      learned LOVO mean held-out F1: 0.626
12
```

288. user (2026-06-11T05:49:28.895Z)

```
Exit code 1
Traceback (most recent call last):
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py", line 27, in
<module>
    from backend.pipeline.validators.base import registry
ModuleNotFoundError: No module named 'backend'
```

289. user (2026-06-11T05:49:40.915Z)

```
usage: main.py [-h] [--video-path VIDEO_PATH] [--output-dir OUTPUT_DIR]
              [--setup] [--debug-clip DEBUG_CLIP] [--server] [--port PORT]
              [--max-frames MAX_FRAMES]
              [--segmenter {motion,gemini,yolo_hybrid,tracknet_hybrid,wasb_hybrid}]
              [--trim-start TRIM_START] [--trim-end TRIM_END]
              [--trim-output TRIM_OUTPUT]
```

Sports Highlight Indexer Orchestrator

options:

```
-h, --help            show this help message and exit
--video-path VIDEO_PATH
                        Local path to the MP4 sports video file
--output-dir OUTPUT_DIR
                        Directory where processed clips/highlights are saved
--setup               Launch dependency checker and setup utility
--debug-clip DEBUG_CLIP
                        Timestamp (in seconds) of a video clip to run detailed
                        diagnostics on
--server              Run the FastAPI local API server
--port PORT           Port to run the FastAPI server on
--max-frames MAX_FRAMES
```

290. user (2026-06-11T05:51:04.855Z)

```
[libx264 @ 00000226c72bcf80] kb/s:3870.01
[aac @ 00000226c72fde00] Qavg: 431.274
exit: 0
86.0160074234009
```

291. user (2026-06-11T05:51:14.141Z)

Command running in background with ID: bia40agg0. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\bia40agg0.output. You will be notified when it completes. To check interim output, use Read on that file path.

292. assistant (2026-06-11T05:51:24.239Z)

Live smoke is running (WASB GPU pass + Gemini handoff ? several minutes). Tailing progress per the working agreement:

293. user (2026-06-11T05:51:25.546Z)

```
<tool_use_error>Blocked: Start-Sleep 75 followed by: Get-Content
"C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\bia40agg0.output" -Tail 12. To wait for a condition, use Monitor with an until-loop (e.g. `until <check>; do sleep 2; done` ? Monitor runs bash). To wait for a command you started, use run_in_background: true. Do not chain shorter sleeps to work around this block.</tool_use_error>
```

294. user (2026-06-11T05:52:18.757Z)

```
0      11:21:44 [INFO] backend.pipeline.segmenters.trajecory_hybrid: WASB WSL env OK:
torch 1.11.0+cu113 cuda True
1      11:21:44 [ERROR] backend.pipeline.detectors.wasb_runner: Failed to stage video
into WSL: cp: cannot create regular file '-/clips/mahadevpura_smoke_180s__05e0e577029d.mp4': No
such file or directory
2      11:21:44 [ERROR] backend.pipeline.segmenters.trajecory_hybrid: WASB produced no
trajectory; aborting.
3      [CLI] Processing video file: output/mahadevpura_smoke_180s.mp4
4      [CLI] Validating...
5
6      =====
```

```

7      VALIDATION SUMMARY
8      =====
9      [PASS] format_check: Video format is valid MP4 container.
10     [PASS] resolution_fps_check: Video resolution and frame rate meet minimum standards.
11     [PASS] pts_continuity_check: PTS timeline is continuous with no abnormal gaps or
backward jumps.
12     [PASS] scene_cut_check: Scene continuity check passed. No abnormal visual cuts detected.
13     [PASS] blur_check: Video sharpness meets focus requirements.
14     [PASS] relevance_check: Video identified as relevant to badminton (badminton court
structures found).
15     =====
16
17     [CLI] Indexing play segments using segmenter: wasb_hybrid...
18     [SUCCESS] Process complete. Indexed 0 play segments inside database:
output\sports_indexer.db
19     [REPORT] Technical report : C:\Users\avidu\Projects\badminton-highlight-
indexer\output\mahadevpura_smoke_180s.report.md
20     [REPORT] Customer summary : C:\Users\avidu\Projects\badminton-highlight-
indexer\output\mahadevpura_smoke_180s.summary.md
21     [REPORT] Machine JSON      : C:\Users\avidu\Projects\badminton-highlight-
indexer\output\mahadevpura_smoke_180s.report.json
22

```

295. user (2026-06-11T05:53:16.068Z)

```

---- when was the quote added ----
8615534 2026-06-07 refactor: adopt approved layered directory structure
---- tracknet staging ----
    """Copy the input video into the WSL filesystem (avoids slow /mnt/c reads).

    Returns the WSL path to the staged copy, or None on failure.
    """
    src_mnt = to_wsl_mnt_path(video_win_path)
    dest = f"{self.cfg.wsl_stage_dir}/{base}"
    cmd = f"mkdir -p {self.cfg.wsl_stage_dir} && cp {shlex.quote(src_mnt)}
{shlex.quote(dest)} && echo OK"
    try:
        res = self._wsl_bash(cmd)
    except subprocess.TimeoutExpired:

```

296. user (2026-06-11T05:53:37.193Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

297. user (2026-06-11T05:53:32.854Z)

Switched to a new branch 'fix/wsl-tilde-quoting'

298. assistant (2026-06-11T05:53:45.898Z)

Now switching both runners to the new helper at every `~`-capable path:

299. user (2026-06-11T05:53:47.772Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

300. user (2026-06-11T05:53:50.121Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

301. user (2026-06-11T05:53:53.361Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

302. user (2026-06-11T05:54:05.054Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

303. user (2026-06-11T05:54:07.412Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

304. user (2026-06-11T05:54:09.834Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

305. assistant (2026-06-11T05:54:33.473Z)

```
"""WSL tilde-quoting regression tests.
```

```
shlex.quote('~/.clips/x.mp4') single-quotes the whole path, so bash never
expands ~ and staging fails with "cannot create regular file '~/.clips/...'".
This broke the live trajectory-hybrid path for the DEFAULT ~/.clips stage dir
(found by the 2026-06-11 wasb_hybrid smoke run; masked until then because
trajectories were generated via manual WSL scripts and tests mock _wsl_bash).
"""
```

```
from unittest.mock import MagicMock, patch
```

```
import pytest
```

```
from backend.pipeline.detectors.base import wsl_tilde_quote
from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
```

```
@pytest.mark.parametrize("path,expected", [
    ("~/.clips/x.mp4", "~/'clips/x.mp4'"),          # leading ~/ stays expandable
    ("~/.clips/has space.mp4", "~/'clips/has space.mp4'"),
    ("~", "~"),                                     # bare home
    ("/mnt/c/Users/x/out", "/mnt/c/Users/x/out"),    # shlex.quote no-op for safe paths
    ("/mnt/c/has space", "'/mnt/c/has space'"),       # ordinary quoting preserved
])
```

```
def test_wsl_tilde_quote(path, expected):
    assert wsl_tilde_quote(path) == expected
```

```
def _proc(rc=0, stdout="OK", stderr=""):
    p = MagicMock()
    p.returncode = rc
    p.stdout = stdout
    p.stderr = stderr
    return p
```

```
def _no_quoted_tilde(cmd: str) -> bool:
    return "'~" not in cmd and "'~'" not in cmd
```

```

def test_wasb_stage_video_cmd_has_no_quoted_tilde():
    r = WasbRunner(WasbConfig()) # defaults: wsl_stage_dir=~/.clips"
    captured = []

    def fake_wsl(cmd):
        captured.append(cmd)
        return _proc()

    with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
        dest = r._stage_video("C:\\vids\\clip.mp4", "clip__abc.mp4")
    assert dest == "~/clips/clip__abc.mp4"
    assert len(captured) == 1
    assert _no_quoted_tilde(captured[0]), captured[0]
    assert "mkdir -p ~/.clips" in captured[0]

def test_tracknet_stage_video_cmd_has_no_quoted_tilde():
    r = TrackNetRunner(TrackNetConfig())
    captured = []

    def fake_wsl(cmd):
        captured.append(cmd)
        return _proc()

    with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
        r._stage_video("C:\\vids\\clip.mp4", "clip__abc.mp4")
    assert captured and _no_quoted_tilde(captured[0]), captured

def test_wasb_run_predict_inference_cmd_has_no_quoted_tilde(tmp_path):
    """The full inference invocation (video/frames/weights/out) must keep ~
    expandable too ? quoting any of them reintroduces the staging bug one
    step later."""
    r = WasbRunner(WasbConfig()) # weights default lives under ~/models
    cmds = []

    def fake_wsl(cmd):
        cmds.append(cmd)
        if "wasb_infer.py" in cmd:
            # simulate the run producing the expected CSV
            (tmp_path / "clip__abcdefabcdef_wasb.csv").write_text("Frame,Visibility,X,Y\n")
        return _proc()

    with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
        out = r.run_predict("C:\\vids\\clip.mp4", str(tmp_path), video_id="abcdefabcdef0000")
    assert out is not None
    infer_cmds = [c for c in cmds if "wasb_infer.py --video" in c or "python wasb_infer.py" in
c]
    assert infer_cmds, cmds
    assert _no_quoted_tilde(infer_cmds[0]), infer_cmds[0]

```

306. user (2026-06-11T05:54:33.484Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_wsl_tilde_quoting.py (file state is current in your context ? no need to Read it back)

307. user (2026-06-11T05:55:07.357Z)

Exit code 1

All checks passed!

Success: no issues found in 76 source files

```

=====
RESULT: FAIL - pytest exit 1
=====

```

308. user (2026-06-11T05:55:17.066Z)

Exit code 1

```
@pytest.mark.parametrize("path,expected", [
    ("~/clips/x.mp4", "~/clips/x.mp4"), # leading ~/ stays expandable
    ("~/clips/has space.mp4", "~/clips/has space.mp4"),
    ("~", "~"), # bare home
    ("/mnt/c/Users/x/out", "/mnt/c/Users/x/out"), # shlex.quote no-op for safe paths
    ("/mnt/c/has space", "'/mnt/c/has space'"), # ordinary quoting preserved
])
def test_wsl_tilde_quote(path, expected):
>     assert wsl_tilde_quote(path) == expected
E     assert '~/clips/x.mp4' == "~/clips/x.mp4"
E
E     - ~/clips/x.mp4'
E     ?   -           -
E     + ~/clips/x.mp4
```

tests\test_wsl_tilde_quoting.py:26: AssertionError

===== short test summary info =====

FAILED tests/test_wsl_tilde_quoting.py::test_wsl_tilde_quote[~/clips/x.mp4-~/clips/x.mp4']

1 failed, 7 passed in 0.12s

309. user (2026-06-11T05:55:29.426Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_wsl_tilde_quoting.py has been updated successfully. (file state is current in your context ? no need to Read it back)

310. user (2026-06-11T05:56:19.941Z)

=====

RESULT: PASS - all tests green

=====

warning: in the working copy of 'tests/test_wsl_tilde_quoting.py', LF will be replaced by CRLF the next time Git touches it

[fix/wsl-tilde-quoting 70ca6fa] fix(detectors): keep ~ expandable in WSL staging/inference commands

4 files changed, 117 insertions(+), 11 deletions(-)

create mode 100644 tests/test_wsl_tilde_quoting.py

remote:

remote: Create a pull request for 'fix/wsl-tilde-quoting' on GitHub by visiting:

remote: <https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/wsl-tilde-quoting>

remote:

branch 'fix/wsl-tilde-quoting' set up to track 'origin/fix/wsl-tilde-quoting'.

To <https://github.com/avidullu/badminton-highlight-indexer.git>

* [new branch] fix/wsl-tilde-quoting -> fix/wsl-tilde-quoting

Warning: 8 uncommitted changes

<https://github.com/avidullu/badminton-highlight-indexer/pull/97>

311. user (2026-06-11T05:56:29.187Z)

Command running in background with ID: b3qbid2hm. Output is being written to:

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\b3qbid2hm.output. You will be notified when it completes. To check interim output, use Read on that file path.

312. user (2026-06-11T05:56:40.889Z)

Monitor started (task bbxes1jxz, timeout 900000ms). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

313. user (2026-06-11T05:58:10.464Z)


```
Syntax sweep + import smoke (no GPU stack)      pass    20s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327011969/job/80730293940
Full test suite (offline, mocked)               pass    1m31s  https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327011969/job/80730293904
Mypy (lenient baseline) pass                    29s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327011969/job/80730293963
Ruff lint      pass      9s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327011969/job/80730293983
Syntax sweep + import smoke (no GPU stack)      pass    20s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327011969/job/80730293940
```

314. assistant (2026-06-11T05:58:20.511Z)

Monitor script failed (wrong path convention for the local bash). Reading the smoke log directly instead:

315. user (2026-06-11T05:58:21.655Z)

```
<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file
has 1 lines.</system-reminder>
```

316. user (2026-06-11T05:58:41.469Z)

Command running in background with ID: b8x6ohc4k. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\b8x6ohc4k.output. You will be notified when it completes. To check interim output, use Read on that file path.

317. assistant (2026-06-11T05:58:58.378Z)

Status while the live smoke runs (WASB frame extraction + GPU inference on the 3-minute slice takes several minutes): [#97](https://github.com/avidullu/badminton-highlight-indexer/pull/97) is CI-green on all four lanes and waiting only on this live verification before I merge it. I'll pick up automatically when the run or the milestone watcher completes.

318. user (2026-06-11T06:05:09.632Z)

```
<task-notification>
<task-id>b3qbid2hm</task-id>
<tool-use-id>toolu_01Hr4pArG8e5aaax9ahHVe4r</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\b3qbid2hm.output</output-file>
<status>completed</status>
<summary>Background command "Re-run live wasb_hybrid smoke with tilde fix" completed (exit code 0)</summary>
</task-notification>
```

319. user (2026-06-11T06:05:20.923Z)

```
1      11:26:58 [INFO] backend.pipeline.segmenters.trajectory_hybrid: WASB WSL env OK:
torch 1.11.0+cu113 cuda True
2      11:26:58 [INFO] backend.pipeline.detectors.wasb_runner: Running WASB inference on
mahadevpura_smoke_180s.mp4 ...
3      11:35:08 [ERROR] backend.pipeline.detectors.wasb_runner: WASB finished but
expected CSV not found at output\wasb\mahadevpura_smoke_180s__05e0e577029d_wasb.csv
4      11:35:08 [ERROR] backend.pipeline.segmenters.trajectory_hybrid: WASB produced no
trajectory; aborting.
5      [CLI] Processing video file: output/mahadevpura_smoke_180s.mp4
6      [CLI] Validating...
7
8      =====
9      VALIDATION SUMMARY
10     =====
11     [PASS] format_check: Video format is valid MP4 container.
```

```

12      [PASS] resolution_fps_check: Video resolution and frame rate meet minimum standards.
13      [PASS] pts_continuity_check: PTS timeline is continuous with no abnormal gaps or
backward jumps.
14      [PASS] scene_cut_check: Scene continuity check passed. No abnormal visual cuts detected.
15      [PASS] blur_check: Video sharpness meets focus requirements.
16      [PASS] relevance_check: Video identified as relevant to badminton (badminton court
structures found).
17      =====
18
19      [CLI] Indexing play segments using segmenter: wasb_hybrid...
20      [SUCCESS] Process complete. Indexed 0 play segments inside database:
output\sports_indexer.db
21      [REPORT] Technical report : C:\Users\avidu\Projects\badminton-highlight-
indexer\output\mahadevpura_smoke_180s.report.md
22      [REPORT] Customer summary : C:\Users\avidu\Projects\badminton-highlight-
indexer\output\mahadevpura_smoke_180s.summary.md
23      [REPORT] Machine JSON      : C:\Users\avidu\Projects\badminton-highlight-
indexer\output\mahadevpura_smoke_180s.report.json
24

```

320. user (2026-06-11T06:06:03.852Z)

```

total 184
drwxr-xr-x 3 avidullu avidullu  4096 Jun 11 06:05 .
drwxr-xr-x 3 avidullu avidullu  4096 Jun 11 06:01 ..
-rw-r--r-- 1 avidullu avidullu 172447 Jun 11 06:05 mahadevpura_smoke_180s__05e0e577029d_wasb.csv
drwxr-xr-x 2 avidullu avidullu  4096 Jun 11 06:05
mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache
---
/home/avidullu/models/WASB-SBDT/src/output/wasb/mahadevpura_smoke_180s__05e0e577029d_wasb.csv

```

321. user (2026-06-11T06:06:37.658Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

322. user (2026-06-11T06:06:38.653Z)

```

118      """Run predict.py on a video. Returns the Windows path to the produced CSV, or
None.
119
120      ``output_win_dir`` is a Windows directory (under the repo's output/) that
121      WSL writes into via its /mnt view, so the Windows process can read the CSV
122      back with no copy. ``video_id`` (file MD5) namespaces the staged video ? and
123      therefore predict.py's ``<stem>_predict.csv`` output ? so two videos that share
124      a filename cannot collide on the output CSV.
125      """
126      if not self.cfg.weights_path:
127          logger.error(
128              "TrackNet weights_path is not set. Set indexing.tracknet.weights_path "
129              "in config.json to the WSL path of your trained TrackNetV4 weights."

```

323. user (2026-06-11T06:06:52.619Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

324. user (2026-06-11T06:07:00.327Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_wsl_tilde_quoting.py has been updated successfully. (file state is current in your context ? no need to Read it back)

325. user (2026-06-11T06:07:41.036Z)

Exit code 1
All checks passed!
Success: no issues found in 76 source files

1 failed, 372 passed in 22.62s
RESULT: FAIL - pytest exit 1

326. user (2026-06-11T06:08:14.055Z)

Exit code 1
E IndexError: list index out of range
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_wsl_tilde_quoting.py:89:
IndexError
FAILED tests/test_wsl_tilde_quoting.py::test_wasb_run_predict_resolves_relative_out_dir
1 failed, 372 passed in 22.25s

327. assistant (2026-06-11T06:08:26.826Z)

The test's predicate matched the script-staging `cp` command (it also contains "wasb_infer.py") instead of the inference invocation ? fixing the selector:

328. user (2026-06-11T06:08:30.967Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_wsl_tilde_quoting.py has been updated successfully. (file state is current in your context ? no need to Read it back)

329. user (2026-06-11T06:09:12.442Z)

373 passed in 22.26s
RESULT: PASS - all tests green
PR97 state: OPEN
warning: in the working copy of 'tests/test_wsl_tilde_quoting.py', LF will be replaced by CRLF the next time Git touches it
[fix/wsl-tilde-quoting 281a81d] fix(detectors): abspath the output dir before /mnt translation
3 files changed, 34 insertions(+)
To https://github.com/avidullu/badminton-highlight-indexer.git
70ca6fa..281a81d fix/wsl-tilde-quoting -> fix/wsl-tilde-quoting

330. assistant (2026-06-11T06:09:22.326Z)

373 green, fix pushed to #97. Launching the third (and hopefully conclusive) live smoke, with CI watch in parallel:

331. user (2026-06-11T06:09:23.787Z)

Command running in background with ID: b8bq6mc9y. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\b8bq6mc9y.output. You will be notified when it completes. To check interim output, use Read on that file path.

332. user (2026-06-11T06:11:14.697Z)

Syntax sweep + import smoke (no GPU stack) pass 18s
<https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327533613/job/80731908222>
Full test suite (offline, mocked) pass 1m42s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327533613/job/80731908234>
Mypy (lenient baseline) pass 29s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327533613/job/80731908263>
Ruff lint pass 8s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327533613/job/80731908229>
Syntax sweep + import smoke (no GPU stack) pass 18s
<https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27327533613/job/80731908222>

333. assistant (2026-06-11T06:11:28.015Z)

#97 is green on all four lanes. Arming the milestone watcher for the smoke run:

334. user (2026-06-11T06:11:31.268Z)

Command running in background with ID: b20v48jf8. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\b20v48jf8.output. You will be notified when it completes. To check interim output, use Read on that file path.

335. assistant (2026-06-11T06:11:42.859Z)

The watcher will wake me at Phase 3 (the Gemini handoff) or on any error; the run itself notifies on completion. WASB's GPU pass takes ~8 minutes, so I'll resume then.

336. user (2026-06-11T12:52:54.237Z)

Hello. Are all the work items done?

337. assistant (2026-06-11T12:53:33.074Z)

Almost ? 8 of 9 threads are done and merged; the last one (the live end-to-end verification gating PR #97) was in flight. Let me check where it landed:

338. user (2026-06-11T12:53:34.068Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

339. user (2026-06-11T12:53:48.613Z)

```
Exit code 1
Name                                     Length
LastWriteTime
----
-----
mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache      6/11/2026 11:43:20 AM
mahadevpura_smoke_180s__05e0e577029d_wasb.csv            172447 6/11/2026 11:43:20 AM
det_raw.jsonl      747258 6/11/2026 11:43:20 AM
manifest.json      312    6/11/2026 11:43:20 AM

--- python proc ---
0
```

340. user (2026-06-11T12:54:08.427Z)

(PowerShell completed with no output)

341. user (2026-06-11T12:54:56.280Z)

```
M docs/DEFERRED_HARDENING_PLAN.md
M docs/UI_ALPHA_EXECUTION_PLAN.md
---
3d84869 ci: bump checkout and setup-python to Node-24 majors (v6)
2c241a8 refactor: collapse the hybrid-segmenter twins into a shared trajectory base
99b84c9 feat(config): warn loudly on unknown config.json keys (dotted paths) (#96)
--- diff vs HEAD (detectors + ci) ---
```

342. user (2026-06-11T12:55:02.098Z)

Exit code 1
: The term '' is not recognized as a name of a cmdlet, function, script file, or

executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

343. user (2026-06-11T12:55:35.174Z)

```
branch: ci/node24-action-bump
* ci/node24-action-bump
  remotes/origin/ci/node24-action-bump
--- pr97 ---
state: OPEN  head: 281a81d
```

344. user (2026-06-11T12:55:39.956Z)

```
videos: [{ 'id': '05e0e577029daafcf425ca70c0c4f0a3', 'status': 'processed' }]
play_segments: 0
candidates: 9
```

345. user (2026-06-11T12:55:59.988Z)

```
stage: validation | status: ok | metrics: [{ 'audience': 'technical', 'key': 'checks_passed',
'unit': None, 'value': 6}, { 'audience': 'technical', 'key': 'checks_total', 'unit': None,
'value': 6}]
stage: segmentation | status: ok | metrics: [{ 'audience': 'technical', 'key':
'candidate_windows', 'unit': None, 'value': 0}, { 'audience': 'both', 'key': 'segment_count',
'unit': None, 'value': 0}, { 'audience': 'technical', 'key': 'total_play_s', 'unit': 's',
'value': 0}]
  details: { "config_default": "gemini", "detector": null, "matches_config_default": false,
"metadata_source": "measured (detector/AI)", "segmenter_actual": "wasb_hybrid",
"segmenter_explicitly_requested": true, "segmenter_requested": "wasb_hybrid", "was_reingest":
true}
stage: ai_handoff | status: ok | metrics: [{ 'audience': 'technical', 'key':
'confirmed_segments', 'unit': None, 'value': 0}]
  details: { "ai_based": true, "api_key_present": true, "model": "gemini-flash-latest",
"oversize_clips_skipped": 0, "provider": "gemini"}
verdict: pass_with_warnings
flags: ['ai_empty_vs_no_rallies', 'reingest_replaced_prior']
```

346. user (2026-06-11T12:56:03.019Z)

(PowerShell completed with no output)

347. user (2026-06-11T12:56:40.799Z)

```
Exit code 1
Name                                         LastWriteTime
----
mahadevpura_smoke_180s.report.json          6/11/2026 11:35:08 AM
mahadevpura_smoke_180s.mp4                  6/11/2026 11:21:04 AM
mahadevpura_smoke_180s__05e0e577029d_wasb.csv 6/11/2026 11:43:20 AM
--- now: 18:26:40 ---
```

348. user (2026-06-11T12:57:29.404Z)

```
M      docs/DEFERRED_HARDENING_PLAN.md
M      docs/UI_ALPHA_EXECUTION_PLAN.md
Your branch is up to date with 'origin/fix/wsl-tilde-quoting'.
Already on 'fix/wsl-tilde-quoting'
281a81d fix(detectors): abspath the output dir before /mnt translation
from backend.pipeline.detectors.base import DetectorRunner, WslCommandMixin, wsl_tilde_quote
_INFER_WIN = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
    cmd = f"mkdir -p {self.cfg.wsl_stage_dir} && cp {shlex.quote(src_mnt)}
{wsl_tilde_quote(dest)} && echo OK"
```

349. user (2026-06-11T12:57:44.763Z)

Command running in background with ID: bljinq5il. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\bljinq5il.output. You will be notified when it completes. To check interim output, use Read on that file path.

350. assistant (2026-06-11T12:57:55.146Z)

While it runs ? checking what the chip session did with its branch:

351. user (2026-06-11T12:57:59.524Z)

```
#98 [MERGED] ci/node24-action-bump: ci: bump checkout and setup-python to Node-24 majors (v6)
#97 [OPEN] fix/wsl-tilde-quoting: fix(detectors): keep ~ expandable in WSL staging/inference
commands
#96 [MERGED] feat/config-unknown-key-warning: feat(config): warn loudly on unknown config.json
keys (dotted paths)
#95 [MERGED] ci/coverage-floor: ci: coverage floor on the full-suite lane (ratchet-only)
#94 [MERGED] fix/db-connection-close: fix(db): actually close SQLite connections (transaction
scope != close)
```

352. user (2026-06-11T12:58:21.210Z)

```
detector: wasb_hybrid | params.detector: wasb_hybrid | weights: ts/wasb_badminton_best.pth.tar |
stats keys: ['active_frame_count', 'mean_velocity', 'peak_velocity', 'track_id_count']
detector: wasb_hybrid | params.detector: wasb_hybrid | weights: ts/wasb_badminton_best.pth.tar |
stats keys: ['active_frame_count', 'mean_velocity', 'peak_velocity', 'track_id_count']
```

353. assistant (2026-06-11T12:58:48.284Z)

Here's the complete status ? **everything is done and merged except one PR that's minutes from closing**:

```
| Item | Status |
|---|---|
| AI-1 hygiene ? AI-6 config warning | ? Merged ([#90](https://github.com/avidullu/badminton-highlight-indexer/pull/90)?[#96](https://github.com/avidullu/badminton-highlight-indexer/pull/96)), all on green CI |
| Bonus: SQLite connection-leak fix | ? Merged ([#94](https://github.com/avidullu/badminton-highlight-indexer/pull/94)) ? found while wiring coverage; 340 ResourceWarnings ? 4 |
| AI-7 twin collapse | ? On master, 13 equivalence tests, mypy quarantine shrunk by two modules ? with one process slip I owe you below |
| Eval checkpoint (offline) | ? LOVO reproduces exactly 0.626 (Mahadevpura Gen-0 0.744 to the third decimal) ? the sweep is numerically inert |
| Eval checkpoint (live) | ? Last step in flight ? details below |
| Node-24 actions chip | ? Your session merged it as
[#98](https://github.com/avidullu/badminton-highlight-indexer/pull/98) |
```

Two things you should know:

1. **Process slip on AI-7:** after merging #96 my script left me on master, and the AI-7 commit was accidentally committed and pushed **directly to master** ? skipping the PR gate. Everything in it was locally verified beforehand and master CI came back green on all four lanes, but it has no PR record. I chose not to force-rewind master (concurrent sessions were live ? and indeed one later flipped this very checkout's branch), so it stands as a confessed convention violation rather than a hidden one.

2. **The live smoke earned its keep** ? it found two real, pre-existing bugs that had silently broken the `*live* `wasb_hybrid`` path (masked for weeks because trajectories were always generated via manual WSL scripts and tests mock the WSL boundary): ``shlex.quote`` was strangling ``~`` expansion in staging (the deferred BACKLOG item, now closed), and the relative ``output/wasb`` dir resolved against the WSL cwd, so an 8-minute GPU run wrote its CSV inside WSL where Windows never looked. Both fixed in [#97](https://github.com/avidullu/badminton-highlight-

indexer/pull/97) (CI green, suite 373, regression tests pin the command shapes). The verification run for the final Gemini phase got killed ~7 hours ago when the session went idle mid-flight; I've relaunched it just now with the warm GPU cache. P1/P2 are already proven live ? healthcheck through the new base, trajectory at the correct path, and 9 candidate rows in the DB with byte-correct detector identity (just verified above).

As soon as this run confirms the Gemini handoff (?5?10 min), I merge #97 and refresh the memory checkpoint ? I'll report back automatically when it lands.

354. user (2026-06-11T13:23:07.400Z)

```
<task-notification>
<task-id>bljinq5il</task-id>
<tool-use-id>toolu_01JKyP95a7pPMAJyZzVGtqMD</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\bljinq5il.output</output-file>
<status>completed</status>
<summary>Background command "Final smoke run with unbuffered file logging" completed (exit code
0)</summary>
</task-notification>
```

355. assistant (2026-06-11T13:23:36.270Z)

Final smoke run completed ? reading the verdict:

356. user (2026-06-11T13:23:39.226Z)

```
18:28:21 [INFO] backend.pipeline.segmenters.trajectory_hybrid: Parsed 5395 trajectory
points (2937 visible).
18:28:21 [INFO] backend.pipeline.segmenters.trajectory_hybrid: Phase 2 (WASB) completed
in 3.2s. Candidate windows: 9
18:28:21 [INFO] backend.pipeline.segmenters.trajectory_hybrid: Phase 3: gemini
validation on 9 candidate windows...
18:29:24 [WARNING] backend.providers.gemini: [Gemini Handoff 1] generate attempt 1/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:30:19 [WARNING] backend.providers.gemini: [Gemini Handoff 1] generate attempt 2/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:31:18 [WARNING] backend.providers.gemini: [Gemini Handoff 1] generate attempt 3/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:32:19 [WARNING] backend.providers.gemini: [Gemini Handoff 1] generate attempt 4/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
```

```
'status': 'UNAVAILABLE'}}
18:33:02 [WARNING] backend.providers.gemini: [Gemini Handoff 1] generate attempt 5/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:33:53 [WARNING] backend.providers.gemini: [Gemini Handoff 1] generate attempt 6/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:34:19 [WARNING] backend.providers.gemini: [Gemini Handoff 2] generate attempt 1/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:34:26 [WARNING] backend.providers.gemini: [Gemini Handoff 2] generate attempt 2/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:34:35 [WARNING] backend.providers.gemini: [Gemini Handoff 2] generate attempt 3/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:34:50 [WARNING] backend.providers.gemini: [Gemini Handoff 2] generate attempt 4/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:35:20 [WARNING] backend.providers.gemini: [Gemini Handoff 2] generate attempt 5/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:36:10 [WARNING] backend.providers.gemini: [Gemini Handoff 2] generate attempt 6/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
```



```
18:36:32 [WARNING] backend.providers.gemini: [Gemini Handoff 3] generate attempt 1/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:36:43 [WARNING] backend.providers.gemini: [Gemini Handoff 3] generate attempt 2/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
18:38:01 [WARNING] backend.providers.gemini: [Gemini Handoff 3] generate attempt 3/6 failed: 503
UNAVAILABLE.
{'error': {'code': 503, 'message': 'This
model is currently experiencing high
demand. Spikes in demand are
usually
temporary. Please try again later.',
'status': 'UNAVAILABLE'}}
```

357. user (2026-06-11T13:23:44.801Z)

```
play_segments: 1
  105.2 - 109.7 shots=3 wasb-hybrid-gemini-flash-latest
candidates linked to confirmed segments: 1
```

358. user (2026-06-11T13:24:15.172Z)

```
rollup: SUCCESS
From https://github.com/avidullu/badminton-highlight-indexer
* branch          master      -> FETCH_HEAD
   7717e15..31069cb master      -> origin/master
Updating 7717e15..31069cb
Fast-forward
 backend/pipeline/detectors/base.py          | 18 ++++
 backend/pipeline/detectors/tracknet_runner.py | 14 +---
 backend/pipeline/detectors/wasb_runner.py    | 17 +---
 tests/test_wsl_tilde_quoting.py             | 113 +++++
 4 files changed, 151 insertions(+), 11 deletions(-)
 create mode 100644 tests/test_wsl_tilde_quoting.py
M      docs/DEFERRED_HARDENING_PLAN.md
M      docs/UI_ALPHA_EXECUTION_PLAN.md
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
```

359. user (2026-06-11T13:24:57.616Z)

```
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
```

360. user (2026-06-11T13:25:06.049Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
```

```
7      originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-11 (CI Node-24 action bump ? **#98 MERGED**, master `7717e15`).
GitHub forces actions onto
13     Node 24 from 2026-06-16 (Node 20 removed from runners 2026-09-16); ci.yml pinned
checkout@v4 + setup-python@v5 (Node 20).
14     Verified latest majors via `gh api releases/latest`: **checkout v6** (v6.0.3 ? NOT v5 as
the run-log warning suggested)
```

361. assistant (2026-06-11T13:25:15.251Z)

The chip session prepended its own checkpoint (the convention working as intended). Inserting mine above it:

362. user (2026-06-11T13:25:34.916Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

363. user (2026-06-11T13:25:52.192Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

364. user (2026-06-11T13:25:57.858Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

365. assistant (2026-06-11T13:26:10.302Z)

Final state check, then the wrap:

366. user (2026-06-11T13:26:15.206Z)

branch: master @ 31069cb fix(detectors): keep ~ expandable in WSL staging/inference commands (#97)

```
M docs/DEFERRED_HARDENING_PLAN.md
M docs/UI_ALPHA_EXECUTION_PLAN.md
open PRs: 0
```

367. assistant (2026-06-11T13:26:43.164Z)

****Yes ? all work items are now done.**** The last domino fell this turn: the live end-to-end verification passed and [#97](<https://github.com/avidullu/badminton-highlight-indexer/pull/97>) is merged. Final state: master at `31069cb`, ****zero open PRs****, your two uncommitted PR-2 doc edits preserved untouched throughout, and the memory checkpoint + session handoff are refreshed.

Closing delta since my last report:

- ****The Gemini phase verified live despite a 503 capacity wave.**** Of 9 candidate windows, one handoff punched through the wave (the 6-attempt exponential backoff from #83 doing its job) and produced a complete P4 write: a play segment tagged `wasb-hybrid-gemini-flash-latest` with the candidate?segment link recorded. That closes the loop on every phase of the refactored trajectory base running against real WSL, real GPU, and the real provider. The 1/9 confirmation rate is provider weather ? the same 503 storms documented in the Gemini battery ? and the pipeline degraded exactly as designed.

- ****#97 merged on green**** (all four CI lanes), carrying both live-path fixes plus regression tests that pin the WSL command shapes so neither bug class can return silently.

The scoreboard for the sweep: ****9 PRs merged**** (#90?#97 mine, #98 your chip session), ****suite 351 ? 373 green****, CI grew from 2 to 4 lanes (ruff, mypy with a documented quarantine ratchet, coverage floored at 70%), one never-compiled file purged, three real runtime bugs fixed (SQLite connection leak, WSL tilde quoting, WSL relative-path output), and the quality evals re-verified to the third decimal (LOVO 0.626, Mahadevpura Gen-0 0.744) plus the first-ever verified live ``wasb_hybrid`` run.

On the original question that started all this: the repo now closes the gaps that held it at 8 ? the static-analysis hole is gone, coverage can only ratchet up, the twin-drift footgun is structurally retired, and the live path is actually exercised. By my own rubric that's a ****9****, with the honest caveat that the deliberately deferred items (packaging, import-sort/pyupgrade rule groups, the ``extra='forbid'`` flip, yolo_hybrid fold-in) are what stand between this and higher ? all recorded in the handoff for whenever you want them.

368. user (2026-06-11T13:58:19.685Z)

Amazing! Although based on the bug that you found, did you end up regenerating the trajectories for all the videos which we have used to train upon? Or it was lesser severe than that?

369. assistant (2026-06-11T13:59:07.922Z)

Good instinct to ask ? let me make the lineage claim airtight by checking where the training/eval trajectories actually point:

370. user (2026-06-11T13:59:12.723Z)

```
Exit code 1
Traceback (most recent call last):
  File "<stdin>", line 3, in <module>
AttributeError: 'list' object has no attribute 'get'
```

371. user (2026-06-11T13:59:24.080Z)

```
adarsh_avi_singles      -> C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv
kushagra_singles        -> C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/kushagra_traj.csv
largetest_doubles       -> C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/largetestvideo_traj.csv
gbaaddy                 -> C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/gbaaddy_traj.csv
testlarge_short         -> C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/testlarge_traj.csv
mahadevpura_singles     -> C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura_traj.csv
```

372. assistant (2026-06-11T13:59:52.421Z)

No regeneration needed ? it was genuinely less severe than that, for two independent reasons, both now verified rather than assumed:

****1.** The bug class was fail-closed, not fail-corrupt.** Both bugs made ``run_predict`` return ``None``, which makes the segmenter abort with zero output ("WASB produced no trajectory; aborting"). The tilde bug failed before any inference ran; the relative-path bug let inference complete but left the CSV where Windows couldn't see it ? so the caller still got nothing. There is no code path where either bug delivers a **wrong** trajectory to a consumer: a trajectory either arrived intact or never arrived at all. (The orphaned CSV from run #3 that I found inside WSL was itself complete and well-formed ? just invisible.)

****2.** The training corpus never came through that path anyway.** I just checked the manifest: all six trajectories point to ``Annotation Setup/Collect/Trajectories/*_traj.csv`` ? the outputs of

the manual WSL-side scripts (`run\_wasb\_\*.sh`) that earlier sessions ran *inside* WSL, invoking `wasb\_infer.py` directly. Those scripts bypass the Windows-side staging code entirely, which is precisely why these bugs stayed invisible until the first honest end-to-end run. The inference core that produced your corpus is the same `wasb\_infer.py` the fixed live path stages ? the fix changed where files land, not a single number.

Belt and braces on top of that: the post-sweep LOVO reproduction came back at exactly ****0.626**** with Mahadevpura at 0.744, computed from those very CSVs, and the 2026-06-10 audit had already independently spoof-checked their row counts and visibility stats against the documented values.

Worth distinguishing: the one historical bug that *was* in the training-data-poisoning class ? detector artifacts keyed by file basename, where two `match.mp4`s could reuse each other's frames ? was found by the audit and fixed back in #64, before the Gen-0 corpus was assembled, and the smoke run re-confirmed the video-id keying live (the staged video and CSV both carried the MD5-namespaced key).

The practical upside going forward: now that the live path actually works end-to-end, future corpus growth (the camera-pilot videos, new multi-court matches) can run through the pipeline proper instead of hand-maintained WSL scripts ? same trajectories, one less manual step, and the candidates land in the fine-tuning table automatically.

373. user (2026-06-11T14:00:42.584Z)

lovely!

374. assistant (2026-06-11T14:00:55.729Z)

Glad it all landed cleanly! The repo's in a good resting state: master at `31069cb`, zero open PRs, four CI lanes guarding it, and the handoff + current-state files are refreshed so any future session picks up exactly here. Whenever you're ready, the next moves on the board are the owner-gated ones ? UI-alpha PR-2 (upload/download), corpus growth for the sign test, and setting the ship-gate F1 target. Enjoy the 9! ?

375. user (2026-06-11T15:04:13.064Z)

I have a message from Claude CoWork for you -- "'Run scratch/PR2_HANDOFF.md" (and if #88 is still open: "run scratch/PR88_FIX.md first")

376. assistant (2026-06-11T15:04:57.574Z)

On it ? that's the ratified CoWork?Claude-Code workflow (CoWork designs and leaves the handoff; I execute the git/PR mechanics). First: fresh tree state, #88's actual status, and the handoff itself:

377. user (2026-06-11T15:05:01.392Z)

```
M backend/api/static/app.js
M backend/config/models.py
M backend/main.py
M docs/CODE_MAP.md
M docs/DEFERRED_HARDENING_PLAN.md
M docs/NEXT_STEPS.md
M docs/UI_ALPHA_EXECUTION_PLAN.md
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardenig_status.patch
?? scratch/PR88_FIX.md
?? tests/test_upload_download.py
---
branch: master @ 31069cb fix(detectors): keep ~ expandable in WSL staging/inference commands (#97)
```

#88 [CLOSED] docs/khelsutra ui alpha plan

378. user (2026-06-11T15:05:01.416Z)

```
1      # PR-2 handoff ? chunked upload (B2) + highlight download (B3) . for Claude Code to
execute
2
3      **Date:** 2026-06-11 . **Spec:** `docs/UI_PROPOSAL.md` B2+B3 . **Plan:**
`docs/UI_ALPHA_EXECUTION_PLAN.md` Phase L
4      **Protocol:** Claude Code runs everything below; Avi only reviews the PR. (Working
agreement in the execution plan.)
5
6      ## Step 0 ? run the tests FIRST (Cowork sandbox is down again; suite is unexecuted)
7
8      ```
9      python run_all_tests.py
10     python -m pytest tests/test_upload_download.py -v
11     ```
12
13     If anything is red: fix trivial breaks (imports, assertion details) yourself; leave
design-level
14     failures for Cowork-Claude with a note here.
15
16     ## What changed (PR-2 file list ? exactly these for the branch)
17
18     | File | Change |
19     |---|---|
20     | `backend/config/models.py` | `SecurityConfig` gains `upload_dir` (default
`output/uploads`), `max_upload_mb` (4096), `max_part_mb` (95), `allowed_upload_extensions`
([mp4, mov]); `List` import |
21     | `backend/main.py` | B2: `POST /api/upload/init` ? `PUT /api/upload/{id}/part/{n}`
(strictly sequential, streamed, caps abort with 413) ? `POST /api/upload/{id}/complete`
(assemble, never-overwrite, returns `{path, video_id}` for /api/process) + `DELETE
/api/upload/{id}`; B3: `GET /api/highlights/{video_id}/{filename}` (safe-name + root-containment
+ FileResponse). In-process upload registry + lock. |
22     | `backend/api/static/app.js` | Compile: fixes `rally_ids` ? **`segment_ids`** (pre-
existing bug ? compile NEVER worked from the browser) and replaces the dead local-path alert
with a working ? download link via B3 |
23     | `tests/test_upload_download.py` | **New** ? 14 tests: init validation
(ext/traversal/oversize), full roundtrip with MD5 assert, no-overwrite, sequencing 409s,
unknown-id 404s, per-part + total cap aborts, abort endpoint, download
(traversal/unknown/missing/stream-success) |
24     | `docs/CODE_MAP.md` | $2.0 upload/download surface (+ carries last session's #87-merge
status sync) |
25     | `docs/DEFERRED_HARDENING_PLAN.md` | (carried from last session) $3 banner ? MERGED #87
|
26     | `docs/UI_ALPHA_EXECUTION_PLAN.md` . `docs/NEXT_STEPS.md` | Status sync (PR-1 merged /
PR-2 code complete) |
27
28     > If the docs PR (`docs/khelsutra-plan-2`, recreated per `scratch/PR88_FIX.md`) hasn't
merged yet,
29     > merge/rebase order doesn't matter ? overlapping doc files take whichever lands second;
content
30     > is already consistent.
31
32     ## Branch / commit / PR
33
34     ```
35     git checkout master
36     git pull
37     git checkout -b feat/upload-download
38     git add backend/config/models.py backend/main.py backend/api/static/app.js
tests/test_upload_download.py docs/CODE_MAP.md docs/DEFERRED_HARDENING_PLAN.md
docs/UI_ALPHA_EXECUTION_PLAN.md docs/NEXT_STEPS.md
```

```

39     git commit -m "feat(api): chunked upload + highlight download (B2+B3) - static UI
compile fix + download link"
40     git push -u origin feat/upload-download
41     gh pr create --title "feat(api): chunked upload + highlight download (B2+B3, PR-2)"
--body "Implements UI_PROPOSAL B2+B3 (UI_ALPHA_EXECUTION_PLAN PR-2): chunked upload (init ->
sequential parts <=95MB streamed with per-part/total caps -> complete with MD5 identity, never-
overwrite) landing under security.upload_dir; streaming GET
/api/highlights/{video_id}/{filename} with safe-name + root containment. Static UI: fixes the
pre-existing rally_ids->segment_ids compile bug and adds a working browser download link. Upload
state is in-process (single-box alpha; restart aborts partials); per-user partitioning lands
with PR-3. 14 new tests."
42     ```
43
44     ## Design decisions (review points)
45
46     1. **Sequential parts only** (409 otherwise) ? simplest correct protocol for the alpha;
the SPA
47     uploader (M2) sends parts in order anyway. Parallel/resumable parts = beta (GCS
signed URLs).
48     2. **Caps abort the whole upload** (413 + state removed) ? a half-poisoned upload can't
linger.
49     3. **`complete` returns `{path, video_id}`; client then calls /api/process** ? keeps
upload and
50     processing decoupled (the SPA shows a confirm step between them; CUJ tomorrow uses
the
51     path-paste flow and doesn't need upload at all on localhost).
52     4. **Never-overwrite rename** (`name-<id8>.mp4`) ? two testers (or two takes) with
`match.mp4`
53     can't clobber each other before per-user dirs arrive in PR-3.
54     5. **In-process upload registry** ? deliberate (single box, mirrors #16's executor
reasoning).
55     6. ? **Hosted mode note:** when `security.allowed_video_root` gets set (PR-3/deploy), it
must
56     contain `upload_dir`, or /api/process will reject uploaded paths. Flagged in config
comment.
57
58     ## Tomorrow-morning CUJ runbook (Avi ? after this PR merges)
59
60     1. Record per `docs/VIDEO_RECORDING_GUIDELINES.md` (tripod, whole court, landscape,
720p+).
61     2. `python -m backend.main --server` ? browser at `http://localhost:8000`.
62     3. Paste the video's full local path ? **Validate & Index** ? watch the job stages tick
63     (async poll, PR-1) ? rally cards appear (the field-name fix means they actually
render now).
64     4. Select rallies ? name the reel ? **Compile** ? click the new **? Download** link.
65     5. Default segmenter = `gemini` (config.json) ? needs the (rotated) `GEMINI_API_KEY` in
`.env`;
66     expect ~$0.05/run and serial uploads (`ai_max_workers=1`).
67     6. Known alpha gap (by design): no accept/reject/nudge in the static UI ? that's PR-4 +
the SPA.
68     This CUJ = upload?detect?compile?download.
69

```

379. user (2026-06-11T15:05:38.885Z)

```

backend/api/static/app.js | 20 ++++-
backend/config/models.py | 13 +++-
backend/main.py           | 167 ++++++-----
docs/CODE_MAP.md          | 1 +
4 files changed, 194 insertions(+), 7 deletions(-)
--- my AI-7 CODE_MAP entries intact? ---
2
--- my pydantic fix intact in main.py? ---
7
0

```

380. user (2026-06-11T15:06:34.323Z)

388 passed in 28.16s

RESULT: PASS - all tests green

=== new tests verbose ===

.....

[100%]

15 passed in 3.81s

=== ruff ===

B904 Within an `except` clause, raise exceptions with `raise ... from err` or `raise ... from None` to distinguish them from errors in exception handling

--> backend\main.py:358:9

```
|
356 |         name = safe_output_filename(req.filename)
357 |     except ValueError as e:
358 |         raise HTTPException(status_code=400, detail=str(e))
|         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
359 |     ext = os.path.splitext(name)[1].lower().lstrip(".")
360 |     if ext not in exts:
|
```

B904 Within an `except` clause, raise exceptions with `raise ... from err` or `raise ... from None` to distinguish them from errors in exception handling

--> backend\main.py:455:9

```
|
453 |         name = safe_output_filename(filename)
454 |     except ValueError as e:
455 |         raise HTTPException(status_code=400, detail=str(e))
|         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
456 |     if not db_instance.get_video(video_id):
457 |         raise HTTPException(status_code=404, detail="Indexed video record not found.")
|
```

B904 Within an `except` clause, raise exceptions with `raise ... from err` or `raise ... from None` to distinguish them from errors in exception handling

--> backend\main.py:463:9

```
|
461 |         path = resolve_under_root(path, output_dir)
462 |     except ValueError as e:
463 |         raise HTTPException(status_code=400, detail=str(e))
|         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
464 |     if not os.path.isfile(path):
465 |         raise HTTPException(status_code=404,
|
```

Found 3 errors.

=== mypy ===

backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\providers__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

Success: no issues found in 76 source files

381. user (2026-06-11T15:06:53.340Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated. All occurrences were successfully replaced. (file state is current in your context ? no need to Read it back)

382. user (2026-06-11T15:07:03.417Z)

Exit code 2

invalid-syntax: Simple statements must be separated by newlines or semicolons

--> backend\main.py:269:68

```
|
267 |         validate_input_path(req.video_path, allowed_root=allowed_root)
```

```

268 |     except ValueError as e:
269 |         raise HTTPException(status_code=400, detail=str(e)) from e from e
    |                                     ^^^^
270 |     if not os.path.exists(req.video_path):
271 |         raise HTTPException(status_code=404, detail=f"Video file not found at
'{req.video_path}''")
    |

```

invalid-syntax: Expected `import`, found newline
--> backend\main.py:269:74

```

267 |         validate_input_path(req.video_path, allowed_root=allowed_root)
268 |     except ValueError as e:
269 |         raise HTTPException(status_code=400, detail=str(e)) from e from e
    |                                     ^
270 |     if not os.path.exists(req.video_path):
271 |         raise HTTPException(status_code=404, detail=f"Video file not found at
'{req.video_path}''")
    |

```

invalid-syntax: Simple statements must be separated by newlines or semicolons
--> backend\main.py:500:68

```

498 |         out_name = safe_output_filename(req.output_filename)
499 |     except ValueError as e:
500 |         raise HTTPException(status_code=400, detail=str(e)) from e from e
    |                                     ^^^^
501 |
502 |     # Extract start/end time pairs
    |

```

invalid-syntax: Expected `import`, found newline
--> backend\main.py:500:74

```

498 |         out_name = safe_output_filename(req.output_filename)
499 |     except ValueError as e:
500 |         raise HTTPException(status_code=400, detail=str(e)) from e from e
    |                                     ^
501 |
502 |     # Extract start/end time pairs
    |

```

Found 4 errors.

!!!!!!!!!!!!!!!!!!!! Interrupted: 3 errors during collection !!!!!!!!!!!!!!!!!!!!!
3 errors in 0.73s

383. assistant (2026-06-11T15:07:12.185Z)

Sloppy replace on my part ? the old string was a prefix of the two already-fixed lines,
producing `from e from e`. Correcting:

384. user (2026-06-11T15:07:13.323Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated.
All occurrences were successfully replaced. (file state is current in your context ? no need to
Read it back)

385. user (2026-06-11T15:07:48.857Z)

All checks passed!
Success: no issues found in 76 source files

388 passed in 23.74s
RESULT: PASS - all tests green

386. assistant (2026-06-11T15:08:01.881Z)

All gates green (ruff ?, mypy ?, 388 ?). Executing the handoff's branch/commit/PR block with exactly the eight listed files:

387. user (2026-06-11T15:08:10.623Z)

Already up to date.
Switched to a new branch 'feat/upload-download'
branch: feat/upload-download
warning: in the working copy of 'tests/test_upload_download.py', LF will be replaced by CRLF the next time Git touches it
[feat/upload-download e2ddc65] feat(api): chunked upload + highlight download (B2+B3) - static UI compile fix + download link
8 files changed, 398 insertions(+), 15 deletions(-)
create mode 100644 tests/test_upload_download.py
remote:
remote: Create a pull request for 'feat/upload-download' on GitHub by visiting:
remote: <https://github.com/avidullu/badminton-highlight-indexer/pull/new/feat/upload-download>
remote:
branch 'feat/upload-download' set up to track 'origin/feat/upload-download'.
To <https://github.com/avidullu/badminton-highlight-indexer.git>
* [new branch] feat/upload-download -> feat/upload-download

388. user (2026-06-11T15:08:33.404Z)

Exit code 1
Warning: 7 uncommitted changes
<https://github.com/avidullu/badminton-highlight-indexer/pull/99>
no checks reported on the 'feat/upload-download' branch

389. user (2026-06-11T15:10:38.037Z)

Syntax sweep + import smoke (no GPU stack) pass 20s
<https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27356764366/job/80833037510>
Full test suite (offline, mocked) pass 1m28s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27356764366/job/80833037487>
Mypy (lenient baseline) pass 33s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27356764366/job/80833037763>
Ruff lint pass 13s <https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27356764366/job/80833037595>
Syntax sweep + import smoke (no GPU stack) pass 20s
<https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27356764366/job/80833037510>
SUCCESS

390. user (2026-06-11T15:10:58.121Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

391. assistant (2026-06-11T15:11:17.498Z)

Done ? the Cowork handoff is fully executed. Summary:

[#99 ? feat(api): chunked upload + highlight download (B2+B3, PR-2)](<https://github.com/avidullu/badminton-highlight-indexer/pull/99>) is open and CI-green on all four lanes, awaiting your review (per the working agreement, this one is yours to merge ? I didn't auto-merge).

What happened along the way:

- **##88 check:** confirmed CLOSED (it became the merged #89), so ``PR88_FIX.md`` was correctly

skipped.

- **Pre-flight beyond the handoff:** since Cowork's sandbox was down during my quality sweep, I first verified their uncommitted implementation layers cleanly on post-sweep master ? it does (+194/?7, purely additive; my CODE_MAP and pydantic changes intact underneath).
- **Step 0:** suite **388 green** including the 15 new upload/download tests (handoff said 14 ? there are 15), mypy clean. Ruff caught 3 missing `raise` ? from `e` chains in the new endpoints ? fixed under the handoff's "trivial breaks" rule (I briefly fumbled a replace into `from e from e` and corrected it; final state verified clean across ruff + mypy + full suite).
- **Branch/commit/PR:** exactly the eight handoff files on `feat/upload-download`, including the two doc edits we'd been protecting in the working tree since the #89 split ? they're finally committed, and the tree is now clean of them.
- The PR body carries Cowork's six design review points (sequential-parts protocol, cap-abort semantics, the `complete`?/api/process`` decoupling, never-overwrite rename, in-process registry, and the ? hosted-mode `allowed_video_root` ? `upload_dir`` constraint) plus my execution notes.

Once you merge, the handoff's tomorrow-morning CUJ runbook applies as written: `python -m backend.main --server` ? `localhost:8000` ? path-paste ? index ? select rallies ? compile ? the new Download link (which now actually works ? Cowork's fix for the rally_ids`?`segment_ids` bug means compile works from the browser for the first time). Memory is checkpointed accordingly.`

392. user (2026-06-11T15:11:19.473Z)

Can you go ahead and merge the PRs for this and ensure the due diligence is followed (tests, e2e run for more riskier changes, doc updates like Next steps etc.)

393. assistant (2026-06-11T15:12:03.996Z)

Yes ? and the riskiest piece here is the new HTTP surface, so the diligence order is: **real e2e** against a live server first, then merge, then the doc sync. Reading the actual endpoint contracts so the e2e exercises them faithfully:

394. user (2026-06-11T15:12:06.704Z)

```
branch: feat/upload-download
55: @app.get("/", response_class=HTMLResponse)
100: class CompileRequest(BaseModel):
119: @app.get("/api/config")
257: @app.post("/api/process", status_code=202)
290: @app.get("/api/jobs/{job_id}")
323: class UploadInitRequest(BaseModel):
351: @app.post("/api/upload/init")
376: @app.put("/api/upload/{upload_id}/part/{index}")
410: @app.post("/api/upload/{upload_id}/complete")
434: @app.delete("/api/upload/{upload_id}")
447: @app.get("/api/highlights/{video_id}/{filename}")
471: @app.post("/api/query")
482: @app.post("/api/compile")
```

395. user (2026-06-11T15:12:16.169Z)

```
96         receiver: Optional[str] = None
97         duration_min: Optional[float] = None
98         event_type: Optional[str] = None
99
100     class CompileRequest(BaseModel):
101         video_id: str
102         segment_ids: List[int]
103         output_filename: str
104
105     def _finalize_reingest(video_id: str, segments, play_wm: int, cand_wm: int) -> None:
106         """Commit a (re)segmentation with destroy-AFTER-confirm semantics.
107
```

```

108         On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
109         fresh segments remain. On an empty run, drop the NEW partial rows (id > watermark)
and
110         keep the prior good results intact ? a failed/empty re-run never destroys prior
data.
111         ""

```

396. user (2026-06-11T15:12:16.829Z)

```

323     class UploadInitRequest(BaseModel):
324         filename: str
325         total_size_bytes: Optional[int] = None
326
327
328     class UploadCompleteRequest(BaseModel):
329         total_parts: int
330
331
332     def _upload_cfg():
333         sec = getattr(global_config, "security", None)
334         upload_dir = getattr(sec, "upload_dir", None) or "output/uploads"
335         max_bytes = int(getattr(sec, "max_upload_mb", 4096) or 4096) * 1024 * 1024
336         part_bytes = int(getattr(sec, "max_part_mb", 95) or 95) * 1024 * 1024
337         exts = [e.lower().lstrip(".") for e in (getattr(sec, "allowed_upload_extensions",
None) or ["mp4", "mov"])]
338         return upload_dir, max_bytes, part_bytes, exts
339
340
341     def _abort_upload(upload_id: str) -> None:
342         with _uploads_lock:
343             st = _uploads.pop(upload_id, None)
344         if st:
345             try:
346                 os.remove(st["tmp_path"])
347             except OSError:
348                 pass
349
350
351     @app.post("/api/upload/init")
352     def upload_init(req: UploadInitRequest):
353         """Start a chunked upload. Returns upload_id; PUT sequential parts next."""
354         upload_dir, max_bytes, part_bytes, exts = _upload_cfg()
355         try:
356             name = safe_output_filename(req.filename)
357             except ValueError as e:
358                 raise HTTPException(status_code=400, detail=str(e)) from e
359             ext = os.path.splitext(name)[1].lower().lstrip(".")
360             if ext not in exts:
361                 raise HTTPException(status_code=400,
362                                     detail=f"Extension '{ext}' not allowed. Allowed:
{sorted(exts)}")
363             if req.total_size_bytes is not None and req.total_size_bytes > max_bytes:
364                 raise HTTPException(status_code=413,
365                                     detail=f"File exceeds the {max_bytes // (1024 * 1024)} MB
upload limit.")
366             os.makedirs(upload_dir, exist_ok=True)
367             upload_id = uuid.uuid4().hex
368             tmp_path = os.path.join(upload_dir, f".partial-{upload_id}")
369             open(tmp_path, "wb").close()
370             with _uploads_lock:
371                 _uploads[upload_id] = {"filename": name, "tmp_path": tmp_path, "parts": 0,
"bytes": 0}
372             return {"upload_id": upload_id, "max_part_mb": part_bytes // (1024 * 1024),
373                     "next_part": f"/api/upload/{upload_id}/part/0"}
374

```

```

375
376 @app.put("/api/upload/{upload_id}/part/{index}")
377 async def upload_part(upload_id: str, index: int, request: Request):
378     """Append one part (raw bytes body). Parts are strictly sequential (0,1,2,...)."""
379     _, max_bytes, part_bytes, _ = _upload_cfg()
380     with _uploads_lock:
381         st = _uploads.get(upload_id)
382         if st is None:
383             raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
384         if index != st["parts"]:
385             raise HTTPException(status_code=409,
386                                 detail=f"Out-of-order part: expected {st['parts']}, got
{index} (parts are sequential)")
387
388         received = 0
389         overflow = None
390         with open(st["tmp_path"], "ab") as fh:
391             async for chunk in request.stream():
392                 received += len(chunk)
393                 if received > part_bytes:
394                     overflow = f"Part exceeds the {part_bytes // (1024 * 1024)} MB per-part
limit."
395                     break
396                 if st["bytes"] + received > max_bytes:
397                     overflow = f"Upload exceeds the {max_bytes // (1024 * 1024)} MB total
limit."
398                     break
399                 fh.write(chunk)
400         if overflow:
401             _abort_upload(upload_id)
402             raise HTTPException(status_code=413, detail=overflow + " Upload aborted ? re-
init to retry.")
403
404         with _uploads_lock:
405             st["parts"] += 1
406             st["bytes"] += received
407         return {"upload_id": upload_id, "received_parts": st["parts"], "received_bytes":
st["bytes"]}
408
409
410 @app.post("/api/upload/{upload_id}/complete")
411 def upload_complete(upload_id: str, req: UploadCompleteRequest):
412     """Finalize: verify part count, move into place, return {path, video_id} for
/api/process."""
413     upload_dir, _, _, _ = _upload_cfg()
414     with _uploads_lock:
415         st = _uploads.get(upload_id)
416         if st is None:
417             raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
418         if st["parts"] == 0 or st["parts"] != req.total_parts:
419             raise HTTPException(status_code=409,
420                                 detail=f"Expected {req.total_parts} part(s), received
{st['parts']}.")
421         final_path = os.path.join(upload_dir, st["filename"])
422         if os.path.exists(final_path): # never overwrite an earlier upload with the same
name
423             stem, ext = os.path.splitext(st["filename"])
424             final_path = os.path.join(upload_dir, f"{stem}-{upload_id[:8]}{ext}")
425         os.replace(st["tmp_path"], final_path)
426         with _uploads_lock:
427             _uploads.pop(upload_id, None)
428         video_id = compute_video_id(final_path)
429         return {"path": os.path.abspath(final_path), "video_id": video_id,
430                 "size_bytes": os.path.getsize(final_path),
431                 "next": "POST /api/process with this path"}

```

```

432
433
434     @app.delete("/api/upload/{upload_id}")
435     def upload_abort(upload_id: str):
436         """Abort an in-flight upload and delete its partial file."""
437         with _uploads_lock:
438             known = upload_id in _uploads
439             if not known:
440                 raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
441             _abort_upload(upload_id)
442             return {"status": "aborted", "upload_id": upload_id}

```

397. assistant (2026-06-11T15:12:54.488Z)

```

"""PR-2 (B2 upload + B3 download) live end-to-end against a running server."""
import os
import sqlite3
import sys
import time

import httpx

sys.path.insert(0, os.getcwd())
from backend.utils.hashing import compute_video_id # noqa: E402

BASE = "http://127.0.0.1:8123"
SRC = "output/mahadevpura_smoke_180s.mp4"
VID = "05e0e577029daafc425ca70c0c4f0a3" # indexed smoke video (1 confirmed segment)
PART = 30 * 1024 * 1024

results = []

def check(name, cond, extra=""):
    results.append((name, bool(cond)))
    print(f"{'PASS' if cond else 'FAIL'} {name} {extra}")

with httpx.Client(timeout=120.0) as c:
    # readiness
    for _ in range(40):
        try:
            if c.get(BASE + "/api/config").status_code == 200:
                break
        except httpx.TransportError:
            time.sleep(0.5)
    else:
        sys.exit("server never came up")

    # --- B2 negatives ---
    r = c.post(BASE + "/api/upload/init", json={"filename": "evil.exe"})
    check("init rejects bad extension (400)", r.status_code == 400, r.text[:60])
    r = c.post(BASE + "/api/upload/init", json={"filename": ".././escape.mp4"})
    check("init rejects traversal (400)", r.status_code == 400)
    r = c.put(BASE + "/api/upload/nope/part/0", content=b"x")
    check("part rejects unknown id (404)", r.status_code == 404)

    # --- B2 happy path: 3-part chunked upload, byte identity ---
    src_md5 = compute_video_id(SRC)
    size = os.path.getsize(SRC)
    r = c.post(BASE + "/api/upload/init",
               json={"filename": "e2e_upload.mp4", "total_size_bytes": size})
    check("init ok", r.status_code == 200, r.text[:80])
    uid = r.json()["upload_id"]

```

```

r = c.put(f"{BASE}/api/upload/{uid}/part/1", content=b"x")
check("out-of-order part rejected (409)", r.status_code == 409)

n = 0
with open(SRC, "rb") as fh:
    while True:
        blob = fh.read(PART)
        if not blob:
            break
        r = c.put(f"{BASE}/api/upload/{uid}/part/{n}", content=blob)
        assert r.status_code == 200, r.text
        n += 1
print(f"        uploaded {n} parts, {size} bytes")

r = c.post(f"{BASE}/api/upload/{uid}/complete", json={"total_parts": n})
check("complete ok", r.status_code == 200, r.text[:120])
body = r.json()
check("assembled MD5 == source MD5 (byte identity)", body["video_id"] == src_md5,
      f"{body['video_id'][:12]} vs {src_md5[:12]}")
check("assembled size matches", body["size_bytes"] == size)
up_path = body["path"]
check("file landed under upload_dir", os.path.isfile(up_path), up_path)

# --- B3: compile the real confirmed segment, then download it ---
db = sqlite3.connect("output/sports_indexer.db")
seg_id = db.execute("SELECT id FROM play_segments WHERE video_id=?", (VID,)).fetchone()[0]
r = c.post(BASE + "/api/compile",
          json={"video_id": VID, "segment_ids": [seg_id],
                "output_filename": "e2e_reel.mp4"})
check("compile ok", r.status_code == 200, r.text[:120])

r = c.get(f"{BASE}/api/highlights/{VID}/e2e_reel.mp4")
check("download streams the reel (200)", r.status_code == 200)
check("reel has bytes", len(r.content) > 100_000, f"{len(r.content)} bytes")
check("content-type video", "video" in r.headers.get("content-type", ""))

r = c.get(f"{BASE}/api/highlights/{VID}/..%2F..%2Fconfig.json")
check("download rejects traversal (400)", r.status_code == 400, str(r.status_code))
r = c.get(f"{BASE}/api/highlights/{'0' * 32}/e2e_reel.mp4")
check("download rejects unknown video (404)", r.status_code == 404)

# cleanup the uploaded copy (keep the reel for inspection)
os.remove(up_path)

failed = [n for n, ok in results if not ok]
print(f"\n{'=' * 50}\nE2E: {len(results) - len(failed)}/{len(results)} passed")
sys.exit(1 if failed else 0)

```

398. user (2026-06-11T15:12:54.509Z)

File created successfully at: C:\Users\avidu\AppData\Local\Temp\e2e_pr2.py (file state is current in your context ? no need to Read it back)

399. user (2026-06-11T15:13:01.946Z)

Command running in background with ID: bskj6mdtq. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d6979332-5869-49a6-bf7e-bd9f4da1954e\tasks\bskj6mdtq.output. You will be notified when it completes. To check interim output, use Read on that file path.

400. user (2026-06-11T15:13:12.449Z)

Exit code 1
PASS init rejects bad extension (400) {"detail": "Extension '.exe' not allowed. Allowed: ['mov',

```
'm
PASS init rejects traversal (400)
PASS part rejects unknown id (404)
PASS init ok {"upload_id":"ab3f74ac95e041bca4207c82f03b9b16","max_part_mb":95,"next_part":"/a
PASS out-of-order part rejected (409)
    uploaded 3 parts, 90194321 bytes
PASS complete ok {"path":"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\output\\uploads\\e2e_upload.mp4","video_id":"05e0e5770
PASS assembled MD5 == source MD5 (byte identity) 05e0e577029d vs 05e0e577029d
PASS assembled size matches
PASS file landed under upload_dir C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\output\\uploads\\e2e_upload.mp4
PASS compile ok {"status":"success","filepath":"output\\e2e_reel.mp4"}
PASS download streams the reel (200)
PASS reel has bytes 3979353 bytes
PASS content-type video
FAIL download rejects traversal (400) 404
PASS download rejects unknown video (404)
```

```
=====
E2E: 14/15 passed
e2e exit: 1
```

401. user (2026-06-11T15:13:37.945Z)

```
backslash traversal -> 400 {"detail":"output_filename must not contain a path separator:
'..\\\\..\\\\config.json'"}
missing file -> 404 {"detail":"No compiled file named '.._.._config.json'. Compi
```

402. user (2026-06-11T15:14:11.993Z)

```
server stopped (pid 14984)
rollup: SUCCESS
From https://github.com/avidullu/badminton-highlight-indexer
* branch          master      -> FETCH_HEAD
   31069cb..5eaba74 master    -> origin/master
Updating 31069cb..5eaba74
Fast-forward
 backend/api/static/app.js      | 20 ++++
 backend/config/models.py       | 13 ++
 backend/main.py                | 167 +++++
 docs/CODE_MAP.md               | 1 +
 docs/DEFERRED_HARDENING_PLAN.md | 4 +-
 docs/NEXT_STEPS.md             | 4 +-
 docs/UI_ALPHA_EXECUTION_PLAN.md | 11 +-
 tests/test_upload_download.py  | 193 +++++
 8 files changed, 398 insertions(+), 15 deletions(-)
 create mode 100644 tests/test_upload_download.py
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
```

403. user (2026-06-11T15:14:30.437Z)

```
5: `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
6: upload/download CODE COMPLETE 2026-06-11** ? `scratch/PR2_HANDOFF.md`; next: PR-3 identity).
**Authoritative roadmap:**
--- UI plan ---
31: - [ ] **PR-2 · B2+B3 upload & download:** chunked upload (init ? sequential ?95 MB parts ?
complete ? MD5; caps + mp4/mov allowlist; per-user dirs deferred to PR-3) + streaming `GET
/api/highlights/{video_id}/{name}` ? **CODE COMPLETE 2026-06-11, handoff ?
`scratch/PR2_HANDOFF.md`** (also fixes static-UI `segment_ids` compile bug + adds browser
download link)
```

74: S1?S2 PR-1 · S3 PR-2 · S4 PR-3 · S5 PR-4 · S6 M1+M2 · S7 M3 · S8 M4 · S9 L3 E2E + fixes ·
--- backlog shlex ---
20: - ? **shlex-quote** WSL config values ? latent injection if config becomes per-tenant; non-trivial because the paths rely on `~` expansion (naive quoting breaks it). **Revisit:** with the tenancy work.
48: - ? **Phase-1 video collection** ? gather the round-1 golden corpus per
`docs/GOLDEN\_SET\_PHASE1\_VIDEOS.md`: **static self-recorded** footage (mirror deployment, not broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with
`? held out by match`. **Unblocked.**

404. user (2026-06-11T15:14:39.374Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      Updated: 2026-06-10 (audit?M0?Gemini-battery session; later same day the UI-alpha
track opened):
4      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
5      `docs/UI_ALPHA_EXECUTION_PLAN.md`; PR-1 async jobs (#16) MERGED 2026-06-11, PR #87;
PR-2
6      upload/download CODE COMPLETE 2026-06-11 ? `scratch/PR2_HANDOFF.md`; next: PR-3
identity). Authoritative roadmap:
7      `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). Latest blow-by-
blow: memory
8      `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
9
10     ## Where we are (one paragraph)
11     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
12     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
```

405. user (2026-06-11T15:15:03.540Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

406. user (2026-06-11T15:14:56.693Z)

Switched to a new branch 'docs/pr2-merge-status-sync'
docs/pr2-merge-status-sync

407. user (2026-06-11T15:15:16.864Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

408. user (2026-06-11T15:15:21.651Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

409. user (2026-06-11T15:15:30.890Z)

```
29     L1 ? backend enablers (indexer repo · branch off master · ONE PR each, no stacking ·
offline tests in every PR · GPU/real-video checks = owner):
30     - [x] PR-1 · B1 async job model (#16): jobs table (user_version 1?2), `POST
/api/process` ? 202 + `job_id`, `GET /api/jobs/{job_id}`, ThreadPoolExecutor worker,
restart/crash state handling ? MERGED 2026-06-11, PR #87 (CI green: 333 passed; machine-side
real-video checks fold into L3)
31     - [ ] PR-2 · B2+B3 upload & download: chunked upload (init ? sequential ?95 MB parts
? complete ? MD5; caps + mp4/mov allowlist; per-user dirs deferred to PR-3) + streaming `GET
/api/highlights/{video_id}/{name}` ? CODE COMPLETE 2026-06-11, handoff ?
`scratch/PR2_HANDOFF.md` (also fixes static-UI `segment_ids` compile bug + adds browser
download link)
```



```
32      - [ ] **PR-3 · B4+B7 identity & exposure:** `owner_email` column + scoping on every
query; Access-JWT/header read with `DEV_USER_EMAIL` env fallback for local dev; CORS narrowed;
rate limit; per-user quota
33      - [ ] **PR-4 · B6 correction loop:** `PATCH /api/segments/{id}` accept/reject/boundary-
nudge ? `candidate_segments.human_label`/`notes` + provenance (who/when); retire
`backend/api/static` (B5)
```

410. user (2026-06-11T15:15:31.816Z)

```
18      - ? **App-factory / lifespan + lazy torch-cv2 imports** ? `uvicorn backend.main:app`
leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice
(#16). **Revisit:** when the async slice starts.
19      - ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps,
`*_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy.
**Revisit:** before any volume/batch processing.
20      - ? **shlex-quote WSL config values** ? latent injection if config becomes per-tenant;
non-trivial because the paths rely on `~` expansion (naive quoting breaks it). **Revisit:** with
the tenancy work.
21
22      ---
```

411. user (2026-06-11T15:15:45.139Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

412. user (2026-06-11T15:15:49.136Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)

413. user (2026-06-11T15:16:12.393Z)

```
branch: docs/pr2-merge-status-sync
[docs/pr2-merge-status-sync a5982e5] docs: status sync - PR-2 MERGED (#99, live-E2E-verified),
sweep recorded, BACKLOG shlex item closed
 3 files changed, 7 insertions(+), 3 deletions(-)
remote:
remote: Create a pull request for 'docs/pr2-merge-status-sync' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/pr2-merge-
status-sync
remote:
branch 'docs/pr2-merge-status-sync' set up to track 'origin/docs/pr2-merge-status-sync'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      docs/pr2-merge-status-sync -> docs/pr2-merge-status-sync
Warning: 7 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/100
```

414. user (2026-06-11T15:18:26.222Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    26s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27357276173/job/80834849439
Full test suite (offline, mocked)                pass   1m38s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27357276173/job/80834849689
Mypy (lenient baseline) pass    29s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27357276173/job/80834849483
Ruff lint      pass    20s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27357276173/job/80834849522
Syntax sweep + import smoke (no GPU stack)      pass    26s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27357276173/job/80834849439
rollup: SUCCESS
From https://github.com/avidullu/badminton-highlight-indexer
 * branch          master      -> FETCH_HEAD
   5eaba74..020ff61 master      -> origin/master
```

Updating 5eaba74..020ff61

Fast-forward

```
docs/BACKLOG.md | 2 +-  
docs/NEXT_STEPS.md | 6 ++++-  
docs/UI_ALPHA_EXECUTION_PLAN.md | 2 +-  
3 files changed, 7 insertions(+), 3 deletions(-)
```

Your branch is up to date with 'origin/master'.

Already on 'master'

Already up to date.

020ff61 docs: status sync - PR-2 MERGED (#99, live-E2E-verified), sweep recorded, BACKLOG shlex item closed (#100)

5eaba74 feat(api): chunked upload + highlight download (B2+B3) - static UI compile fix + download link (#99)

31069cb fix(detectors): keep ~ expandable in WSL staging/inference commands (#97)

415. assistant (2026-06-11T15:18:42.841Z)

All merged. Final memory sync to reflect the true end-state:

416. user (2026-06-11T15:18:51.884Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

417. assistant (2026-06-11T15:19:10.047Z)

Done ? both PRs merged with the full diligence chain. Master is at `020ff61`, zero open PRs.

****What merged:****

- [#99 ? chunked upload + highlight download (B2+B3, PR-2)](<https://github.com/avidullu/badminton-highlight-indexer/pull/99>)
- [#100 ? docs status sync](<https://github.com/avidullu/badminton-highlight-indexer/pull/100>)

****Due diligence performed, in order:****

1. ****Tests**** ? full suite 388 green locally (including the 15 new upload/download tests) plus all four CI lanes green on both PRs.
2. ****Live E2E for the risky change**** (the new HTTP surface) ? real server on `:8123`, real artifacts:
 - ****Upload:**** the 86MB smoke video sent in 3 sequential parts, reassembled to ****MD5 byte-identity**** with the source (the returned `video\_id` matched `compute\_video\_id` of the original exactly).
 - ****Negative paths:**** out-of-order part ? 409, bad extension and traversal filenames ? 400, unknown upload id ? 404.
 - ****Download hardening, both layers:**** backslash-encoded traversal reaches the handler and gets the safe-name 400; `%2F`-encoded traversal never even reaches it (router 404). The one e2e "failure" was my test expecting 400 where the router answers first ? defense verified intact at both layers.
 - ****The actual CUJ tail:**** compiled the real confirmed rally from this morning's smoke run into a 3.9MB reel and streamed it back through `GET /api/highlights/...` ? so the `segment\_ids` compile fix and the download link are proven against real data, not just mocks.
3. ****Doc updates**** (#100) ? `NEXT\_STEPS.md` header now shows PR-2 merged plus a summary of the #90?#98 quality sweep; `UI\_ALPHA\_EXECUTION\_PLAN.md` has PR-2 checked off with the E2E evidence inline; and `BACKLOG.md`'s shlex-quote WSL item is formally closed by #97 (with the residual tenancy-era note kept honest).
4. ****Memory**** ? current-state checkpoint updated to the merged reality.

Your tomorrow-morning CUJ runbook is now fully unblocked: `python -m backend.main --server` ? paste path ? index ? select rallies ? compile ? the ? download link that, as of today, actually works. The one thing to remember when PR-3 (identity) lands: `security.allowed\_video\_root` must contain `upload\_dir`, or `/api/process` will reject uploaded paths ? it's flagged in the plan.

418. user (2026-06-13T15:15:07.917Z)

can you sync to remote head and refresh the meory and handoff of this repo?

419. assistant (2026-06-13T15:15:08.909Z)

There's an issue with the selected model (claude-fable-5). It may not exist or you may not have access to it.